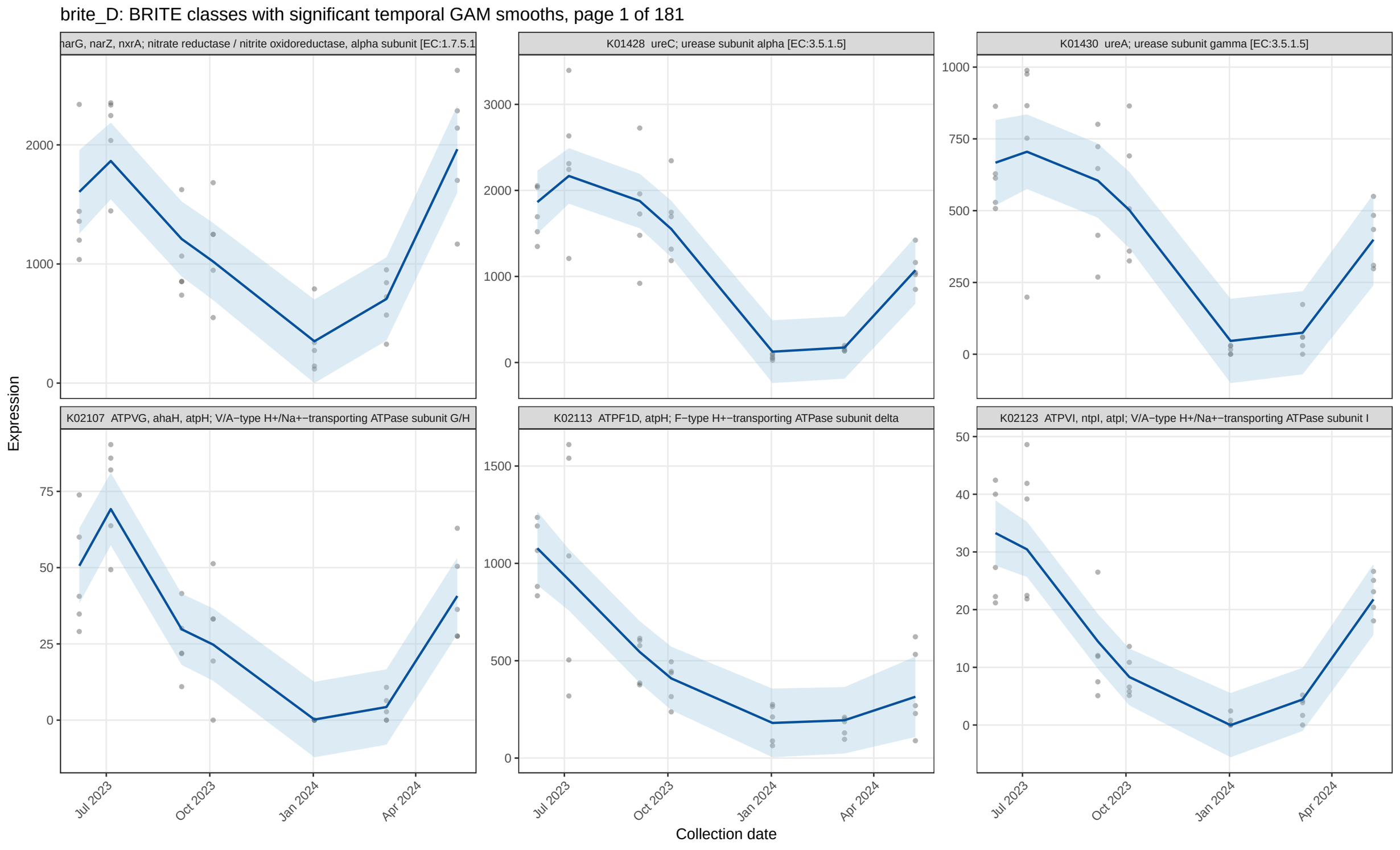
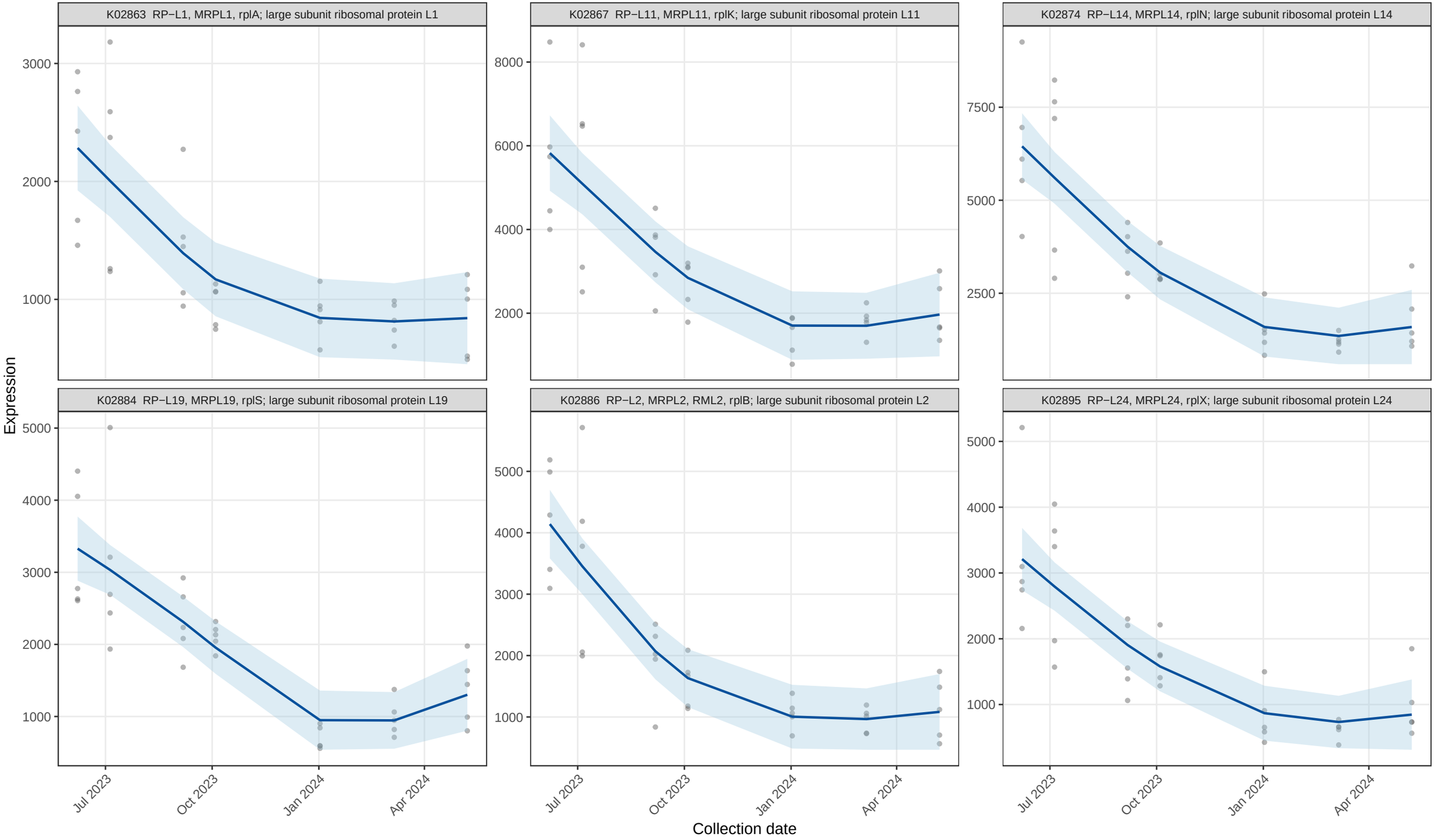
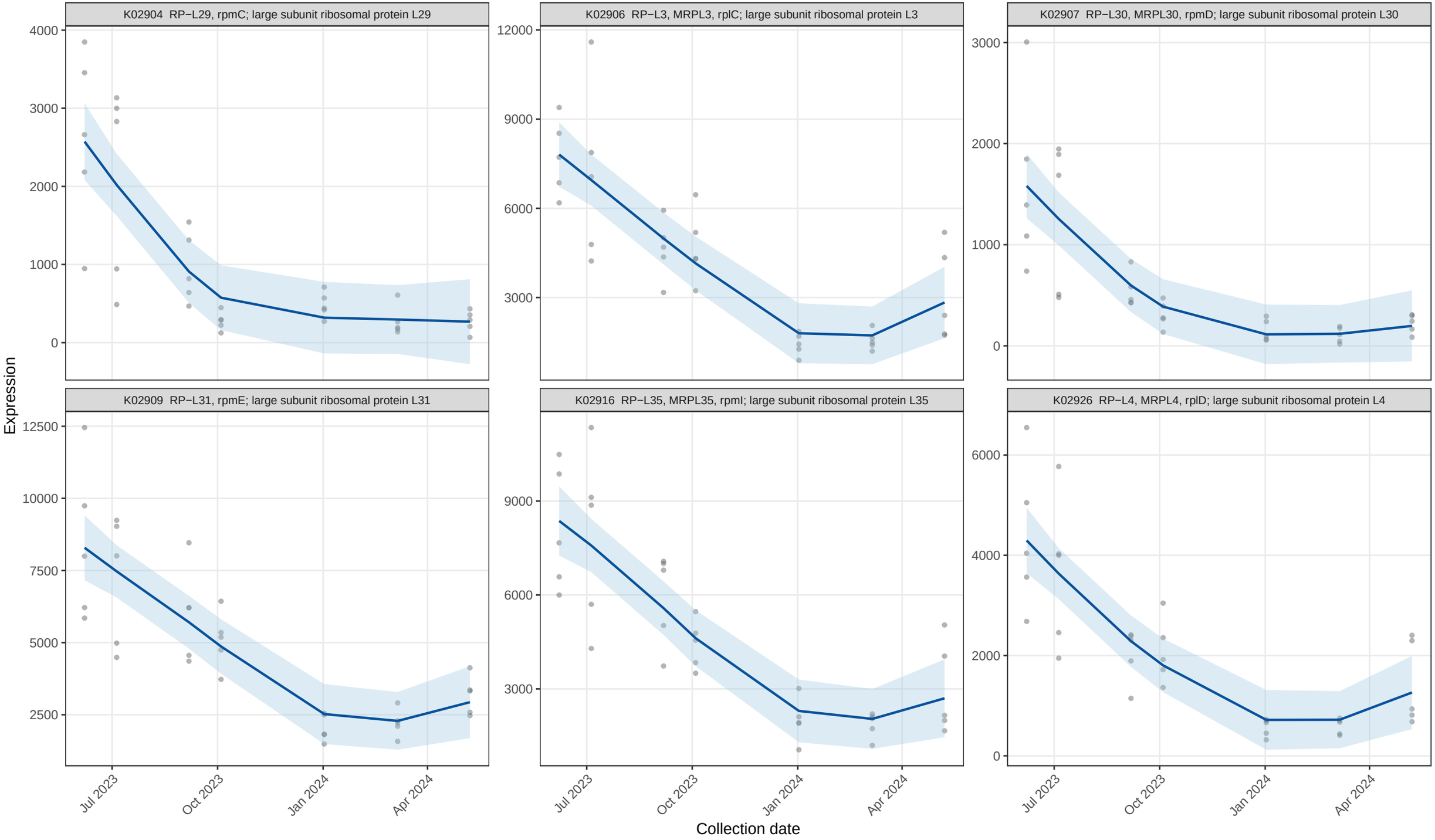
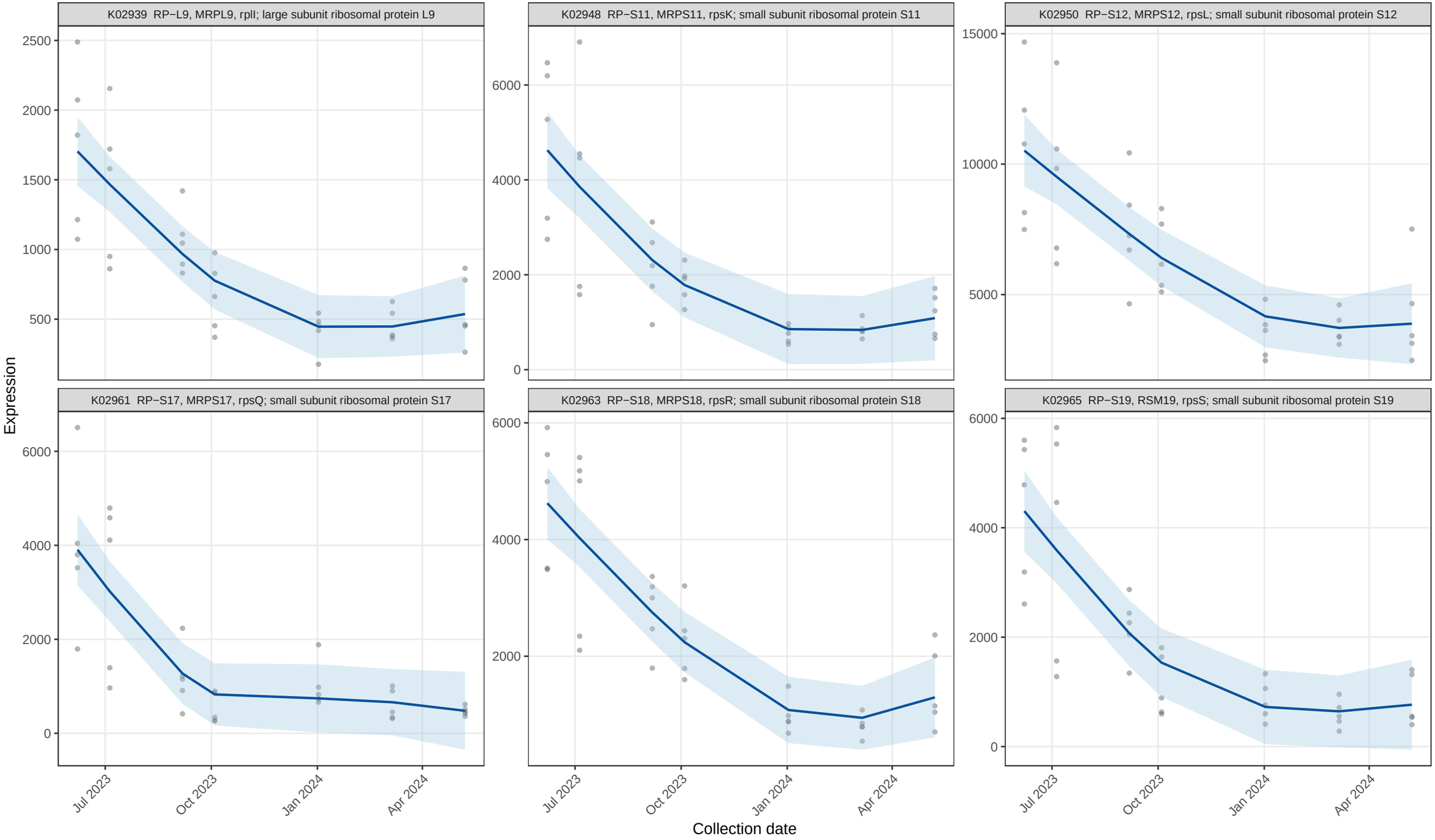
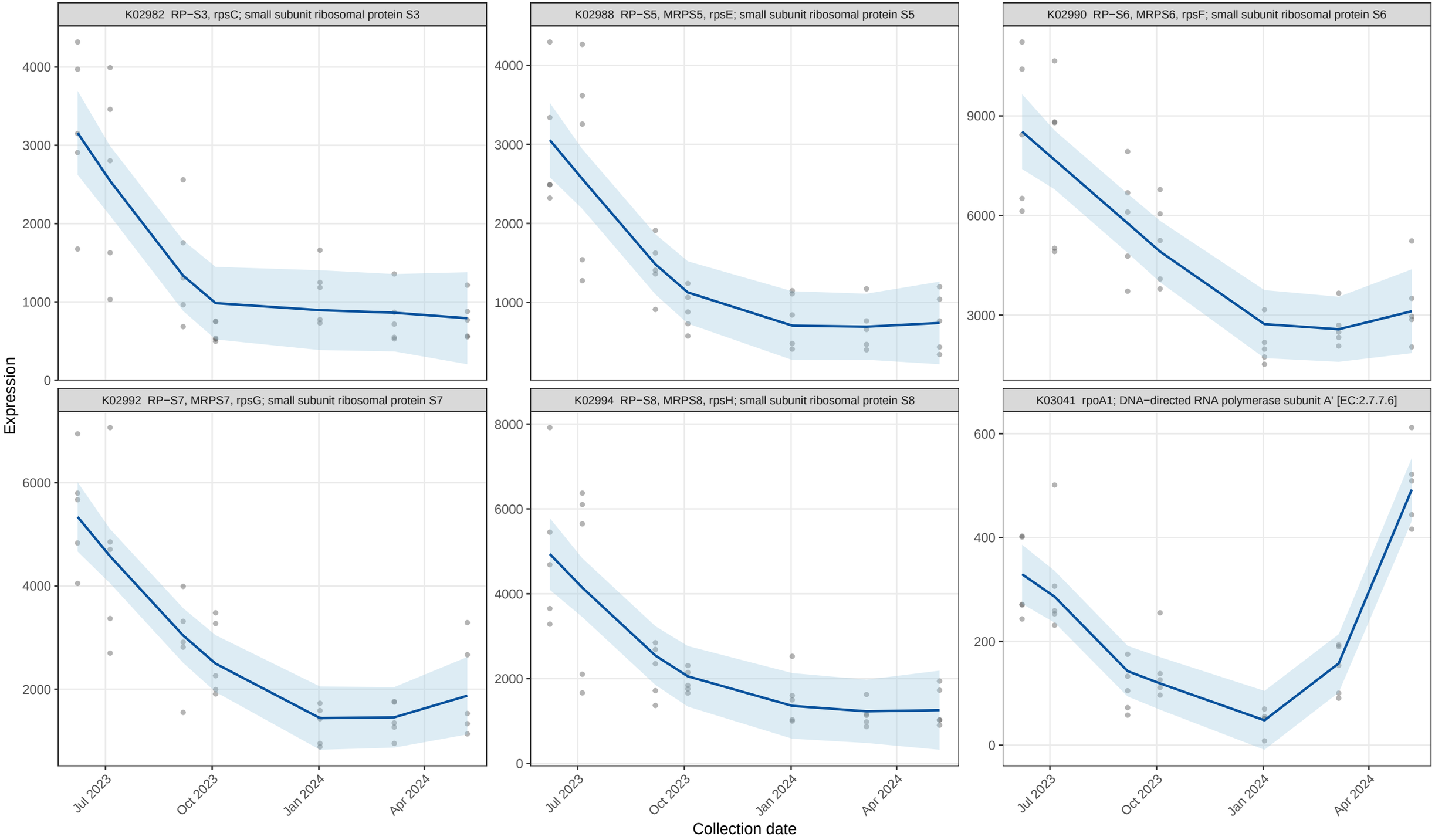

brite_D: BRITE classes with significant temporal GAM smooths, page 1 of 181
narG, narZ, nxrA; nitrate reductase / nitrite oxidoreductase, alpha subunit [EC:1.7.5.1
K01428 ureC; urease subunit alpha [EC:3.5.1.5]
K01430 ureA; urease subunit gamma [EC:3.5.1.5]
K02107 ATPVG, ahaH, atpH; V/A−type H+/Na+−transporting ATPase subunit G/H
K02113 ATPF1D, atpH; F−type H+−transporting ATPase subunit delta
K02123 ATPVI, ntpI, atpI; V/A−type H+/Na+−transporting ATPase subunit I
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

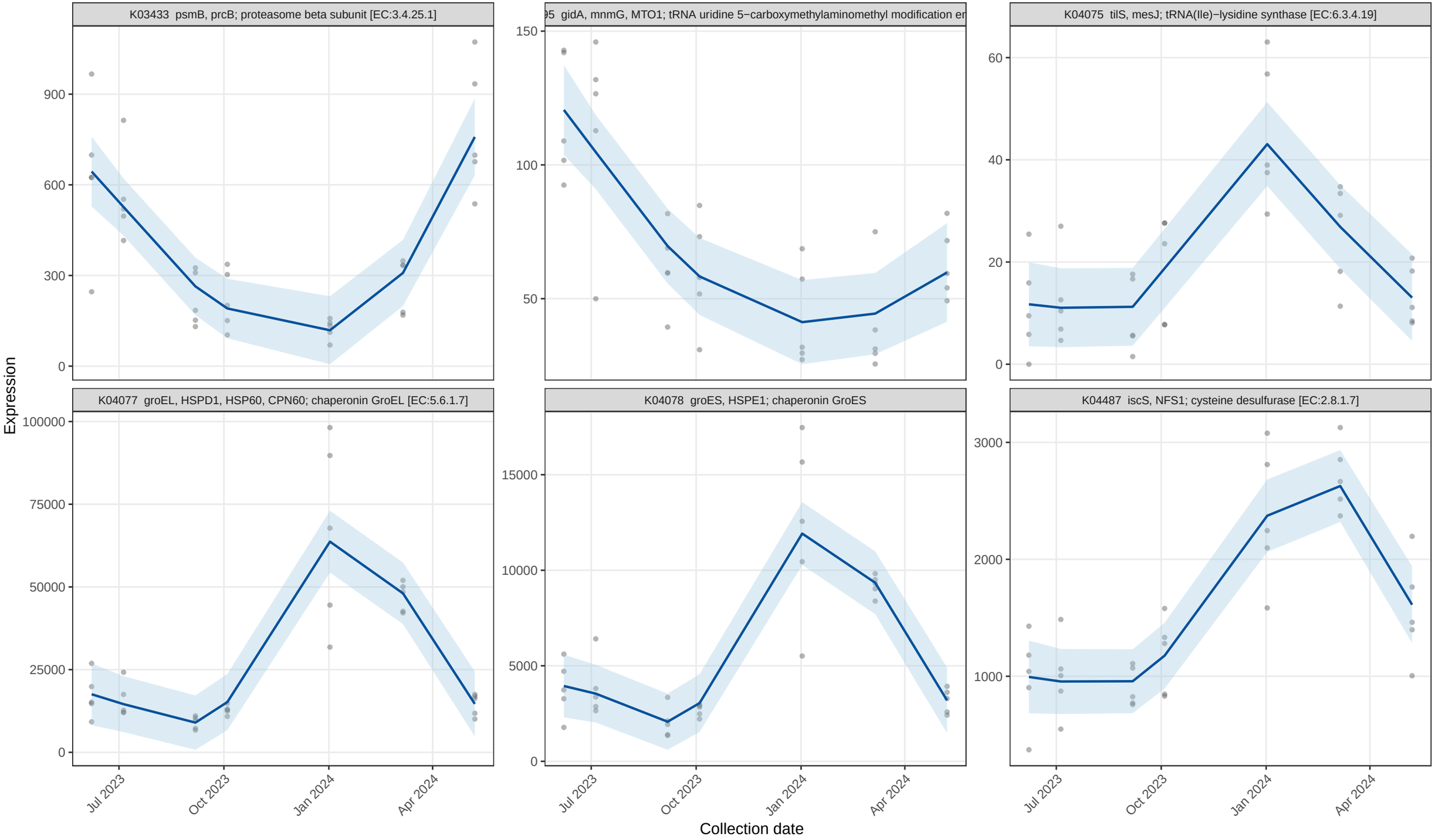

K02863 RP−L1, MRPL1, rplA; large subunit ribosomal protein L1
K02867 RP−L11, MRPL11, rplK; large subunit ribosomal protein L11
K02874 RP−L14, MRPL14, rplN; large subunit ribosomal protein L14
K02884 RP−L19, MRPL19, rplS; large subunit ribosomal protein L19
K02886 RP−L2, MRPL2, RML2, rplB; large subunit ribosomal protein L2
K02895 RP−L24, MRPL24, rplX; large subunit ribosomal protein L24
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

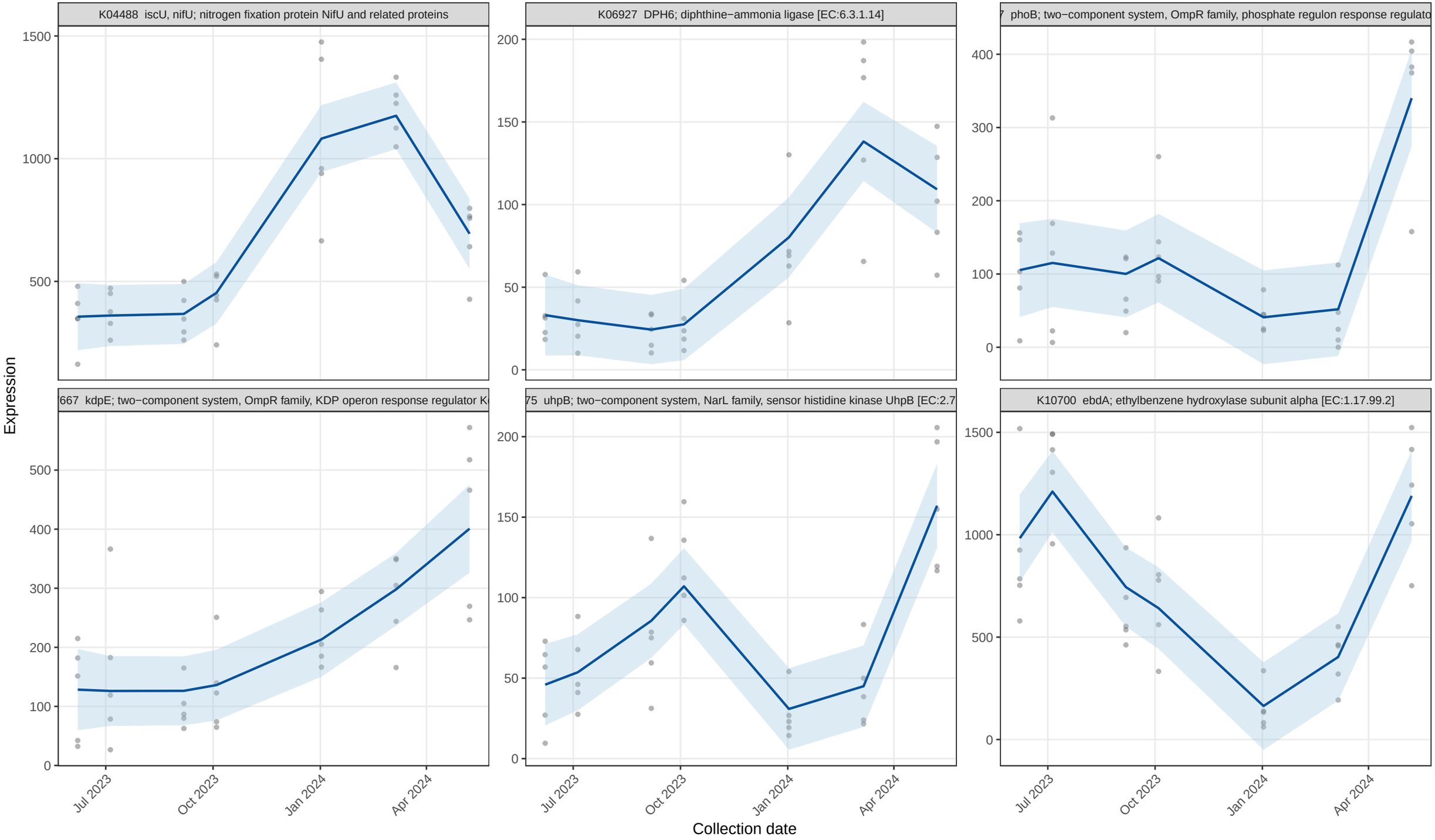
brite_D: BRITE classes with significant temporal GAM smooths, page 3 of 181

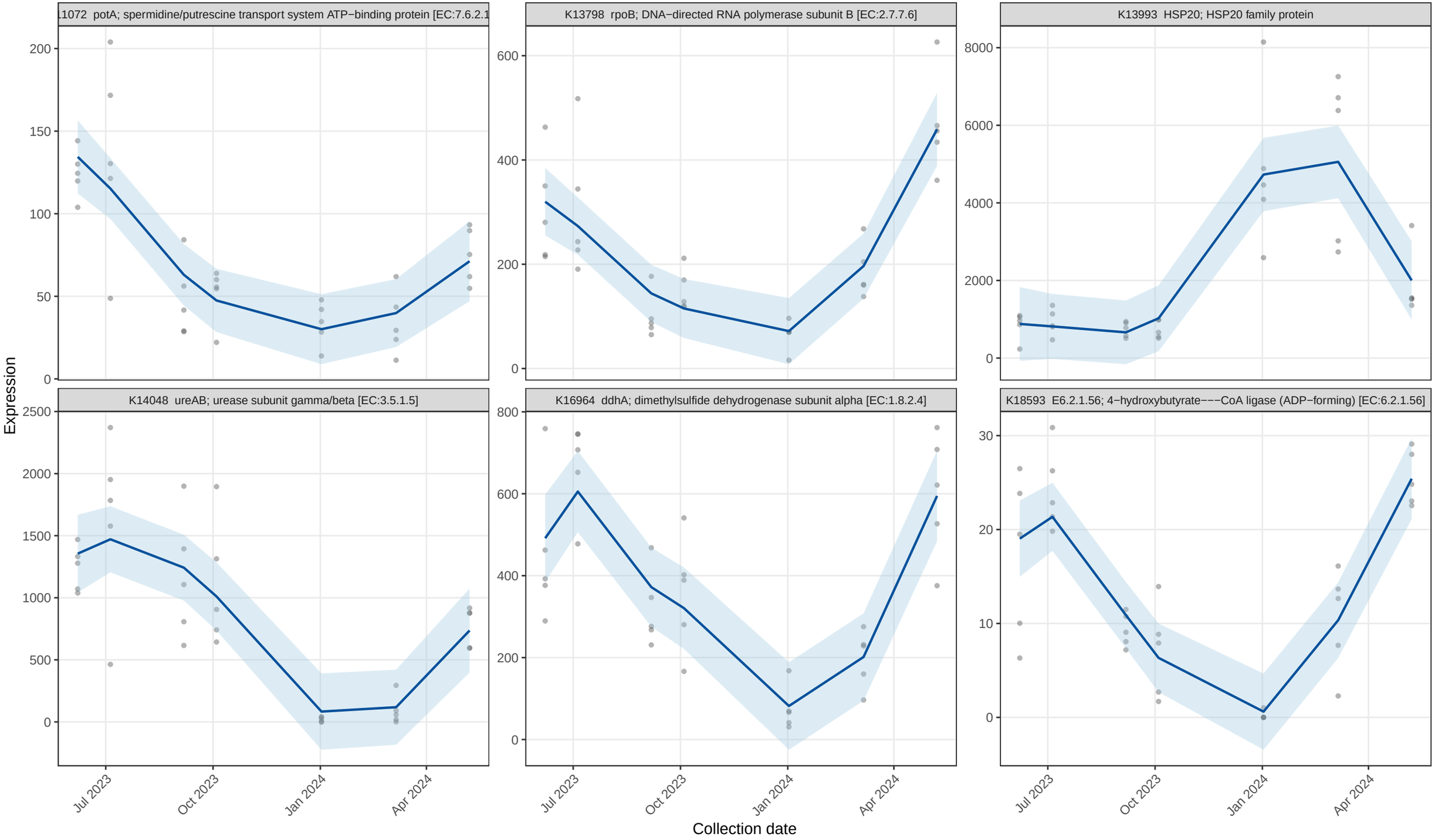


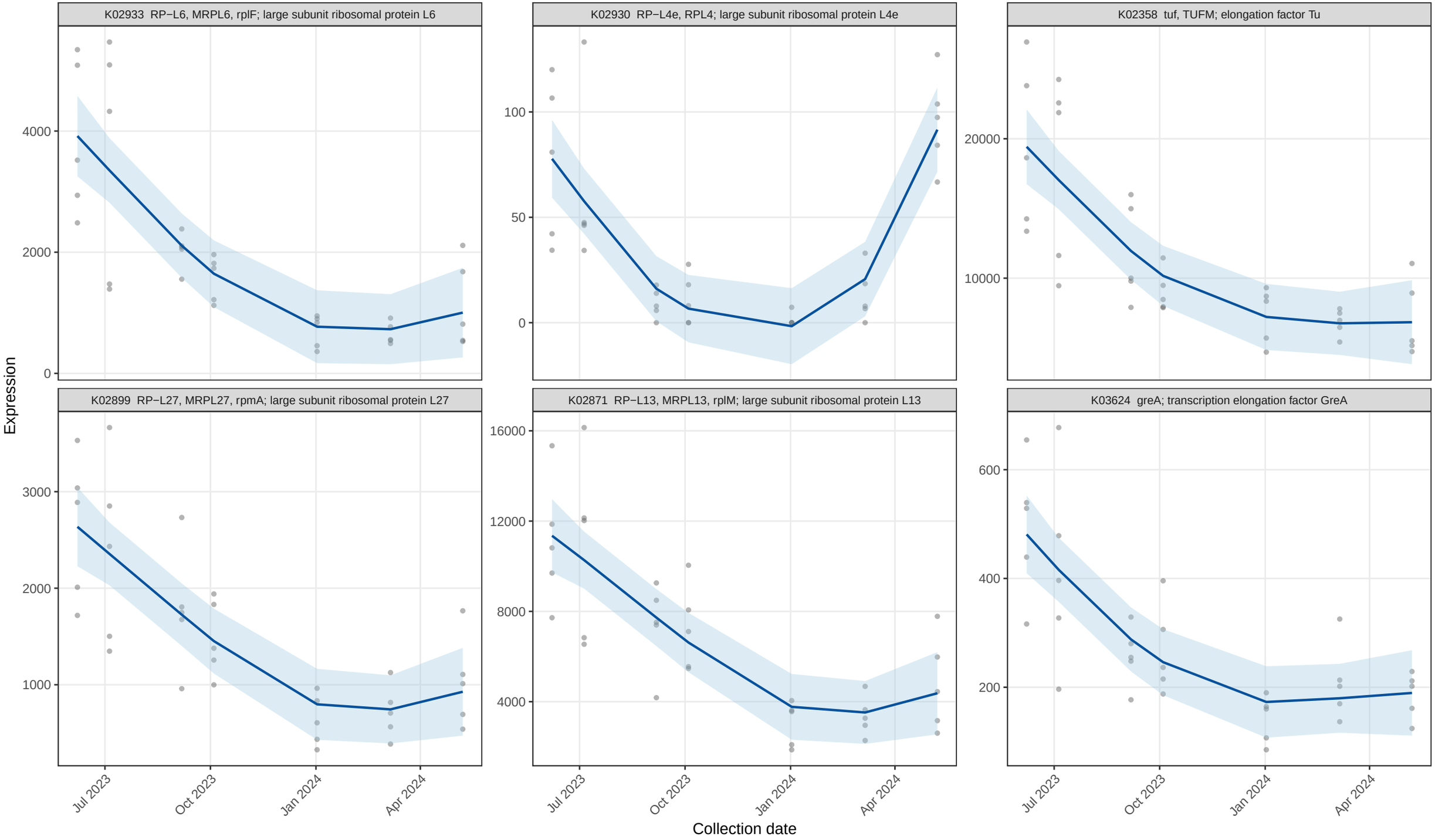

K02939 RP−L9, MRPL9, rplI; large subunit ribosomal protein L9
K02948 RP−S11, MRPS11, rpsK; small subunit ribosomal protein S11
K02950 RP−S12, MRPS12, rpsL; small subunit ribosomal protein S12
K02961 RP−S17, MRPS17, rpsQ; small subunit ribosomal protein S17
K02963 RP−S18, MRPS18, rpsR; small subunit ribosomal protein S18
K02965 RP−S19, RSM19, rpsS; small subunit ribosomal protein S19
Expression
Collection date
Jul 2023
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Jan 2024
Apr 2024

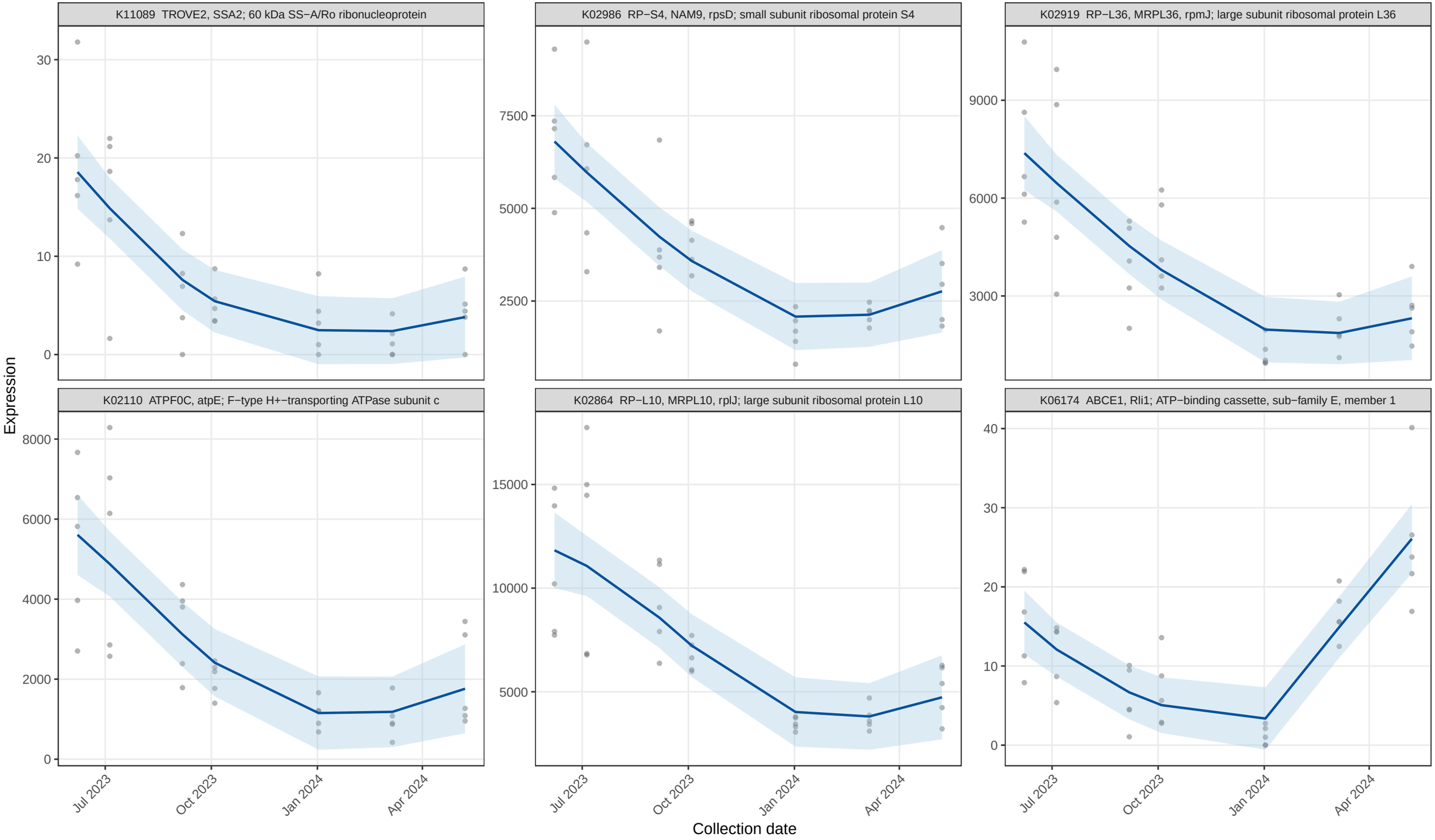

K02982 RP−S3, rpsC; small subunit ribosomal protein S3
K02988 RP−S5, MRPS5, rpsE; small subunit ribosomal protein S5
K02990 RP−S6, MRPS6, rpsF; small subunit ribosomal protein S6
K02992 RP−S7, MRPS7, rpsG; small subunit ribosomal protein S7
K02994 RP−S8, MRPS8, rpsH; small subunit ribosomal protein S8
K03041 rpoA1; DNA−directed RNA polymerase subunit A' [EC:2.7.7.6]
Expression
Collection date
Jul 2023
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Jan 2024
Apr 2024

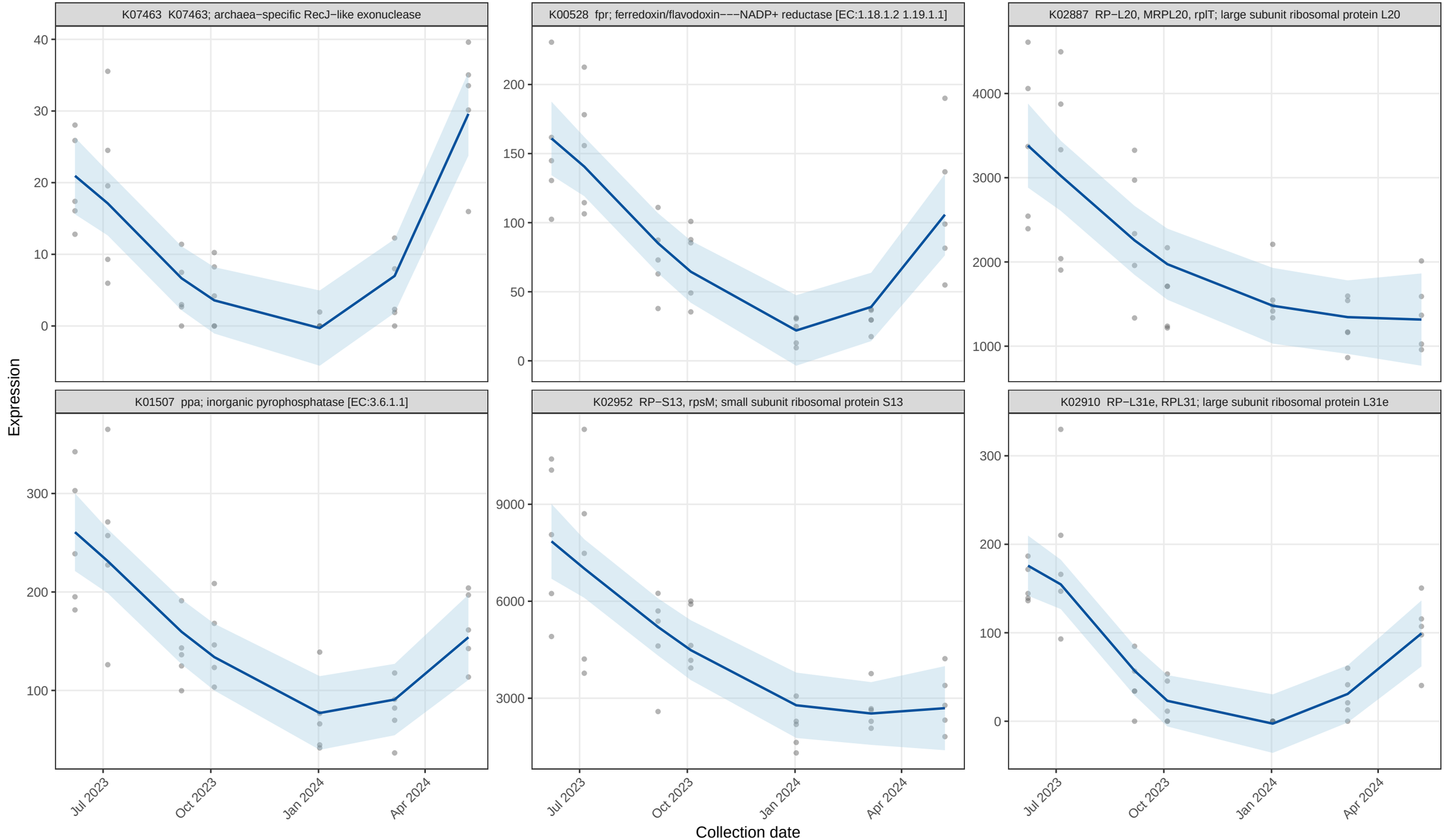

K03433 psmB, prcB; proteasome beta subunit [EC:3.4.25.1]
95 gidA, mnmG, MTO1; tRNA uridine 5−carboxymethylaminomethyl modification er
K04075 tilS, mesJ; tRNA(Ile)−lysidine synthase [EC:6.3.4.19]
K04077 groEL, HSPD1, HSP60, CPN60; chaperonin GroEL [EC:5.6.1.7]
K04078 groES, HSPE1; chaperonin GroES
K04487 iscS, NFS1; cysteine desulfurase [EC:2.8.1.7]
Expression
Collection date
Jul 2023
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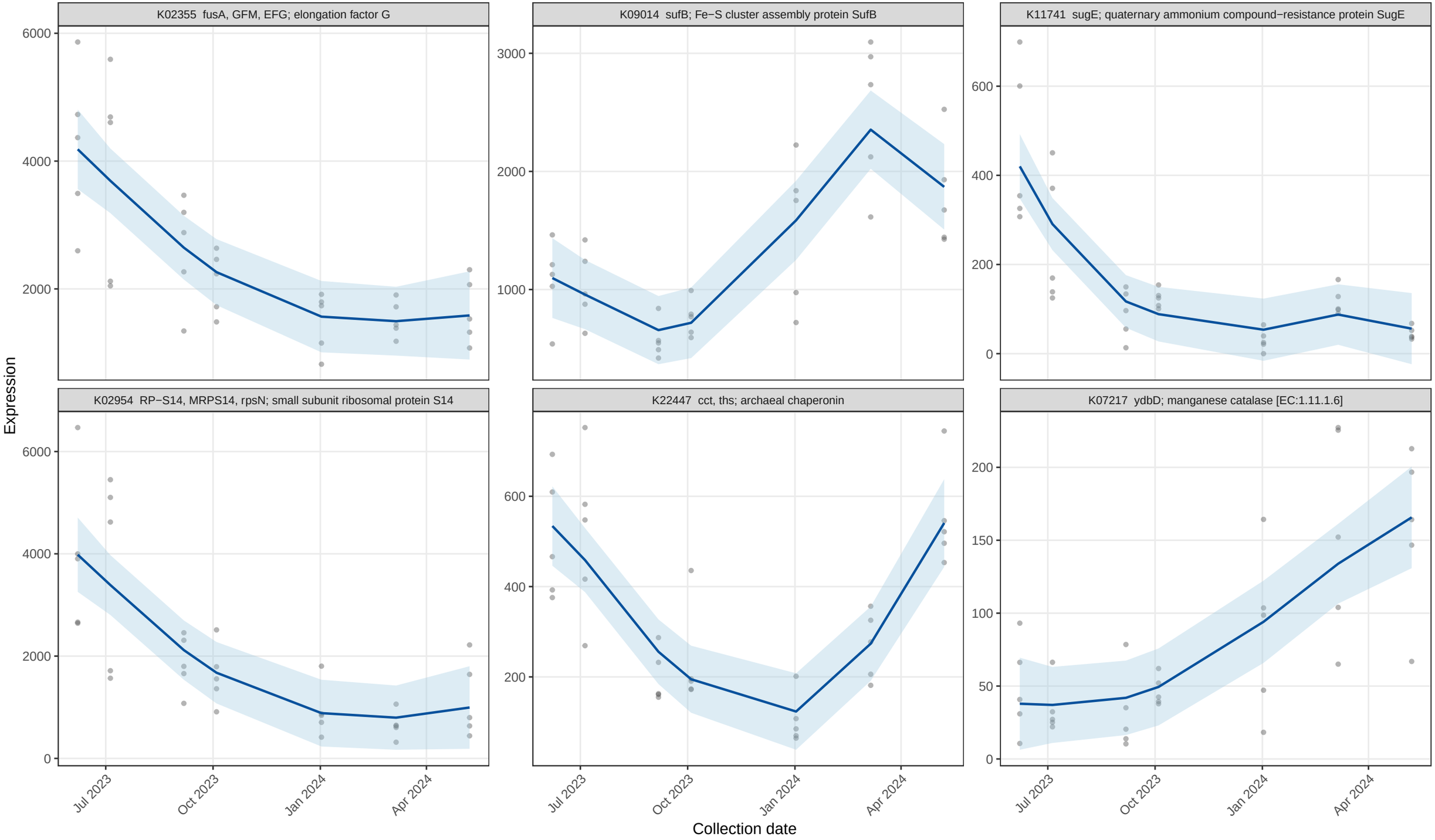

K04488 iscU, nifU; nitrogen fixation protein NifU and related proteins
K06927 DPH6; diphthine−ammonia ligase [EC:6.3.1.14]
7 phoB; two−component system, OmpR family, phosphate regulon response regulato
667 kdpE; two−component system, OmpR family, KDP operon response regulator K
75 uhpB; two−component system, NarL family, sensor histidine kinase UhpB [EC:2.7
K10700 ebdA; ethylbenzene hydroxylase subunit alpha [EC:1.17.99.2]
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

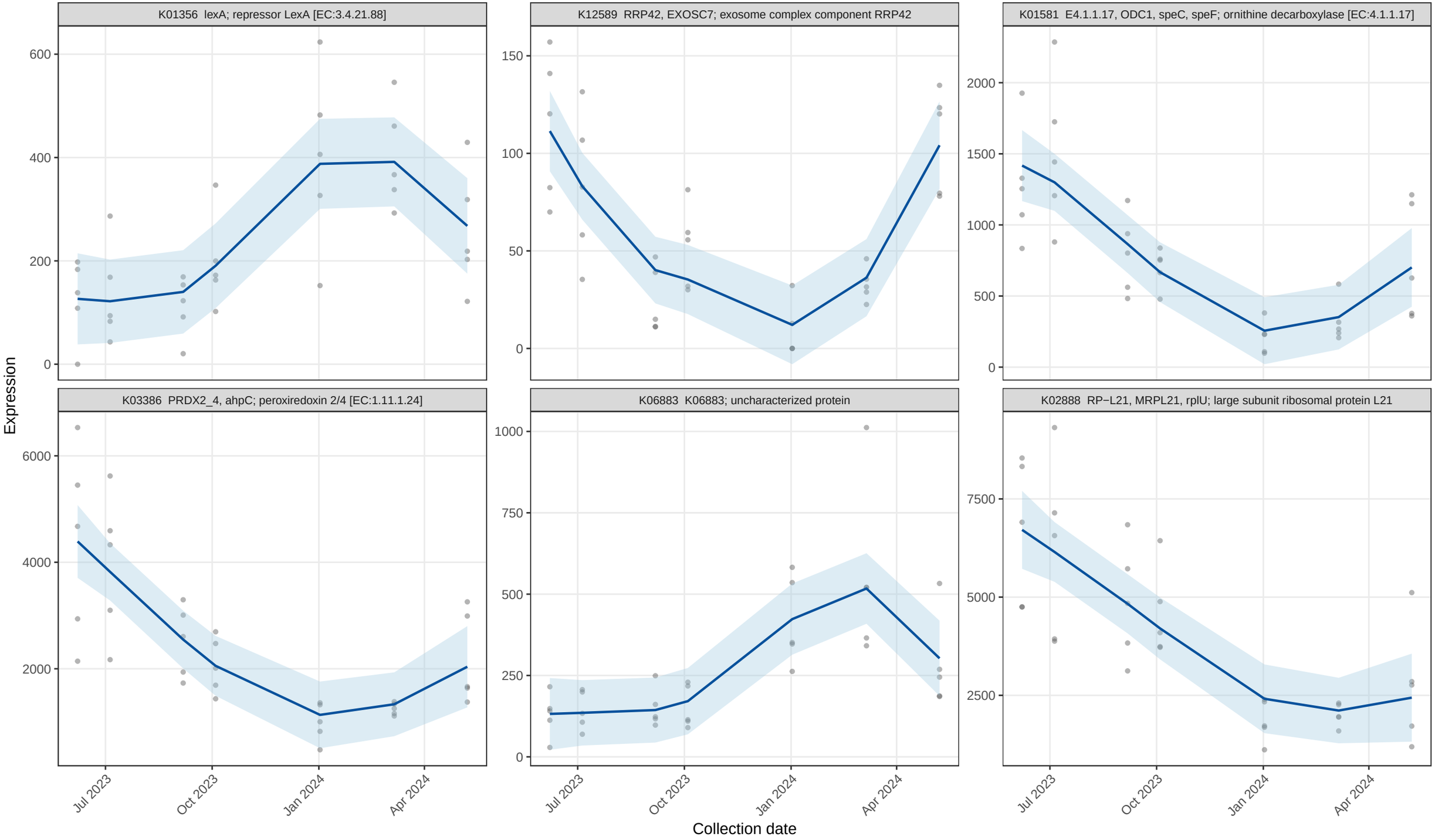

K01072 potA; spermidine/putrescine transport system ATP−binding protein [EC:7.6.2.1]
K13798 rpoB; DNA−directed RNA polymerase subunit B [EC:2.7.7.6]
K13993 HSP20; HSP20 family protein
K14048 ureAB; urease subunit gamma/beta [EC:3.5.1.5]
K16964 ddhA; dimethylsulfide dehydrogenase subunit alpha [EC:1.8.2.4]
K18593 E6.2.1.56; 4−hydroxybutyrate−−−CoA ligase (ADP−forming) [EC:6.2.1.56]
Expression
Collection date
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Apr 2024

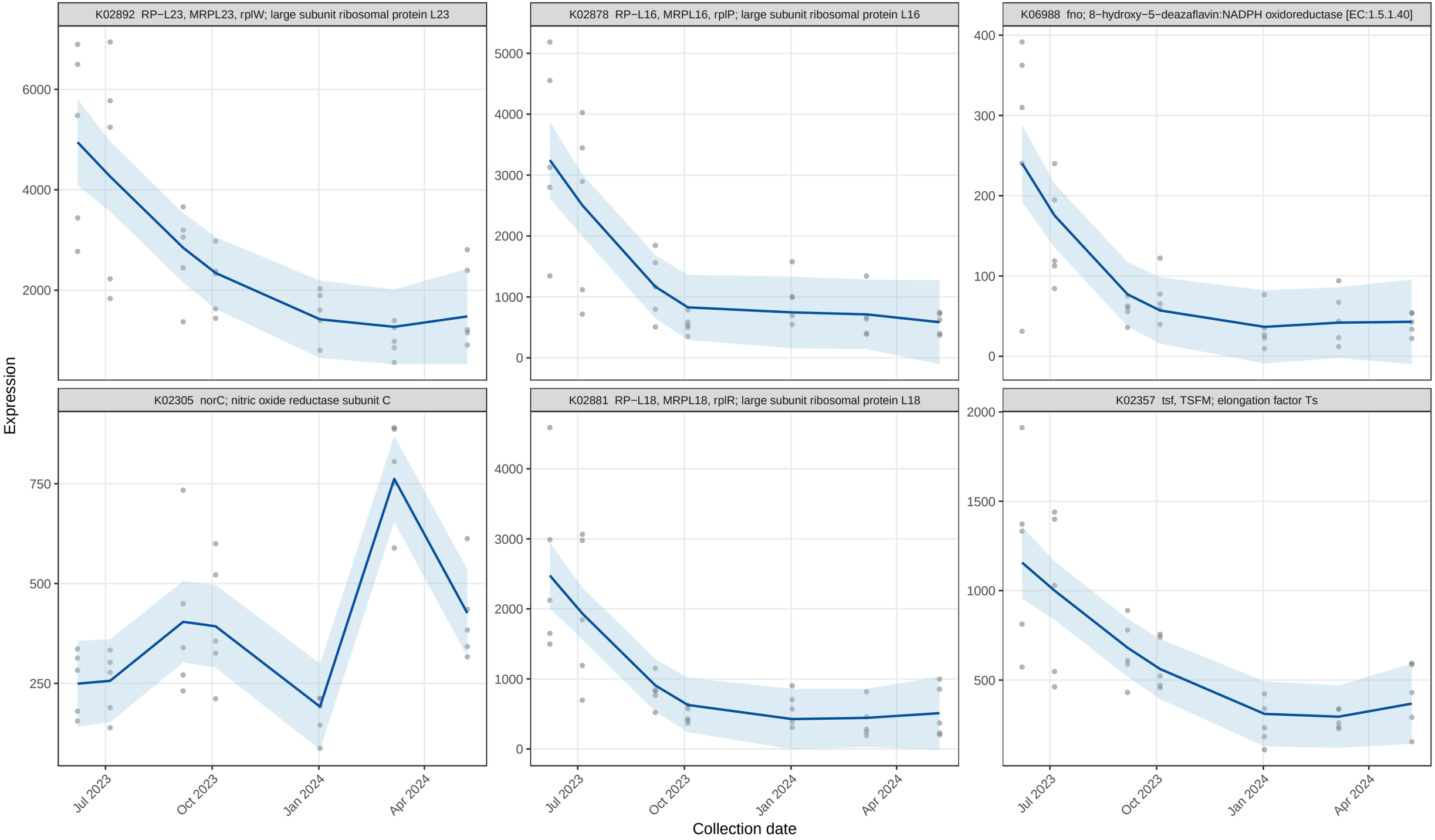

K02933 RP−L6, MRPL6, rplF; large subunit ribosomal protein L6
K02930 RP−L4e, RPL4; large subunit ribosomal protein L4e
K02358 tuf, TUFM; elongation factor Tu
K02899 RP−L27, MRPL27, rpmA; large subunit ribosomal protein L27
K02871 RP−L13, MRPL13, rplM; large subunit ribosomal protein L13
K03624 greA; transcription elongation factor GreA
Expression
Collection date
Jul 2023
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Apr 2024

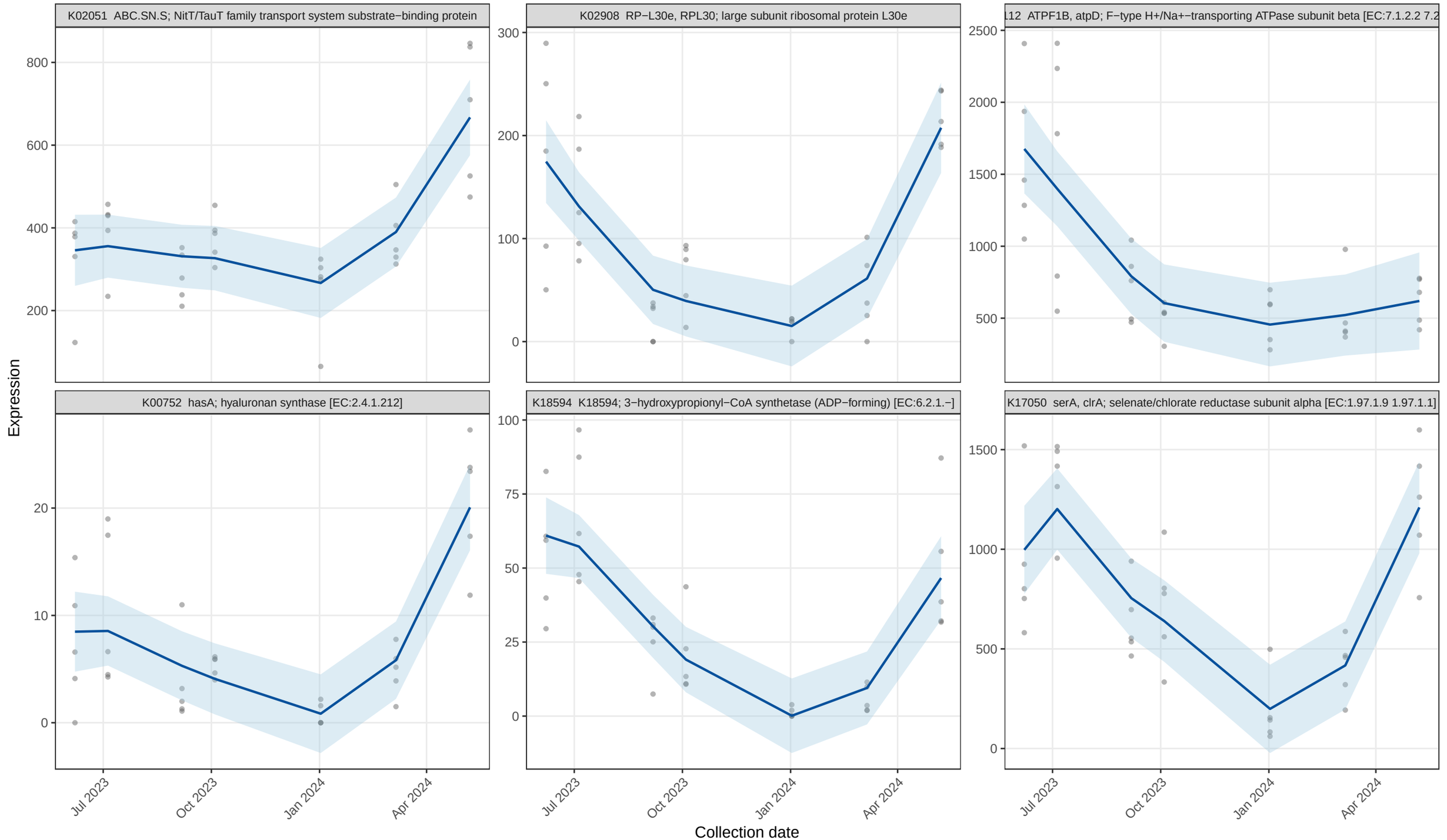

K11089 TROVE2, SSA2; 60 kDa SS−A/Ro ribonucleoprotein
K02986 RP−S4, NAM9, rpsD; small subunit ribosomal protein S4
K02919 RP−L36, MRPL36, rpmJ; large subunit ribosomal protein L36
K02110 ATPF0C, atpE; F−type H+−transporting ATPase subunit c
K02864 RP−L10, MRPL10, rplJ; large subunit ribosomal protein L10
K06174 ABCE1, Rli1; ATP−binding cassette, sub−family E, member 1
Expression
Collection date
Jul 2023
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Apr 2024

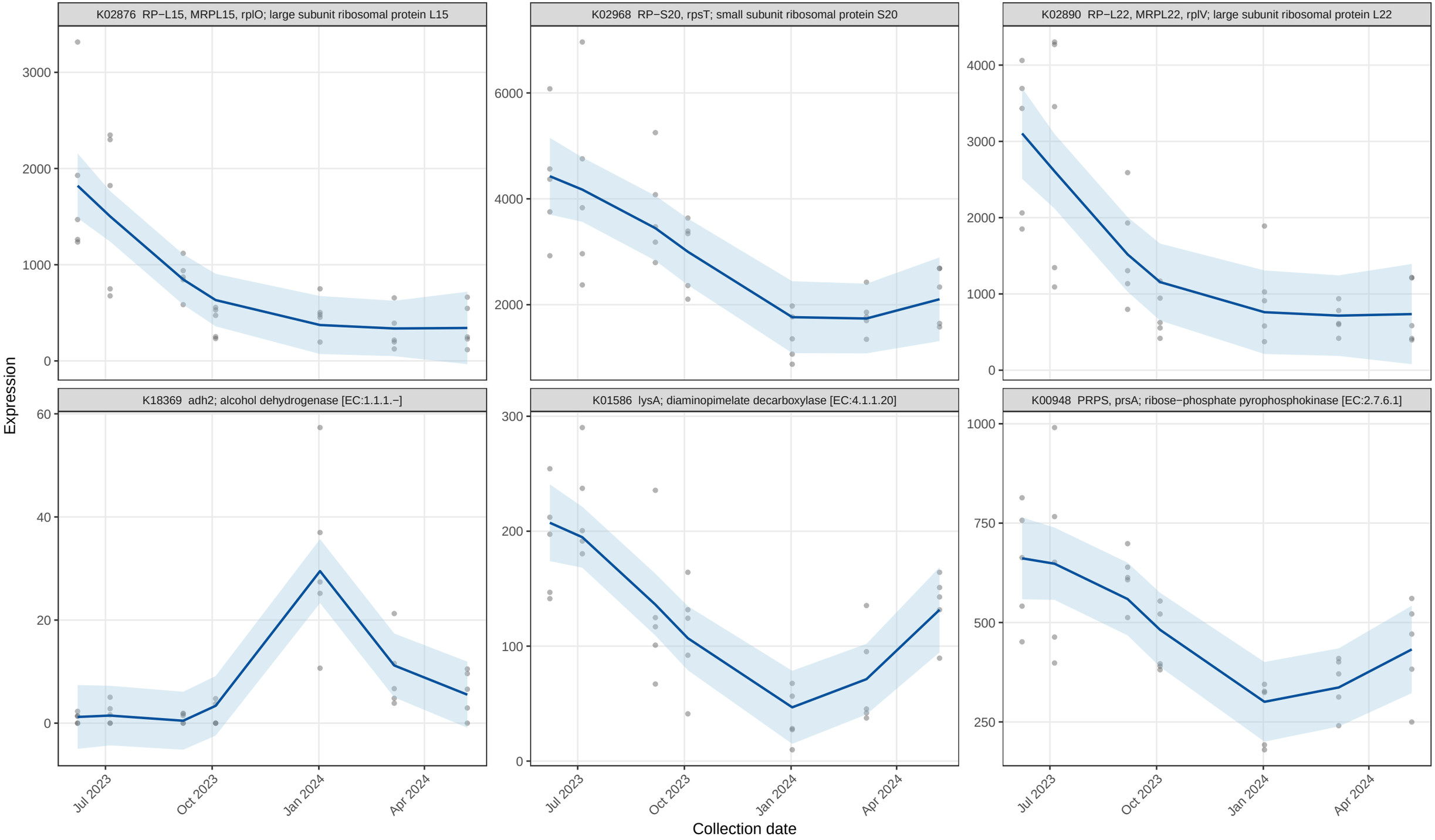

K07463 K07463; archaea−specific RecJ−like exonuclease
K00528 fpr; ferredoxin/flavodoxin−−−NADP+ reductase [EC:1.18.1.2 1.19.1.1]
K02887 RP−L20, MRPL20, rplT; large subunit ribosomal protein L20
K01507 ppa; inorganic pyrophosphatase [EC:3.6.1.1]
K02952 RP−S13, rpsM; small subunit ribosomal protein S13
K02910 RP−L31e, RPL31; large subunit ribosomal protein L31e
Expression
Collection date
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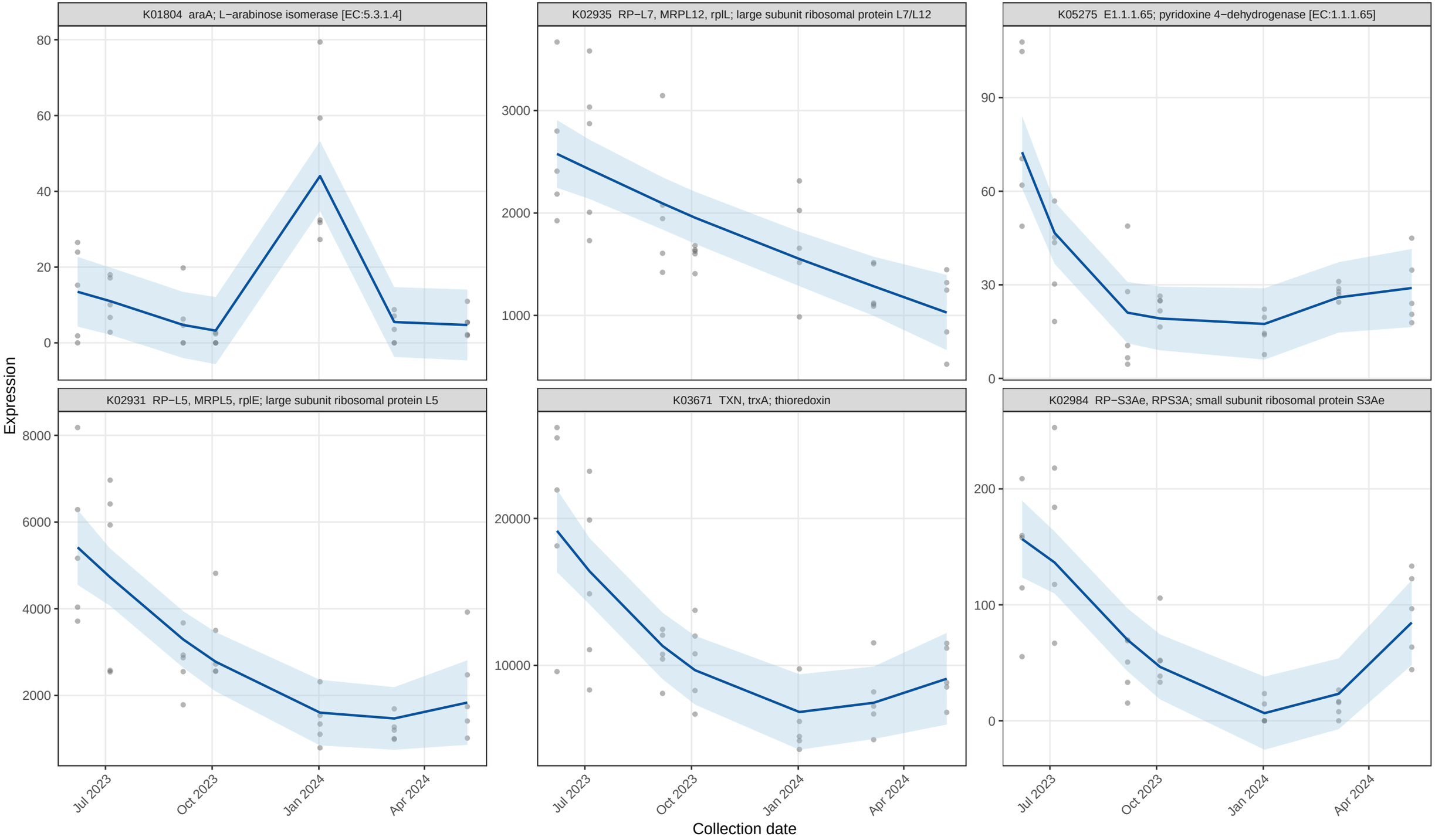

K02355 fusA, GFM, EFG; elongation factor G
K09014 sufB; Fe−S cluster assembly protein SufB
K11741 sugE; quaternary ammonium compound−resistance protein SugE
K02954 RP−S14, MRPS14, rpsN; small subunit ribosomal protein S14
K22447 cct, ths; archaeal chaperonin
K07217 ydbD; manganese catalase [EC:1.11.1.6]
Expression
Collection date
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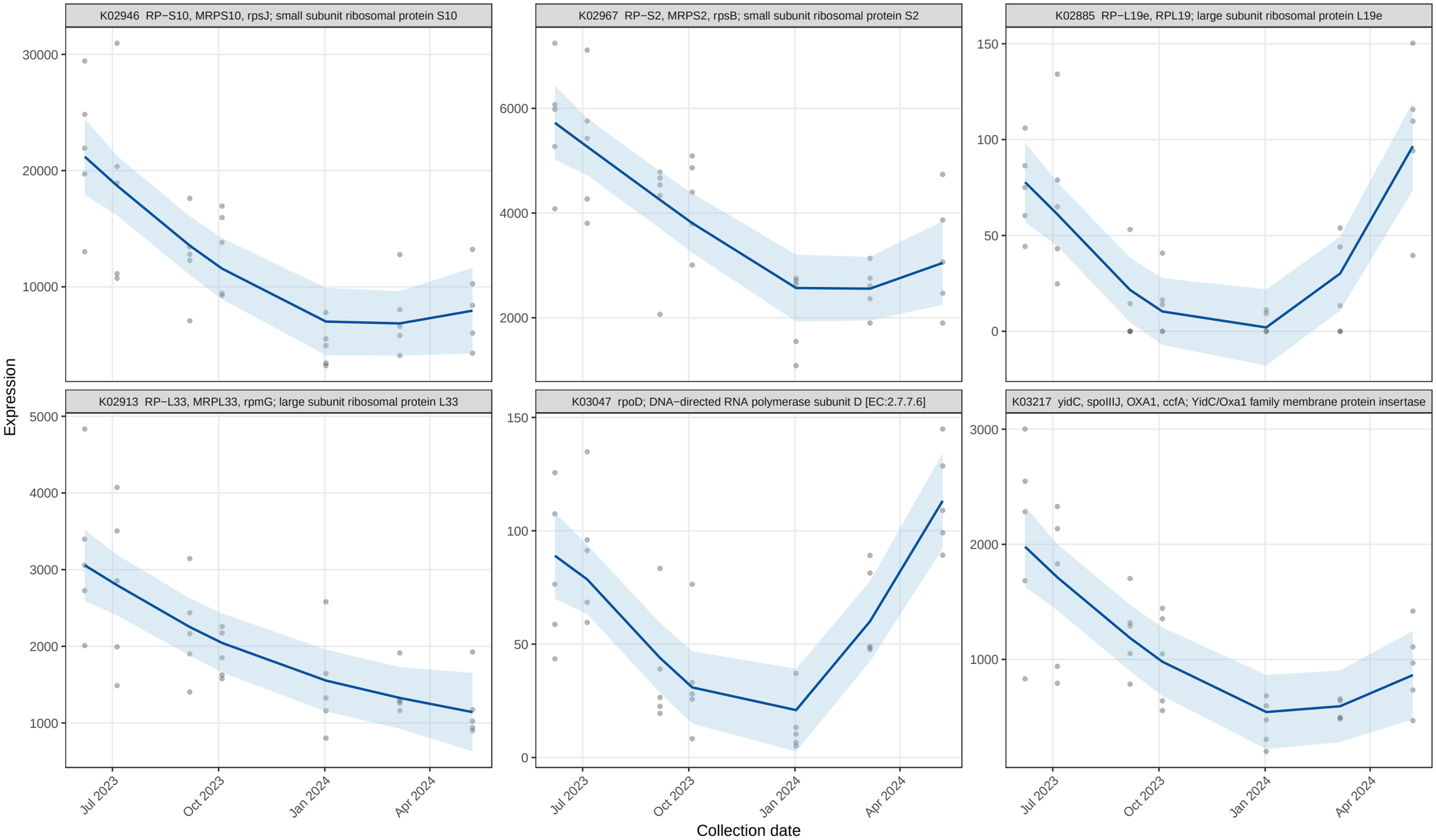

K01356 lexA; repressor LexA [EC:3.4.21.88]
K12589 RRP42, EXOSC7; exosome complex component RRP42
K01581 E4.1.1.17, ODC1, speC, speF; ornithine decarboxylase [EC:4.1.1.17]
K03386 PRDX2_4, ahpC; peroxiredoxin 2/4 [EC:1.11.1.24]
K06883 K06883; uncharacterized protein
K02888 RP−L21, MRPL21, rplU; large subunit ribosomal protein L21
Expression
Collection date
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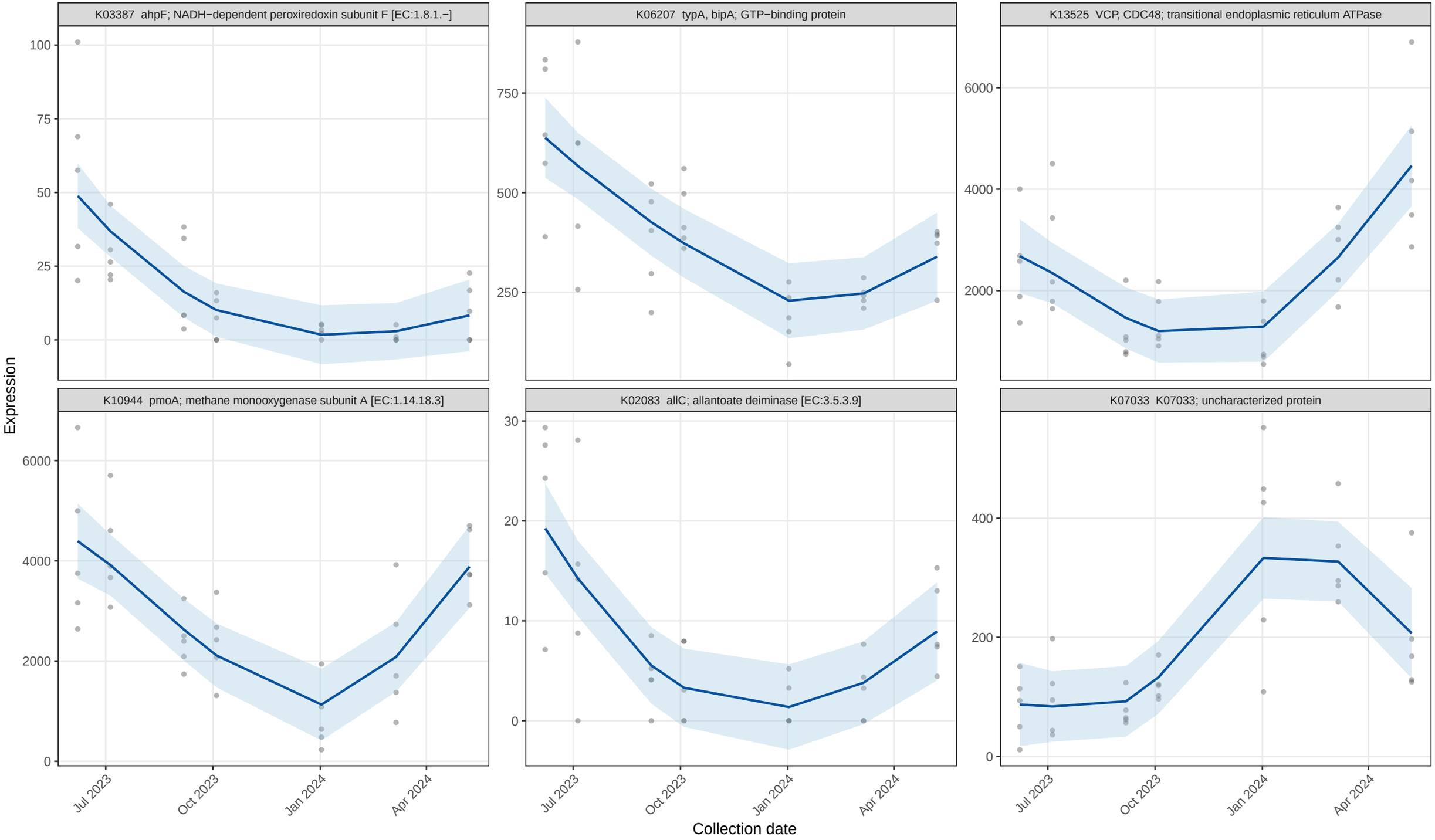

K02892 RP−L23, MRPL23, rplW; large subunit ribosomal protein L23
K02878 RP−L16, MRPL16, rplP; large subunit ribosomal protein L16
K06988 fno; 8−hydroxy−5−deazaflavin:NADPH oxidoreductase [EC:1.5.1.40]
K02305 norC; nitric oxide reductase subunit C
K02881 RP−L18, MRPL18, rplR; large subunit ribosomal protein L18
K02357 tsf, TSFM; elongation factor Ts
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

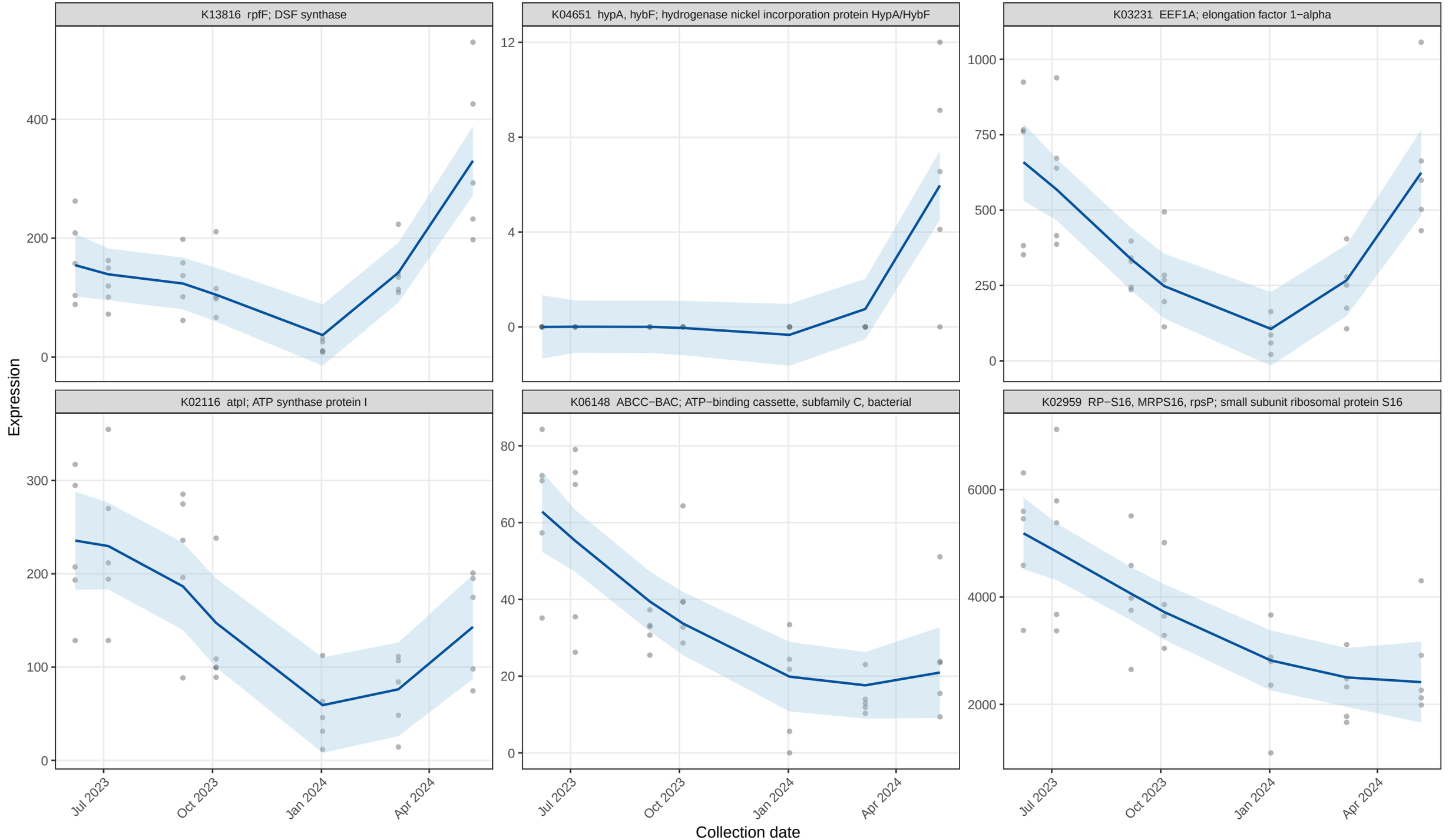

K02051 ABC.SN.S; NitT/TauT family transport system substrate−binding protein
K02908 RP−L30e, RPL30; large subunit ribosomal protein L30e
12 ATPF1B, atpD; F−type H+/Na+−transporting ATPase subunit beta [EC:7.1.2.2 7.2
K00752 hasA; hyaluronan synthase [EC:2.4.1.212]
K18594 K18594; 3−hydroxypropionyl−CoA synthetase (ADP−forming) [EC:6.2.1.−]
K17050 serA, clrA; selenate/chlorate reductase subunit alpha [EC:1.97.1.9 1.97.1.1]
Expression
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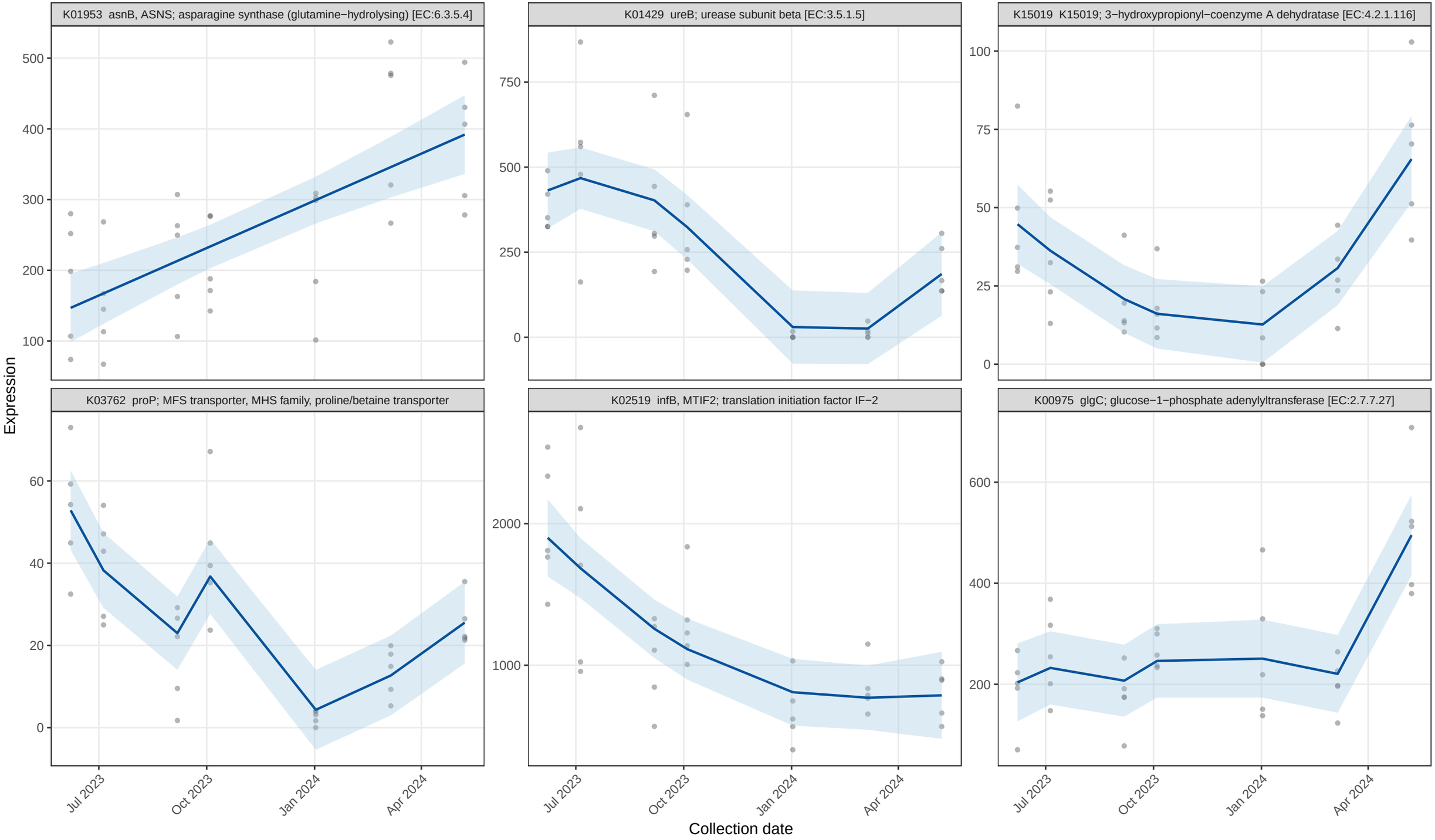

K02876 RP−L15, MRPL15, rplO; large subunit ribosomal protein L15
K02968 RP−S20, rpsT; small subunit ribosomal protein S20
K02890 RP−L22, MRPL22, rplV; large subunit ribosomal protein L22
K18369 adh2; alcohol dehydrogenase [EC:1.1.1.−]
K01586 lysA; diaminopimelate decarboxylase [EC:4.1.1.20]
K00948 PRPS, prsA; ribose−phosphate pyrophosphokinase [EC:2.7.6.1]
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

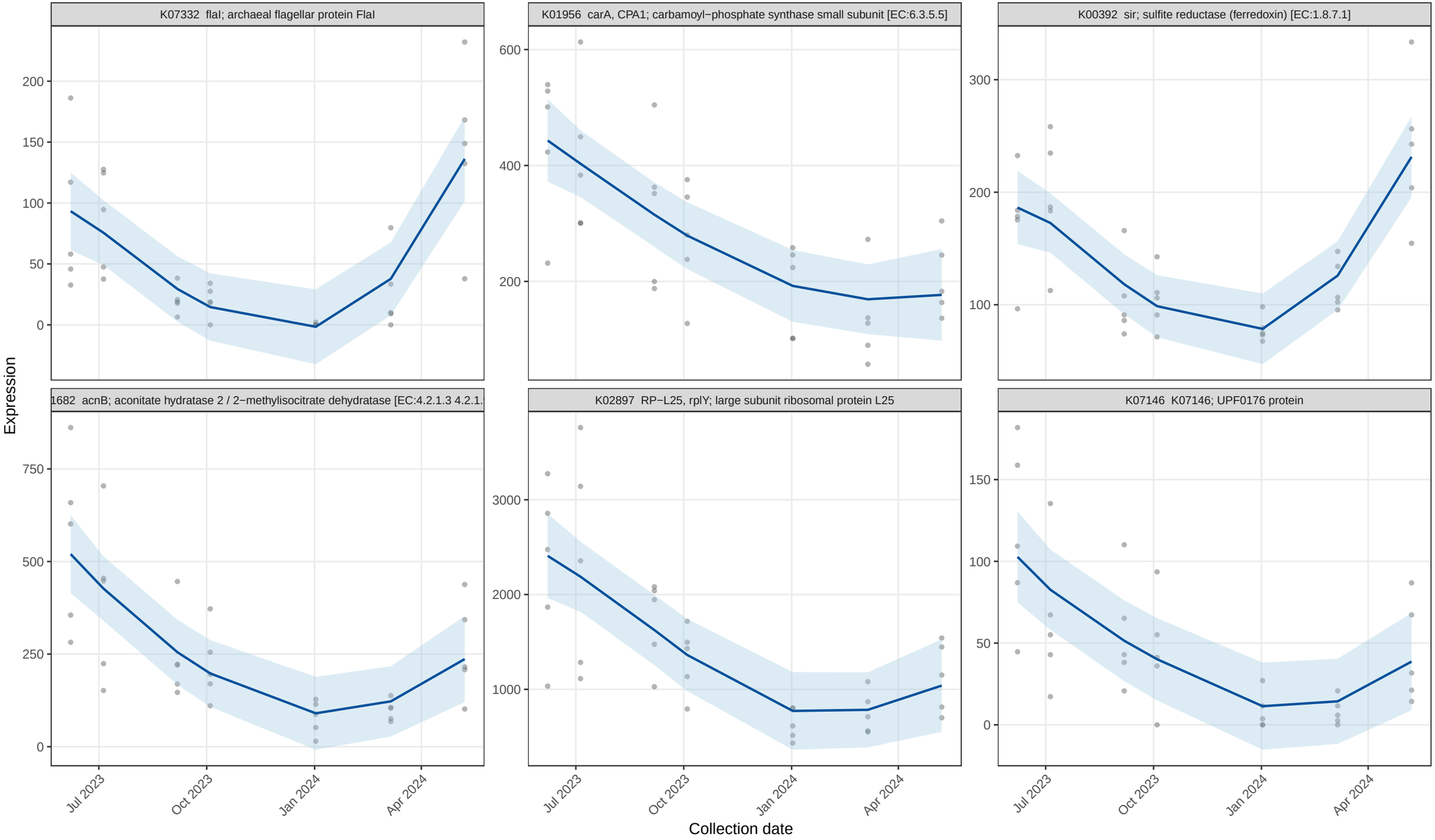

K01804 araA; L−arabinose isomerase [EC:5.3.1.4]
K02935 RP−L7, MRPL12, rplL; large subunit ribosomal protein L7/L12
K05275 E1.1.1.65; pyridoxine 4−dehydrogenase [EC:1.1.1.65]
K02931 RP−L5, MRPL5, rplE; large subunit ribosomal protein L5
K03671 TXN, trxA; thioredoxin
K02984 RP−S3Ae, RPS3A; small subunit ribosomal protein S3Ae
Expression
Collection date
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Apr 2024

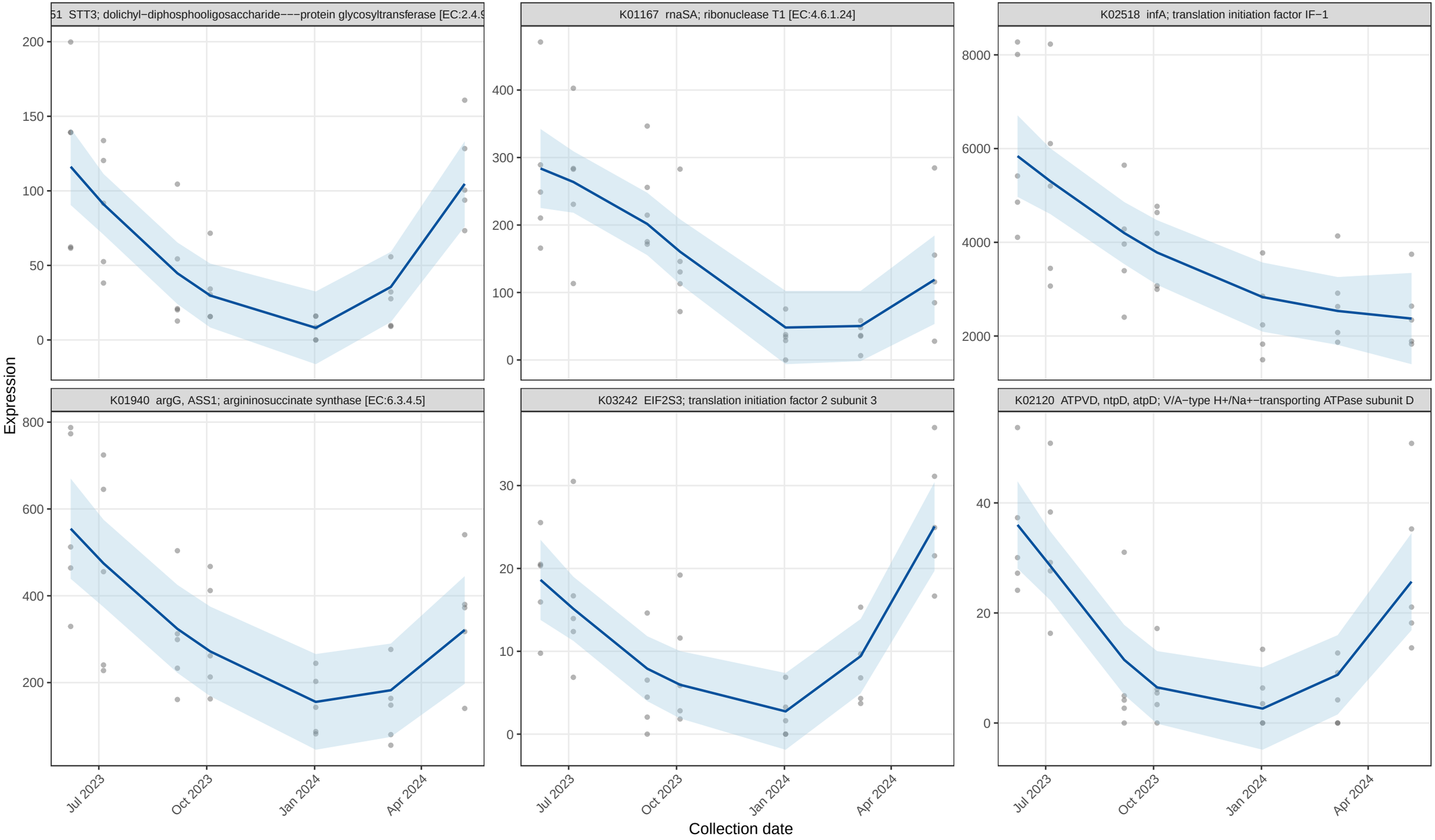

K02946 RP−S10, MRPS10, rpsJ; small subunit ribosomal protein S10
K02967 RP−S2, MRPS2, rpsB; small subunit ribosomal protein S2
K02885 RP−L19e, RPL19; large subunit ribosomal protein L19e
K02913 RP−L33, MRPL33, rpmG; large subunit ribosomal protein L33
K03047 rpoD; DNA−directed RNA polymerase subunit D [EC:2.7.7.6]
K03217 yidC, spoIIIJ, OXA1, ccfA; YidC/Oxa1 family membrane protein insertase
Expression
Collection date
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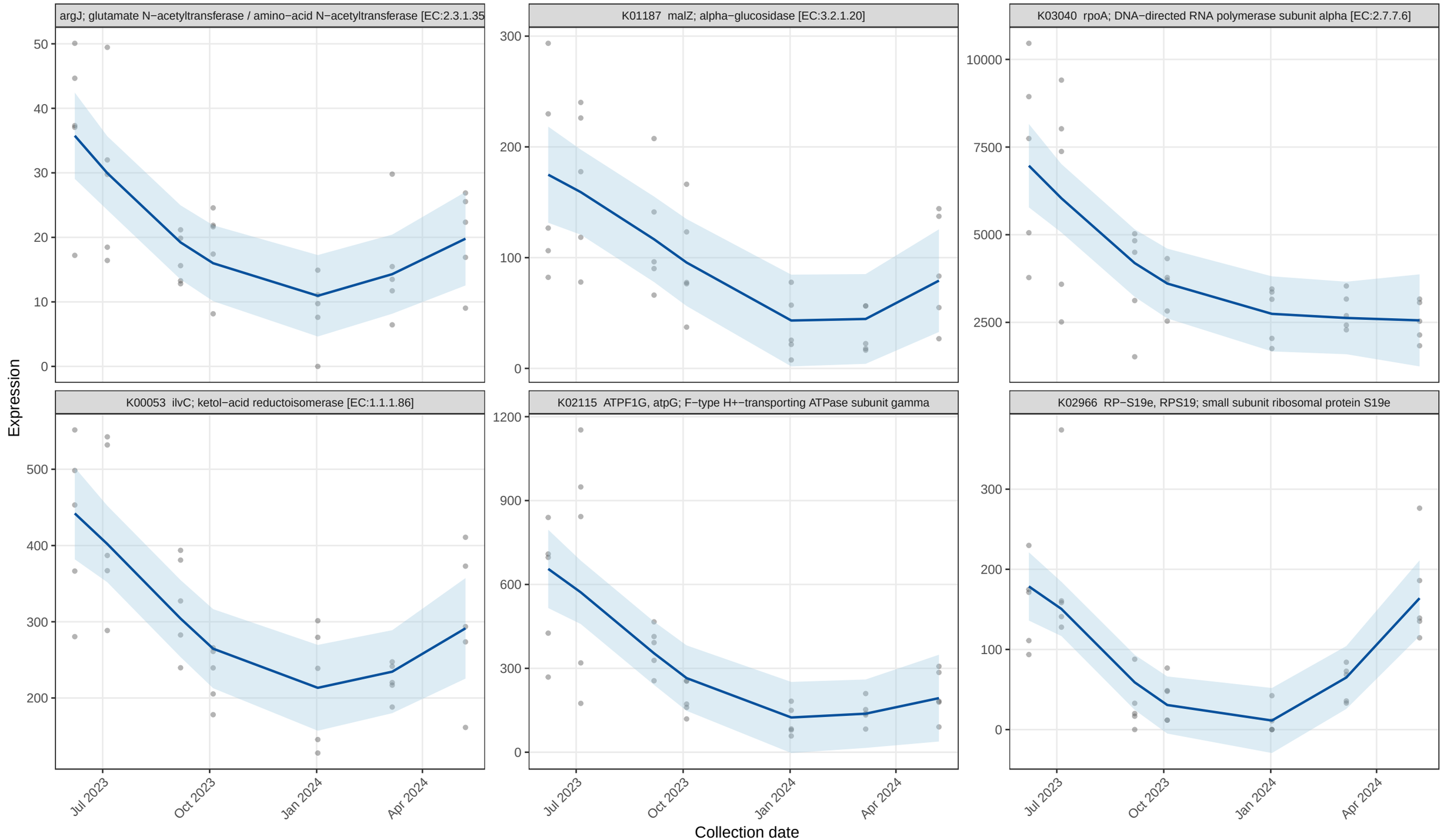

K03387 ahpF; NADH−dependent peroxiredoxin subunit F [EC:1.8.1.−]
K06207 typA, bipA; GTP−binding protein
K13525 VCP, CDC48; transitional endoplasmic reticulum ATPase
K10944 pmoA; methane monooxygenase subunit A [EC:1.14.18.3]
K02083 allC; allantoate deiminase [EC:3.5.3.9]
K07033 K07033; uncharacterized protein
Expression
Collection date
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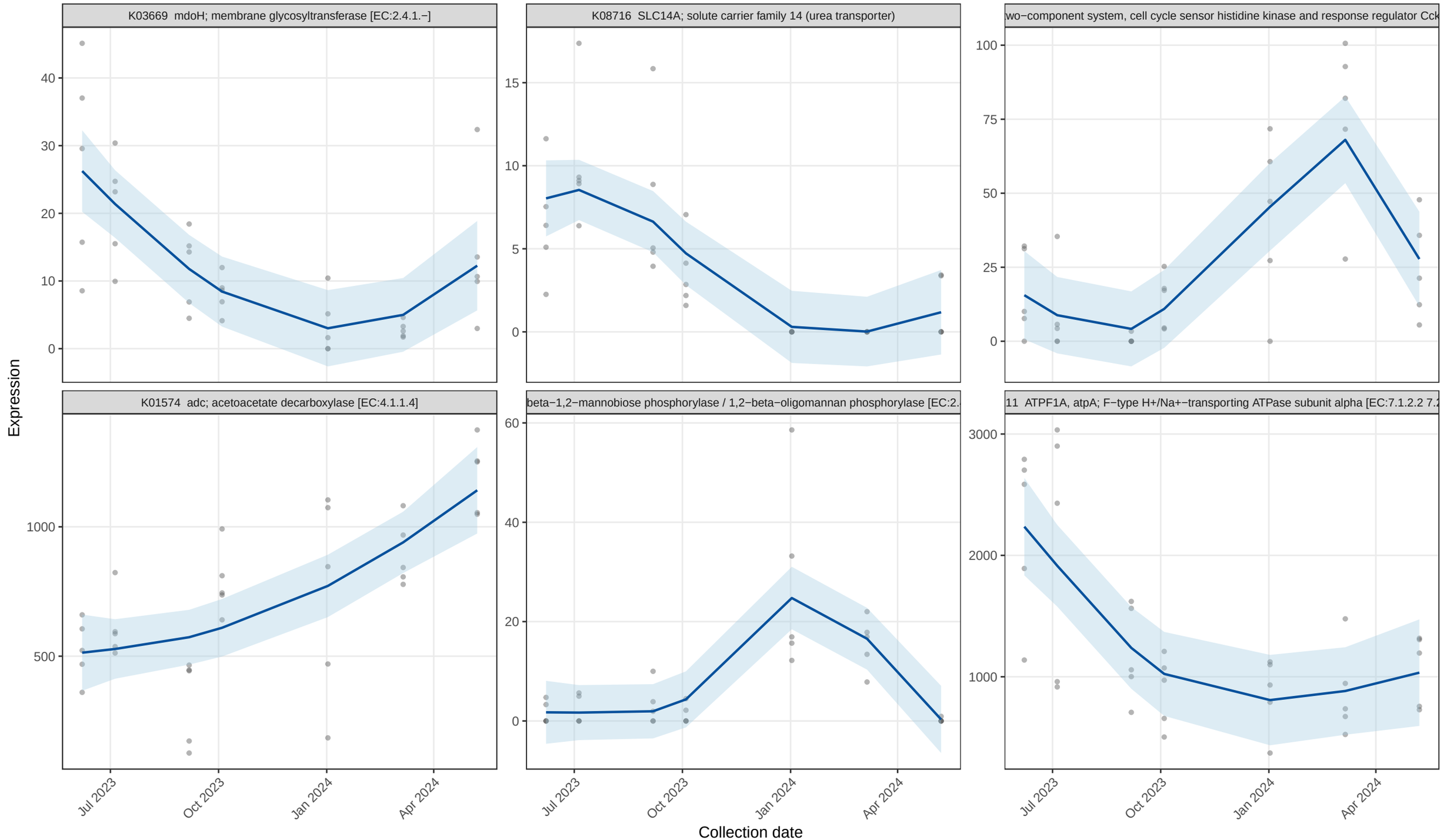

K13816 rpfF; DSF synthase
K04651 hypA, hybF; hydrogenase nickel incorporation protein HypA/HybF
K03231 EEF1A; elongation factor 1−alpha
K02116 atpI; ATP synthase protein I
K06148 ABCC−BAC; ATP−binding cassette, subfamily C, bacterial
K02959 RP−S16, MRPS16, rpsP; small subunit ribosomal protein S16
Expression
Collection date
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Apr 2024

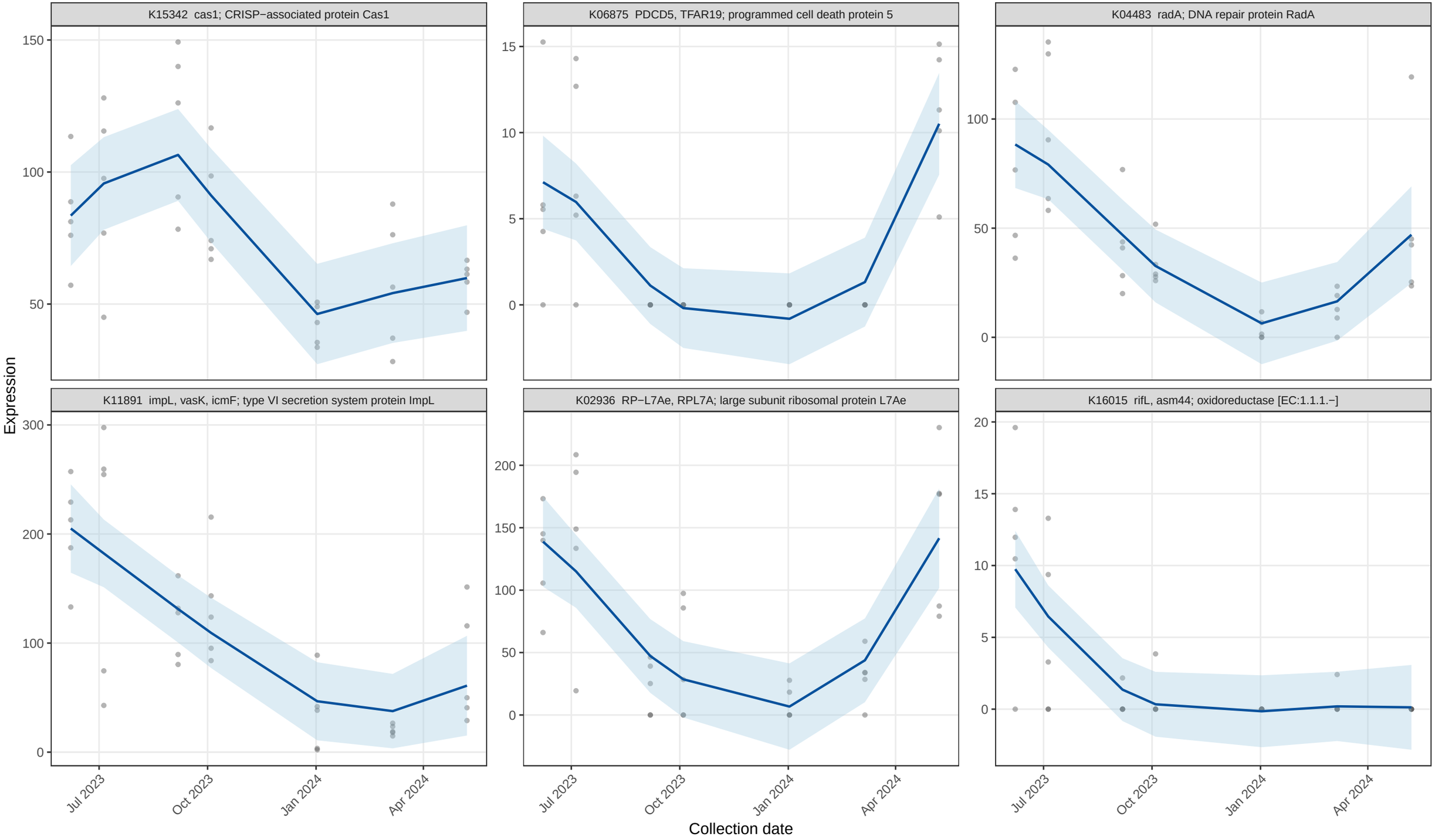

K01953 asnB, ASNS; asparagine synthase (glutamine−hydrolysing) [EC:6.3.5.4]
K01429 ureB; urease subunit beta [EC:3.5.1.5]
K15019 K15019; 3−hydroxypropionyl−coenzyme A dehydratase [EC:4.2.1.116]
K03762 proP; MFS transporter, MHS family, proline/betaine transporter
K02519 infB, MTIF2; translation initiation factor IF−2
K00975 glgC; glucose−1−phosphate adenylyltransferase [EC:2.7.7.27]
Expression
Collection date
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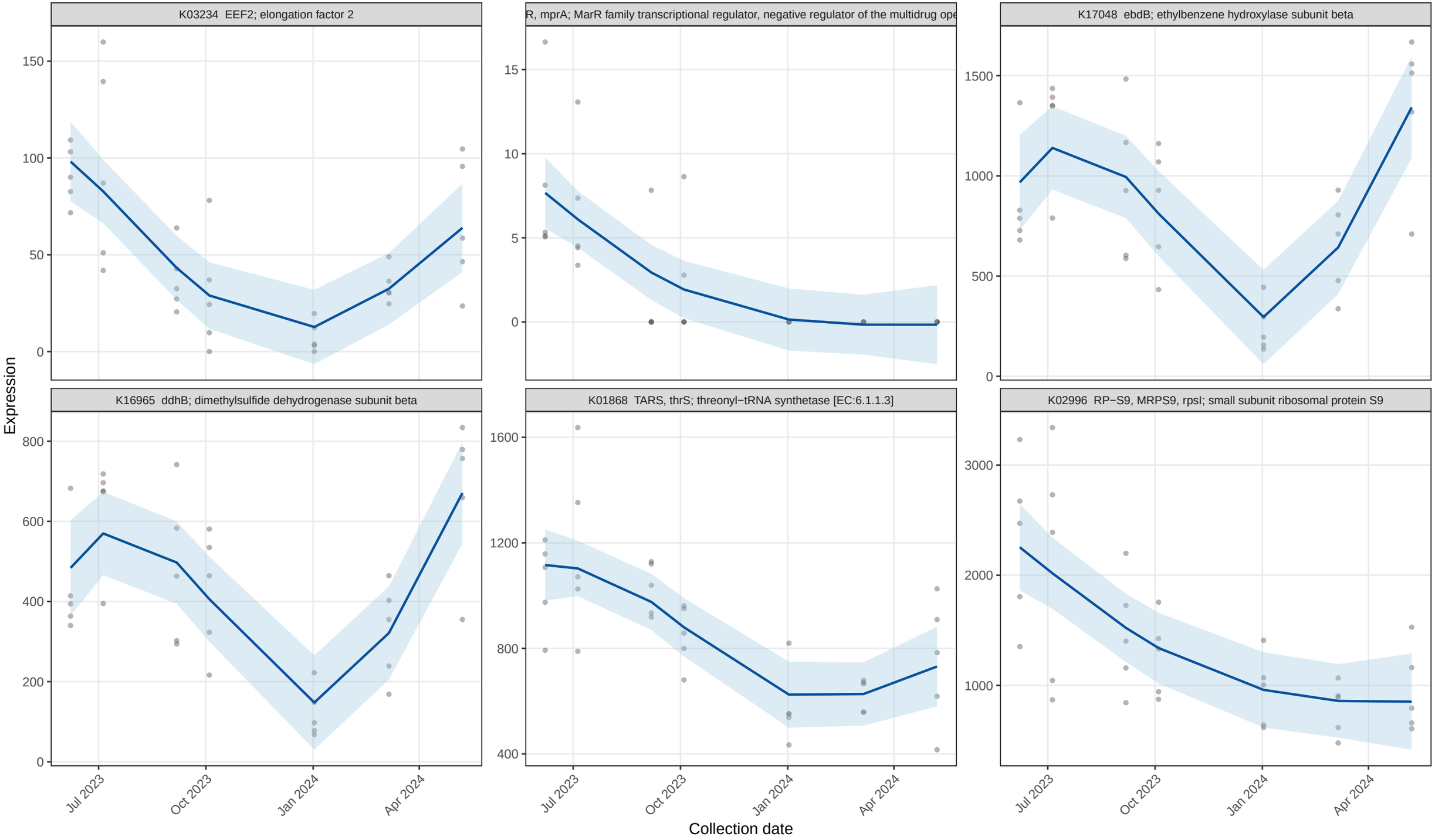

K07332 flaI; archaeal flagellar protein FlaI
K01956 carA, CPA1; carbamoyl−phosphate synthase small subunit [EC:6.3.5.5]
K00392 sir; sulfite reductase (ferredoxin) [EC:1.8.7.1]
1682 acnB; aconitate hydratase 2 / 2−methylisocitrate dehydratase [EC:4.2.1.3 4.2.1.
K02897 RP−L25, rplY; large subunit ribosomal protein L25
K07146 K07146; UPF0176 protein
Expression
Collection date
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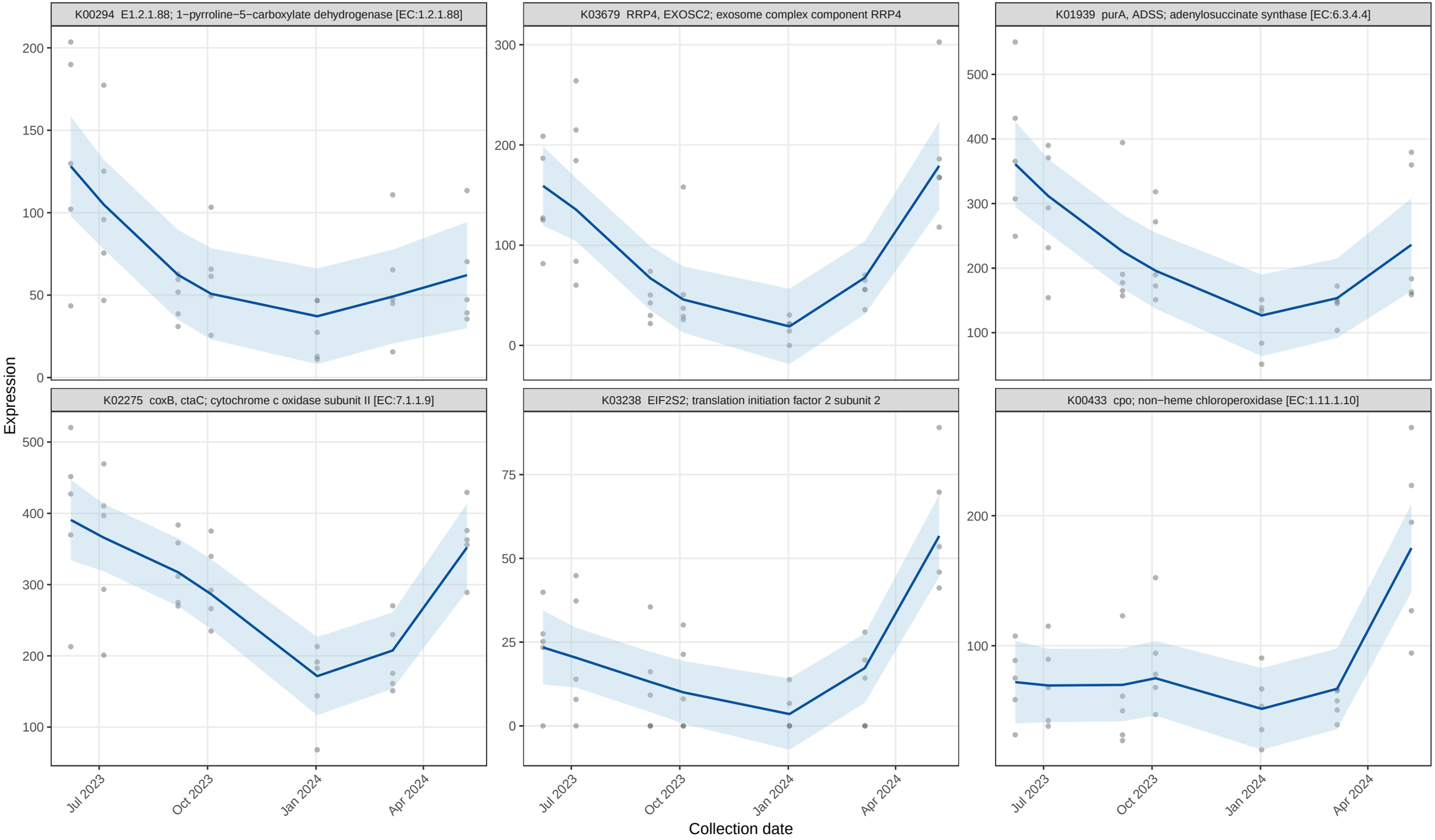

51 STT3; dolichyl−diphosphooligosaccharide−−−protein glycosyltransferase [EC:2.4.9
K01167 rnaSA; ribonuclease T1 [EC:4.6.1.24]
K02518 infA; translation initiation factor IF−1
K01940 argG, ASS1; argininosuccinate synthase [EC:6.3.4.5]
K03242 EIF2S3; translation initiation factor 2 subunit 3
K02120 ATPVD, ntpD, atpD; V/A−type H+/Na+−transporting ATPase subunit D
Expression
Collection date
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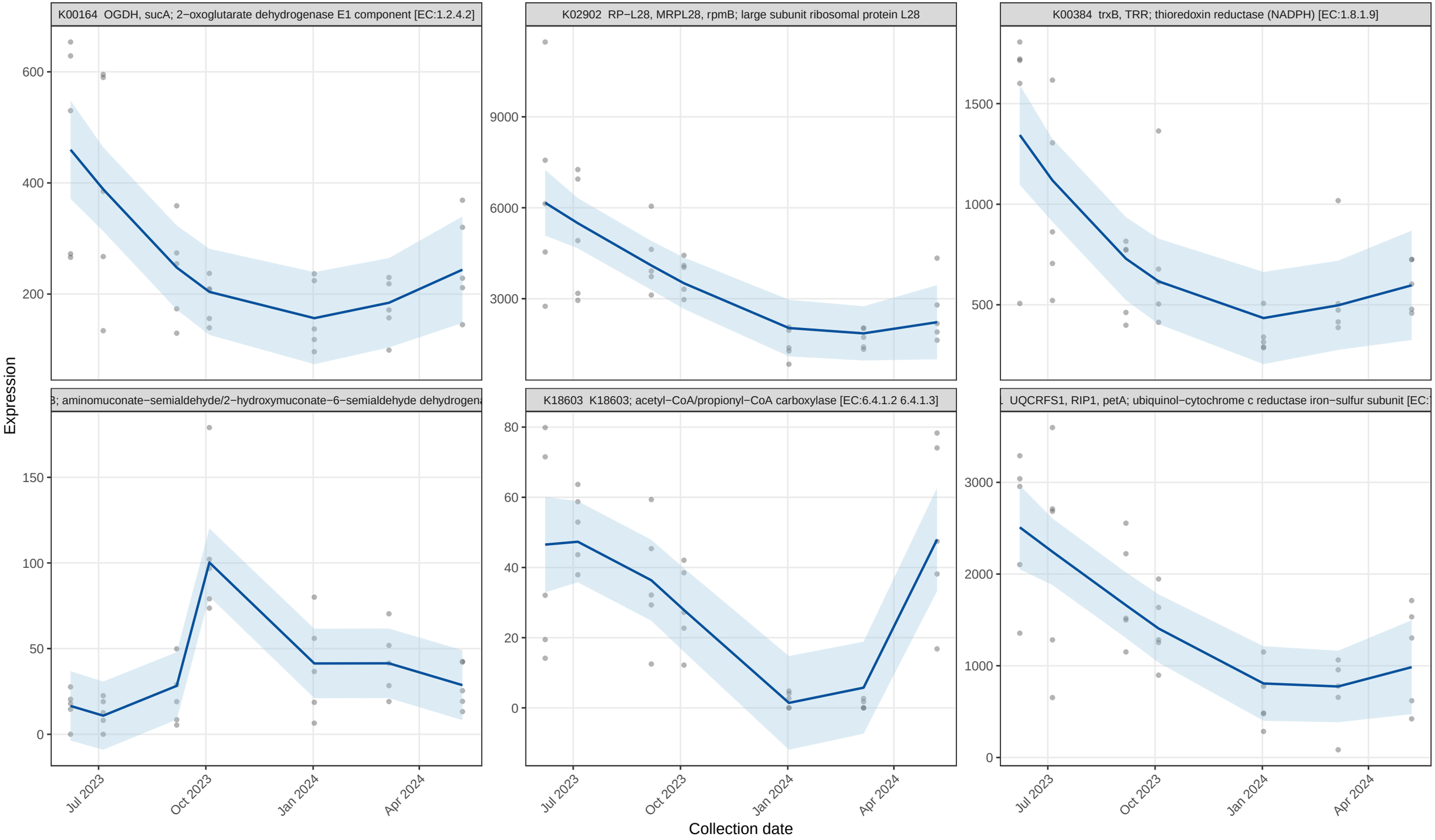

argJ; glutamate N−acetyltransferase / amino−acid N−acetyltransferase [EC:2.3.1.35]
K01187 malZ; alpha−glucosidase [EC:3.2.1.20]
K03040 rpoA; DNA−directed RNA polymerase subunit alpha [EC:2.7.7.6]
K00053 ilvC; ketol−acid reductoisomerase [EC:1.1.1.86]
K02115 ATPF1G, atpG; F−type H+−transporting ATPase subunit gamma
K02966 RP−S19e, RPS19; small subunit ribosomal protein S19e
Expression
Collection date
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Apr 2024

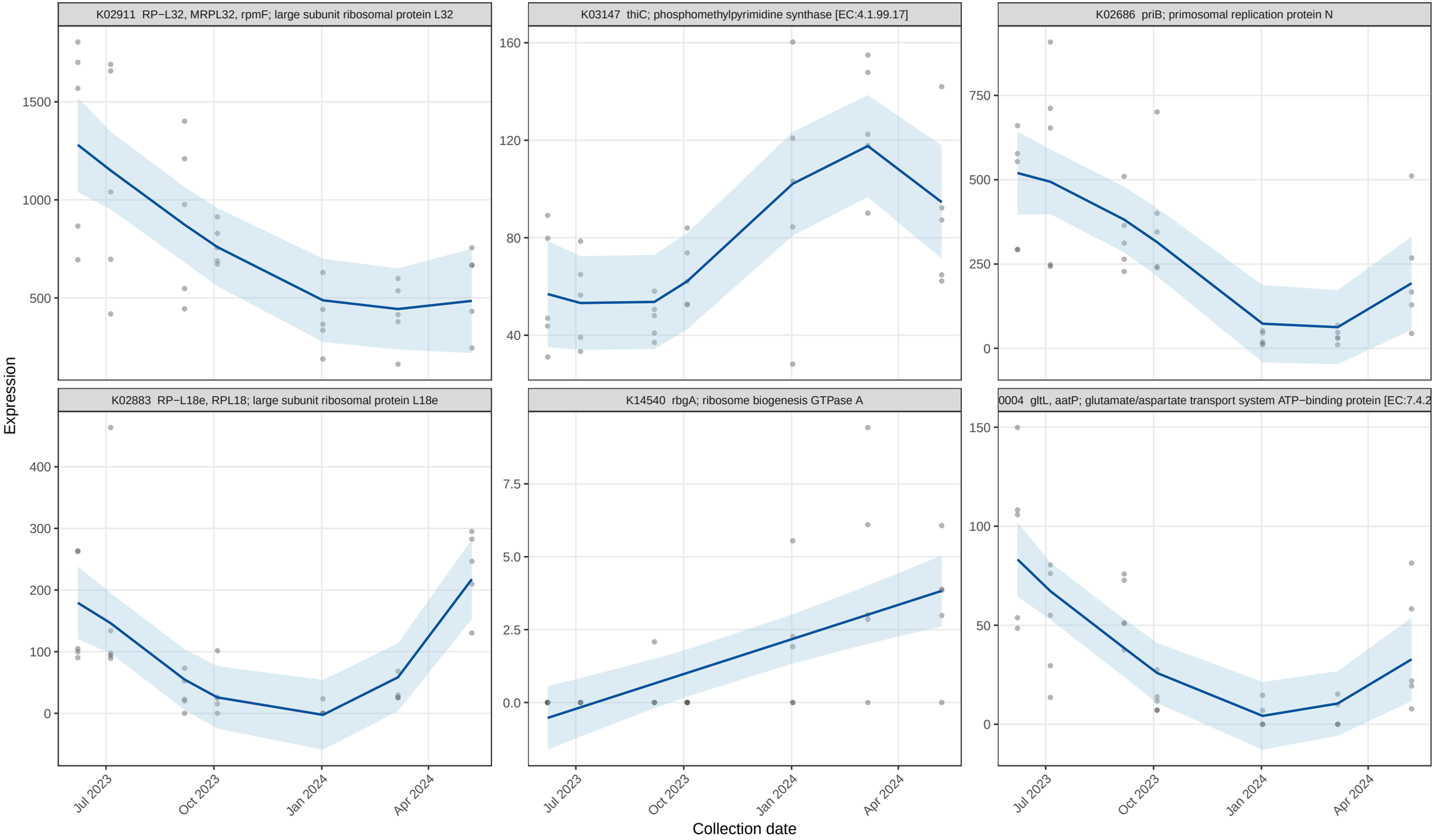

K03669 mdoH; membrane glycosyltransferase [EC:2.4.1.−]
K08716 SLC14A; solute carrier family 14 (urea transporter)
wo−component system, cell cycle sensor histidine kinase and response regulator Cck
K01574 adc; acetoacetate decarboxylase [EC:4.1.1.4]
beta−1,2−mannobiose phosphorylase / 1,2−beta−oligomannan phosphorylase [EC:2.
11 ATPF1A, atpA; F−type H+/Na+−transporting ATPase subunit alpha [EC:7.1.2.2 7.2
Expression
Collection date
Jul 2023
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Jan 2024
Apr 2024

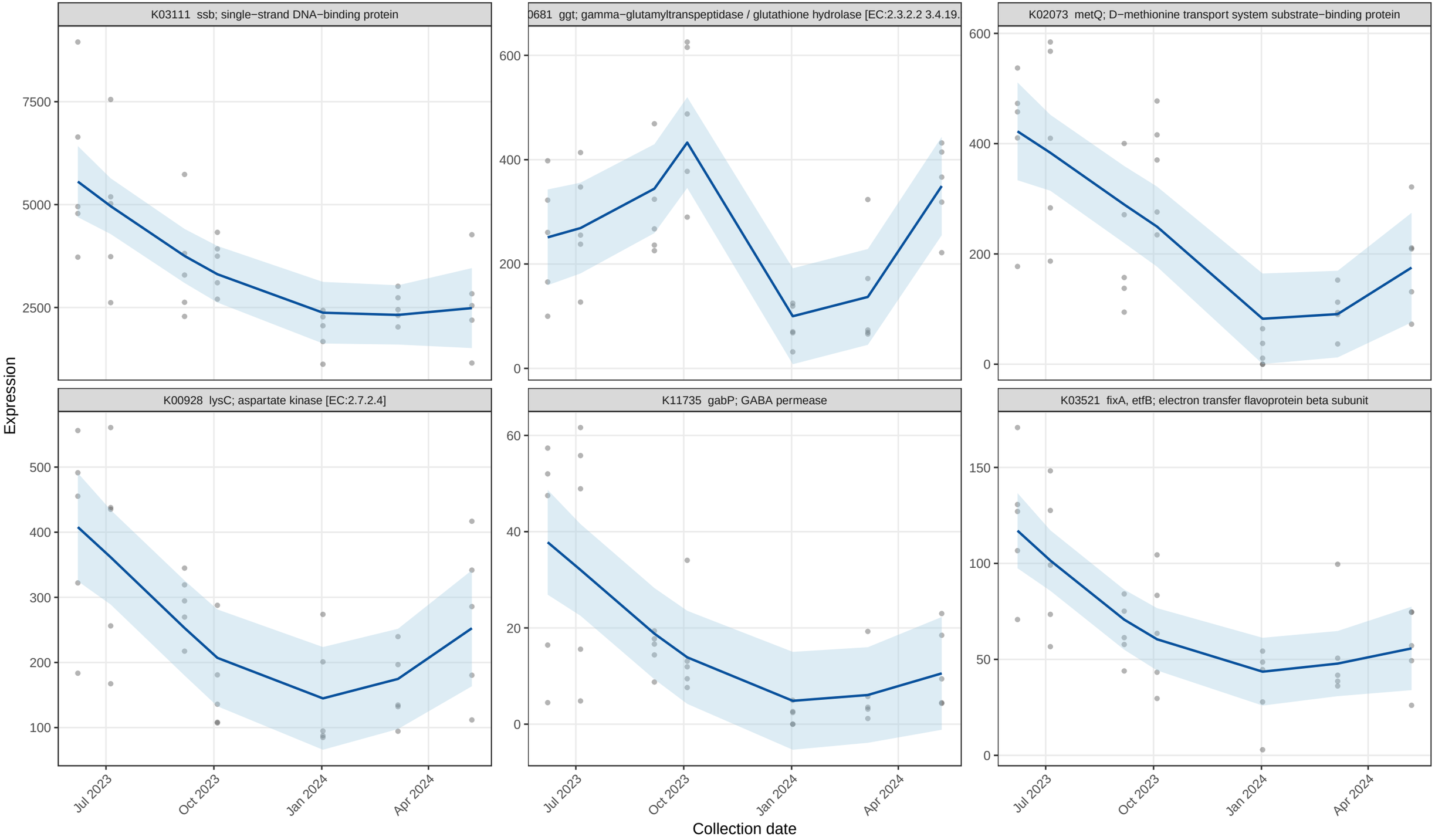

K15342 cas1; CRISP−associated protein Cas1
K06875 PDCD5, TFAR19; programmed cell death protein 5
K04483 radA; DNA repair protein RadA
K11891 impL, vasK, icmF; type VI secretion system protein ImpL
K02936 RP−L7Ae, RPL7A; large subunit ribosomal protein L7Ae
K16015 rifL, asm44; oxidoreductase [EC:1.1.1.−]
Expression
Collection date
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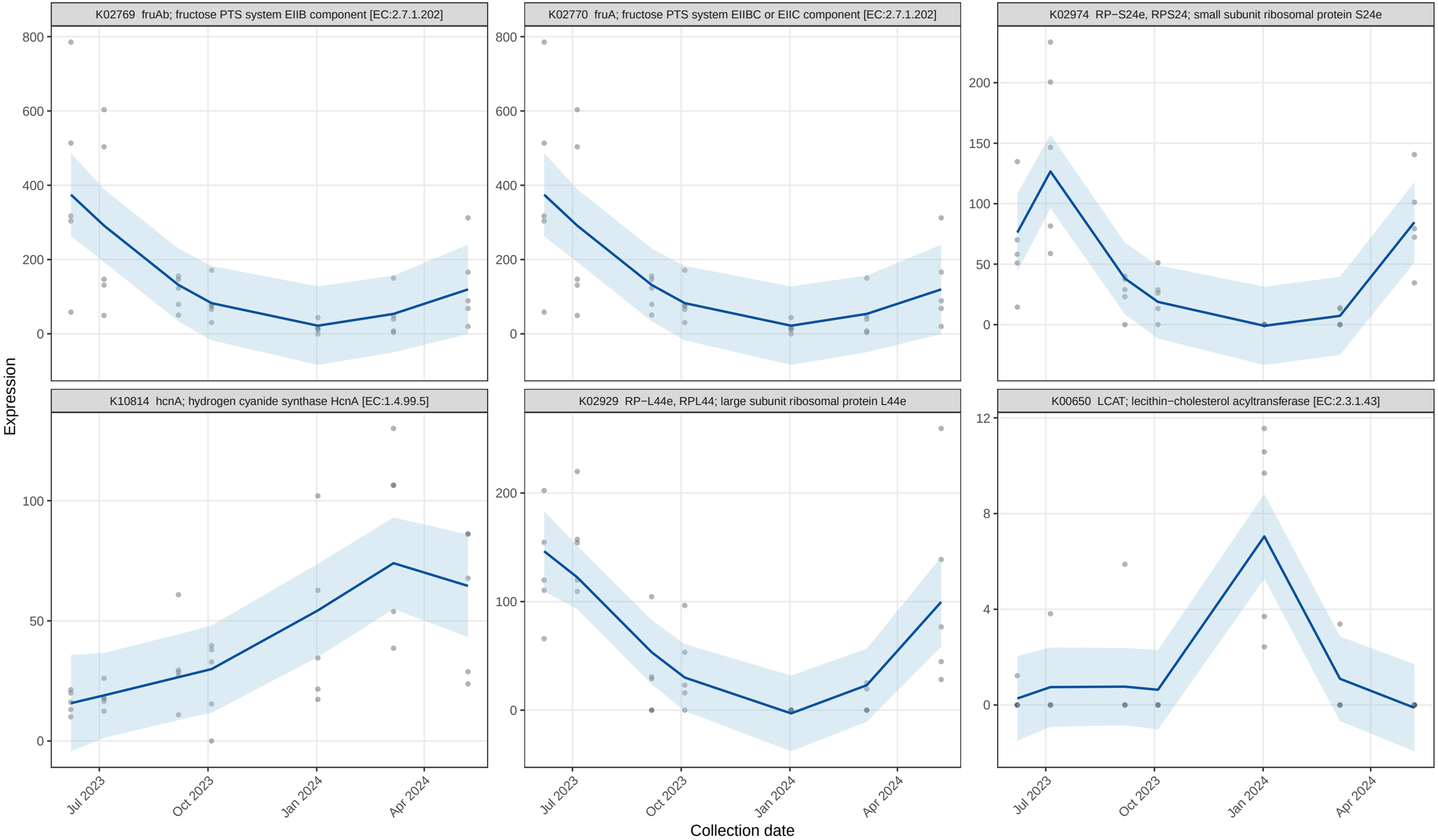

K03234 EEF2; elongation factor 2
R, mprA; MarR family transcriptional regulator, negative regulator of the multidrug ope
K17048 ebdB; ethylbenzene hydroxylase subunit beta
K16965 ddhB; dimethylsulfide dehydrogenase subunit beta
K01868 TARS, thrS; threonyl−tRNA synthetase [EC:6.1.1.3]
K02996 RP−S9, MRPS9, rpsI; small subunit ribosomal protein S9
Expression
Collection date
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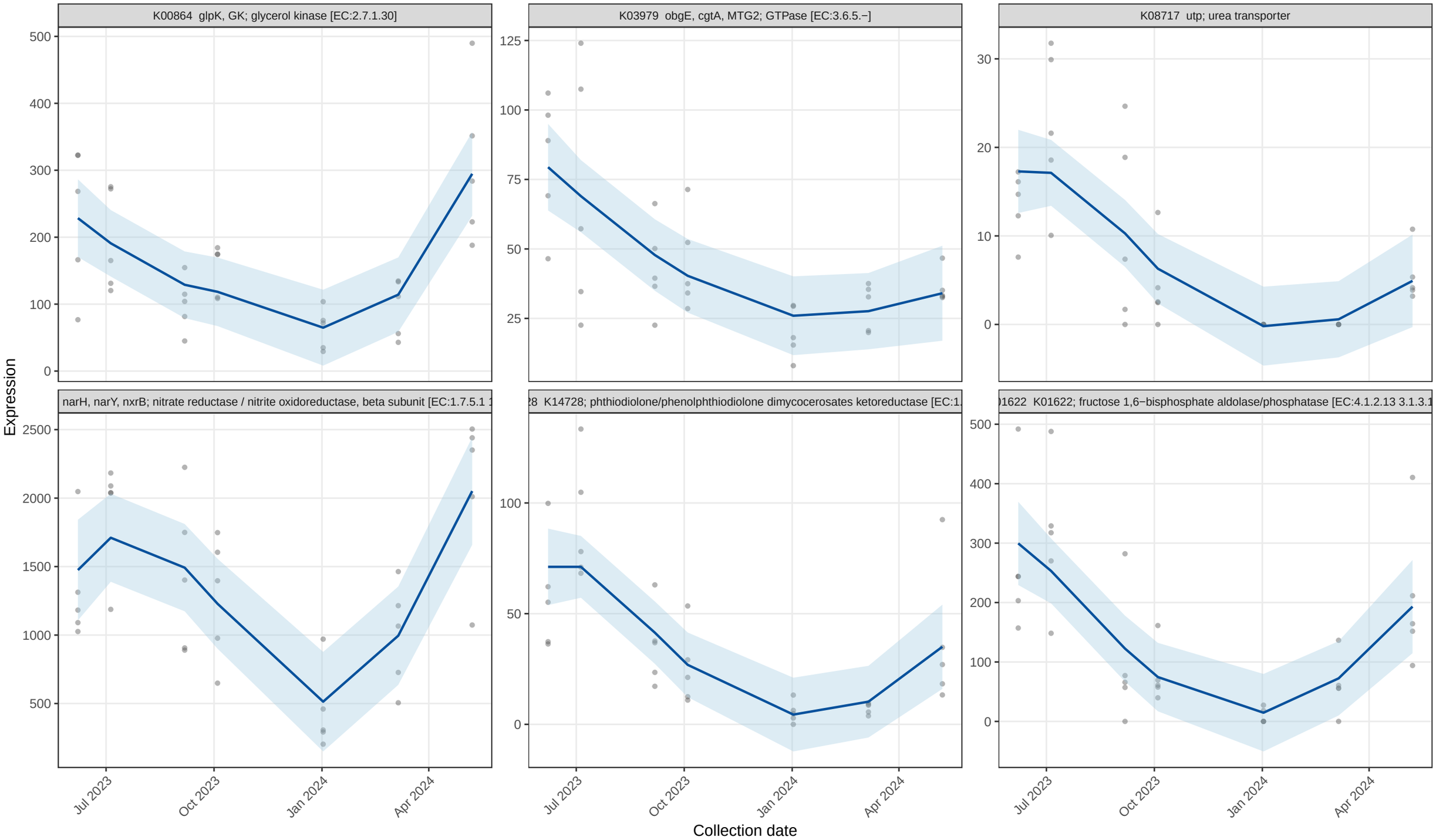

K00294 E1.2.1.88; 1−pyrroline−5−carboxylate dehydrogenase [EC:1.2.1.88]
K03679 RRP4, EXOSC2; exosome complex component RRP4
K01939 purA, ADSS; adenylosuccinate synthase [EC:6.3.4.4]
K02275 coxB, ctaC; cytochrome c oxidase subunit II [EC:7.1.1.9]
K03238 EIF2S2; translation initiation factor 2 subunit 2
K00433 cpo; non−heme chloroperoxidase [EC:1.11.1.10]
Expression
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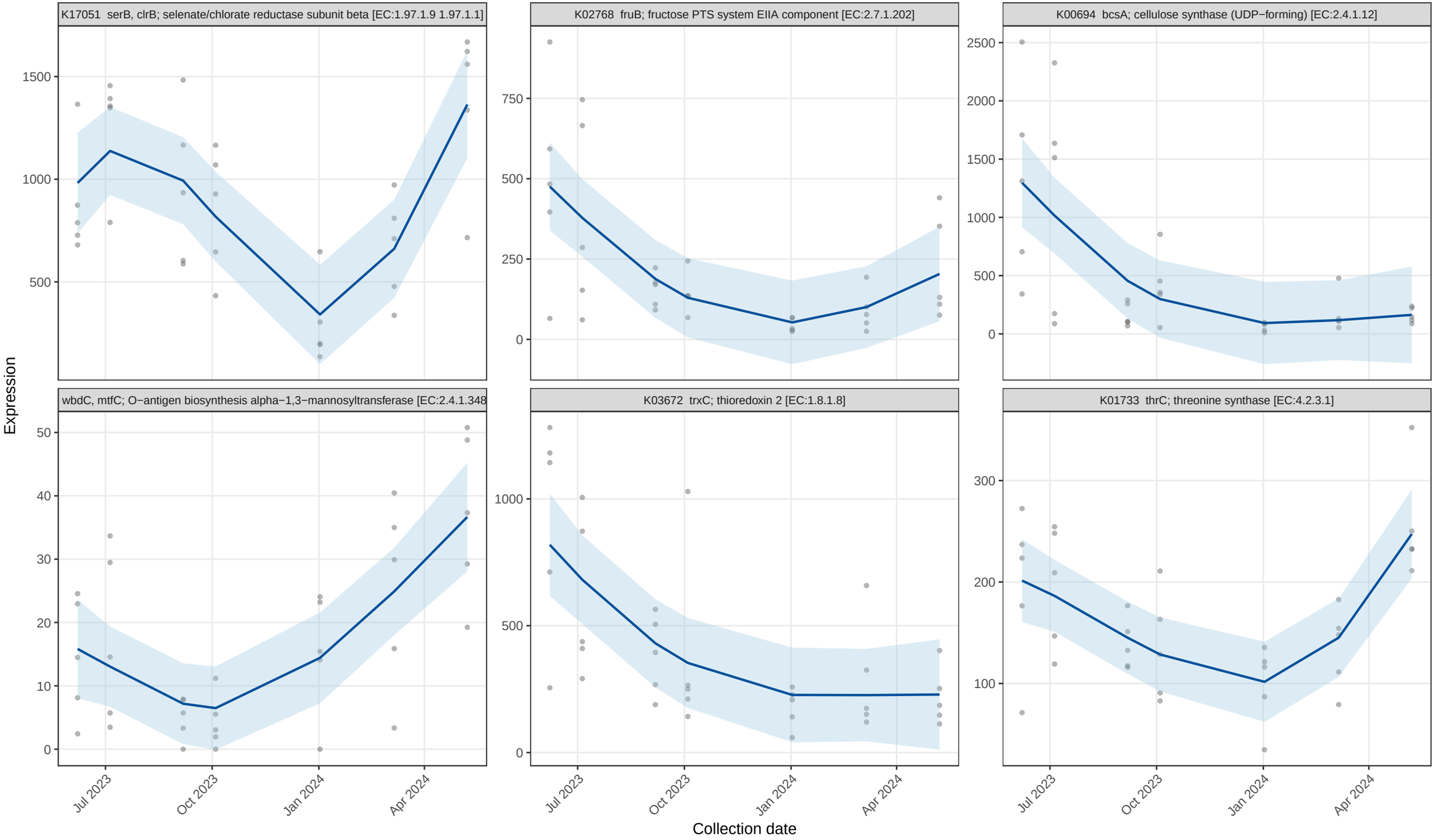

K00164 OGDH, sucA; 2−oxoglutarate dehydrogenase E1 component [EC:1.2.4.2]
K02902 RP−L28, MRPL28, rpmB; large subunit ribosomal protein L28
K00384 trxB, TRR; thioredoxin reductase (NADPH) [EC:1.8.1.9]
; aminomuconate−semialdehyde/2−hydroxymuconate−6−semialdehyde dehydrogena
K18603 K18603; acetyl−CoA/propionyl−CoA carboxylase [EC:6.4.1.2 6.4.1.3]
UQCRFS1, RIP1, petA; ubiquinol−cytochrome c reductase iron−sulfur subunit [EC:
Expression
Collection date
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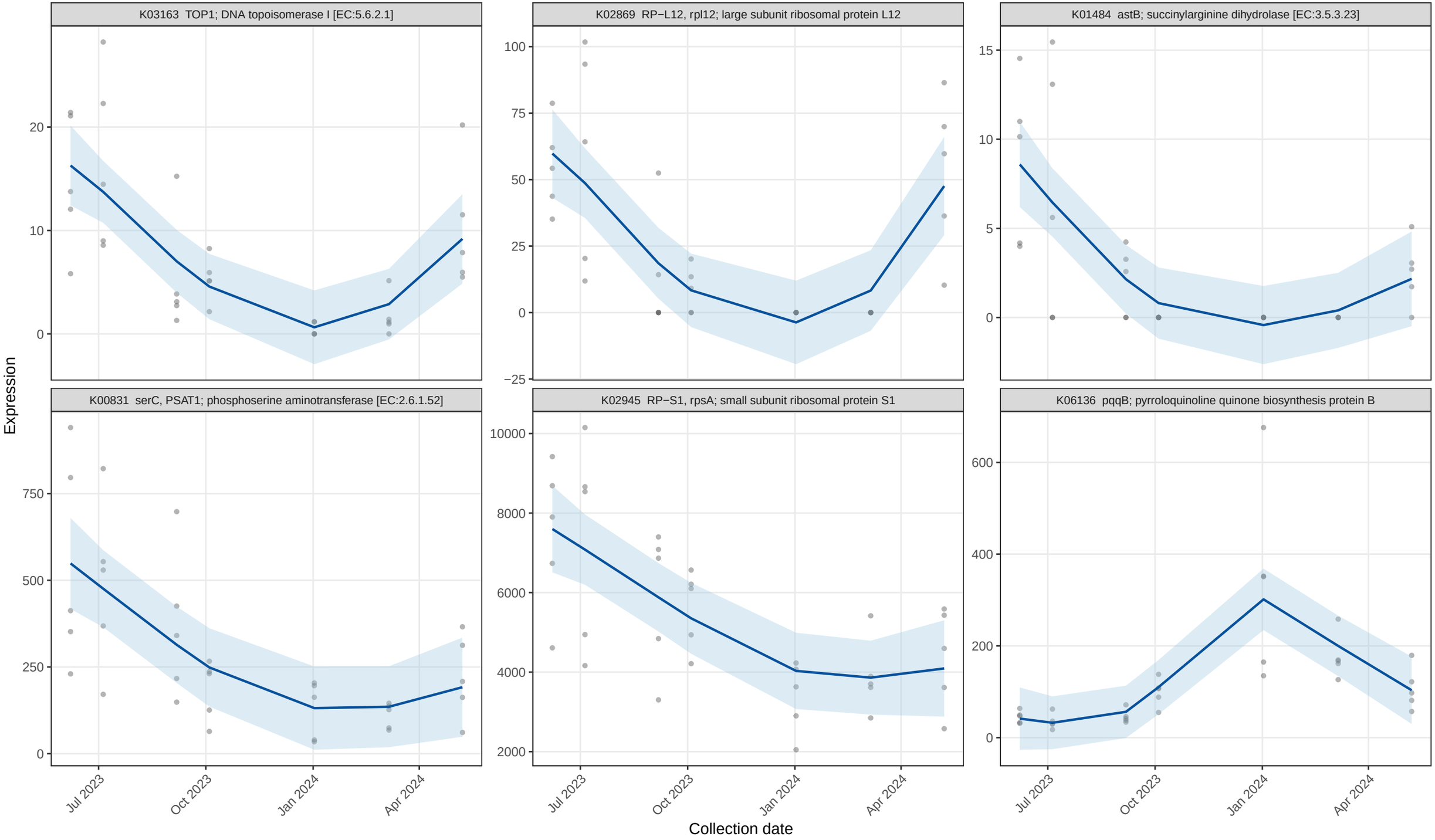

K02911 RP−L32, MRPL32, rpmF; large subunit ribosomal protein L32
K03147 thiC; phosphomethylpyrimidine synthase [EC:4.1.99.17]
K02686 priB; primosomal replication protein N
K02883 RP−L18e, RPL18; large subunit ribosomal protein L18e
K14540 rbgA; ribosome biogenesis GTPase A
0004 gltL, aatP; glutamate/aspartate transport system ATP−binding protein [EC:7.4.2
Expression
Collection date
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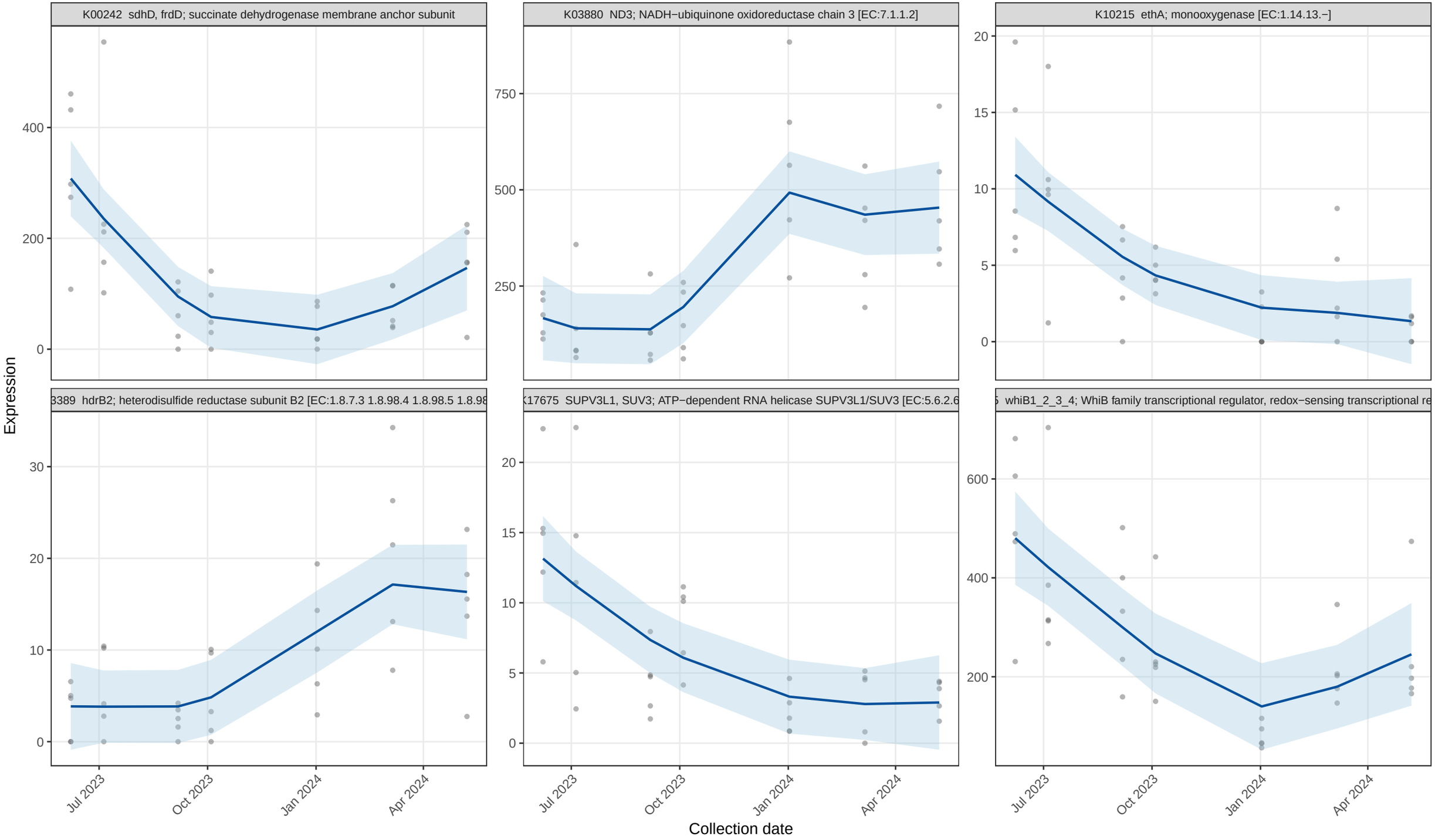

K03111 ssb; single−strand DNA−binding protein
0681 ggt; gamma−glutamyltranspeptidase / glutathione hydrolase [EC:2.3.2.2 3.4.19.
K02073 metQ; D−methionine transport system substrate−binding protein
K00928 lysC; aspartate kinase [EC:2.7.2.4]
K11735 gabP; GABA permease
K03521 fixA, etfB; electron transfer flavoprotein beta subunit
Expression
Collection date
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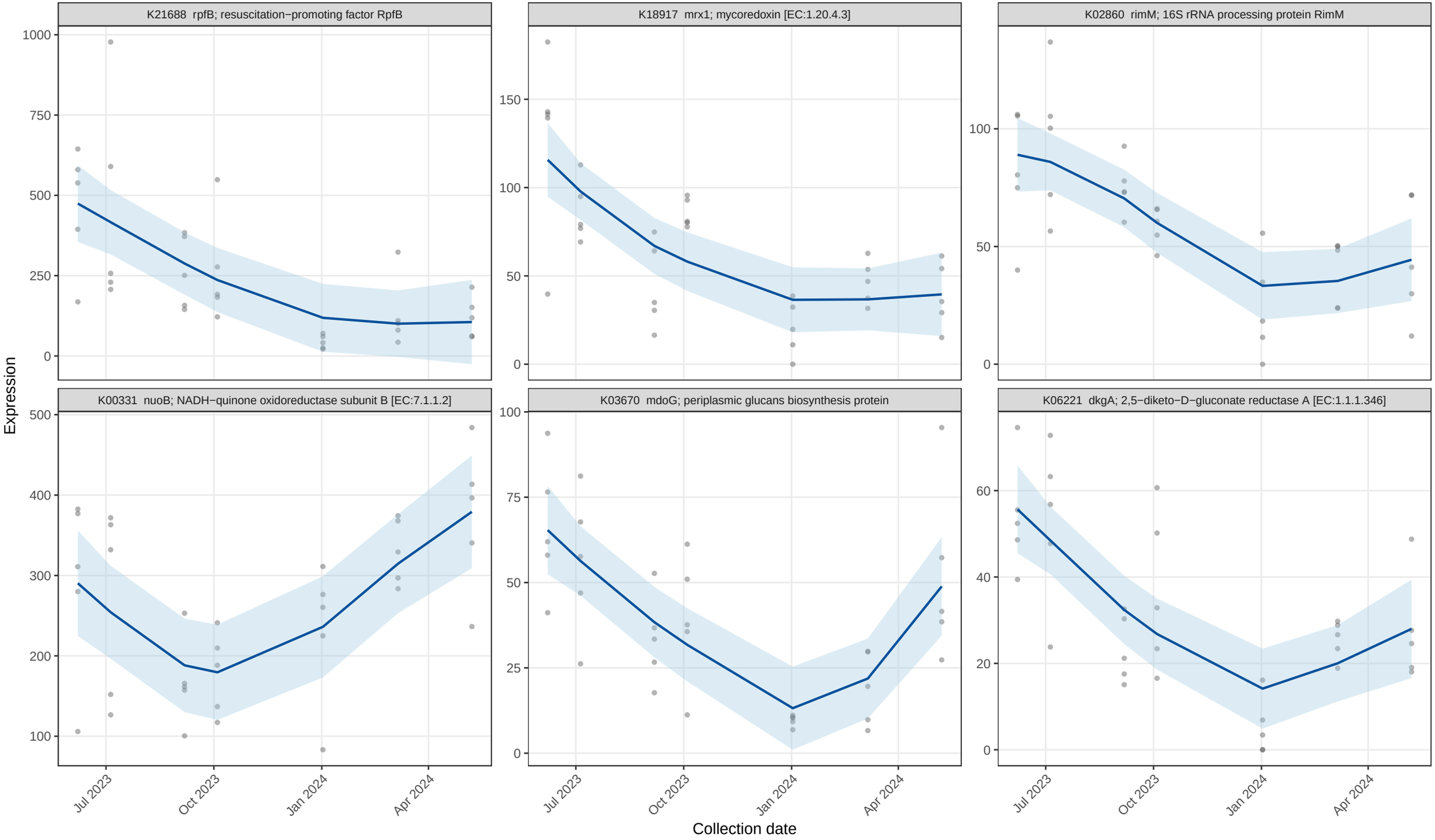

K02769 fruAb; fructose PTS system EIIB component [EC:2.7.1.202]
K02770 fruA; fructose PTS system EIIBC or EIIC component [EC:2.7.1.202]
K02974 RP−S24e, RPS24; small subunit ribosomal protein S24e
K10814 hcnA; hydrogen cyanide synthase HcnA [EC:1.4.99.5]
K02929 RP−L44e, RPL44; large subunit ribosomal protein L44e
K00650 LCAT; lecithin−cholesterol acyltransferase [EC:2.3.1.43]
Expression
Collection date
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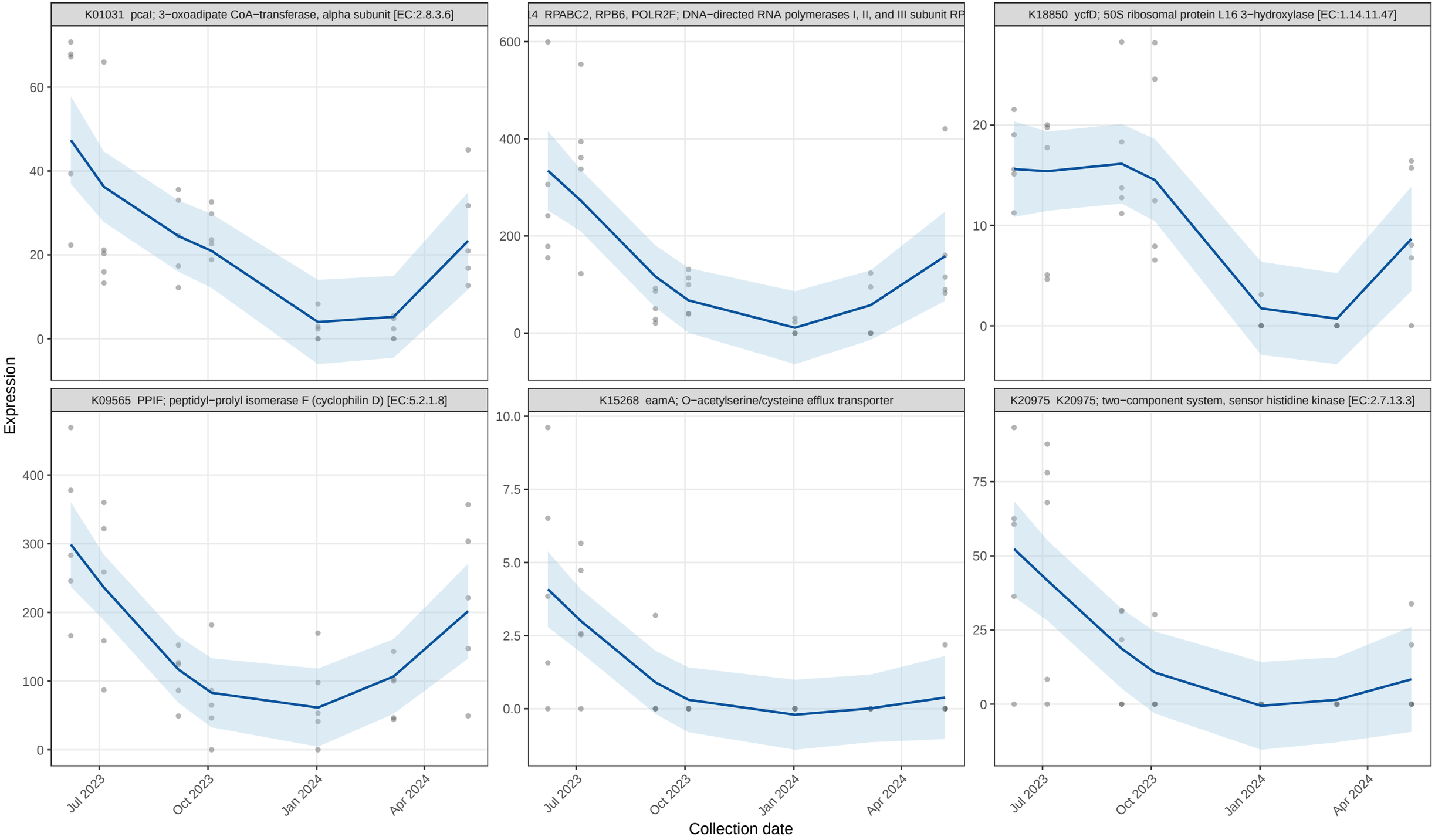

K00864 glpK, GK; glycerol kinase [EC:2.7.1.30]
K03979 obgE, cgtA, MTG2; GTPase [EC:3.6.5.−]
K08717 utp; urea transporter
narH, narY, nxrB; nitrate reductase / nitrite oxidoreductase, beta subunit [EC:1.7.5.1
28 K14728; phthiodiolone/phenolphthiodiolone dimycocerosates ketoreductase [EC:1.
01622 K01622; fructose 1,6−bisphosphate aldolase/phosphatase [EC:4.1.2.13 3.1.3.1
Expression
Collection date
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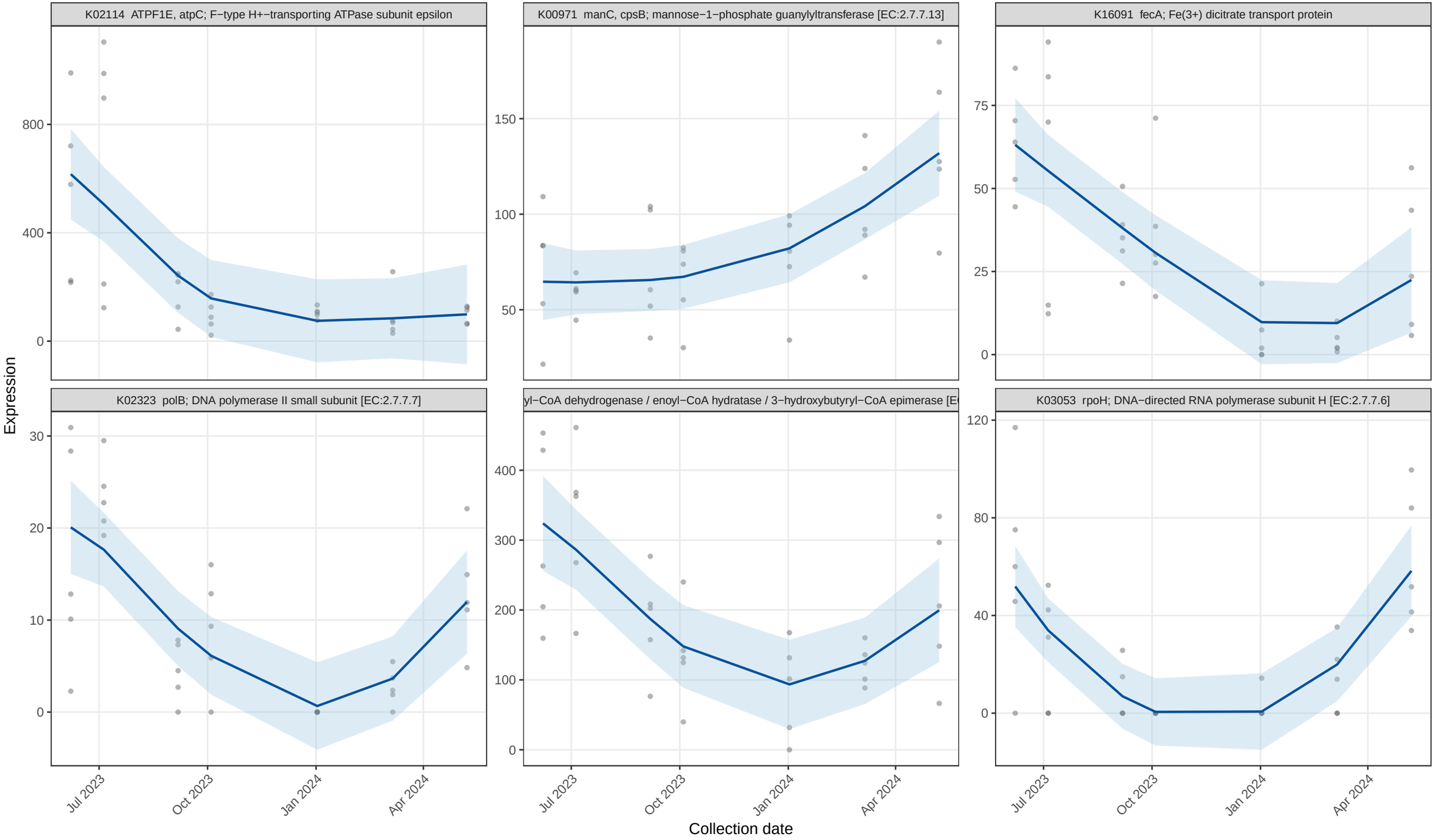

K17051 serB, clrB; selenate/chlorate reductase subunit beta [EC:1.97.1.9 1.97.1.1]
K02768 fruB; fructose PTS system EIIA component [EC:2.7.1.202]
K00694 bcsA; cellulose synthase (UDP−forming) [EC:2.4.1.12]
wbdC, mtfC; O−antigen biosynthesis alpha−1,3−mannosyltransferase [EC:2.4.1.348
K03672 trxC; thioredoxin 2 [EC:1.8.1.8]
K01733 thrC; threonine synthase [EC:4.2.3.1]
Expression
Collection date
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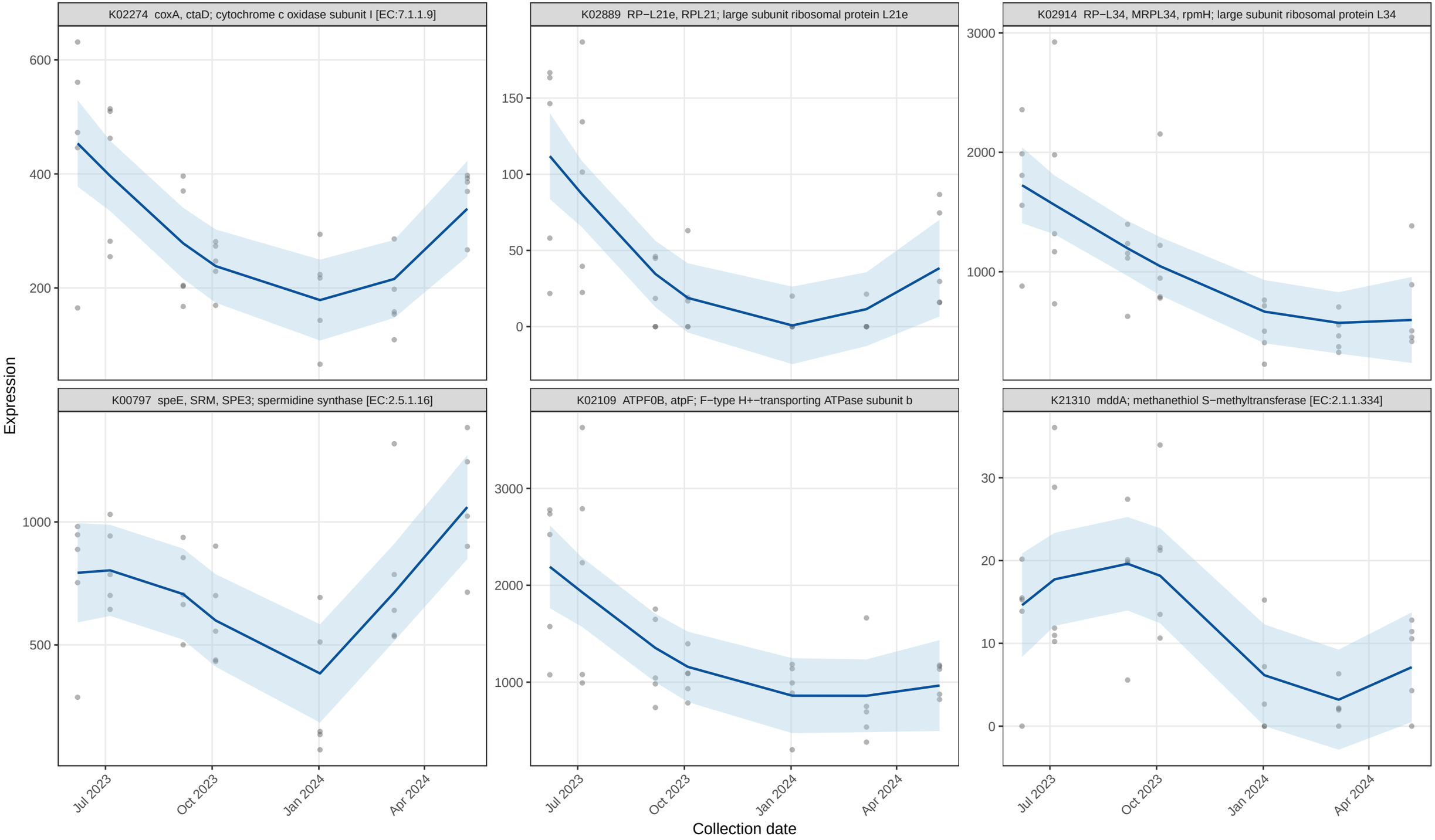

K03163 TOP1; DNA topoisomerase I [EC:5.6.2.1]
K02869 RP−L12, rpl12; large subunit ribosomal protein L12
K01484 astB; succinylarginine dihydrolase [EC:3.5.3.23]
K00831 serC, PSAT1; phosphoserine aminotransferase [EC:2.6.1.52]
K02945 RP−S1, rpsA; small subunit ribosomal protein S1
K06136 pqqB; pyrroloquinoline quinone biosynthesis protein B
Expression
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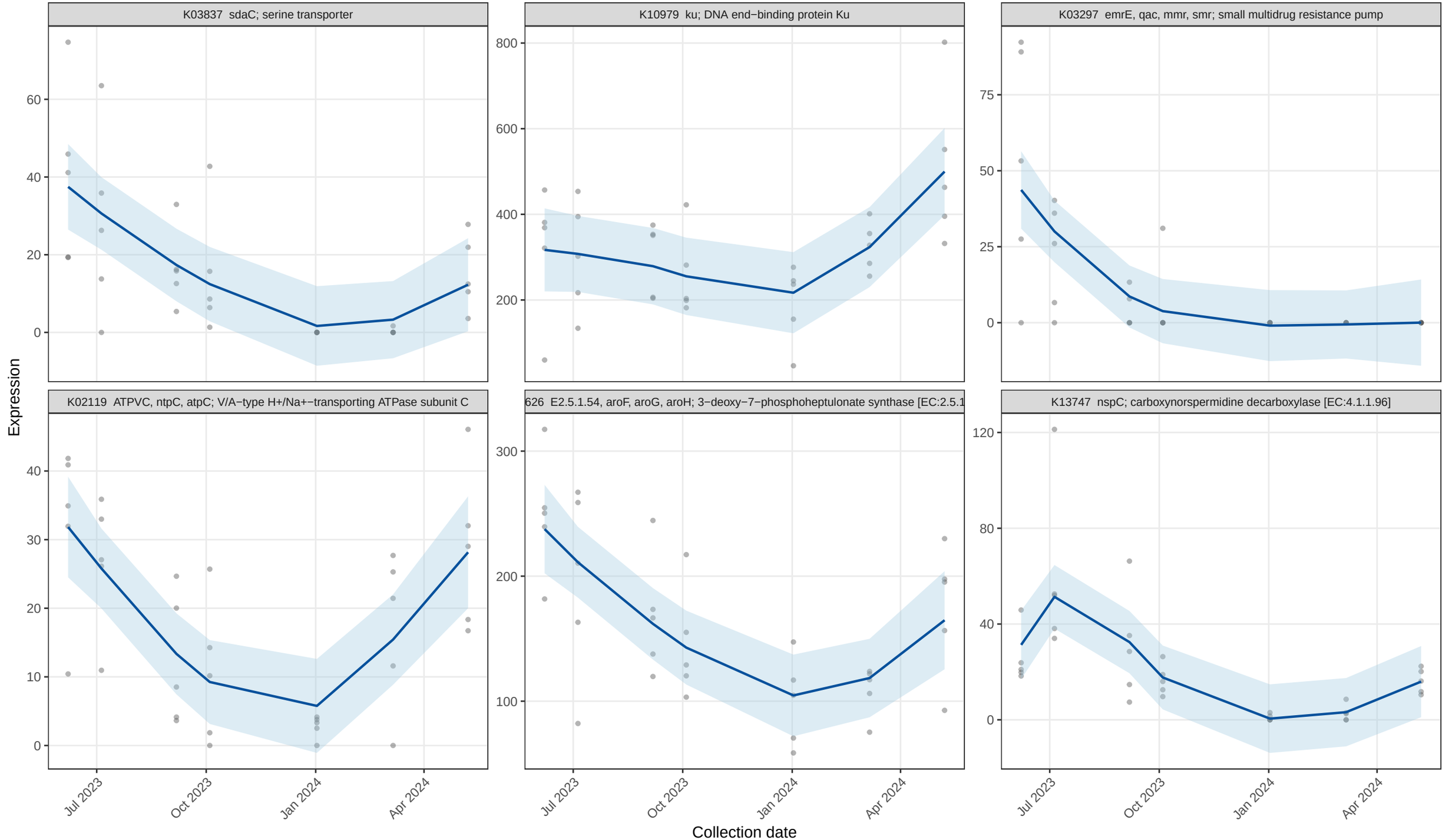

K00242 sdhD, frdD; succinate dehydrogenase membrane anchor subunit
K03880 ND3; NADH−ubiquinone oxidoreductase chain 3 [EC:7.1.1.2]
K10215 ethA; monooxygenase [EC:1.14.13.−]
3389 hdrB2; heterodisulfide reductase subunit B2 [EC:1.8.7.3 1.8.98.4 1.8.98.5 1.8.98
K17675 SUPV3L1, SUV3; ATP−dependent RNA helicase SUPV3L1/SUV3 [EC:5.6.2.6
whiB1_2_3_4; WhiB family transcriptional regulator, redox−sensing transcriptional re
Expression
Collection date
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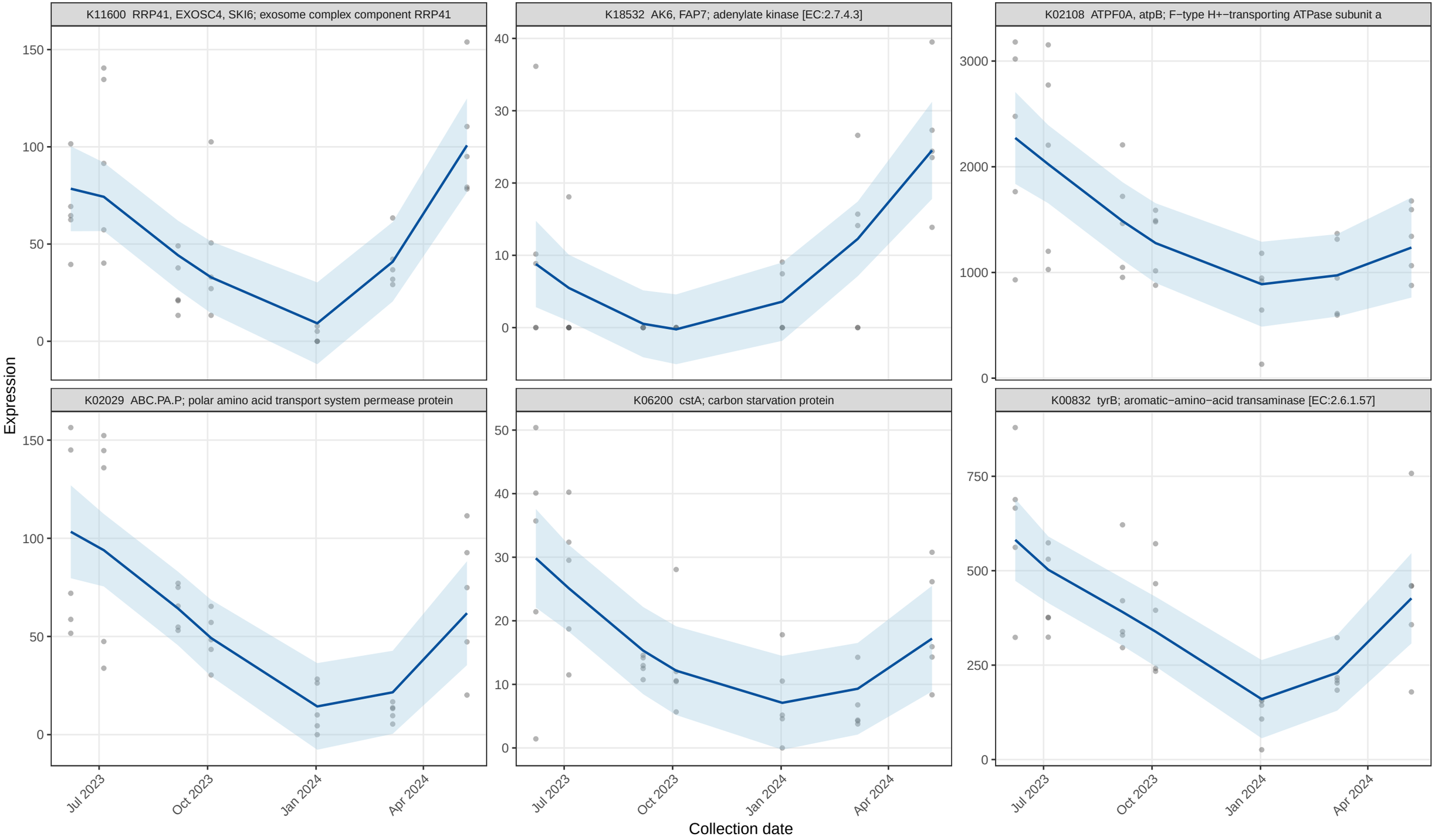

K21688 rpfB; resuscitation−promoting factor RpfB
K18917 mrx1; mycoredoxin [EC:1.20.4.3]
K02860 rimM; 16S rRNA processing protein RimM
K00331 nuoB; NADH−quinone oxidoreductase subunit B [EC:7.1.1.2]
K03670 mdoG; periplasmic glucans biosynthesis protein
K06221 dkgA; 2,5−diketo−D−gluconate reductase A [EC:1.1.1.346]
Expression
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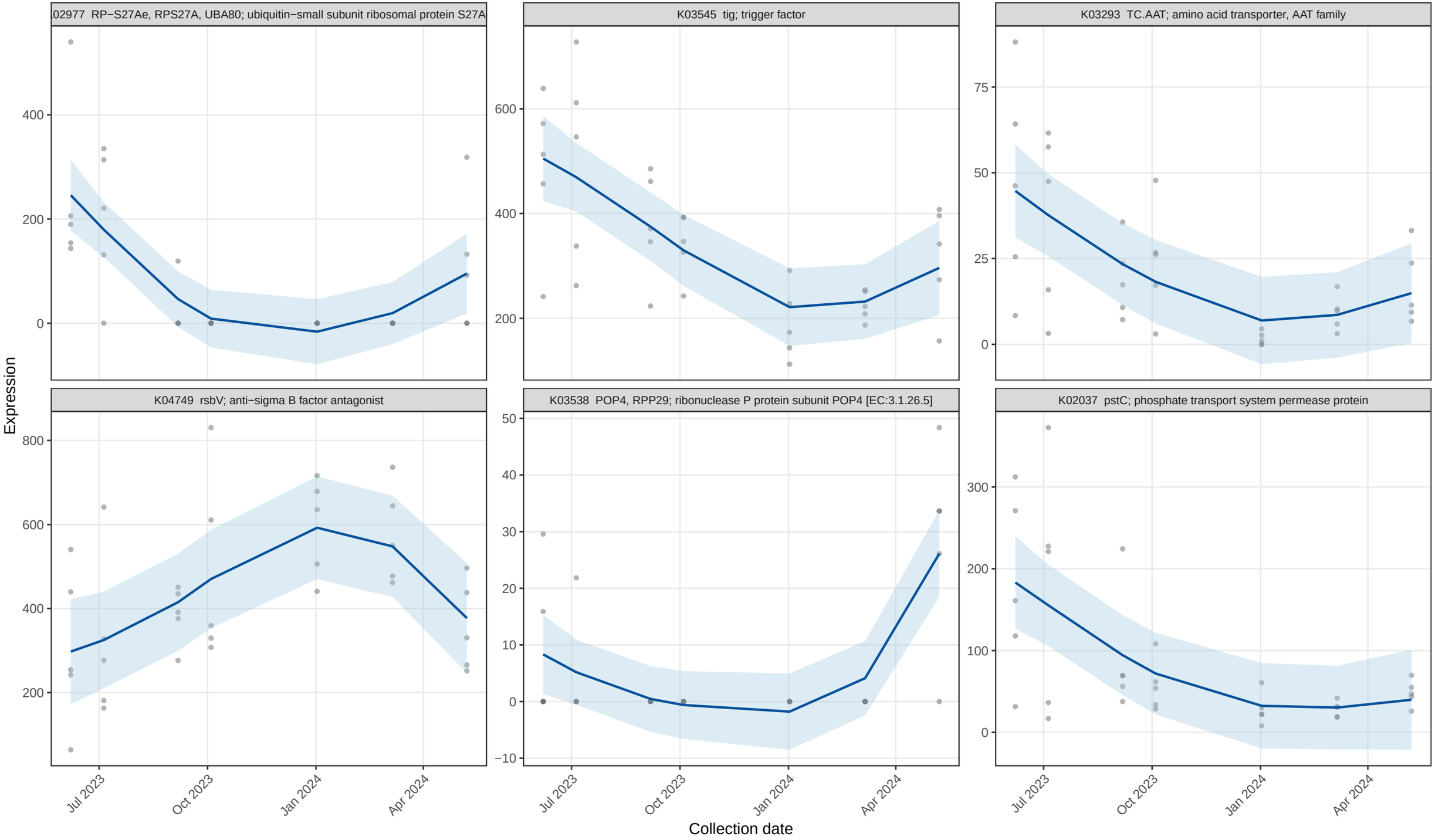

K01031 pcaI; 3−oxoadipate CoA−transferase, alpha subunit [EC:2.8.3.6]
L4 RPABC2, RPB6, POLR2F; DNA−directed RNA polymerases I, II, and III subunit RP
K18850 ycfD; 50S ribosomal protein L16 3−hydroxylase [EC:1.14.11.47]
K09565 PPIF; peptidyl−prolyl isomerase F (cyclophilin D) [EC:5.2.1.8]
K15268 eamA; O−acetylserine/cysteine efflux transporter
K20975 K20975; two−component system, sensor histidine kinase [EC:2.7.13.3]
Expression
Collection date
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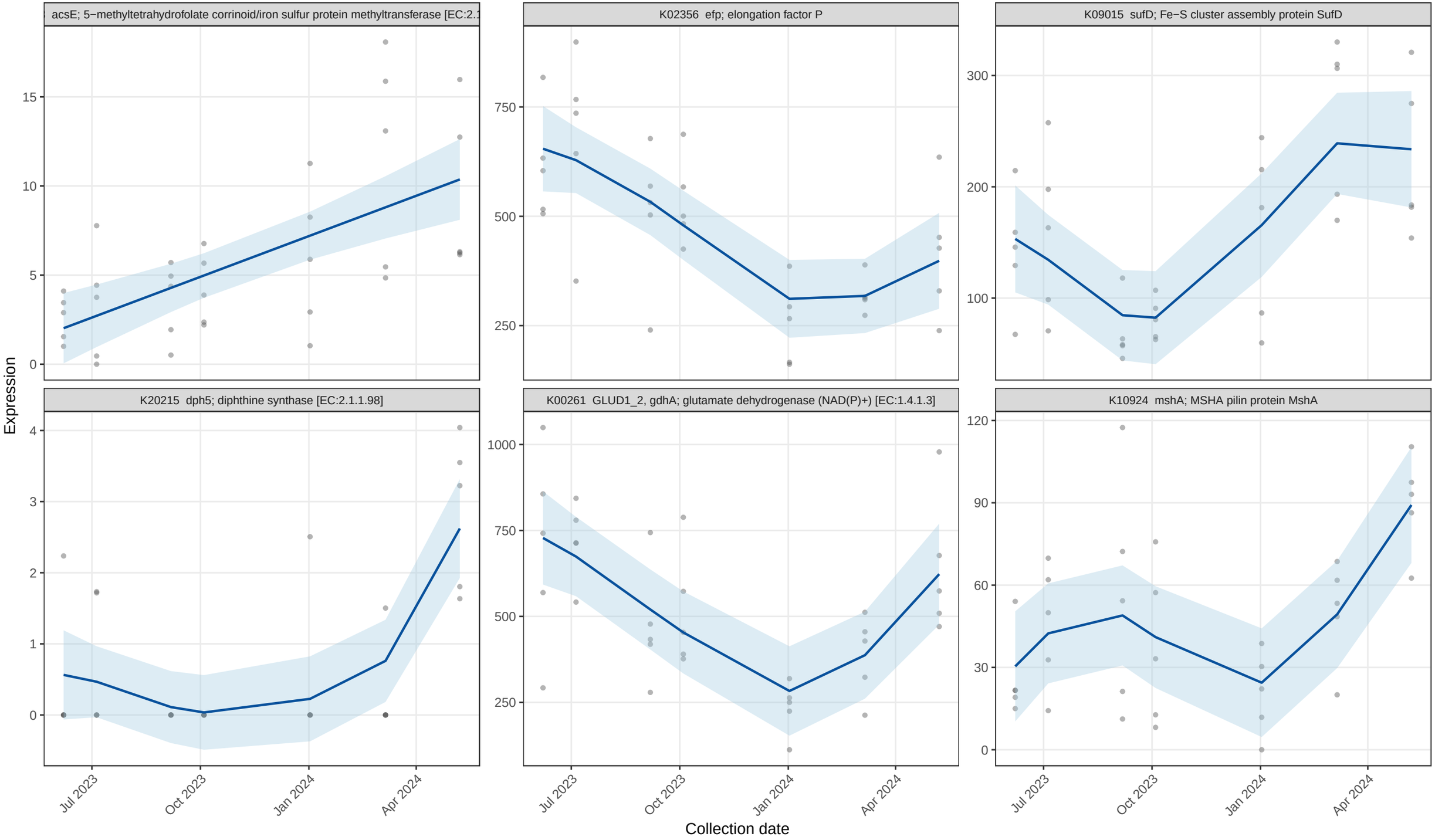

K02114 ATPF1E, atpC; F−type H+−transporting ATPase subunit epsilon
K00971 manC, cpsB; mannose−1−phosphate guanylyltransferase [EC:2.7.7.13]
K16091 fecA; Fe(3+) dicitrate transport protein
K02323 polB; DNA polymerase II small subunit [EC:2.7.7.7]
yl−CoA dehydrogenase / enoyl−CoA hydratase / 3−hydroxybutyryl−CoA epimerase [E
K03053 rpoH; DNA−directed RNA polymerase subunit H [EC:2.7.7.6]
Expression
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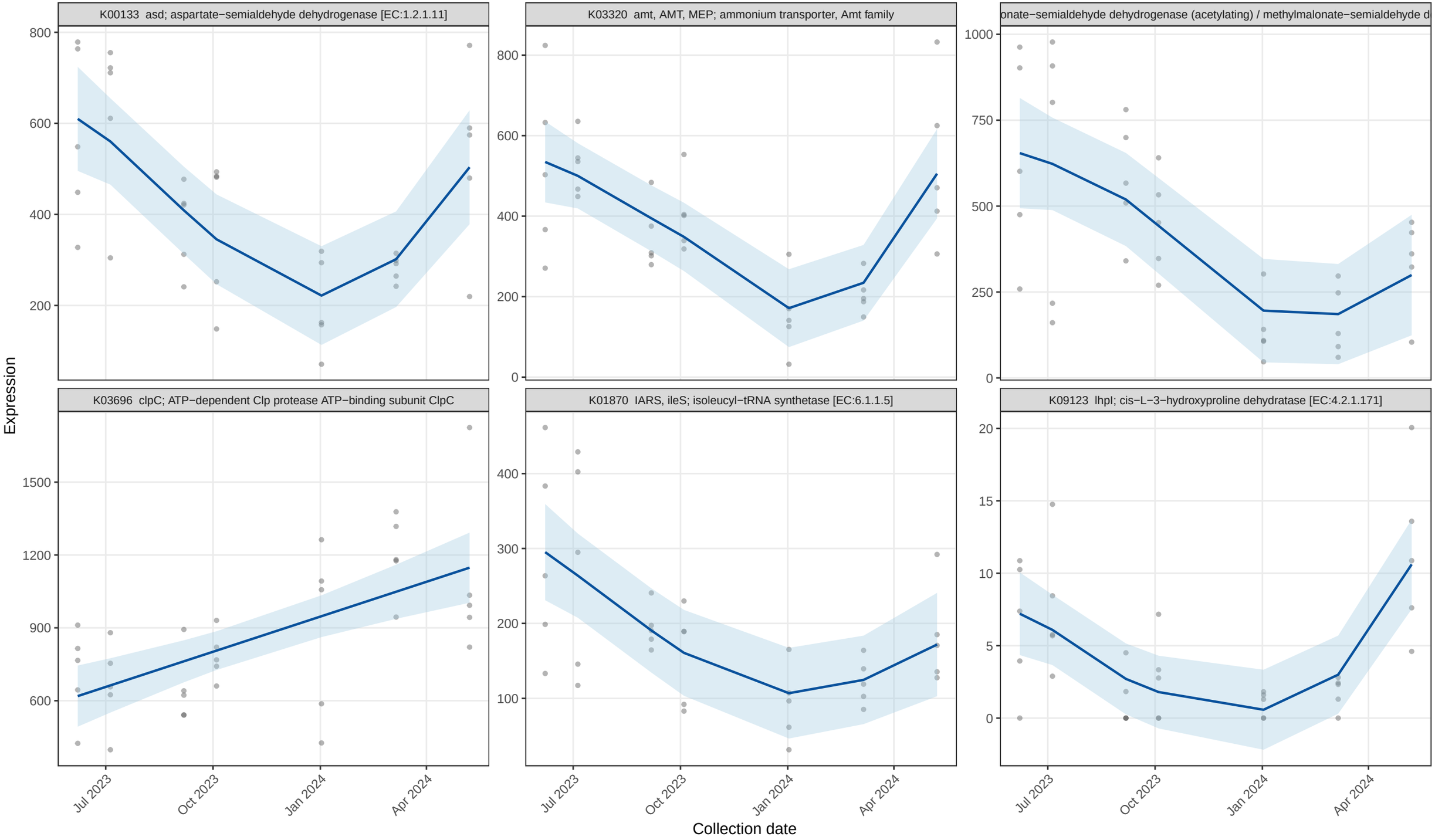

K02274 coxA, ctaD; cytochrome c oxidase subunit I [EC:7.1.1.9]
K02889 RP−L21e, RPL21; large subunit ribosomal protein L21e
K02914 RP−L34, MRPL34, rpmH; large subunit ribosomal protein L34
K00797 speE, SRM, SPE3; spermidine synthase [EC:2.5.1.16]
K02109 ATPF0B, atpF; F−type H+−transporting ATPase subunit b
K21310 mddA; methanethiol S−methyltransferase [EC:2.1.1.334]
Expression
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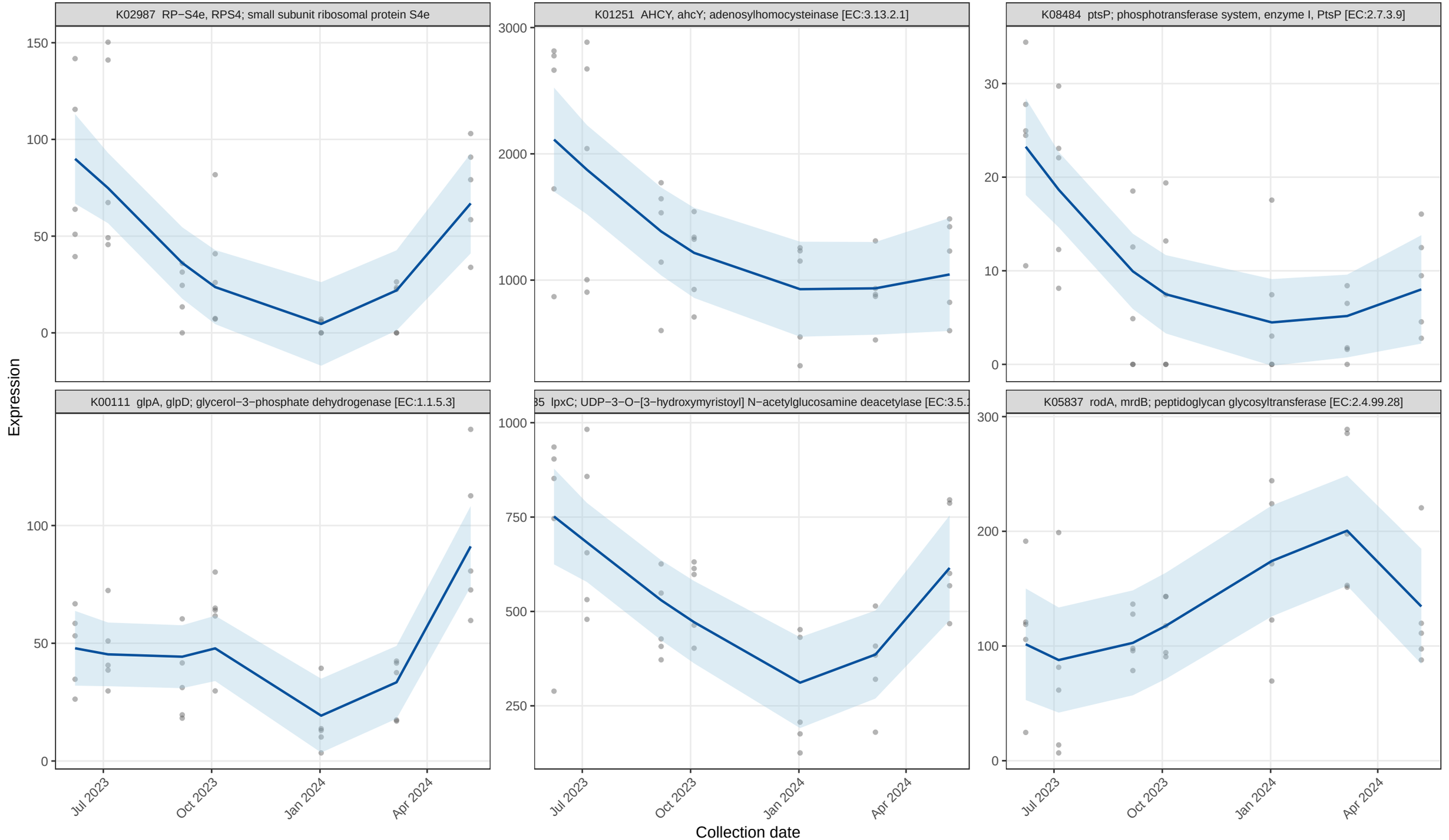

K03837 sdaC; serine transporter
K10979 ku; DNA end−binding protein Ku
K03297 emrE, qac, mmr, smr; small multidrug resistance pump
K02119 ATPVC, ntpC, atpC; V/A−type H+/Na+−transporting ATPase subunit C
626 E2.5.1.54, aroF, aroG, aroH; 3−deoxy−7−phosphoheptulonate synthase [EC:2.5.1
K13747 nspC; carboxynorspermidine decarboxylase [EC:4.1.1.96]
Expression
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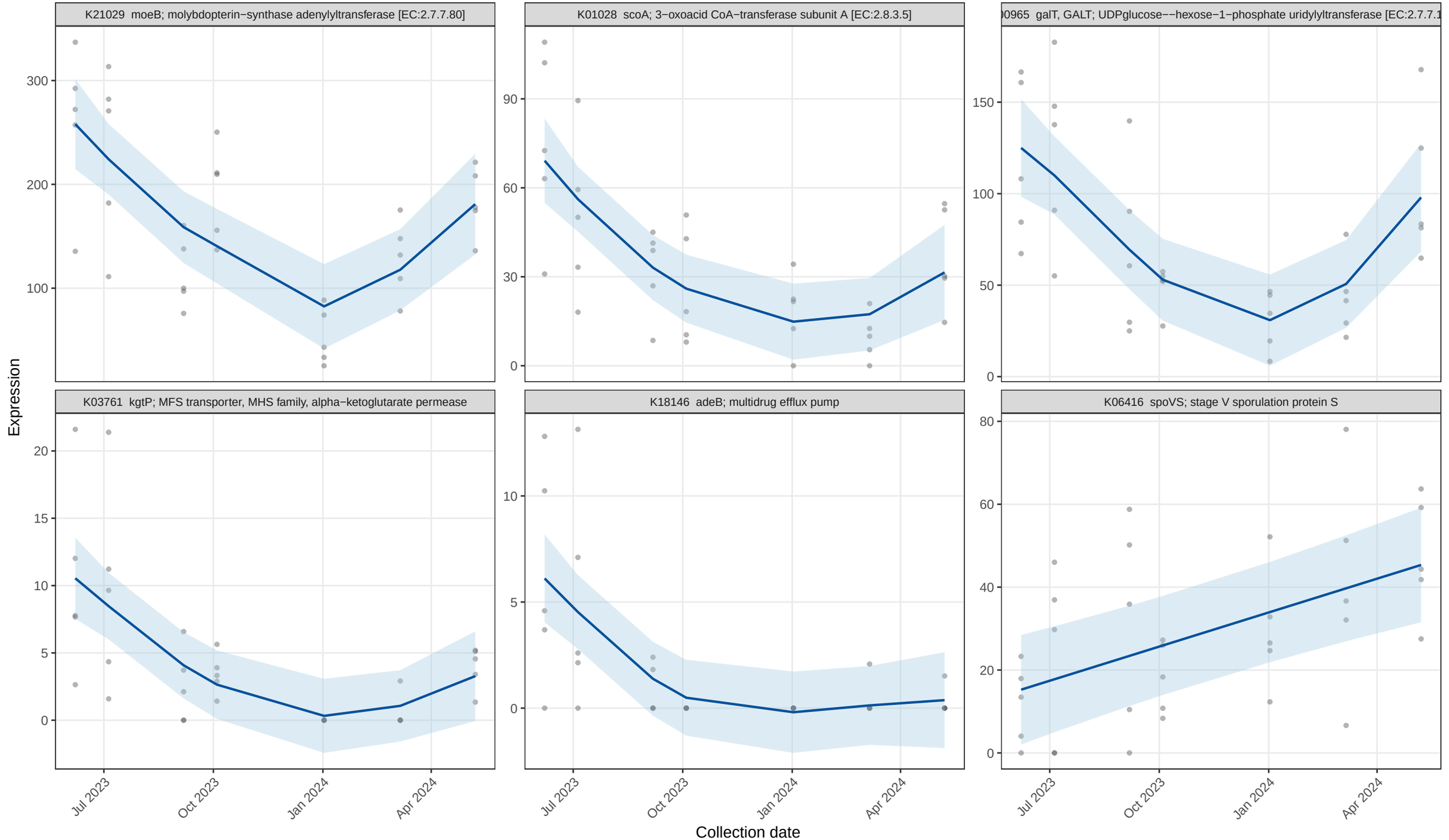

K11600 RRP41, EXOSC4, SKI6; exosome complex component RRP41
K18532 AK6, FAP7; adenylate kinase [EC:2.7.4.3]
K02108 ATPF0A, atpB; F−type H+−transporting ATPase subunit a
K02029 ABC.PA.P; polar amino acid transport system permease protein
K06200 cstA; carbon starvation protein
K00832 tyrB; aromatic−amino−acid transaminase [EC:2.6.1.57]
Expression
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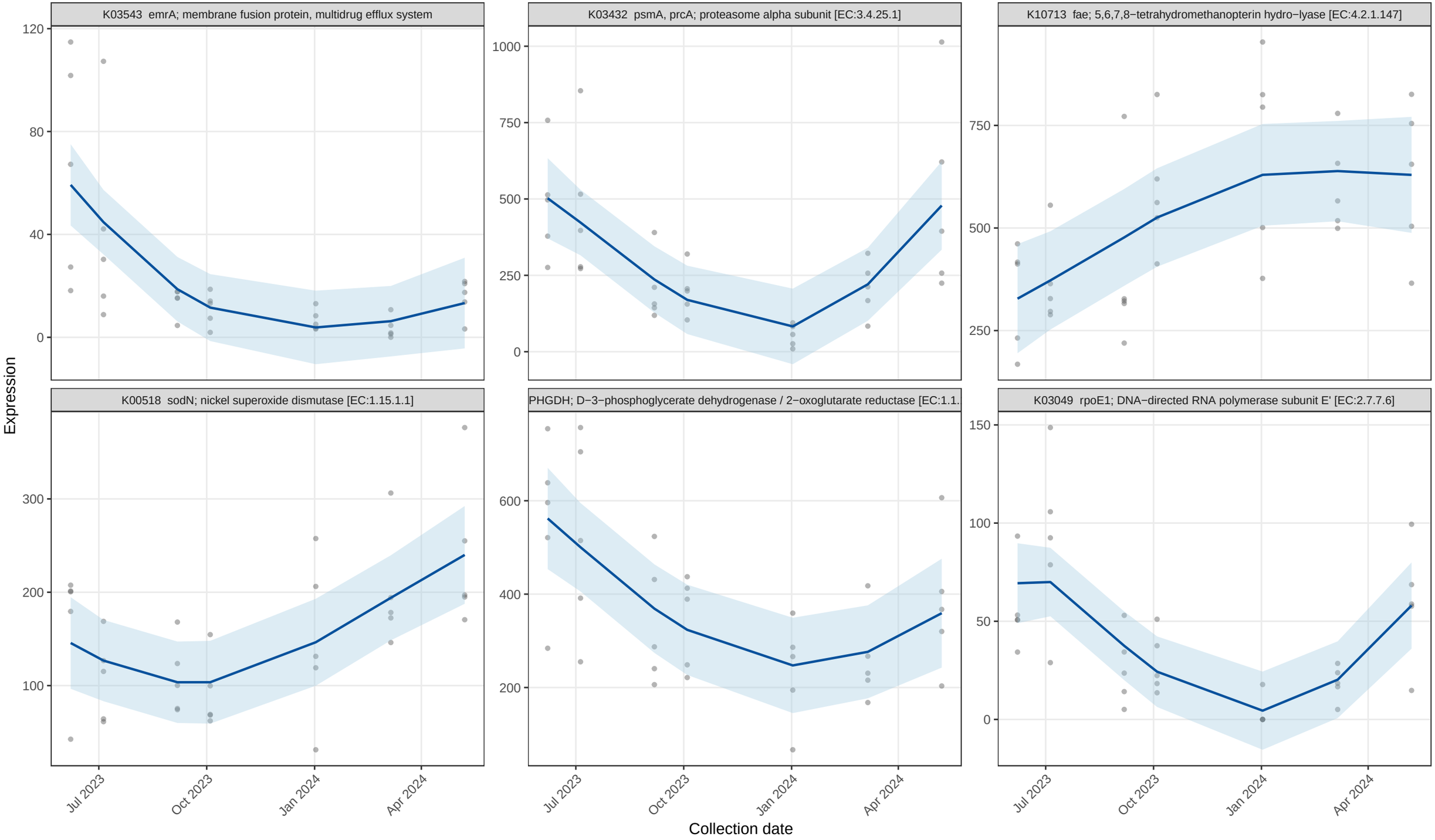

02977 RP−S27Ae, RPS27A, UBA80; ubiquitin−small subunit ribosomal protein S27A
K03545 tig; trigger factor
K03293 TC.AAT; amino acid transporter, AAT family
K04749 rsbV; anti−sigma B factor antagonist
K03538 POP4, RPP29; ribonuclease P protein subunit POP4 [EC:3.1.26.5]
K02037 pstC; phosphate transport system permease protein
Expression
Collection date
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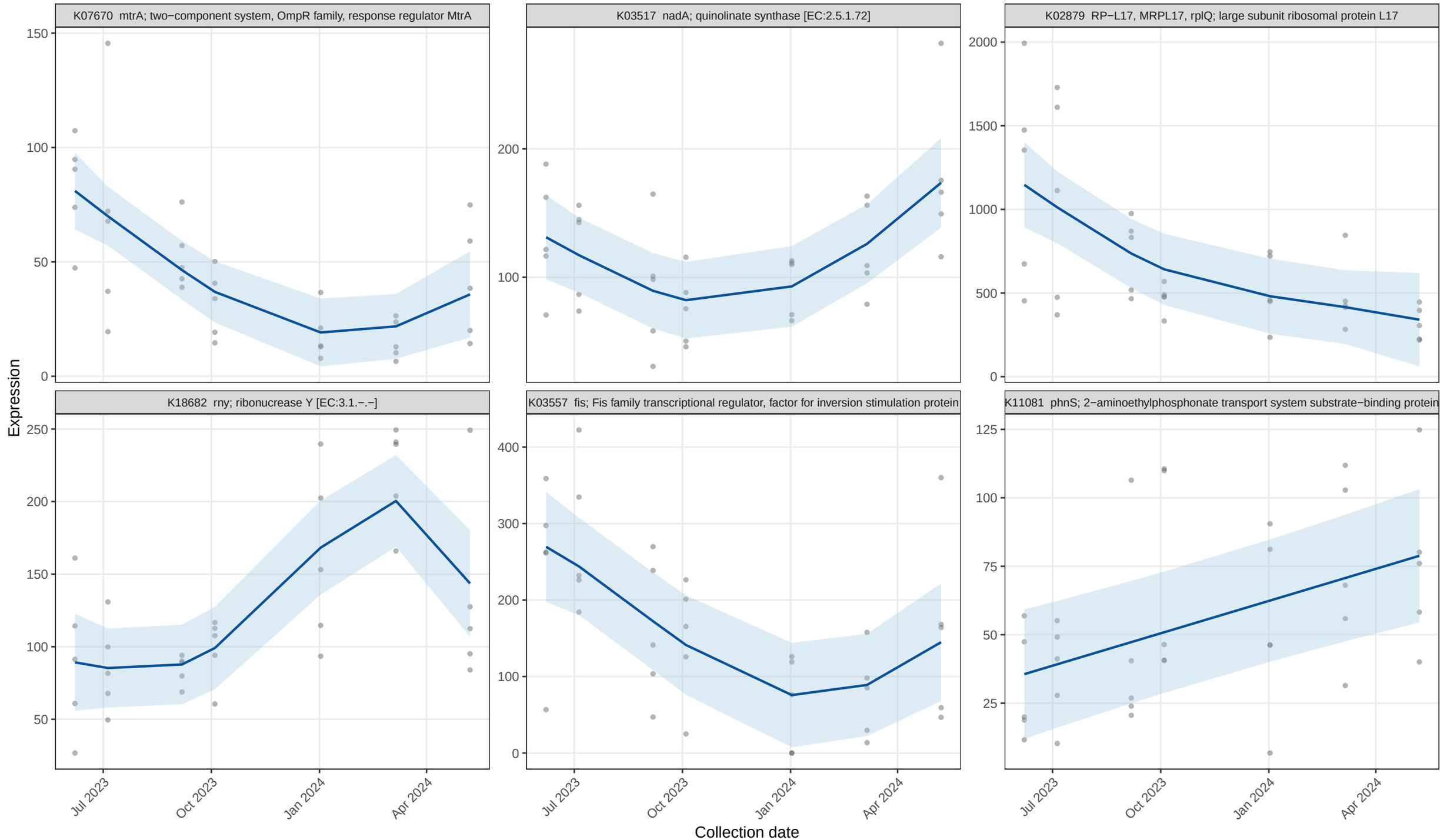

acsE; 5−methyltetrahydrofolate corrinoid/iron sulfur protein methyltransferase [EC:2.1
K02356 efp; elongation factor P
K09015 sufD; Fe−S cluster assembly protein SufD
K20215 dph5; diphthine synthase [EC:2.1.1.98]
K00261 GLUD1_2, gdhA; glutamate dehydrogenase (NAD(P)+) [EC:1.4.1.3]
K10924 mshA; MSHA pilin protein MshA
Expression
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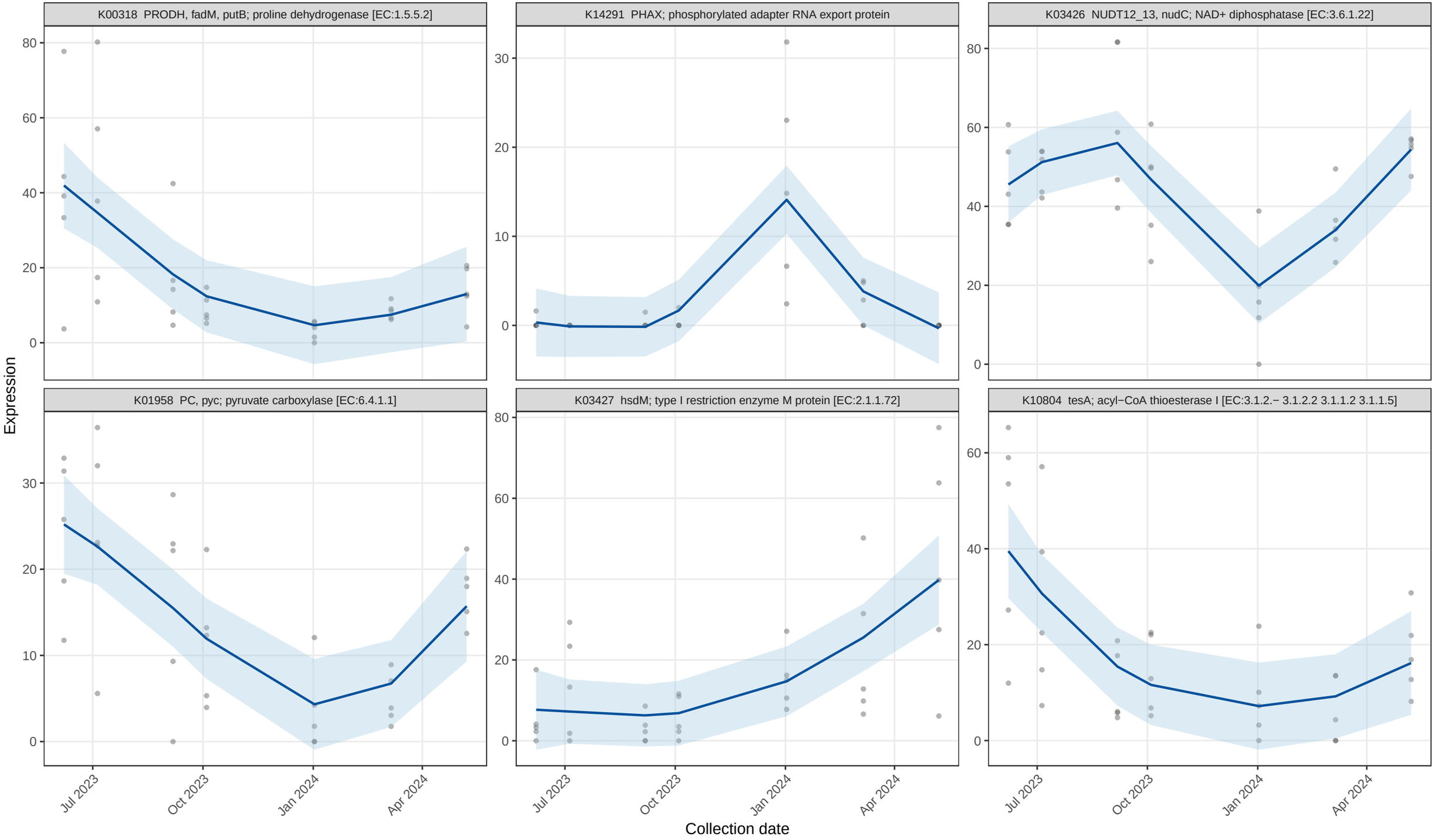

K00133 asd; aspartate−semialdehyde dehydrogenase [EC:1.2.1.11]
K03320 amt, AMT, MEP; ammonium transporter, Amt family
onate−semialdehyde dehydrogenase (acetylating) / methylmalonate−semialdehyde d
K03696 clpC; ATP−dependent Clp protease ATP−binding subunit ClpC
K01870 IARS, ileS; isoleucyl−tRNA synthetase [EC:6.1.1.5]
K09123 lhpI; cis−L−3−hydroxyproline dehydratase [EC:4.2.1.171]
Expression
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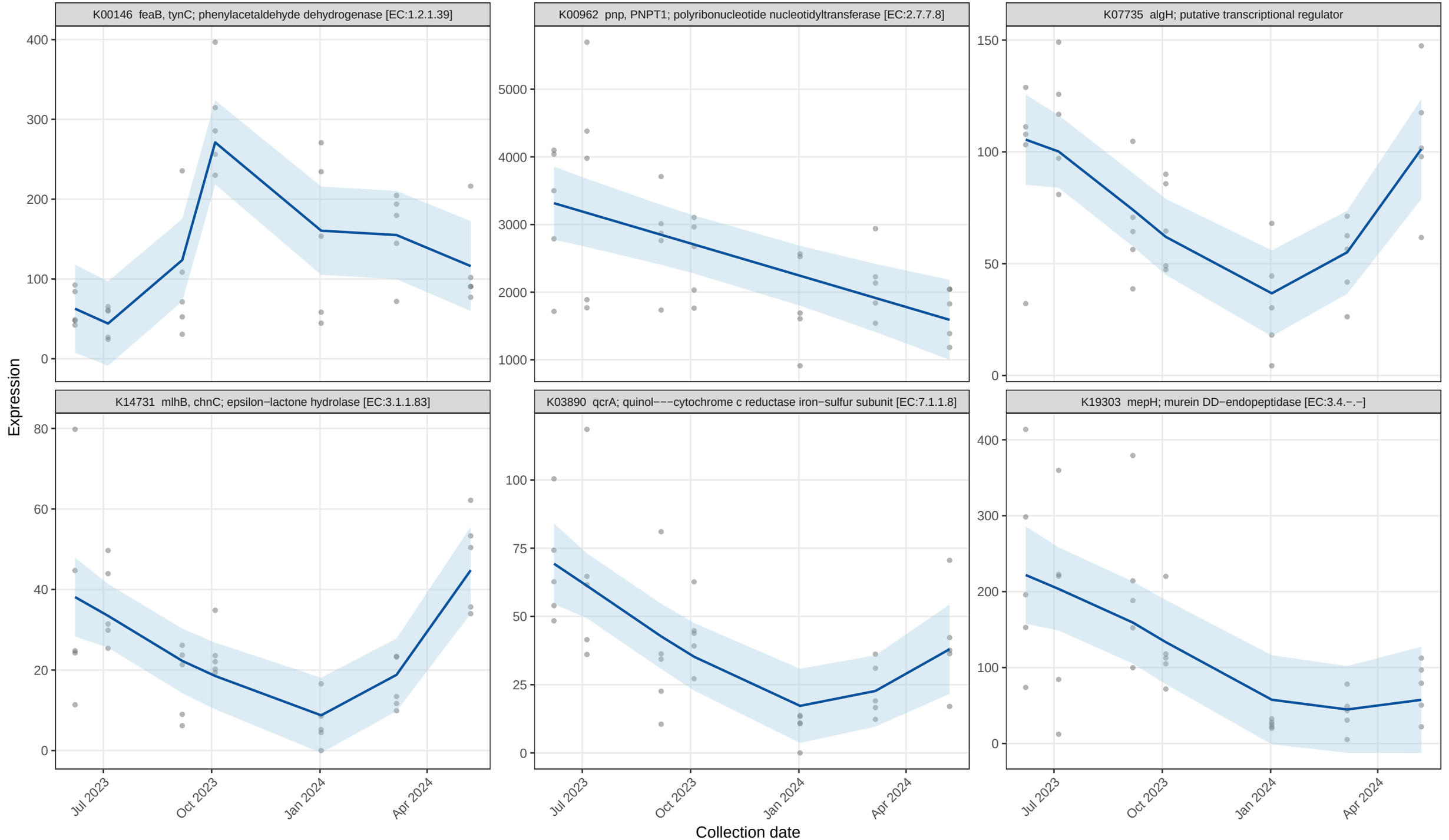

K02987 RP−S4e, RPS4; small subunit ribosomal protein S4e
K01251 AHCY, ahcY; adenosylhomocysteinase [EC:3.13.2.1]
K08484 ptsP; phosphotransferase system, enzyme I, PtsP [EC:2.7.3.9]
K00111 glpA, glpD; glycerol−3−phosphate dehydrogenase [EC:1.1.5.3]
lpxC; UDP−3−O−[3−hydroxymyristoyl] N−acetylglucosamine deacetylase [EC:3.5.
K05837 rodA, mrdB; peptidoglycan glycosyltransferase [EC:2.4.99.28]
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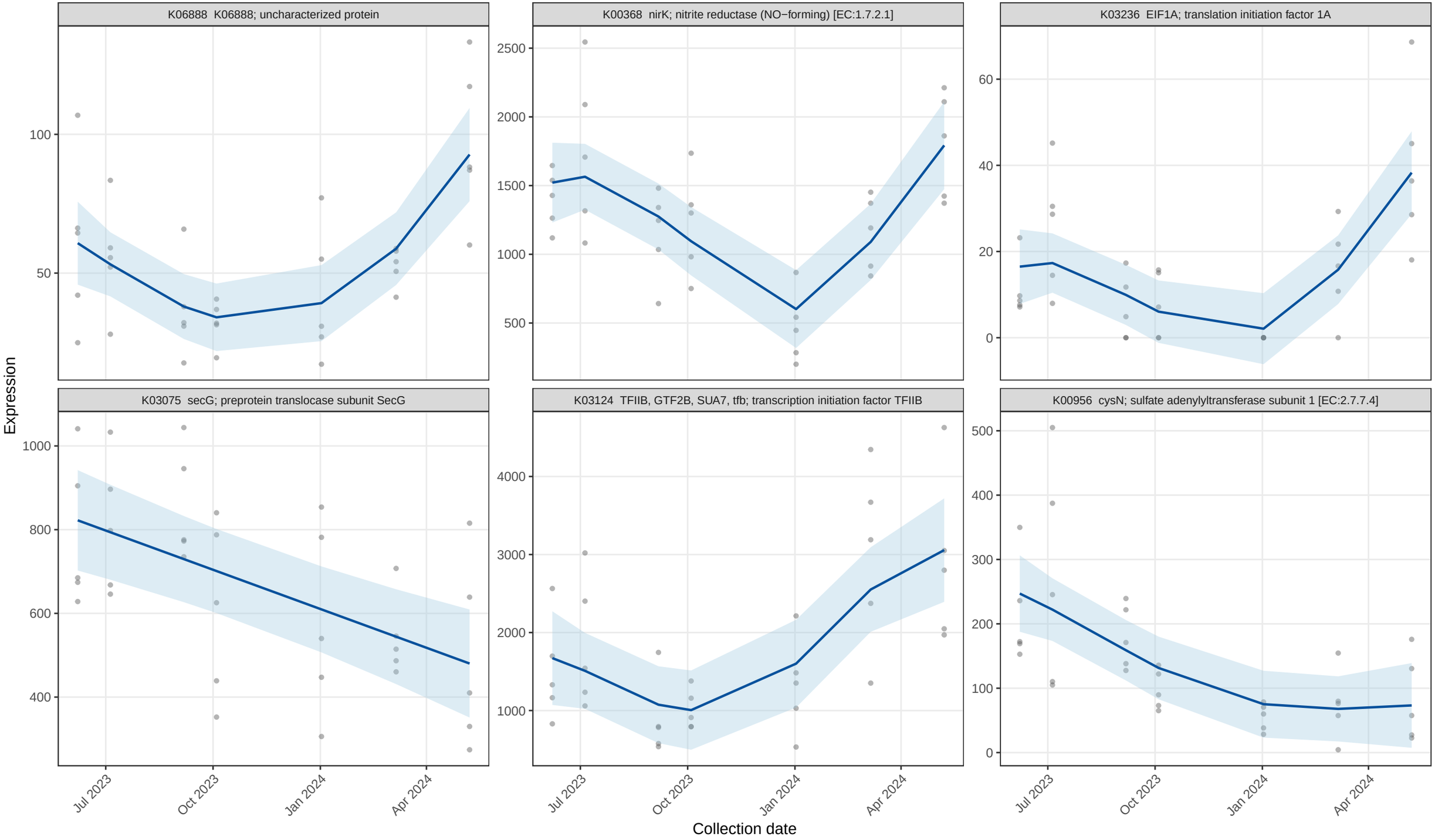

K21029 moeB; molybdopterin−synthase adenylyltransferase [EC:2.7.7.80]
K01028 scoA; 3−oxoacid CoA−transferase subunit A [EC:2.8.3.5]
0965 galT, GALT; UDPglucose−−hexose−1−phosphate uridylyltransferase [EC:2.7.7.1
K03761 kgtP; MFS transporter, MHS family, alpha−ketoglutarate permease
K18146 adeB; multidrug efflux pump
K06416 spoVS; stage V sporulation protein S
Expression
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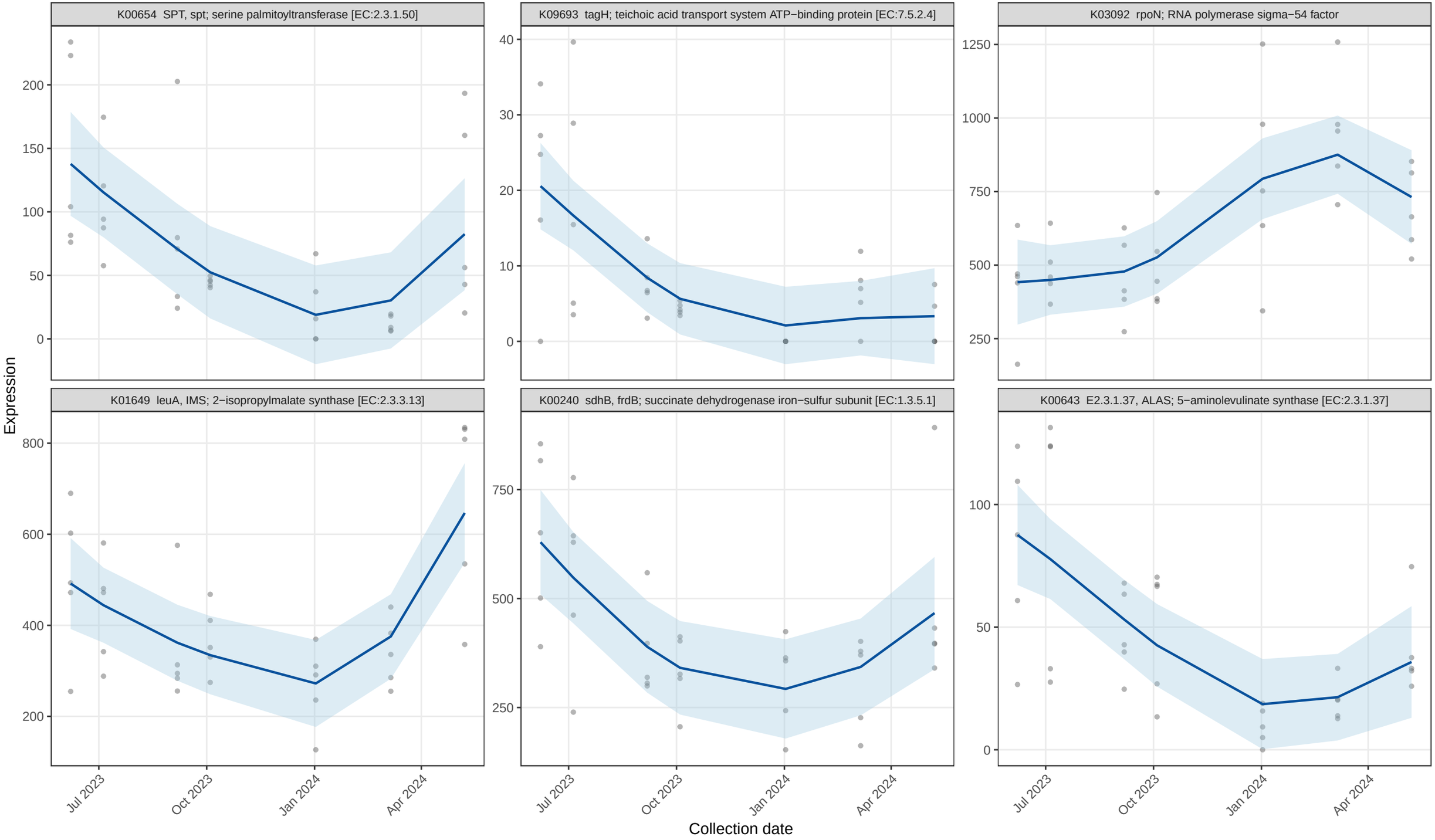

K03543 emrA; membrane fusion protein, multidrug efflux system
K03432 psmA, prcA; proteasome alpha subunit [EC:3.4.25.1]
K10713 fae; 5,6,7,8−tetrahydromethanopterin hydro−lyase [EC:4.2.1.147]
K00518 sodN; nickel superoxide dismutase [EC:1.15.1.1]
PHGDH; D−3−phosphoglycerate dehydrogenase / 2−oxoglutarate reductase [EC:1.1.
K03049 rpoE1; DNA−directed RNA polymerase subunit E' [EC:2.7.7.6]
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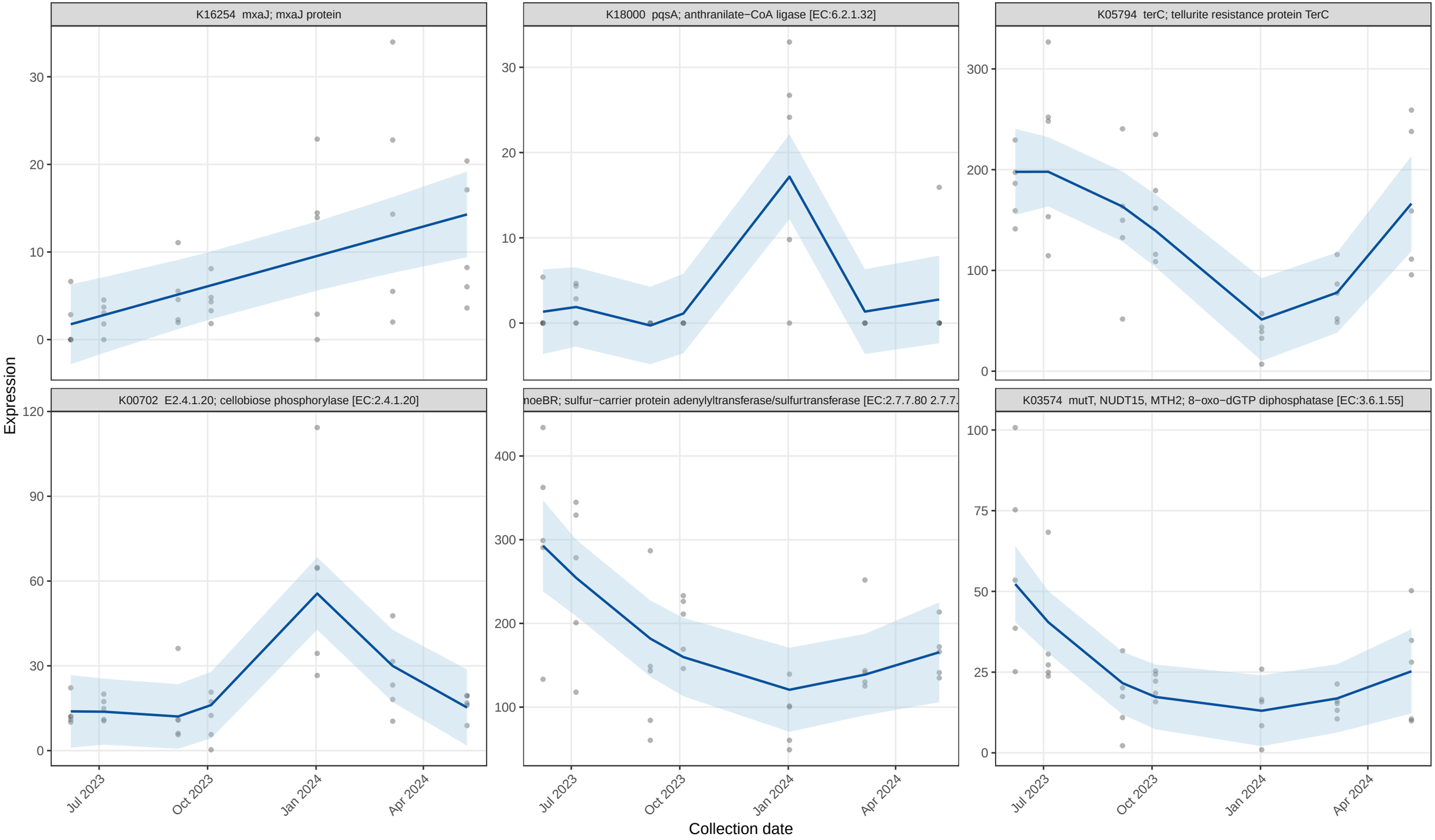

K07670 mtrA; two−component system, OmpR family, response regulator MtrA
K03517 nadA; quinolinate synthase [EC:2.5.1.72]
K02879 RP−L17, MRPL17, rplQ; large subunit ribosomal protein L17
K18682 rny; ribonucrease Y [EC:3.1.−.−]
K03557 fis; Fis family transcriptional regulator, factor for inversion stimulation protein
K11081 phnS; 2−aminoethylphosphonate transport system substrate−binding protein
Expression
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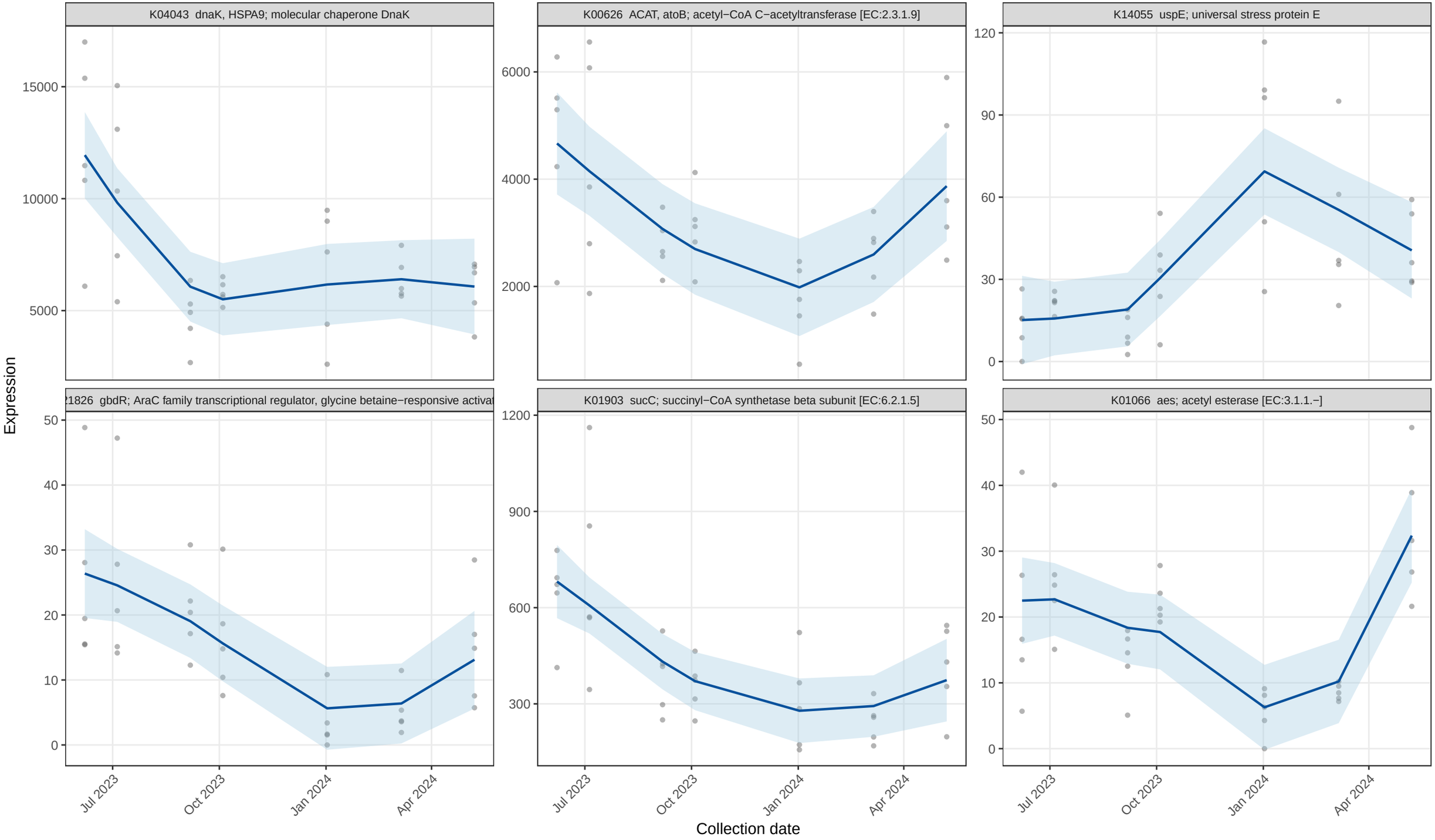

K00318 PRODH, fadM, putB; proline dehydrogenase [EC:1.5.5.2]
K14291 PHAX; phosphorylated adapter RNA export protein
K03426 NUDT12_13, nudC; NAD+ diphosphatase [EC:3.6.1.22]
K01958 PC, pyc; pyruvate carboxylase [EC:6.4.1.1]
K03427 hsdM; type I restriction enzyme M protein [EC:2.1.1.72]
K10804 tesA; acyl−CoA thioesterase I [EC:3.1.2.− 3.1.2.2 3.1.1.2 3.1.1.5]
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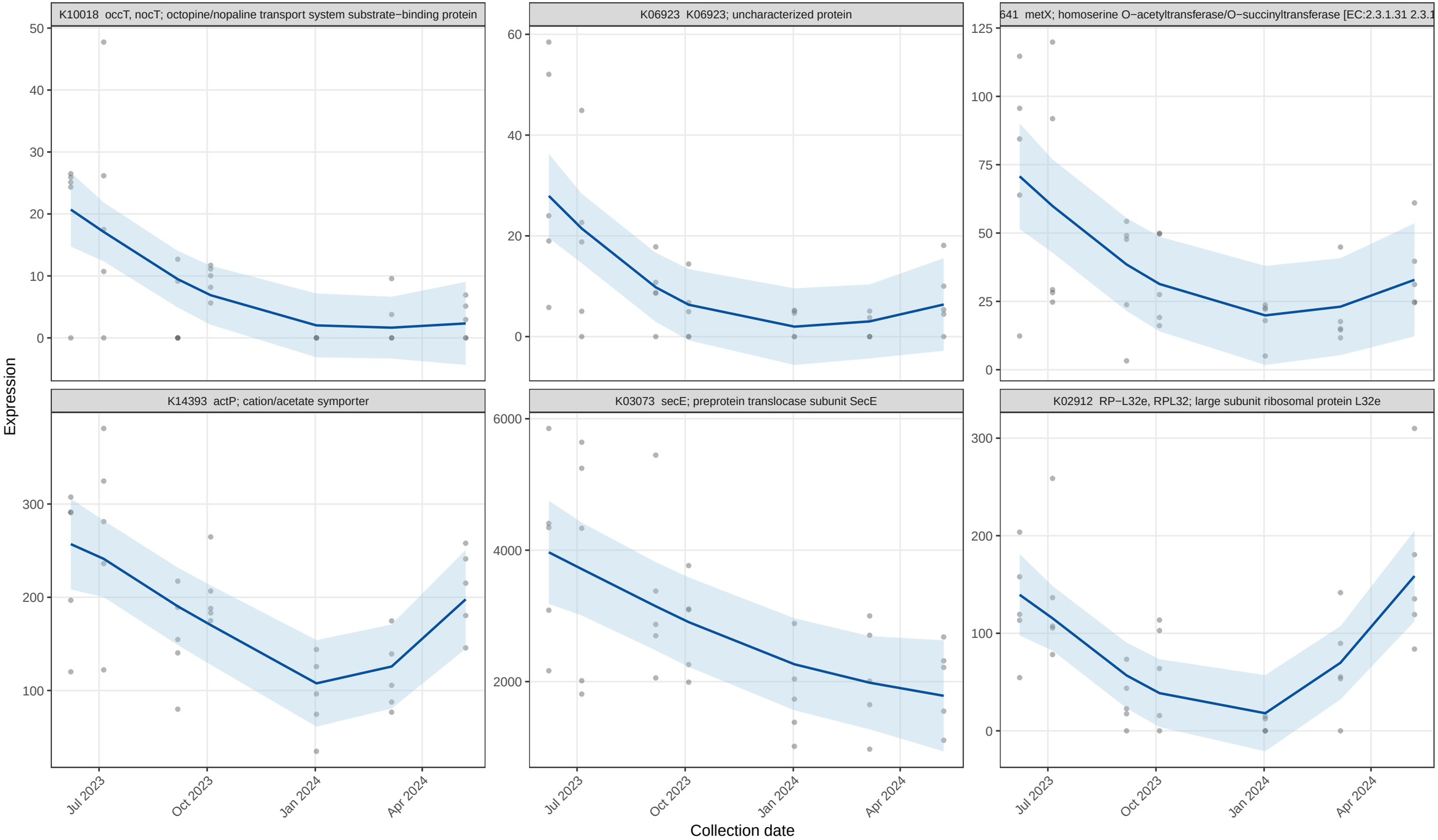

K00146 feaB, tynC; phenylacetaldehyde dehydrogenase [EC:1.2.1.39]
K00962 pnp, PNPT1; polyribonucleotide nucleotidyltransferase [EC:2.7.7.8]
K07735 algH; putative transcriptional regulator
K14731 mlhB, chnC; epsilon−lactone hydrolase [EC:3.1.1.83]
K03890 qcrA; quinol−−−cytochrome c reductase iron−sulfur subunit [EC:7.1.1.8]
K19303 mepH; murein DD−endopeptidase [EC:3.4.−.−]
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

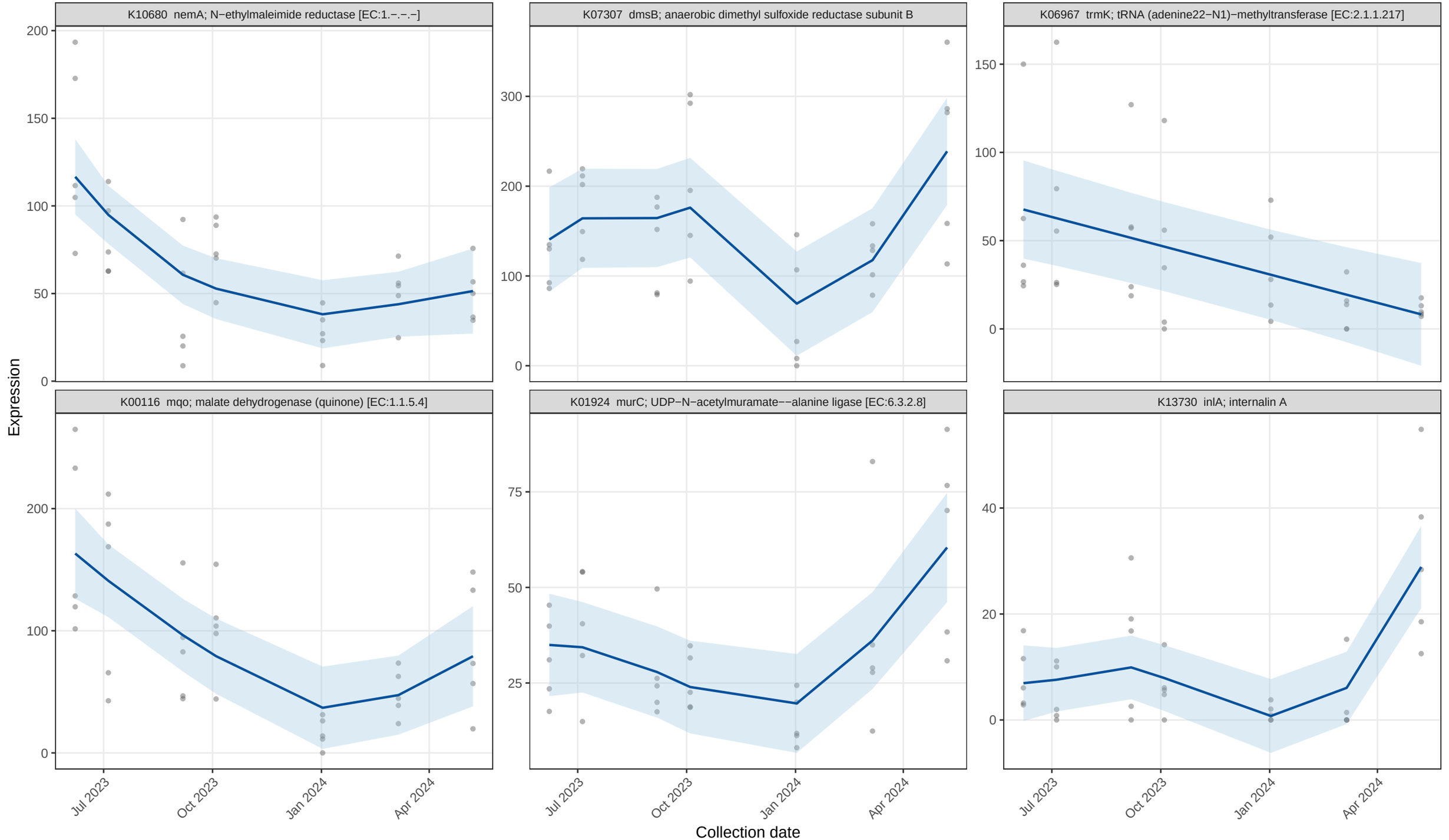

K06888 K06888; uncharacterized protein
K00368 nirK; nitrite reductase (NO−forming) [EC:1.7.2.1]
K03236 EIF1A; translation initiation factor 1A
K03075 secG; preprotein translocase subunit SecG
K03124 TFIIB, GTF2B, SUA7, tfb; transcription initiation factor TFIIB
K00956 cysN; sulfate adenylyltransferase subunit 1 [EC:2.7.7.4]
Expression
Collection date
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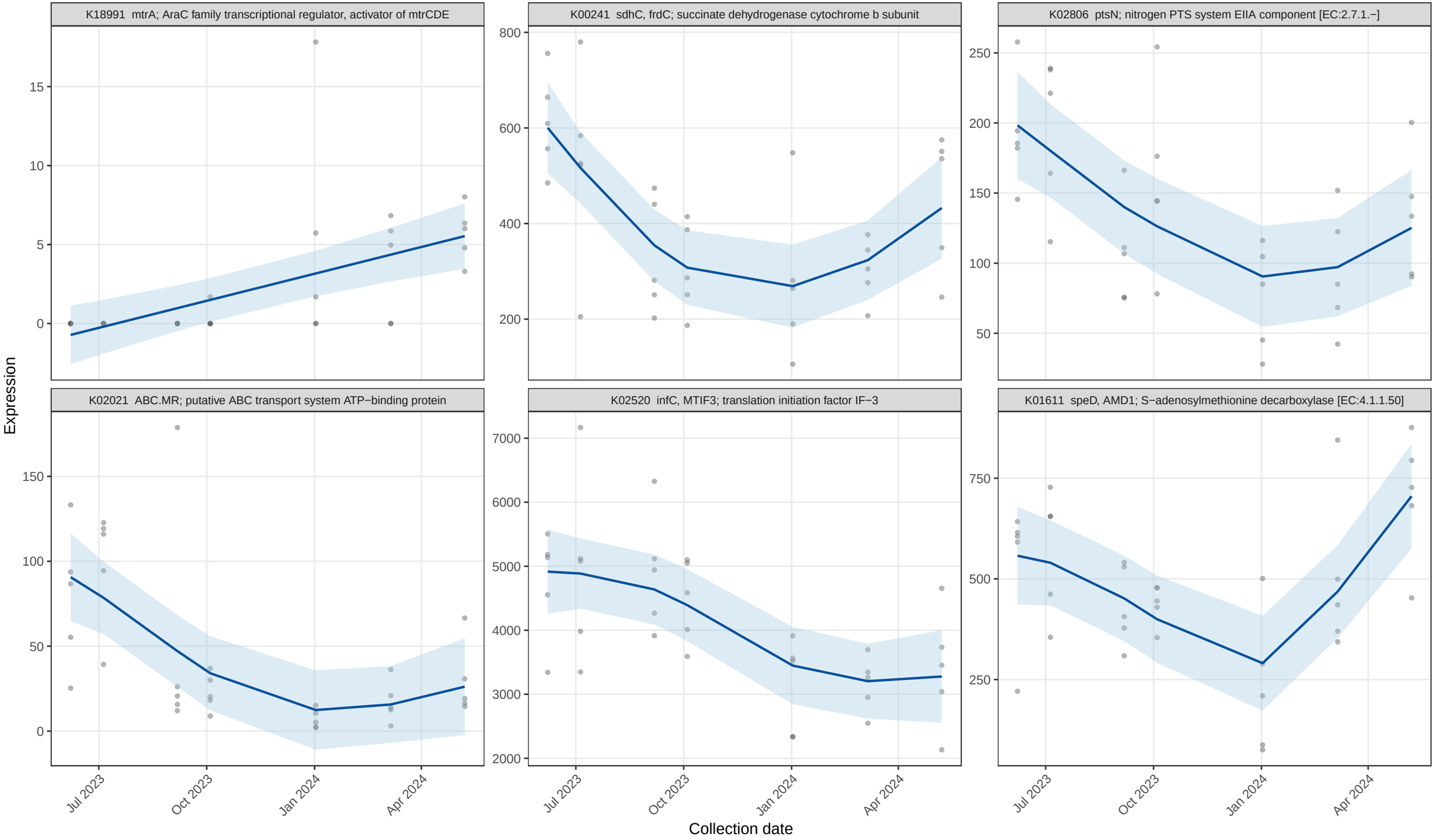

K00654 SPT, spt; serine palmitoyltransferase [EC:2.3.1.50]
K09693 tagH; teichoic acid transport system ATP−binding protein [EC:7.5.2.4]
K03092 rpoN; RNA polymerase sigma−54 factor
K01649 leuA, IMS; 2−isopropylmalate synthase [EC:2.3.3.13]
K00240 sdhB, frdB; succinate dehydrogenase iron−sulfur subunit [EC:1.3.5.1]
K00643 E2.3.1.37, ALAS; 5−aminolevulinate synthase [EC:2.3.1.37]
Expression
Jul 2023
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Apr 2024
Collection date

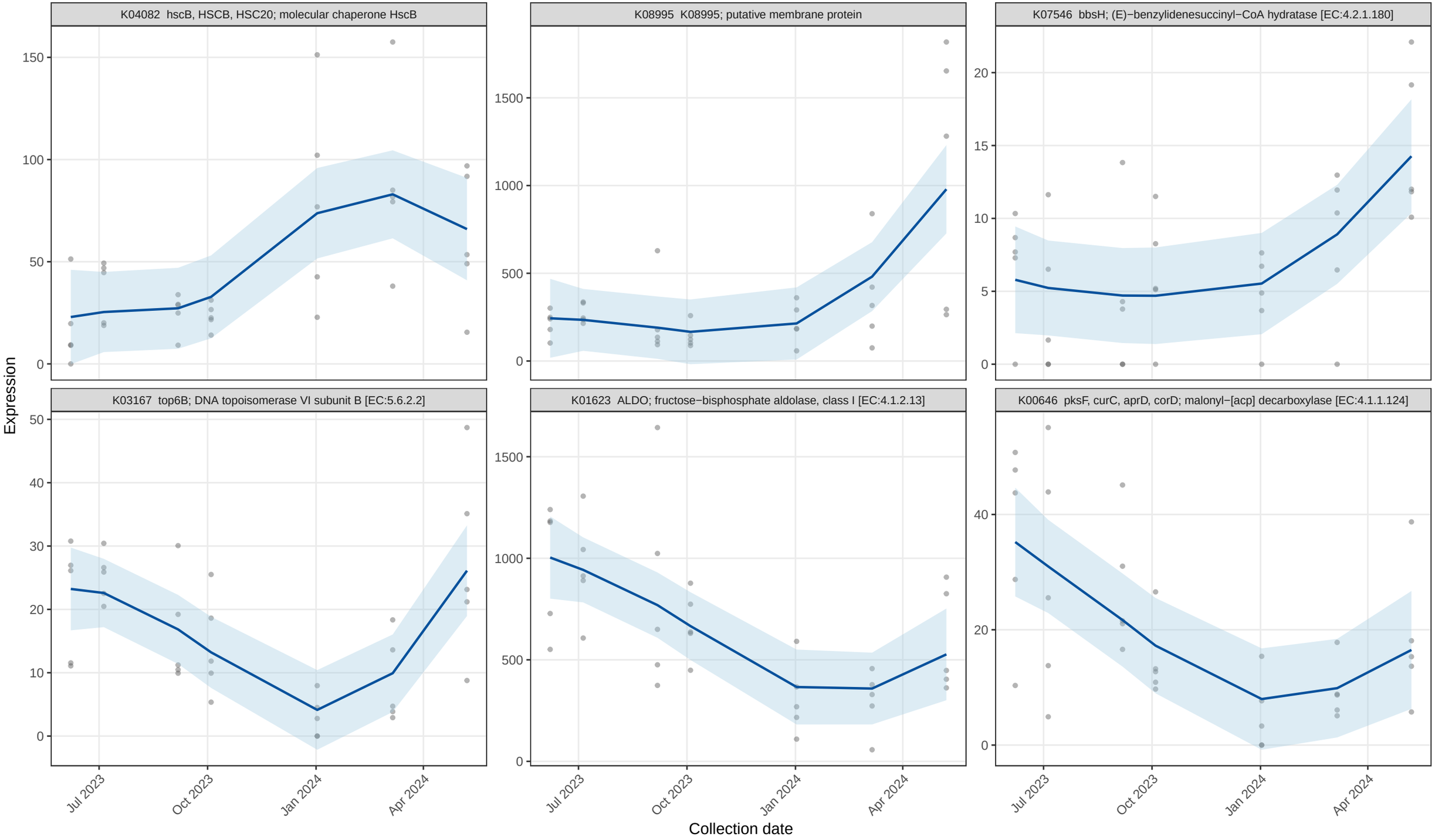

K16254 mxaJ; mxaJ protein
K18000 pqsA; anthranilate−CoA ligase [EC:6.2.1.32]
K05794 terC; tellurite resistance protein TerC
K00702 E2.4.1.20; cellobiose phosphorylase [EC:2.4.1.20]
noeBR; sulfur−carrier protein adenylyltransferase/sulfurtransferase [EC:2.7.7.80 2.7.7.
K03574 mutT, NUDT15, MTH2; 8−oxo−dGTP diphosphatase [EC:3.6.1.55]
Expression
Collection date
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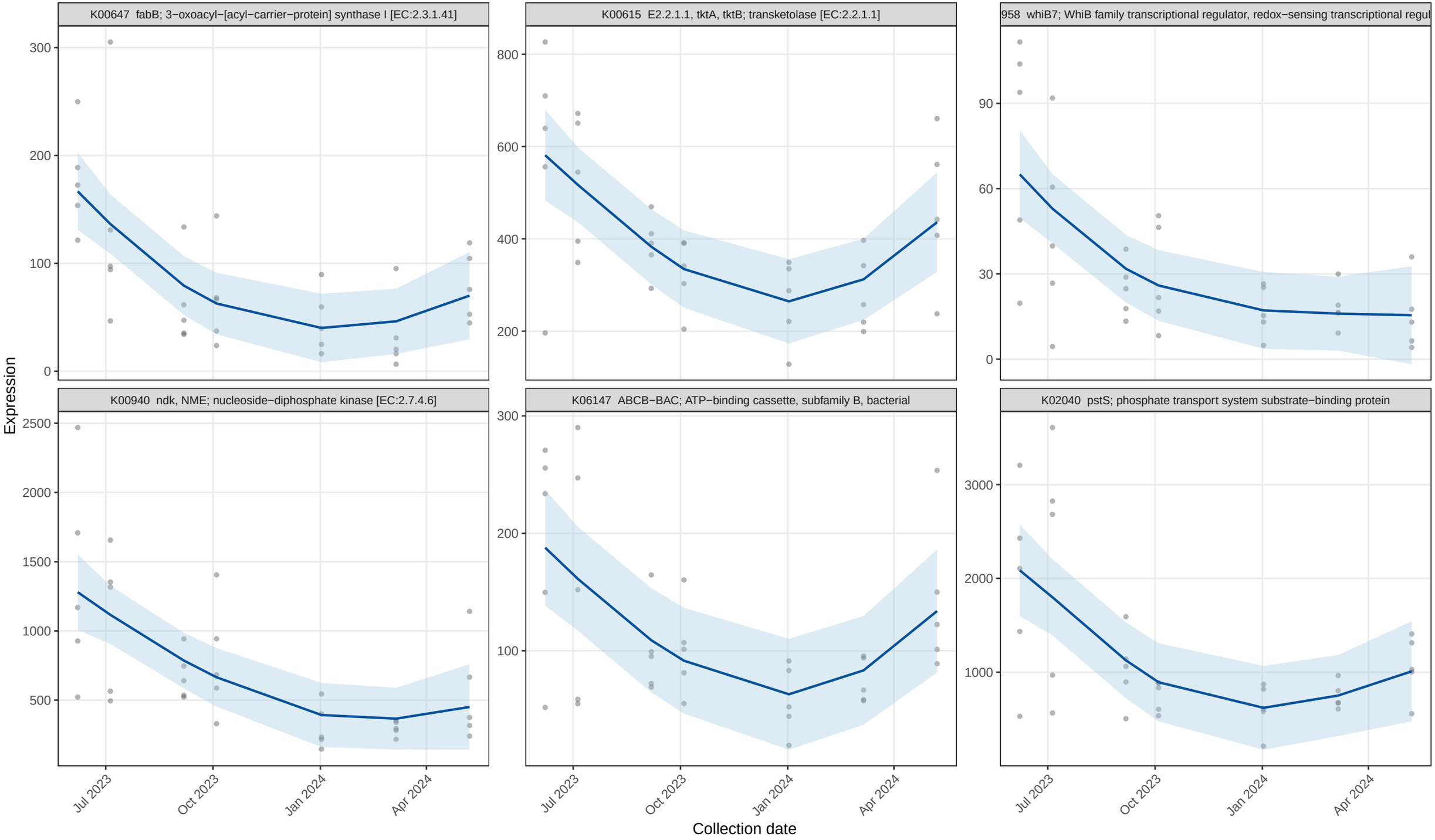

K04043 dnaK, HSPA9; molecular chaperone DnaK
K00626 ACAT, atoB; acetyl−CoA C−acetyltransferase [EC:2.3.1.9]
K14055 uspE; universal stress protein E
1826 gbdR; AraC family transcriptional regulator, glycine betaine−responsive activat
K01903 sucC; succinyl−CoA synthetase beta subunit [EC:6.2.1.5]
K01066 aes; acetyl esterase [EC:3.1.1.−]
Expression
Collection date
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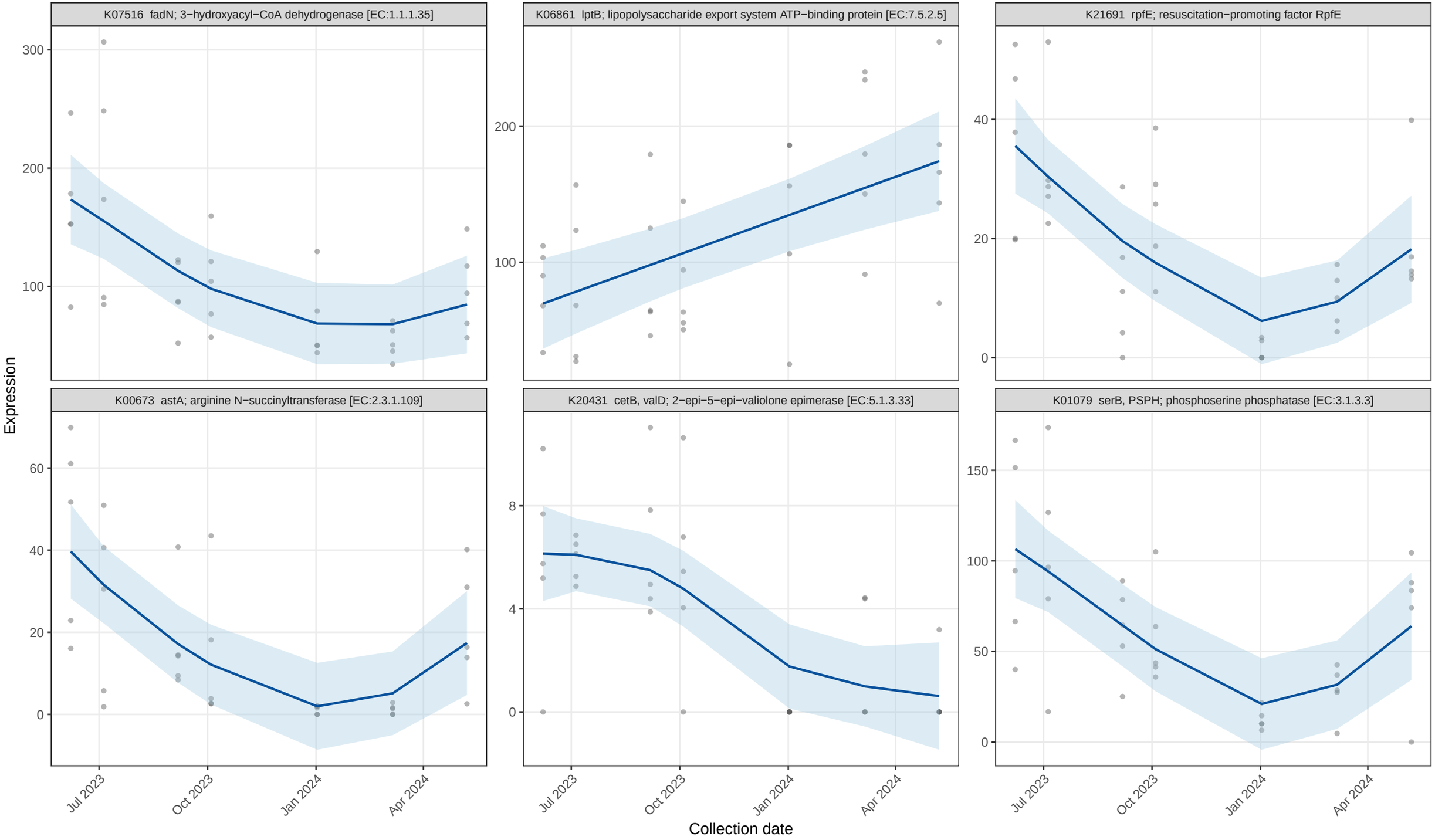

K10018 occT, nocT; octopine/nopaline transport system substrate−binding protein
K06923 K06923; uncharacterized protein
641 metX; homoserine O−acetyltransferase/O−succinyltransferase [EC:2.3.1.31 2.3.1
K14393 actP; cation/acetate symporter
K03073 secE; preprotein translocase subunit SecE
K02912 RP−L32e, RPL32; large subunit ribosomal protein L32e
Expression
Collection date
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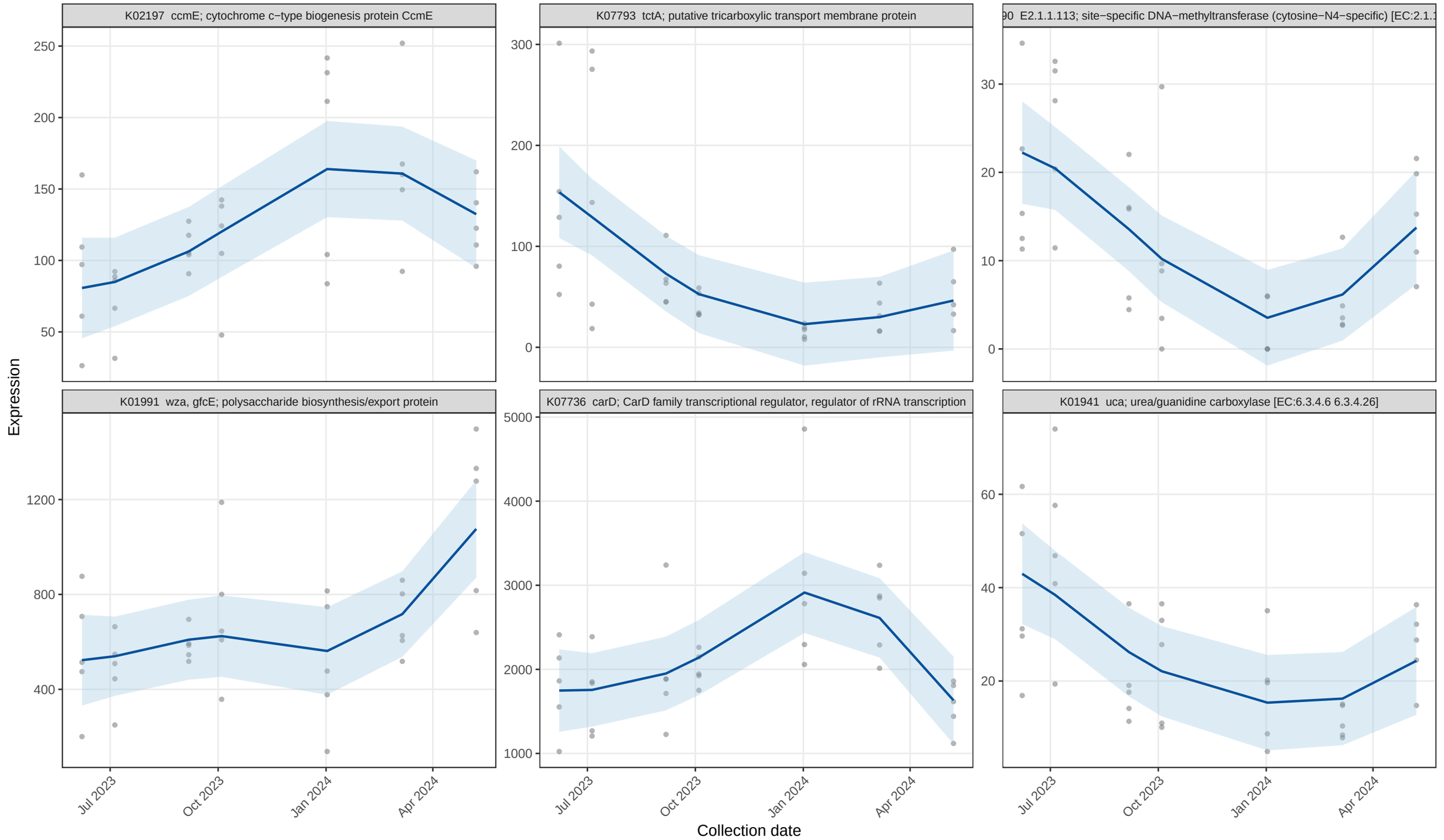

K10680 nemA; N−ethylmaleimide reductase [EC:1.−.−.−]
K07307 dmsB; anaerobic dimethyl sulfoxide reductase subunit B
K06967 trmK; tRNA (adenine22−N1)−methyltransferase [EC:2.1.1.217]
K00116 mqo; malate dehydrogenase (quinone) [EC:1.1.5.4]
K01924 murC; UDP−N−acetylmuramate−−alanine ligase [EC:6.3.2.8]
K13730 inlA; internalin A
Expression
Collection date
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Apr 2024

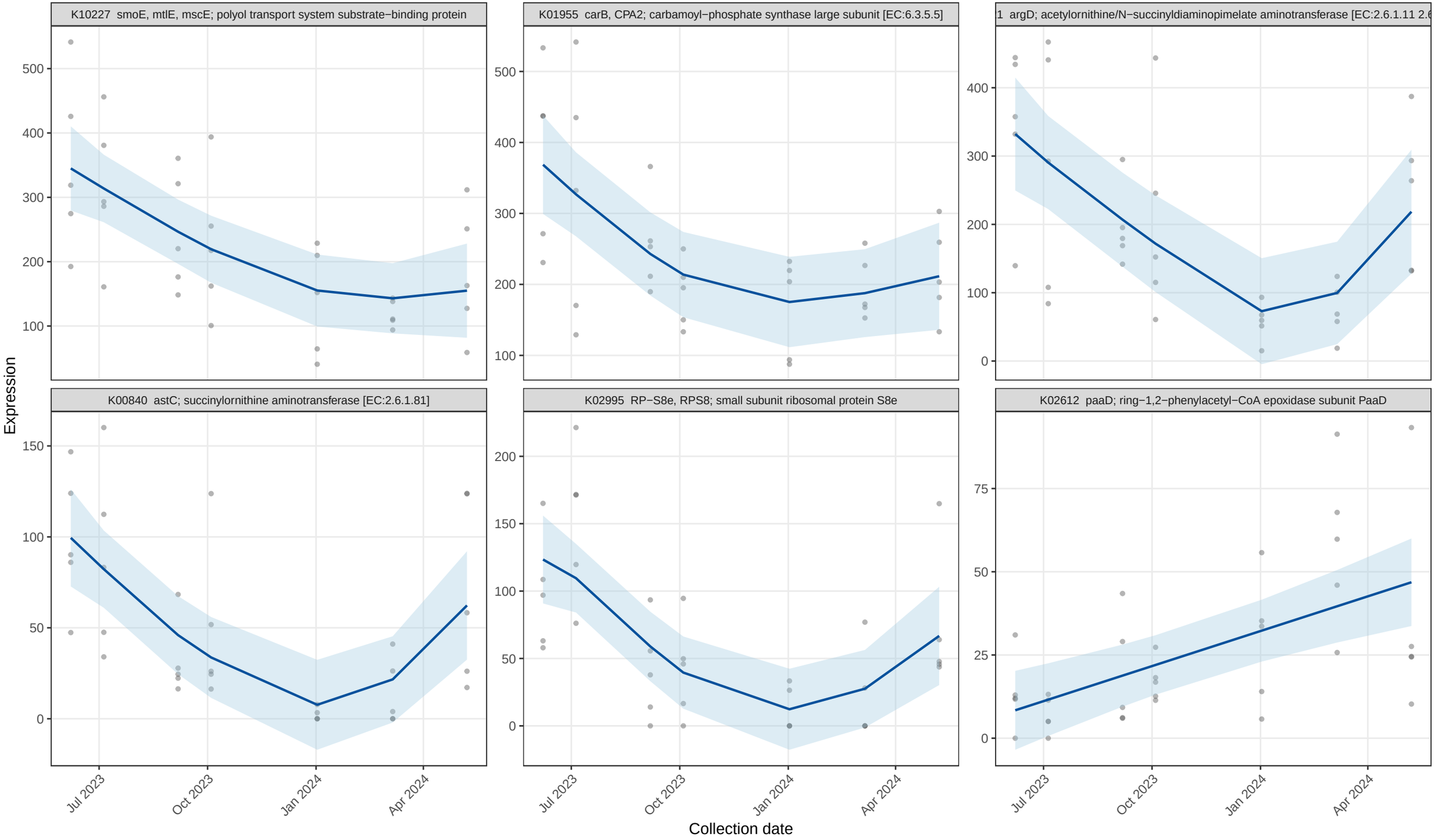

K18991 mtrA; AraC family transcriptional regulator, activator of mtrCDE
K00241 sdhC, frdC; succinate dehydrogenase cytochrome b subunit
K02806 ptsN; nitrogen PTS system EIIA component [EC:2.7.1.−]
K02021 ABC.MR; putative ABC transport system ATP−binding protein
K02520 infC, MTIF3; translation initiation factor IF−3
K01611 speD, AMD1; S−adenosylmethionine decarboxylase [EC:4.1.1.50]
Expression
Collection date
Jul 2023
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Apr 2024

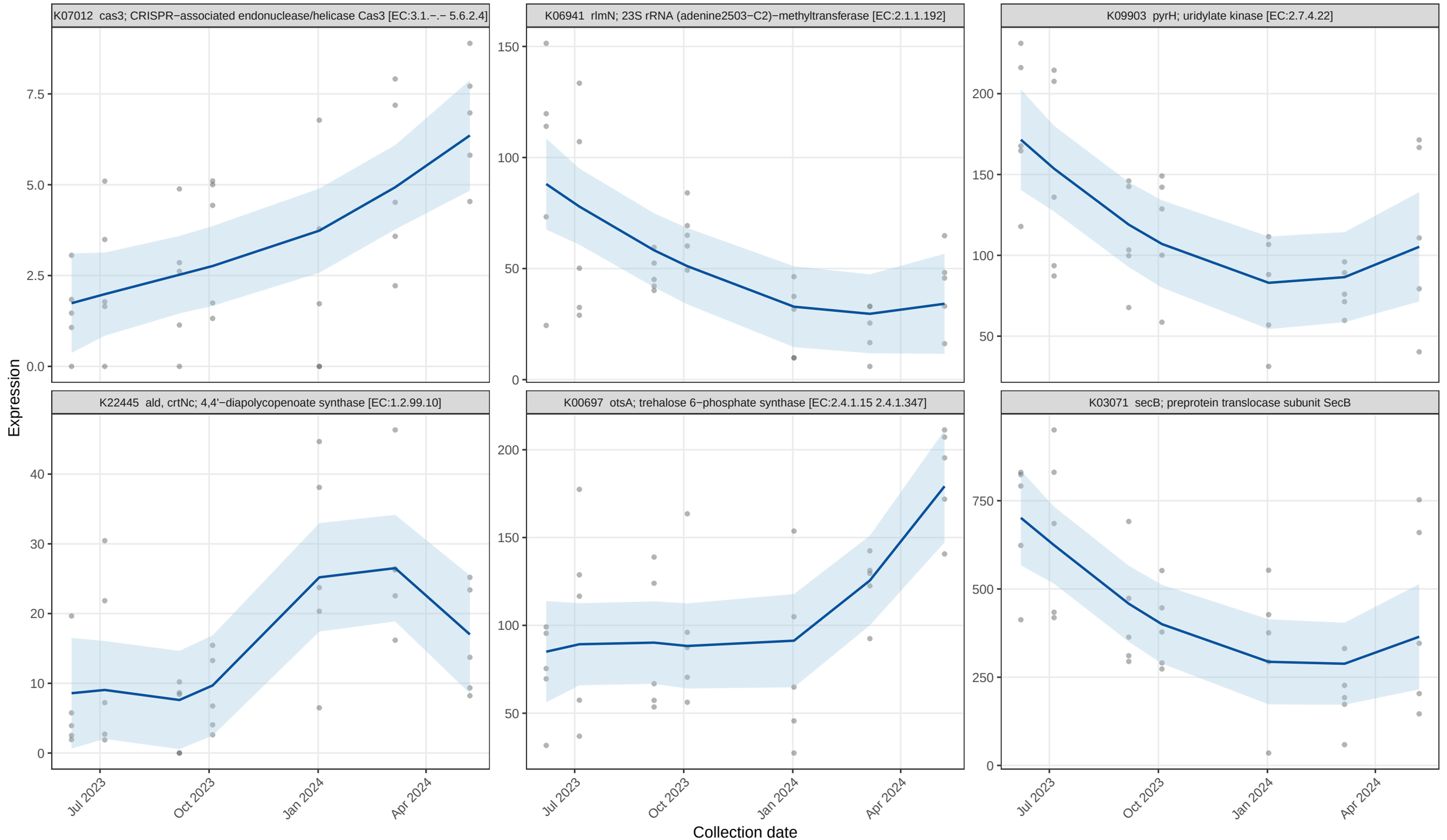

K04082 hscB, HSCB, HSC20; molecular chaperone HscB
K08995 K08995; putative membrane protein
K07546 bbsH; (E)−benzylidenesuccinyl−CoA hydratase [EC:4.2.1.180]
K03167 top6B; DNA topoisomerase VI subunit B [EC:5.6.2.2]
K01623 ALDO; fructose−bisphosphate aldolase, class I [EC:4.1.2.13]
K00646 pksF, curC, aprD, corD; malonyl−[acp] decarboxylase [EC:4.1.1.124]
Expression
Collection date
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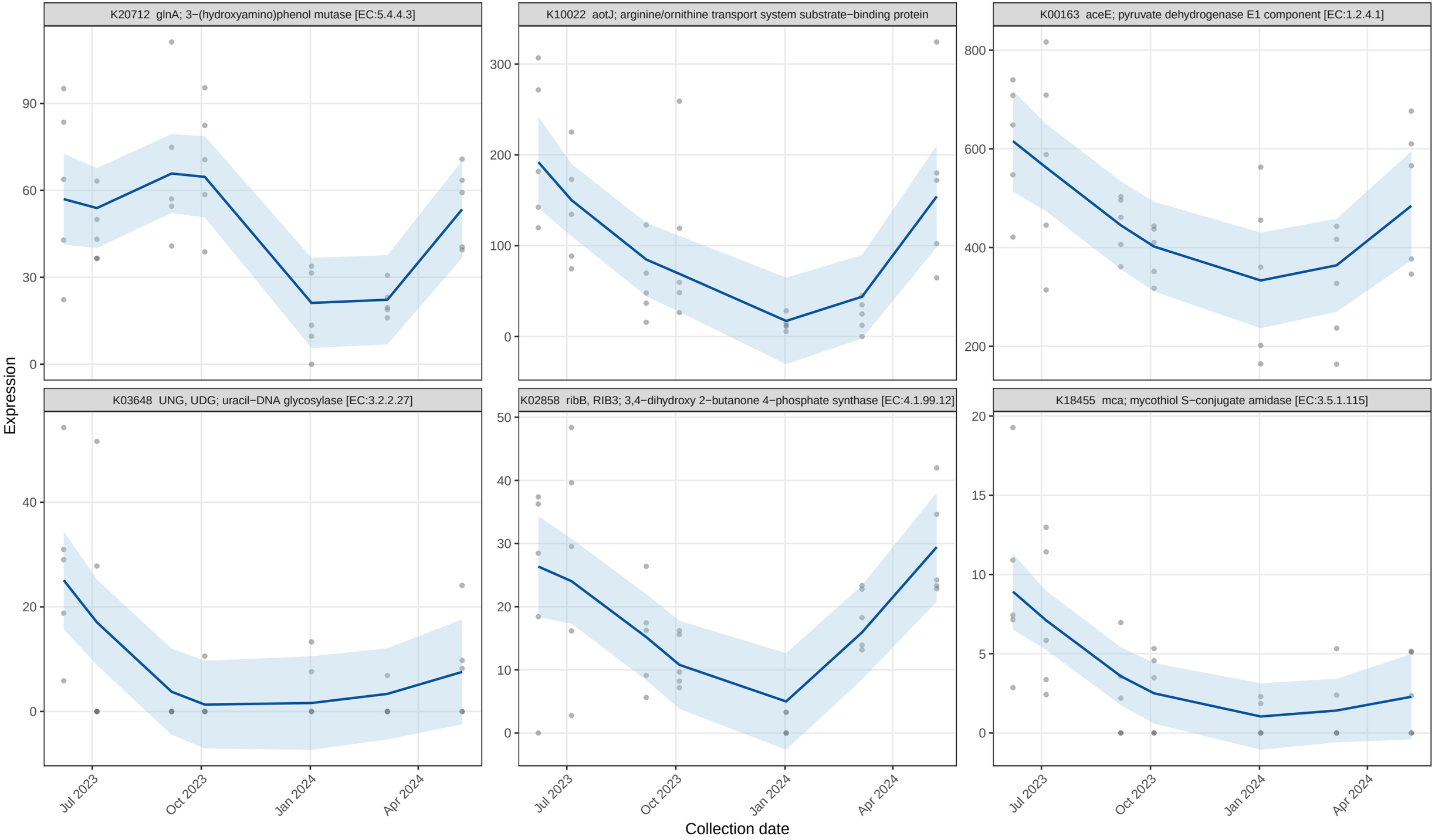

K00647 fabB; 3−oxoacyl−[acyl−carrier−protein] synthase I [EC:2.3.1.41]
K00615 E2.2.1.1, tktA, tktB; transketolase [EC:2.2.1.1]
958 whiB7; WhiB family transcriptional regulator, redox−sensing transcriptional regul
K00940 ndk, NME; nucleoside−diphosphate kinase [EC:2.7.4.6]
K06147 ABCB−BAC; ATP−binding cassette, subfamily B, bacterial
K02040 pstS; phosphate transport system substrate−binding protein
Expression
Collection date
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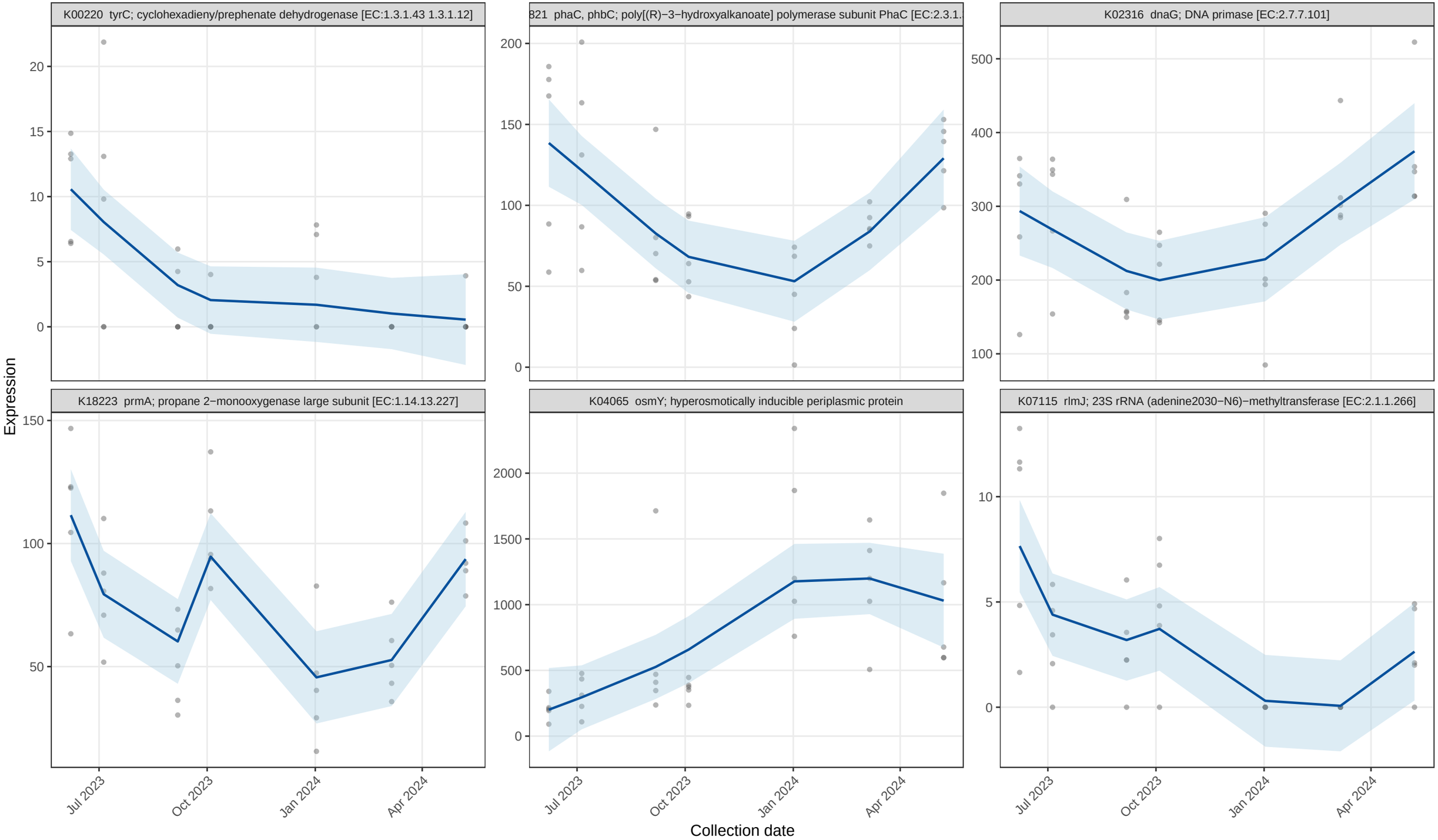

K07516 fadN; 3−hydroxyacyl−CoA dehydrogenase [EC:1.1.1.35]
K06861 lptB; lipopolysaccharide export system ATP−binding protein [EC:7.5.2.5]
K21691 rpfE; resuscitation−promoting factor RpfE
K00673 astA; arginine N−succinyltransferase [EC:2.3.1.109]
K20431 cetB, valD; 2−epi−5−epi−valiolone epimerase [EC:5.1.3.33]
K01079 serB, PSPH; phosphoserine phosphatase [EC:3.1.3.3]
Expression
Collection date
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Apr 2024

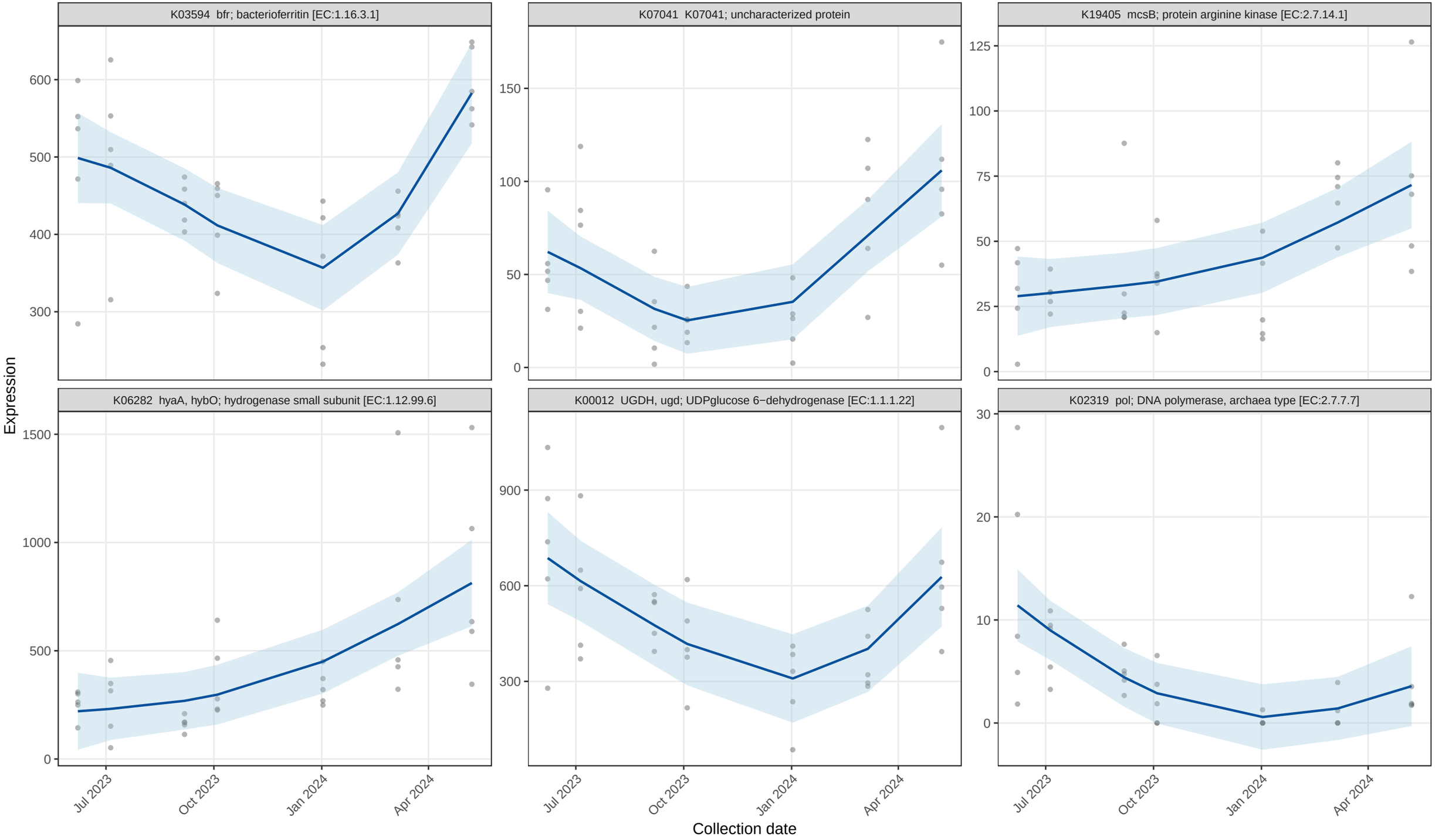

K02197 ccmE; cytochrome c−type biogenesis protein CcmE
K07793 tctA; putative tricarboxylic transport membrane protein
90 E2.1.1.113; site−specific DNA−methyltransferase (cytosine−N4−specific) [EC:2.1.1
K01991 wza, gfcE; polysaccharide biosynthesis/export protein
K07736 carD; CarD family transcriptional regulator, regulator of rRNA transcription
K01941 uca; urea/guanidine carboxylase [EC:6.3.4.6 6.3.4.26]
Expression
Jul 2023
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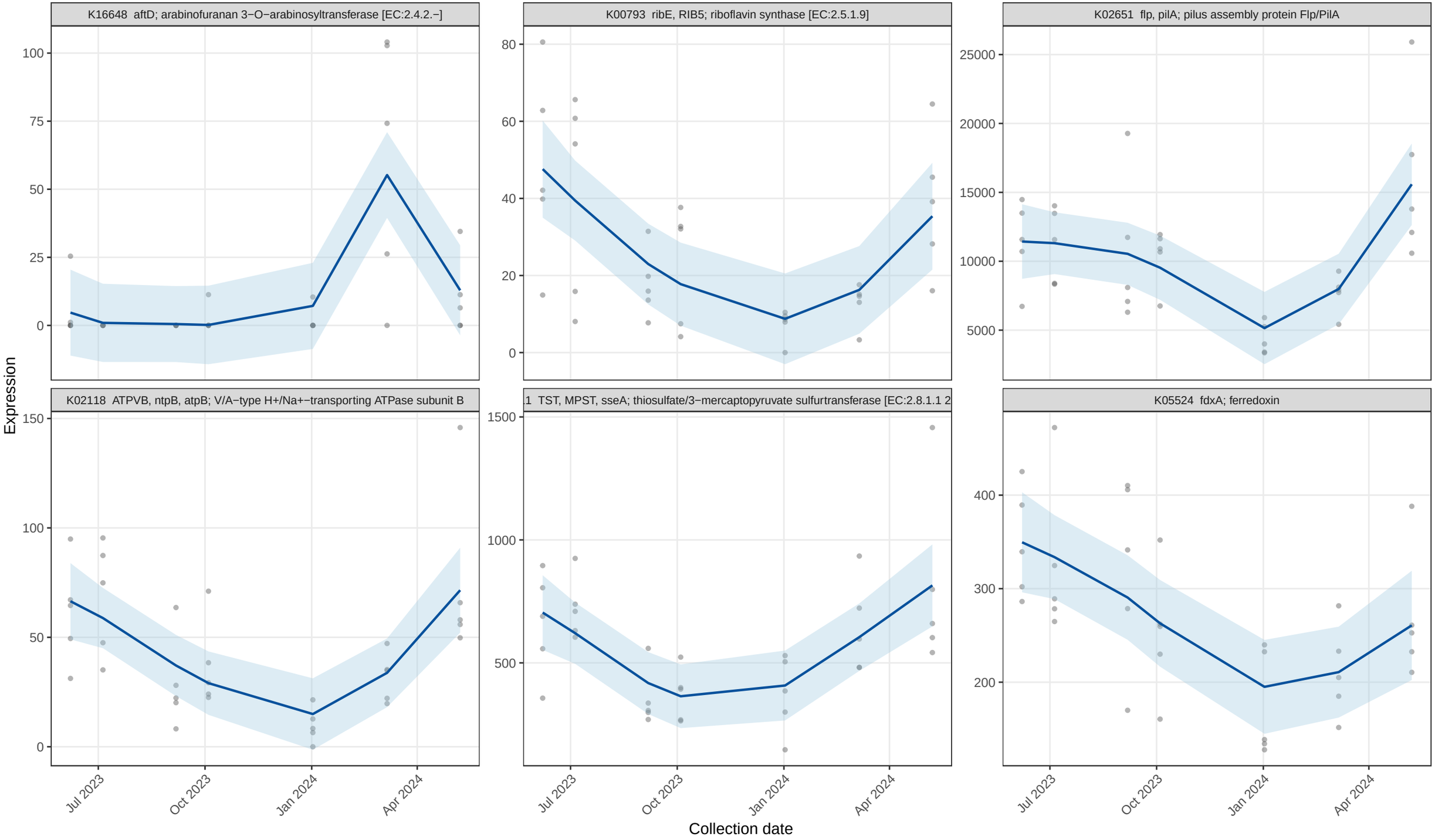

K10227 smoE, mtlE, mscE; polyol transport system substrate−binding protein
K01955 carB, CPA2; carbamoyl−phosphate synthase large subunit [EC:6.3.5.5]
1 argD; acetylornithine/N−succinyldiaminopimelate aminotransferase [EC:2.6.1.11 2.6
K00840 astC; succinylornithine aminotransferase [EC:2.6.1.81]
K02995 RP−S8e, RPS8; small subunit ribosomal protein S8e
K02612 paaD; ring−1,2−phenylacetyl−CoA epoxidase subunit PaaD
Expression
Collection date
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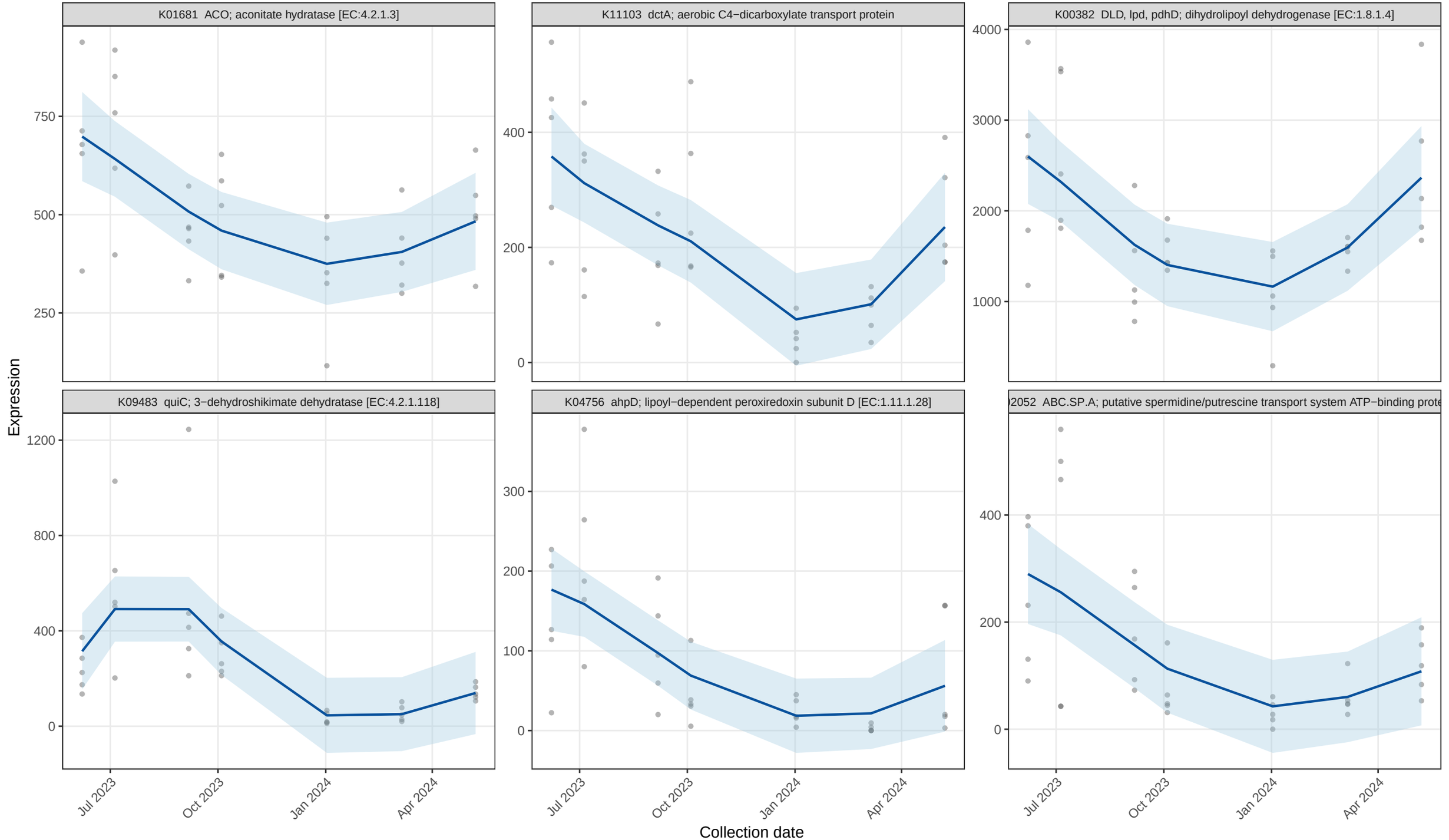

K07012 cas3; CRISPR−associated endonuclease/helicase Cas3 [EC:3.1.−.− 5.6.2.4]
K06941 rlmN; 23S rRNA (adenine2503−C2)−methyltransferase [EC:2.1.1.192]
K09903 pyrH; uridylate kinase [EC:2.7.4.22]
K22445 ald, crtNc; 4,4'−diapolycopenoate synthase [EC:1.2.99.10]
K00697 otsA; trehalose 6−phosphate synthase [EC:2.4.1.15 2.4.1.347]
K03071 secB; preprotein translocase subunit SecB
Expression
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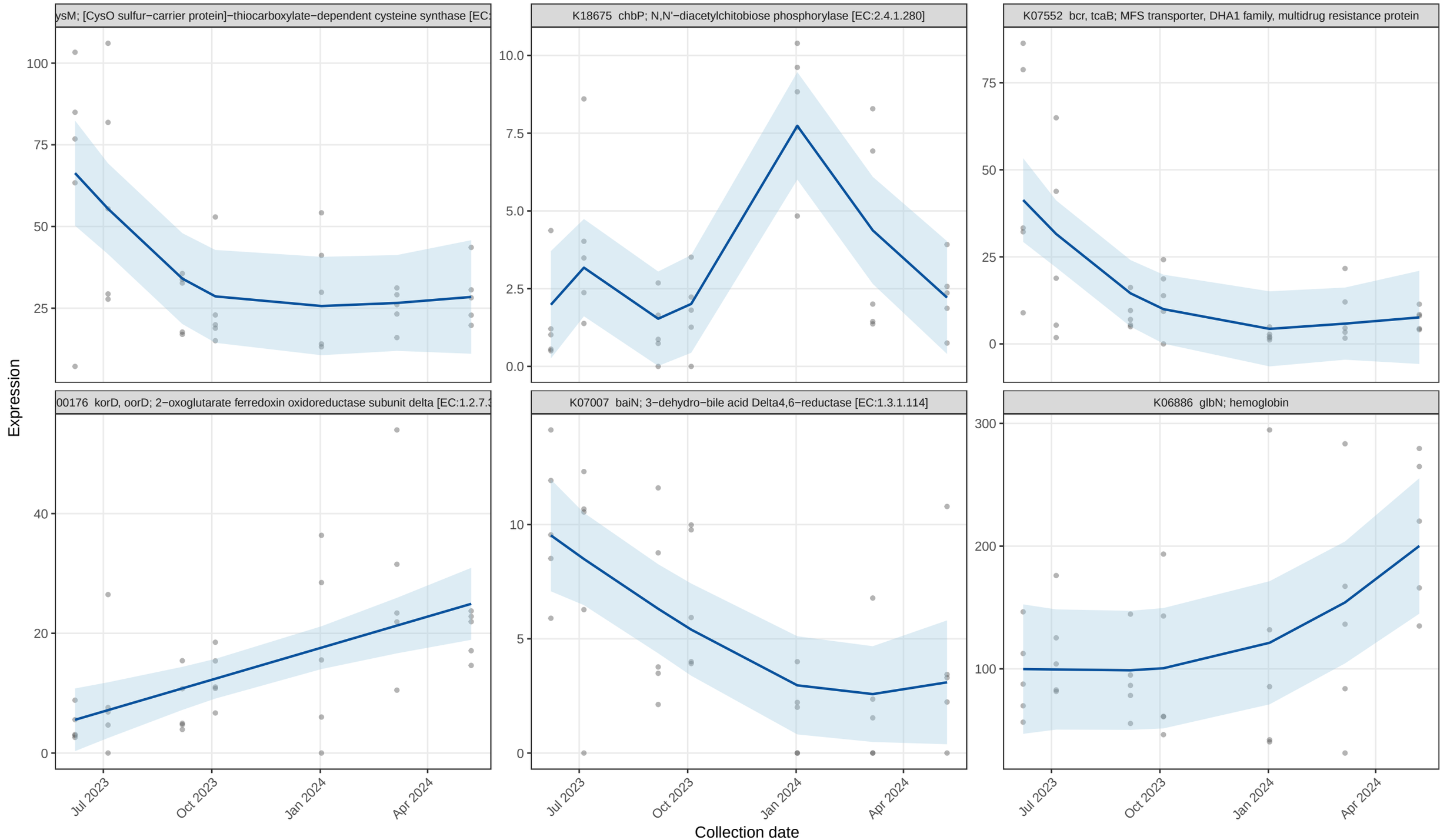

K20712 glnA; 3−(hydroxyamino)phenol mutase [EC:5.4.4.3]
K10022 aotJ; arginine/ornithine transport system substrate−binding protein
K00163 aceE; pyruvate dehydrogenase E1 component [EC:1.2.4.1]
K03648 UNG, UDG; uracil−DNA glycosylase [EC:3.2.2.27]
K02858 ribB, RIB3; 3,4−dihydroxy 2−butanone 4−phosphate synthase [EC:4.1.99.12]
K18455 mca; mycothiol S−conjugate amidase [EC:3.5.1.115]
Expression
Collection date
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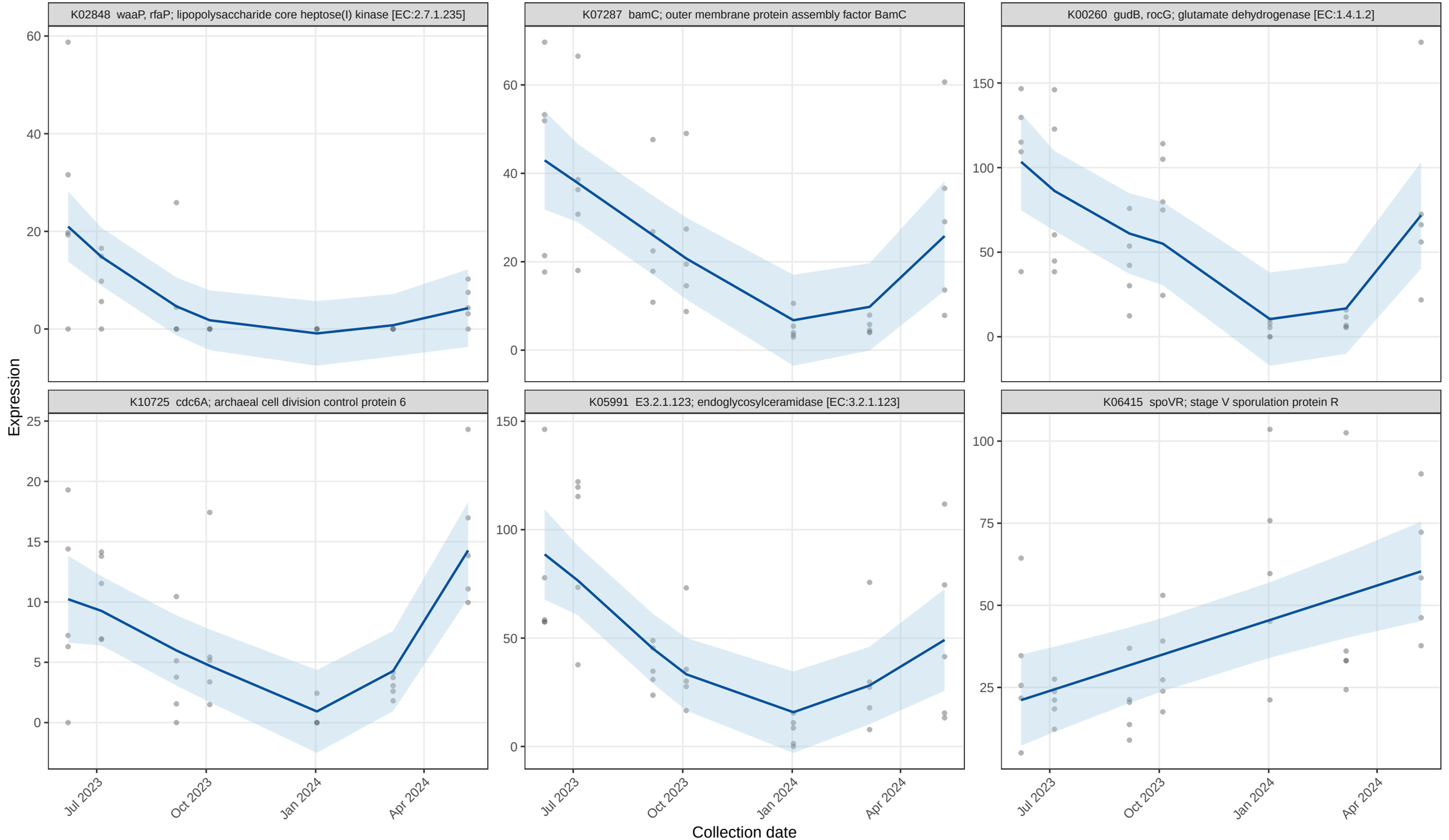

K00220 tyrC; cyclohexadieny/prephenate dehydrogenase [EC:1.3.1.43 1.3.1.12]
821 phaC, phbC; poly[(R)−3−hydroxyalkanoate] polymerase subunit PhaC [EC:2.3.1.
K02316 dnaG; DNA primase [EC:2.7.7.101]
K18223 prmA; propane 2−monooxygenase large subunit [EC:1.14.13.227]
K04065 osmY; hyperosmotically inducible periplasmic protein
K07115 rlmJ; 23S rRNA (adenine2030−N6)−methyltransferase [EC:2.1.1.266]
Expression
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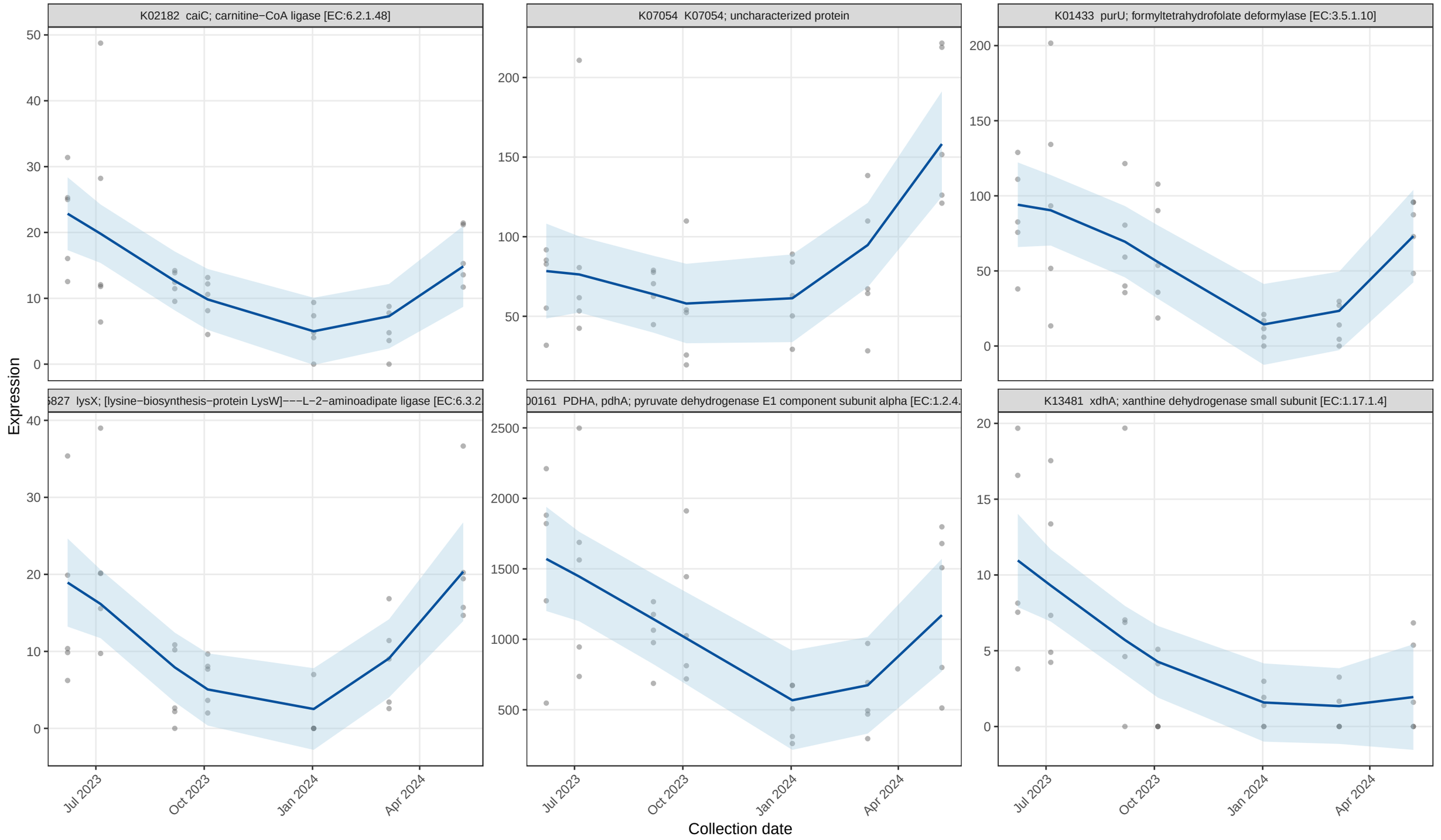

K03594 bfr; bacterioferritin [EC:1.16.3.1]
K07041 K07041; uncharacterized protein
K19405 mcsB; protein arginine kinase [EC:2.7.14.1]
K06282 hyaA, hybO; hydrogenase small subunit [EC:1.12.99.6]
K00012 UGDH, ugd; UDPglucose 6−dehydrogenase [EC:1.1.1.22]
K02319 pol; DNA polymerase, archaea type [EC:2.7.7.7]
Expression
Collection date
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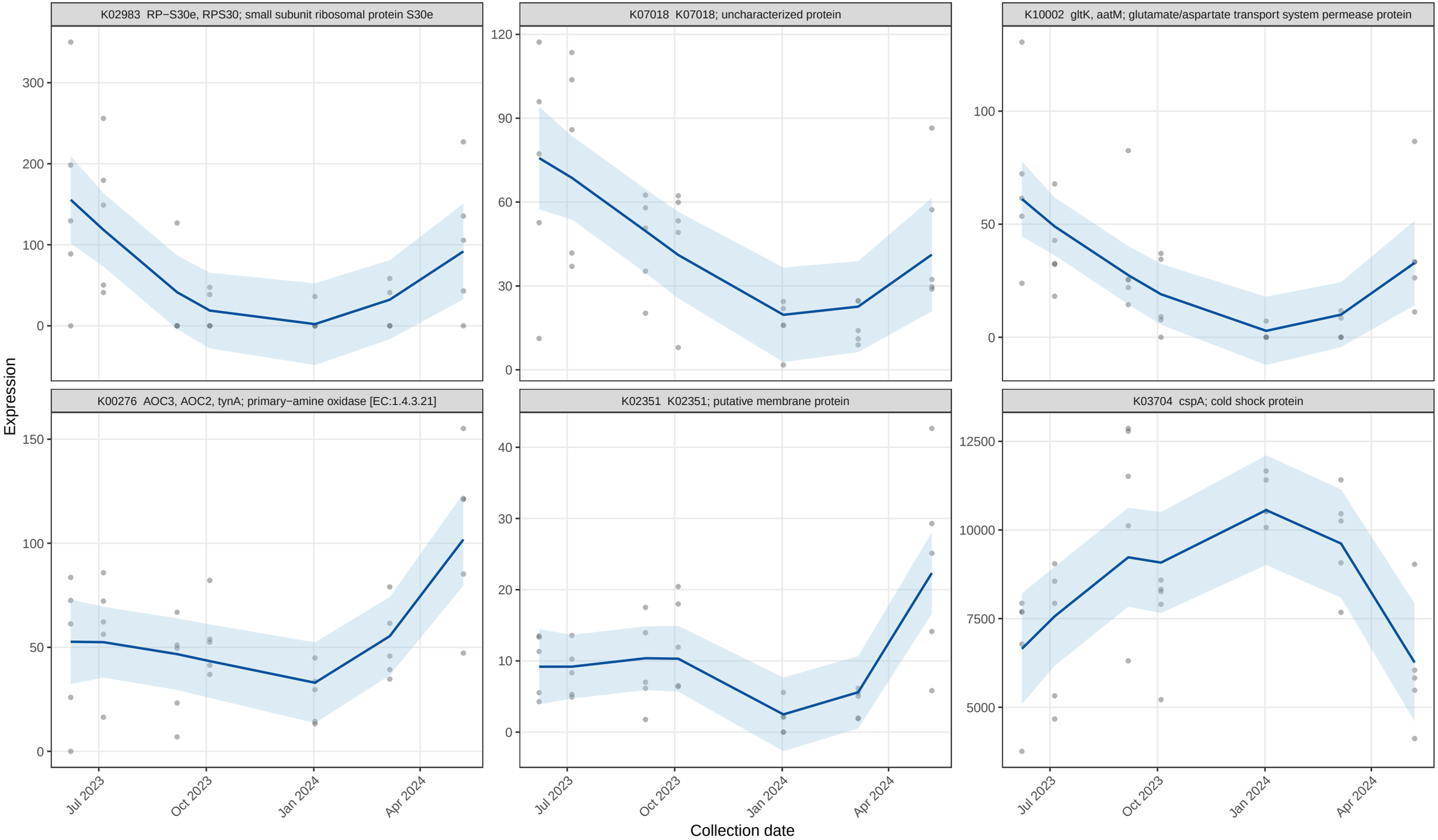

K16648 aftD; arabinofuranan 3−O−arabinosyltransferase [EC:2.4.2.−]
K00793 ribE, RIB5; riboflavin synthase [EC:2.5.1.9]
K02651 flp, pilA; pilus assembly protein Flp/PilA
K02118 ATPVB, ntpB, atpB; V/A−type H+/Na+−transporting ATPase subunit B
1 TST, MPST, sseA; thiosulfate/3−mercaptopyruvate sulfurtransferase [EC:2.8.1.1 2
K05524 fdxA; ferredoxin
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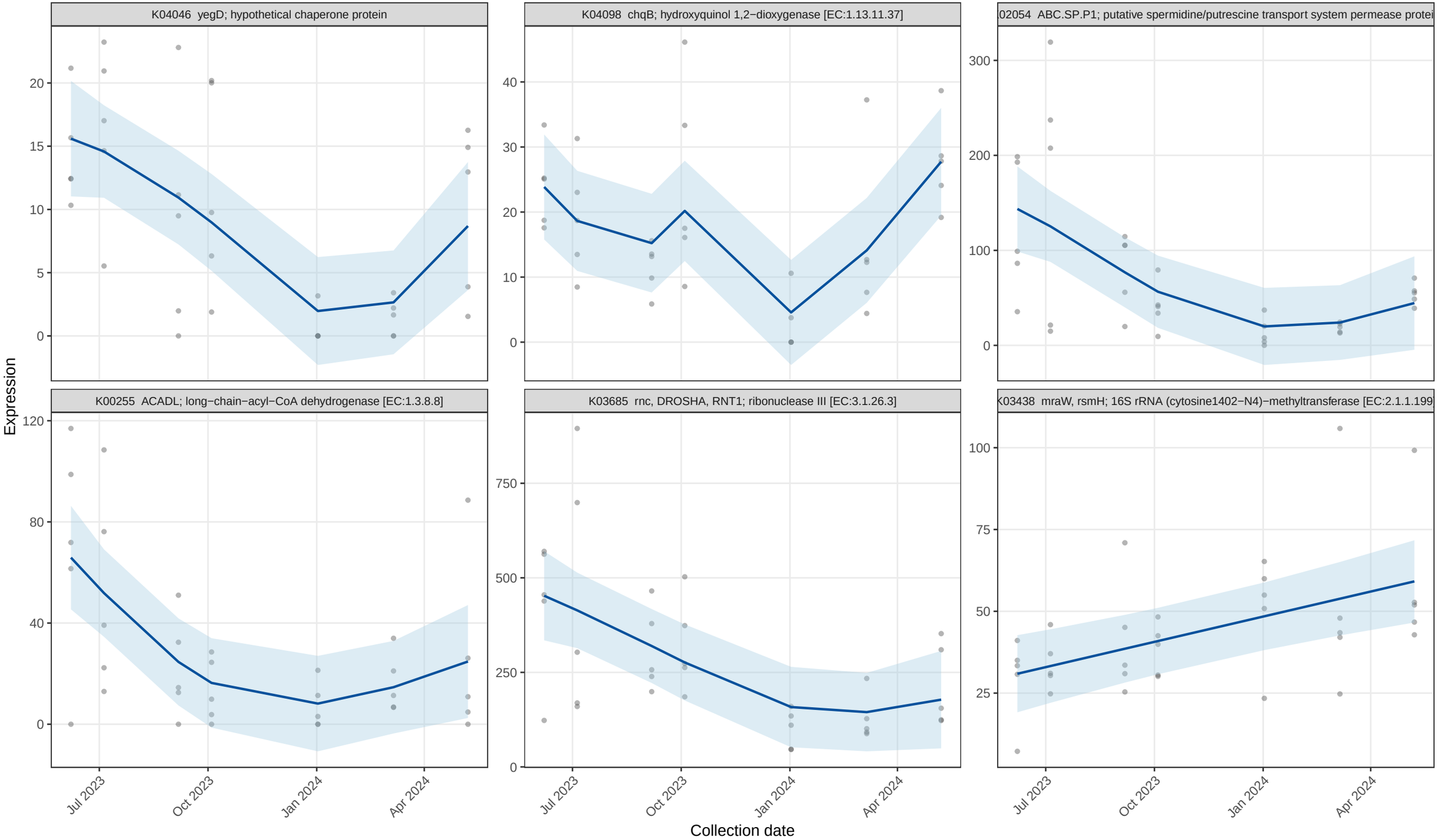

K01681 ACO; aconitate hydratase [EC:4.2.1.3]
K11103 dctA; aerobic C4−dicarboxylate transport protein
K00382 DLD, lpd, pdhD; dihydrolipoyl dehydrogenase [EC:1.8.1.4]
K09483 quiC; 3−dehydroshikimate dehydratase [EC:4.2.1.118]
K04756 ahpD; lipoyl−dependent peroxiredoxin subunit D [EC:1.11.1.28]
2052 ABC.SP.A; putative spermidine/putrescine transport system ATP−binding prote
Expression
Collection date
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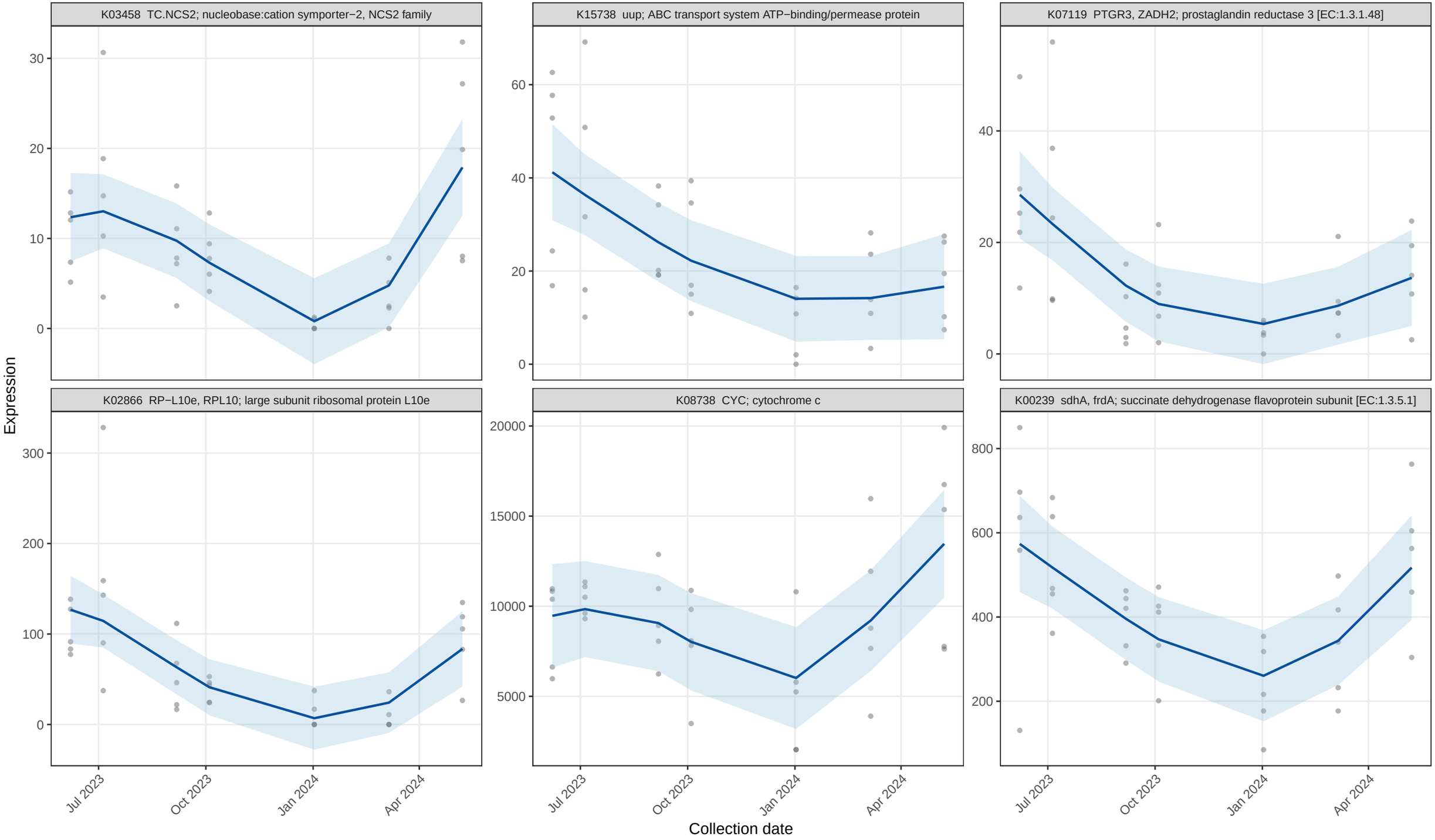

ysM; [CysO sulfur−carrier protein]−thiocarboxylate−dependent cysteine synthase [EC:
K18675 chbP; N,N'−diacetylchitobiose phosphorylase [EC:2.4.1.280]
K07552 bcr, tcaB; MFS transporter, DHA1 family, multidrug resistance protein
00176 korD, oorD; 2−oxoglutarate ferredoxin oxidoreductase subunit delta [EC:1.2.7.
K07007 baiN; 3−dehydro−bile acid Delta4,6−reductase [EC:1.3.1.114]
K06886 glbN; hemoglobin
Expression
Collection date
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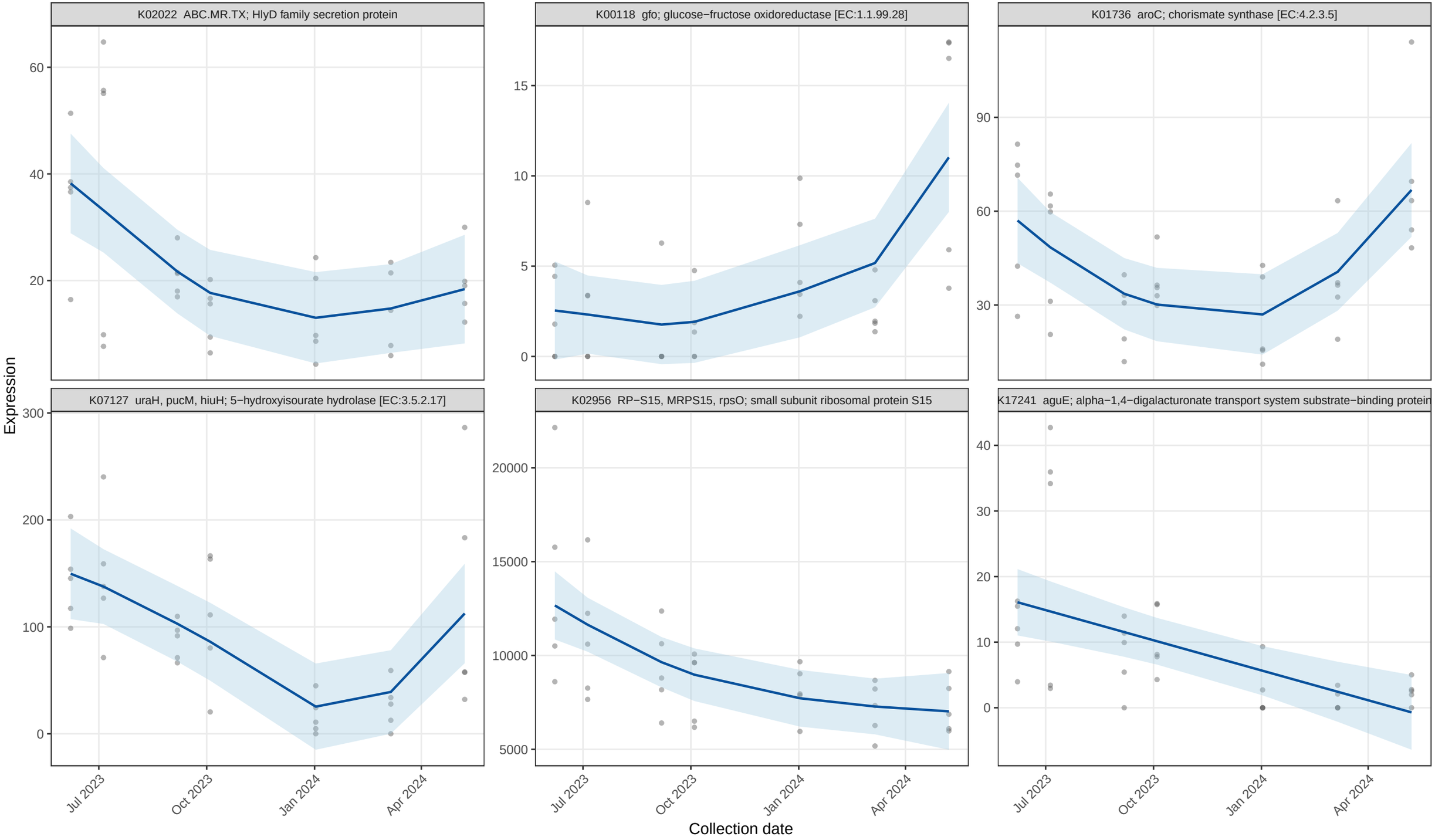

K02848 waaP, rfaP; lipopolysaccharide core heptose(I) kinase [EC:2.7.1.235]
K07287 bamC; outer membrane protein assembly factor BamC
K00260 gudB, rocG; glutamate dehydrogenase [EC:1.4.1.2]
K10725 cdc6A; archaeal cell division control protein 6
K05991 E3.2.1.123; endoglycosylceramidase [EC:3.2.1.123]
K06415 spoVR; stage V sporulation protein R
Expression
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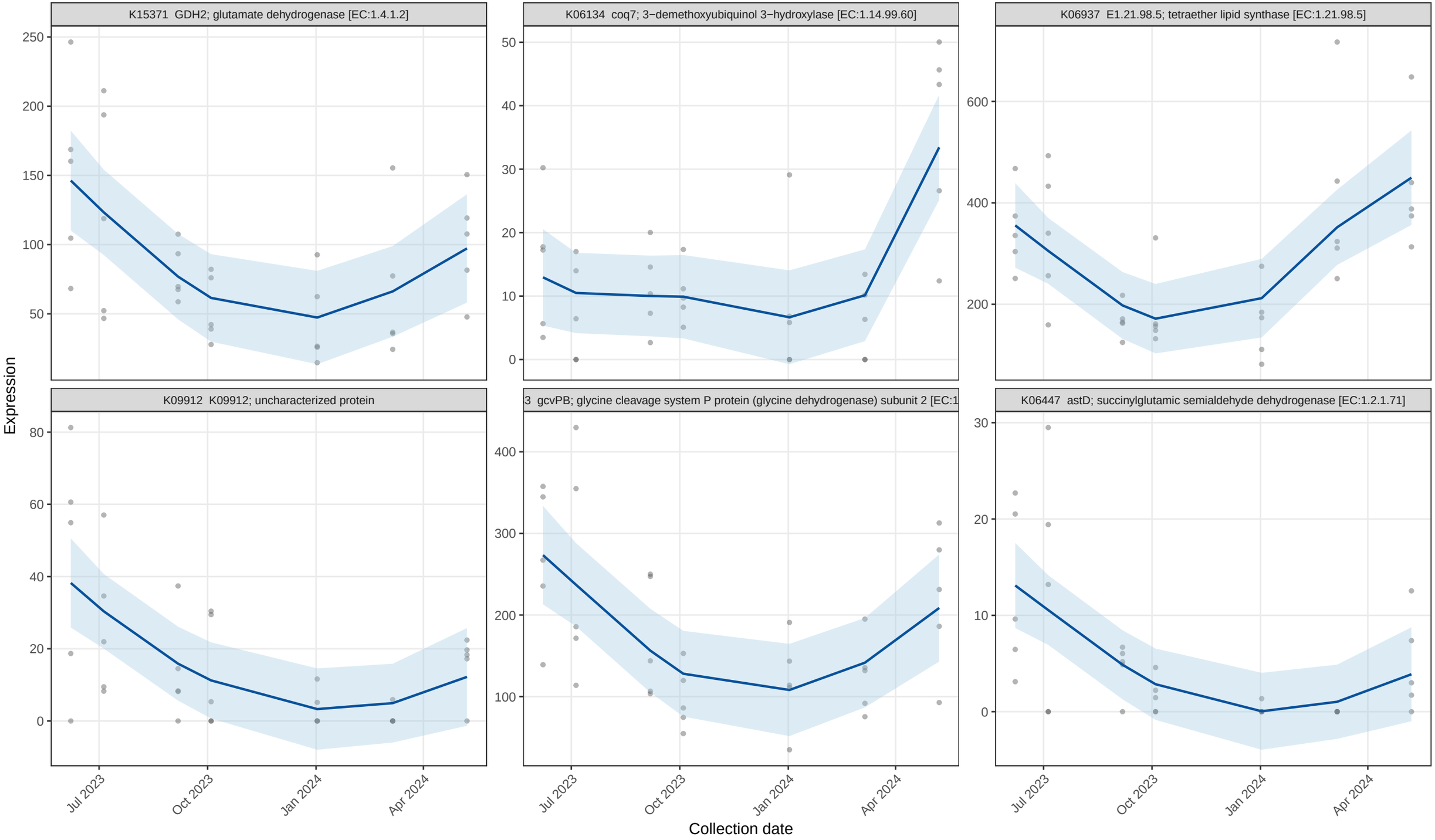

K02182 caiC; carnitine−CoA ligase [EC:6.2.1.48]
K07054 K07054; uncharacterized protein
K01433 purU; formyltetrahydrofolate deformylase [EC:3.5.1.10]
5827 lysX; [lysine−biosynthesis−protein LysW]−−−L−2−aminoadipate ligase [EC:6.3.2
00161 PDHA, pdhA; pyruvate dehydrogenase E1 component subunit alpha [EC:1.2.4.
K13481 xdhA; xanthine dehydrogenase small subunit [EC:1.17.1.4]
Expression
Collection date
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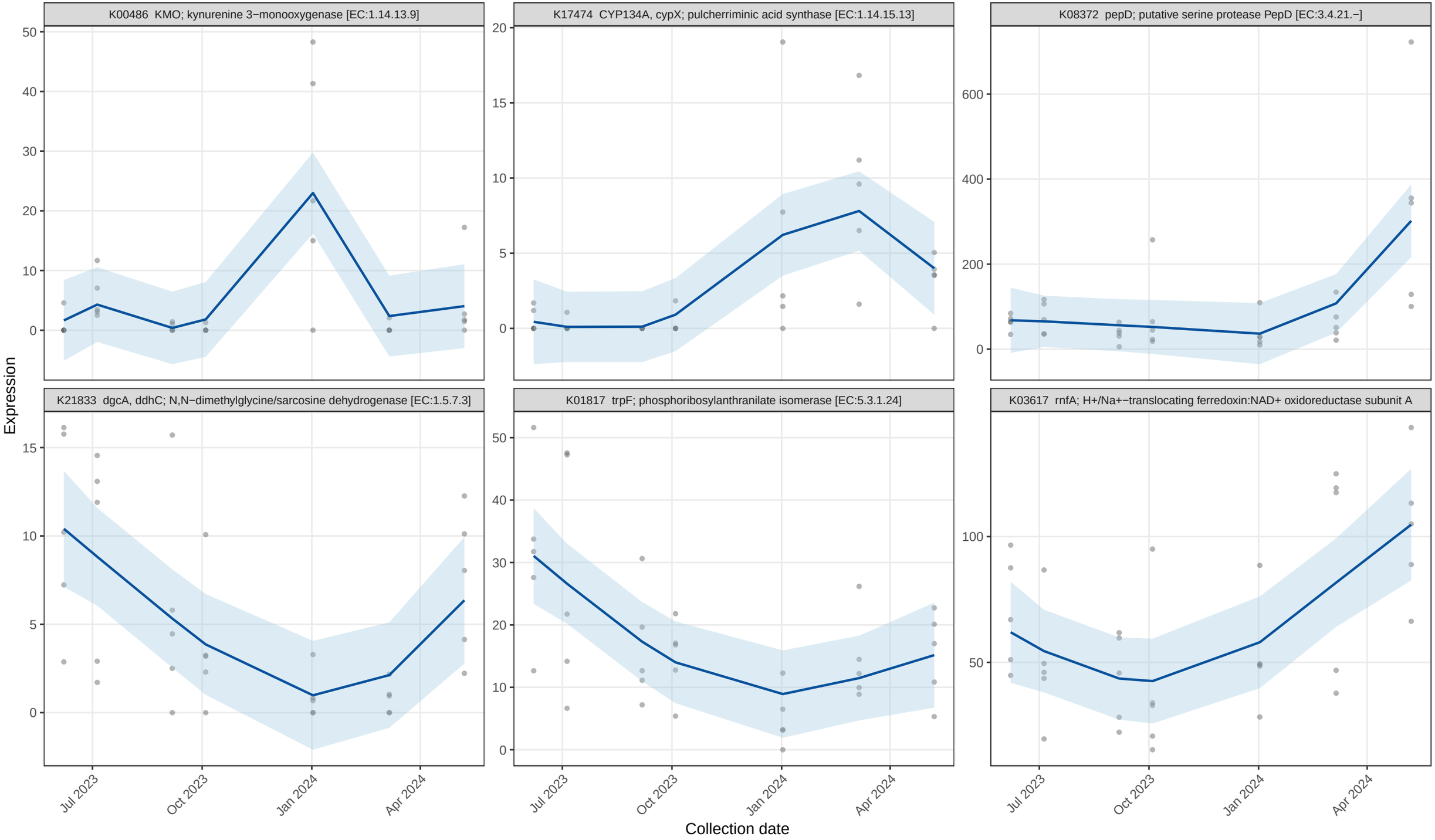

K02983 RP−S30e, RPS30; small subunit ribosomal protein S30e
K07018 K07018; uncharacterized protein
K10002 gltK, aatM; glutamate/aspartate transport system permease protein
K00276 AOC3, AOC2, tynA; primary−amine oxidase [EC:1.4.3.21]
K02351 K02351; putative membrane protein
K03704 cspA; cold shock protein
Expression
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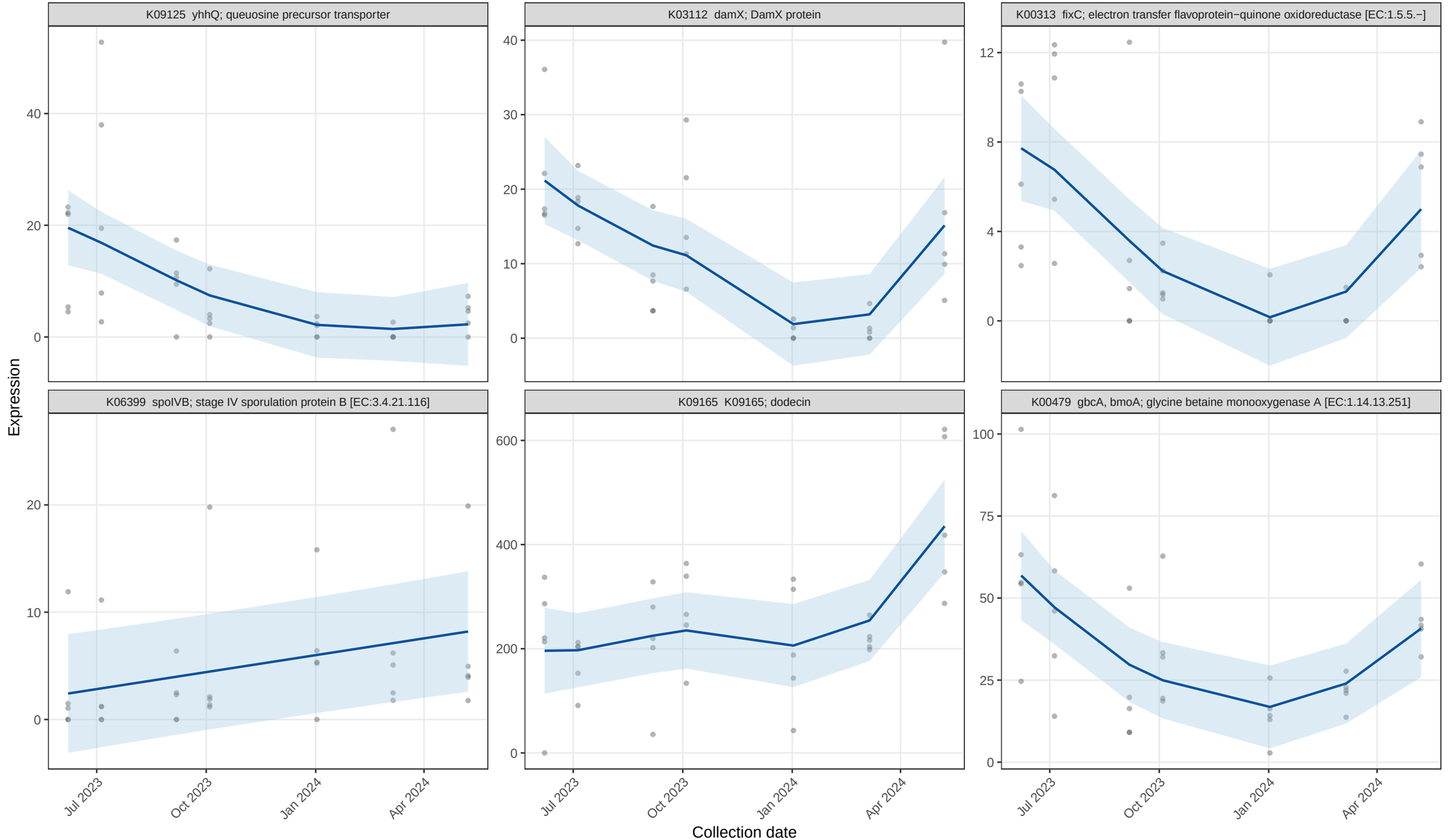

K04046 yegD; hypothetical chaperone protein
K04098 chqB; hydroxyquinol 1,2−dioxygenase [EC:1.13.11.37]
02054 ABC.SP.P1; putative spermidine/putrescine transport system permease protei
K00255 ACADL; long−chain−acyl−CoA dehydrogenase [EC:1.3.8.8]
K03685 rnc, DROSHA, RNT1; ribonuclease III [EC:3.1.26.3]
K03438 mraW, rsmH; 16S rRNA (cytosine1402−N4)−methyltransferase [EC:2.1.1.199
Expression
Collection date
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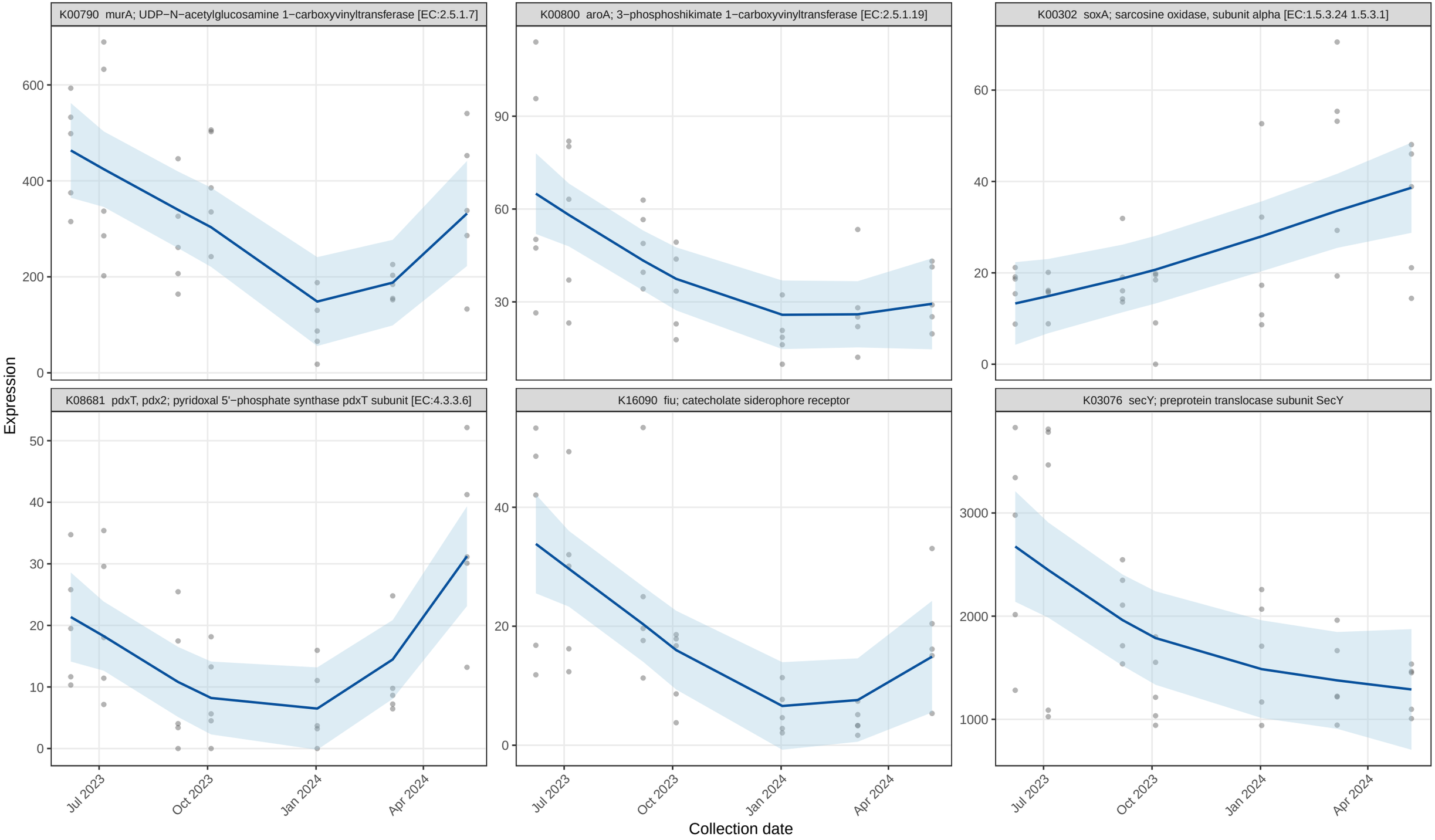

K03458 TC.NCS2; nucleobase:cation symporter−2, NCS2 family
K15738 uup; ABC transport system ATP−binding/permease protein
K07119 PTGR3, ZADH2; prostaglandin reductase 3 [EC:1.3.1.48]
K02866 RP−L10e, RPL10; large subunit ribosomal protein L10e
K08738 CYC; cytochrome c
K00239 sdhA, frdA; succinate dehydrogenase flavoprotein subunit [EC:1.3.5.1]
Expression
Collection date
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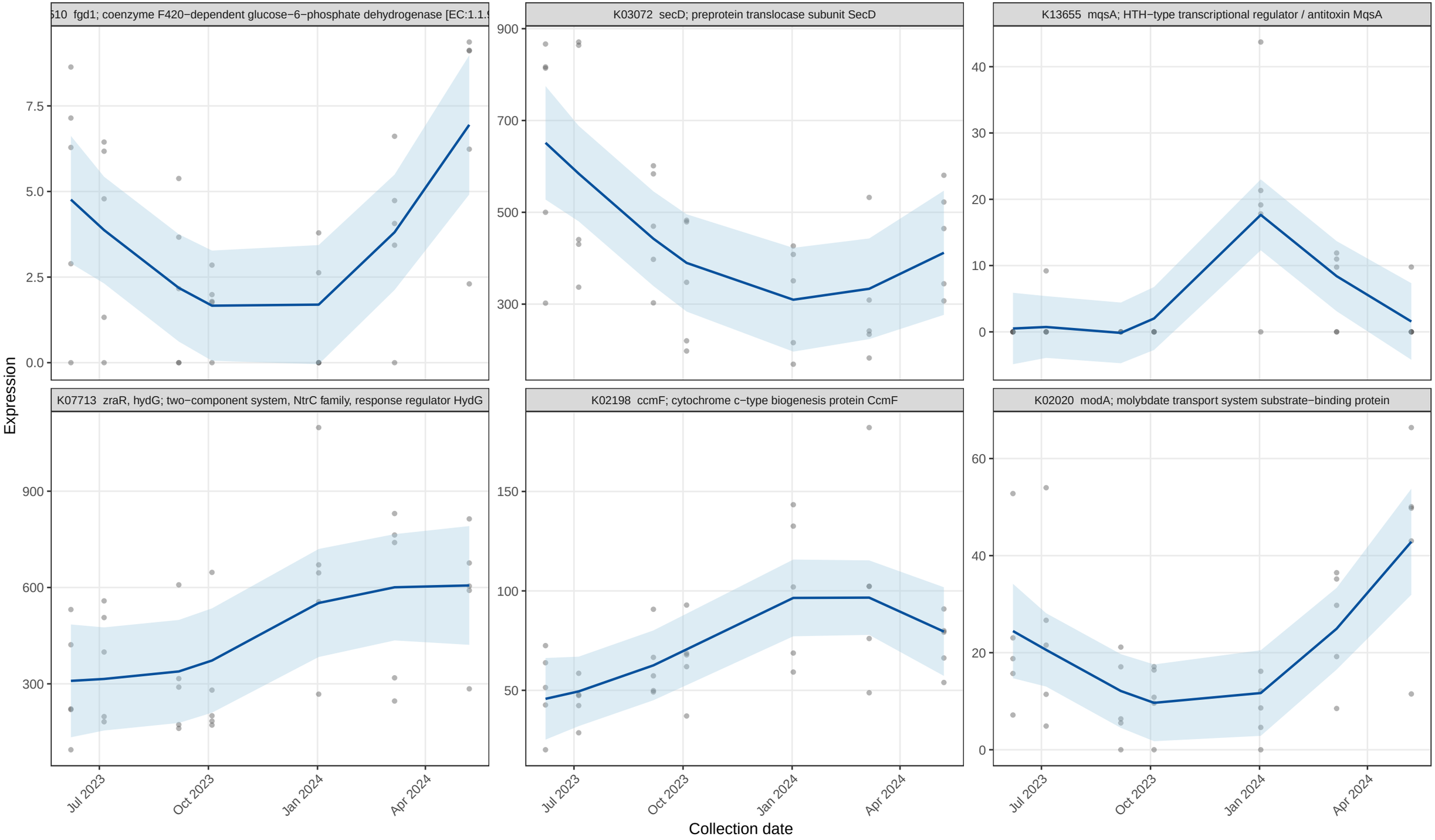

K02022 ABC.MR.TX; HlyD family secretion protein
K00118 gfo; glucose−fructose oxidoreductase [EC:1.1.99.28]
K01736 aroC; chorismate synthase [EC:4.2.3.5]
K07127 uraH, pucM, hiuH; 5−hydroxyisourate hydrolase [EC:3.5.2.17]
K02956 RP−S15, MRPS15, rpsO; small subunit ribosomal protein S15
K17241 aguE; alpha−1,4−digalacturonate transport system substrate−binding protein
Expression
Collection date
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Apr 2024

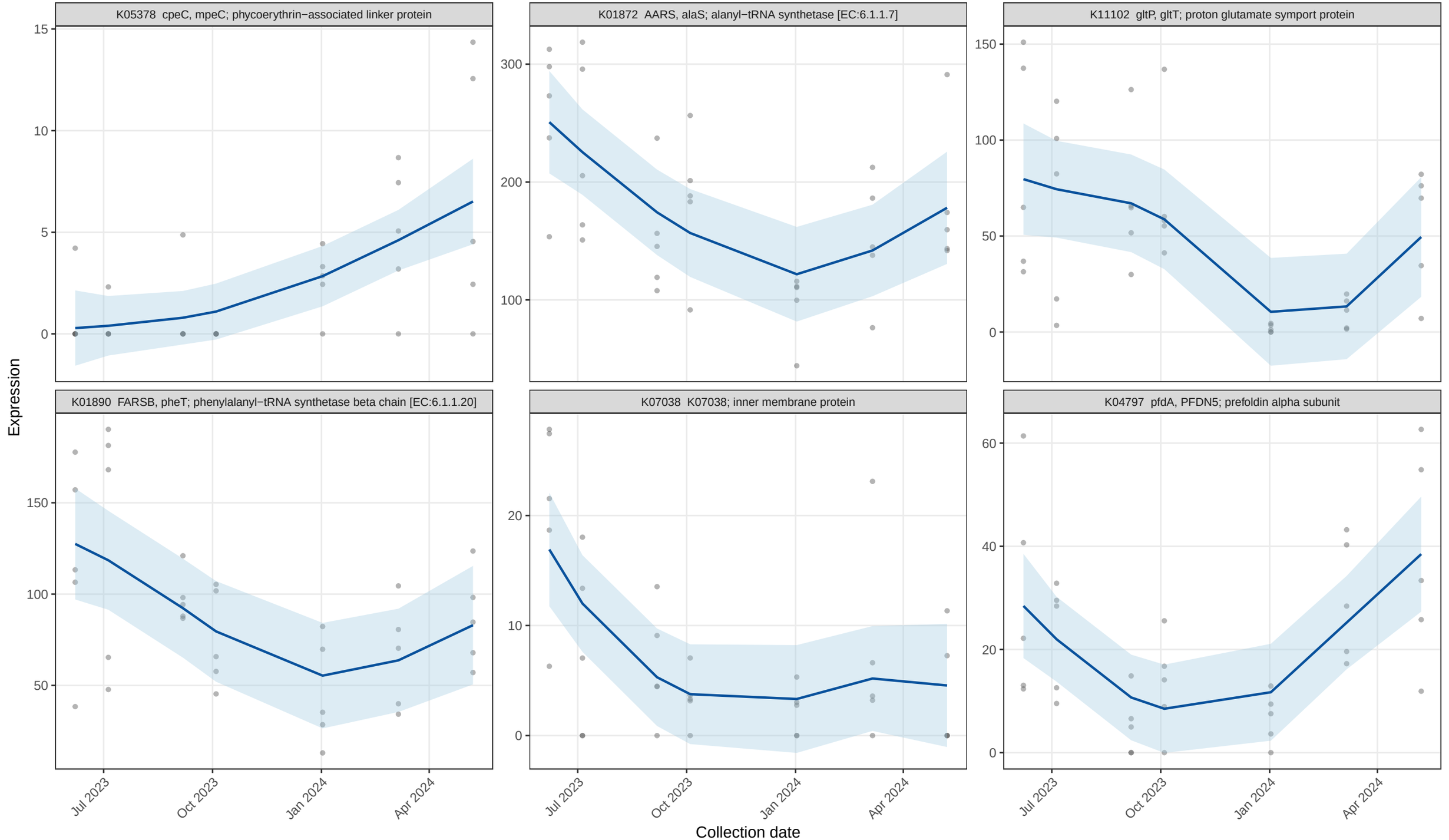

K15371 GDH2; glutamate dehydrogenase [EC:1.4.1.2]
K06134 coq7; 3−demethoxyubiquinol 3−hydroxylase [EC:1.14.99.60]
K06937 E1.21.98.5; tetraether lipid synthase [EC:1.21.98.5]
K09912 K09912; uncharacterized protein
3 gcvPB; glycine cleavage system P protein (glycine dehydrogenase) subunit 2 [EC:1
K06447 astD; succinylglutamic semialdehyde dehydrogenase [EC:1.2.1.71]
Expression
Jul 2023
Oct 2023
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Apr 2024
Collection date

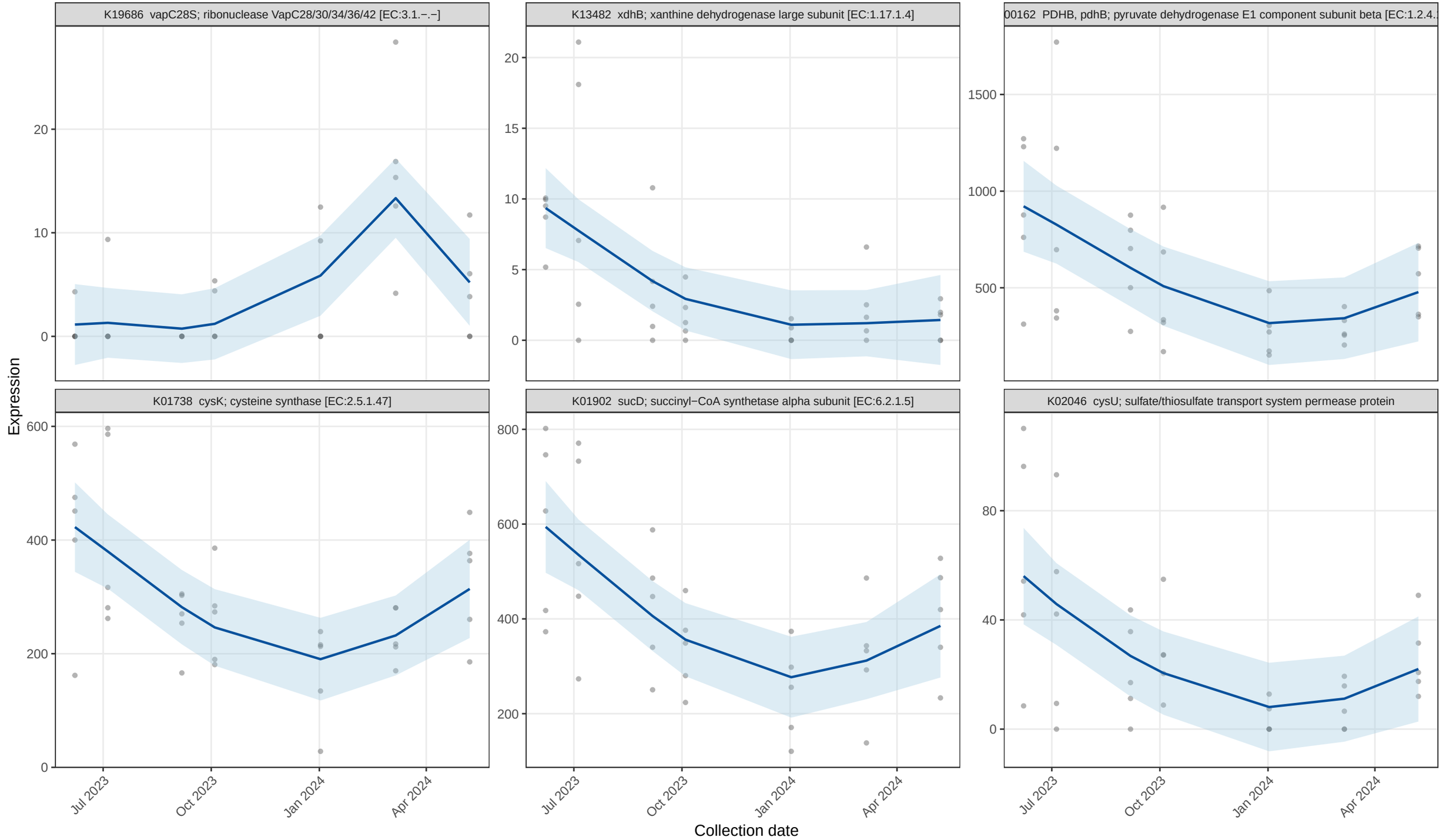

K00486 KMO; kynurenine 3−monooxygenase [EC:1.14.13.9]
K17474 CYP134A, cypX; pulcherriminic acid synthase [EC:1.14.15.13]
K08372 pepD; putative serine protease PepD [EC:3.4.21.−]
K21833 dgcA, ddhC; N,N−dimethylglycine/sarcosine dehydrogenase [EC:1.5.7.3]
K01817 trpF; phosphoribosylanthranilate isomerase [EC:5.3.1.24]
K03617 rnfA; H+/Na+−translocating ferredoxin:NAD+ oxidoreductase subunit A
Expression
Collection date
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Apr 2024

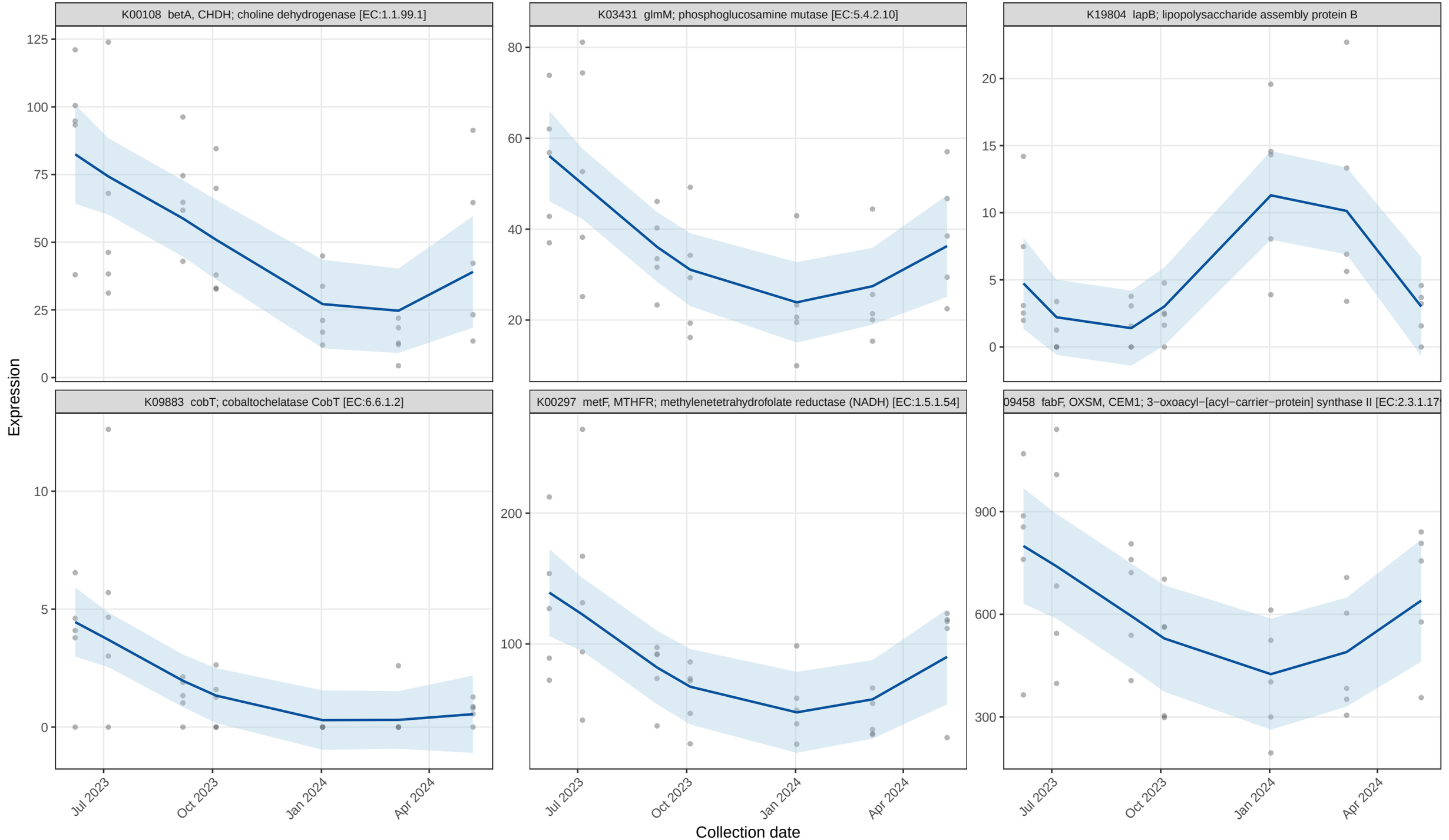

K09125 yhhQ; queuosine precursor transporter
K03112 damX; DamX protein
K00313 fixC; electron transfer flavoprotein−quinone oxidoreductase [EC:1.5.5.−]
K06399 spoIVB; stage IV sporulation protein B [EC:3.4.21.116]
K09165 K09165; dodecin
K00479 gbcA, bmoA; glycine betaine monooxygenase A [EC:1.14.13.251]
Expression
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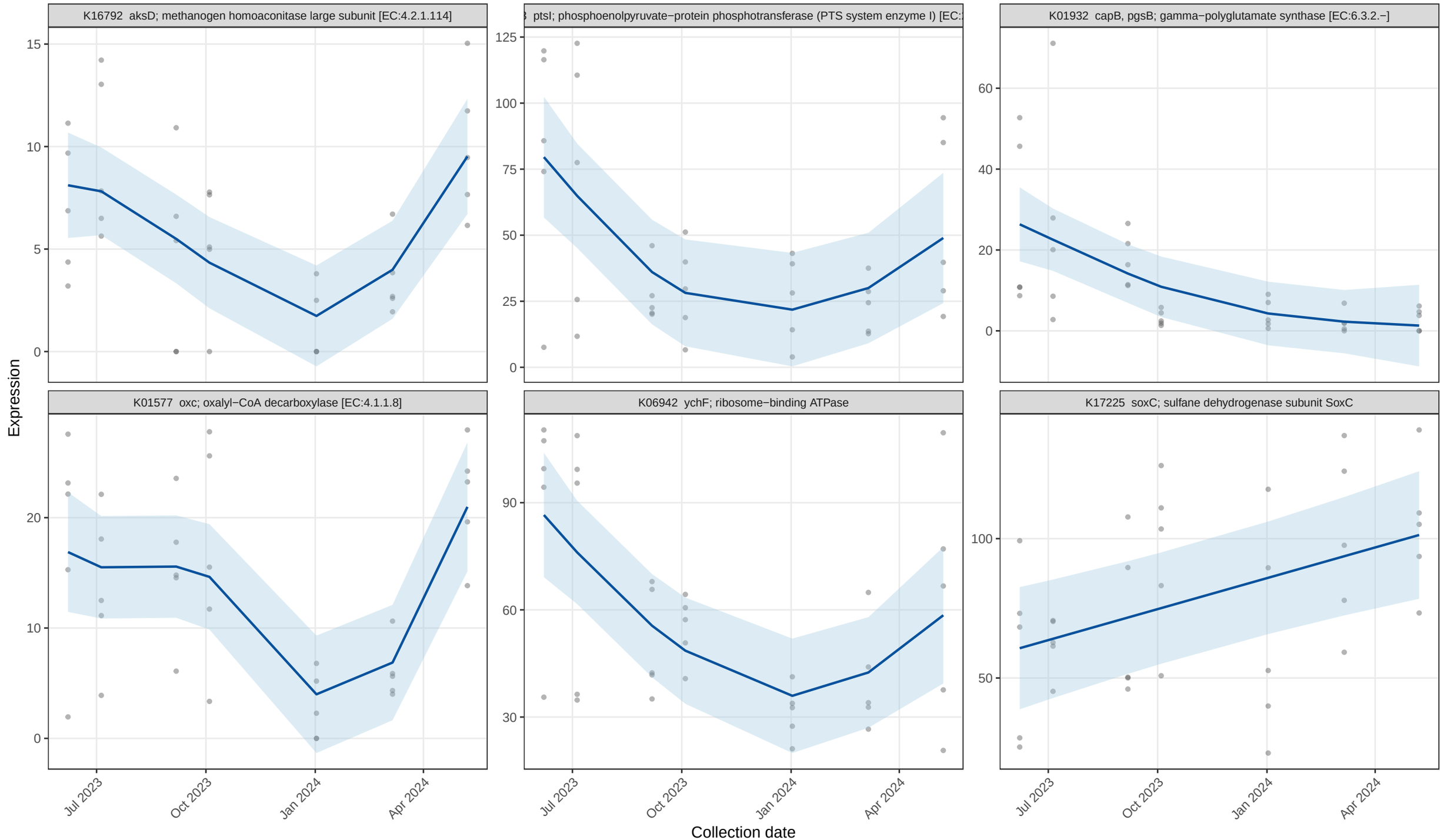

K00790 murA; UDP−N−acetylglucosamine 1−carboxyvinyltransferase [EC:2.5.1.7]
K00800 aroA; 3−phosphoshikimate 1−carboxyvinyltransferase [EC:2.5.1.19]
K00302 soxA; sarcosine oxidase, subunit alpha [EC:1.5.3.24 1.5.3.1]
K08681 pdxT, pdx2; pyridoxal 5'−phosphate synthase pdxT subunit [EC:4.3.3.6]
K16090 fiu; catecholate siderophore receptor
K03076 secY; preprotein translocase subunit SecY
Expression
Collection date
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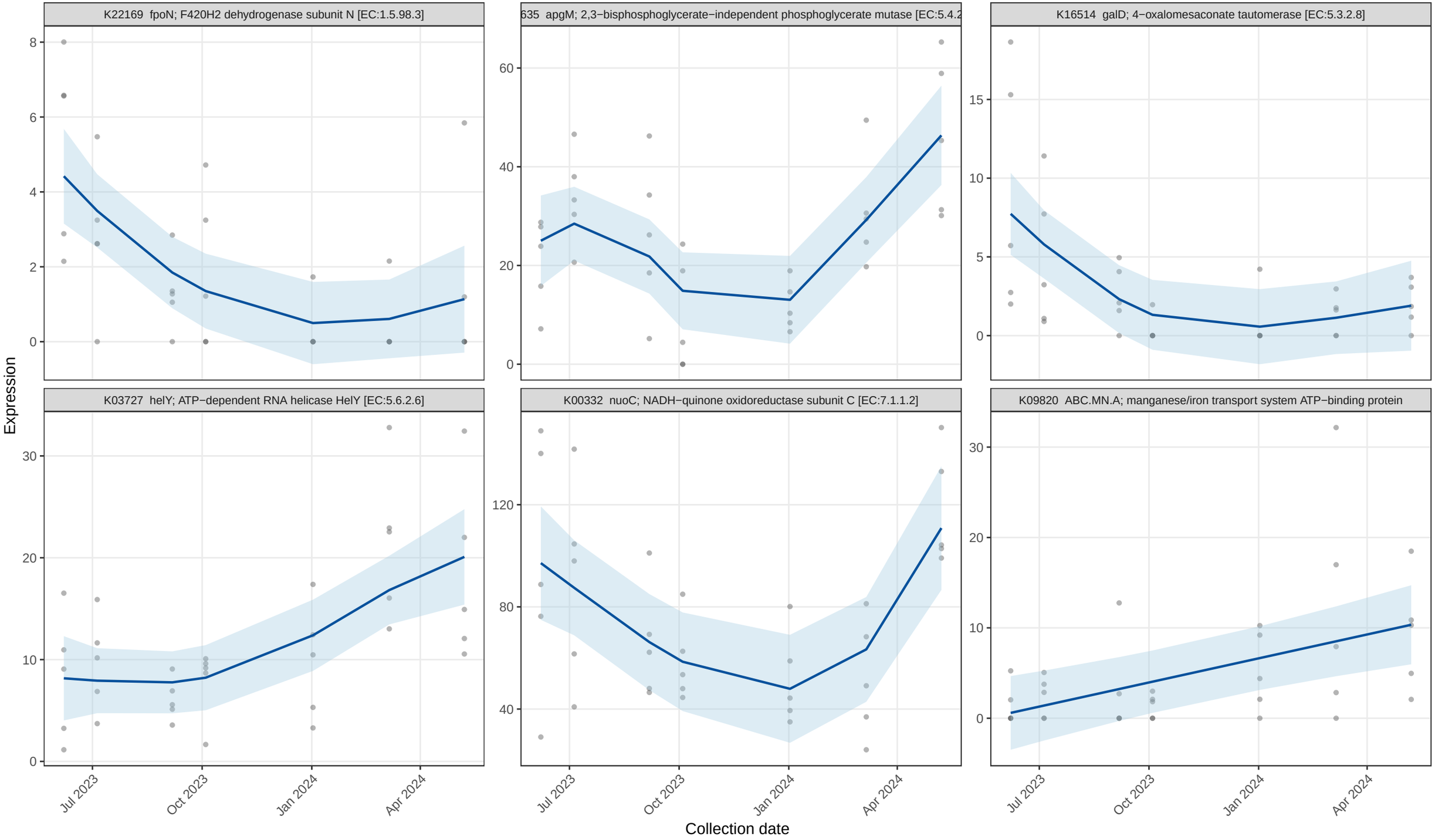

510 fgd1; coenzyme F420−dependent glucose−6−phosphate dehydrogenase [EC:1.1.
K03072 secD; preprotein translocase subunit SecD
K13655 mqsA; HTH−type transcriptional regulator / antitoxin MqsA
K07713 zraR, hydG; two−component system, NtrC family, response regulator HydG
K02198 ccmF; cytochrome c−type biogenesis protein CcmF
K02020 modA; molybdate transport system substrate−binding protein
Expression
Collection date
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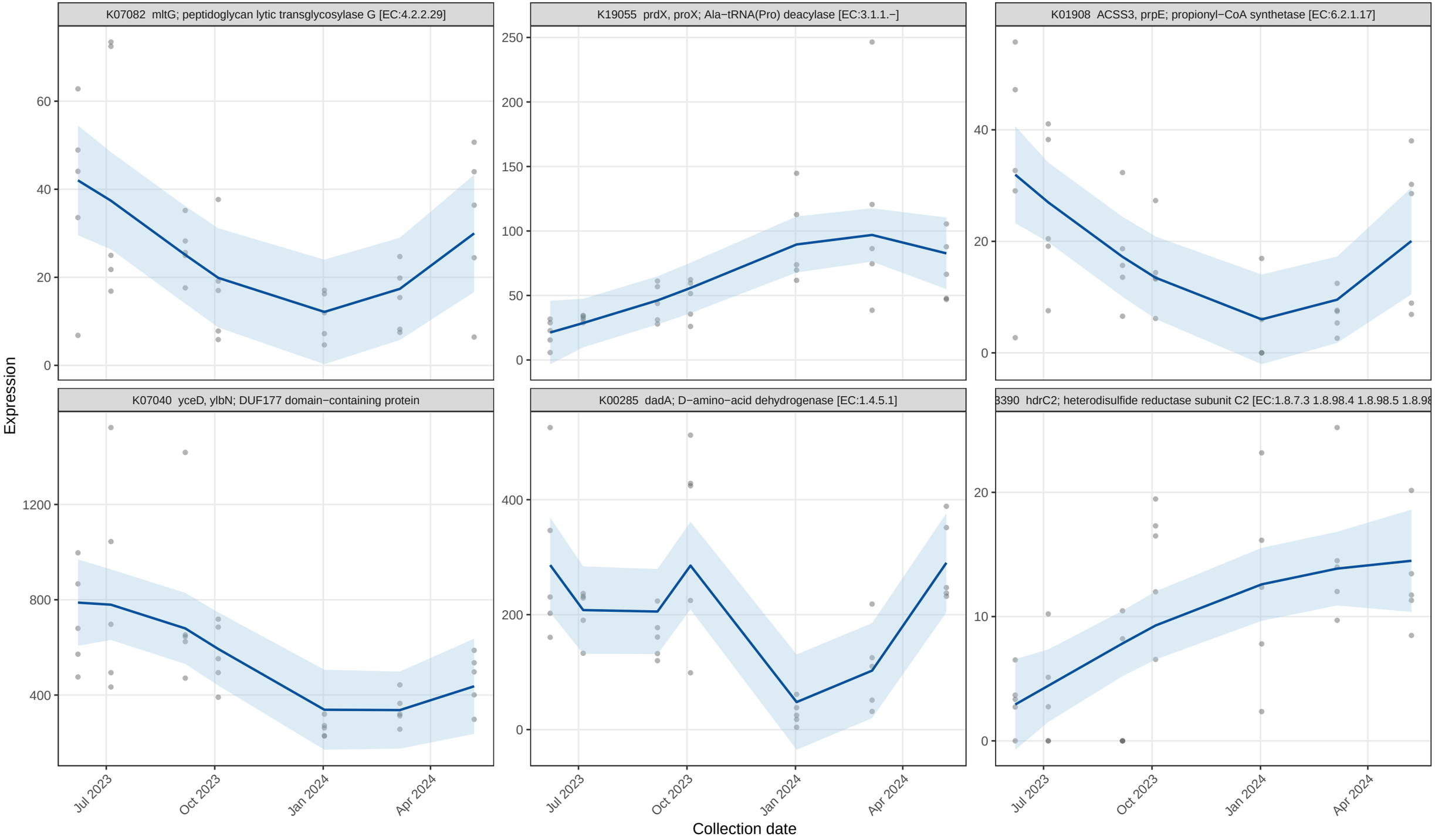

K05378 cpeC, mpeC; phycoerythrin−associated linker protein
K01872 AARS, alaS; alanyl−tRNA synthetase [EC:6.1.1.7]
K11102 gltP, gltT; proton glutamate symport protein
K01890 FARSB, pheT; phenylalanyl−tRNA synthetase beta chain [EC:6.1.1.20]
K07038 K07038; inner membrane protein
K04797 pfdA, PFDN5; prefoldin alpha subunit
Expression
Collection date
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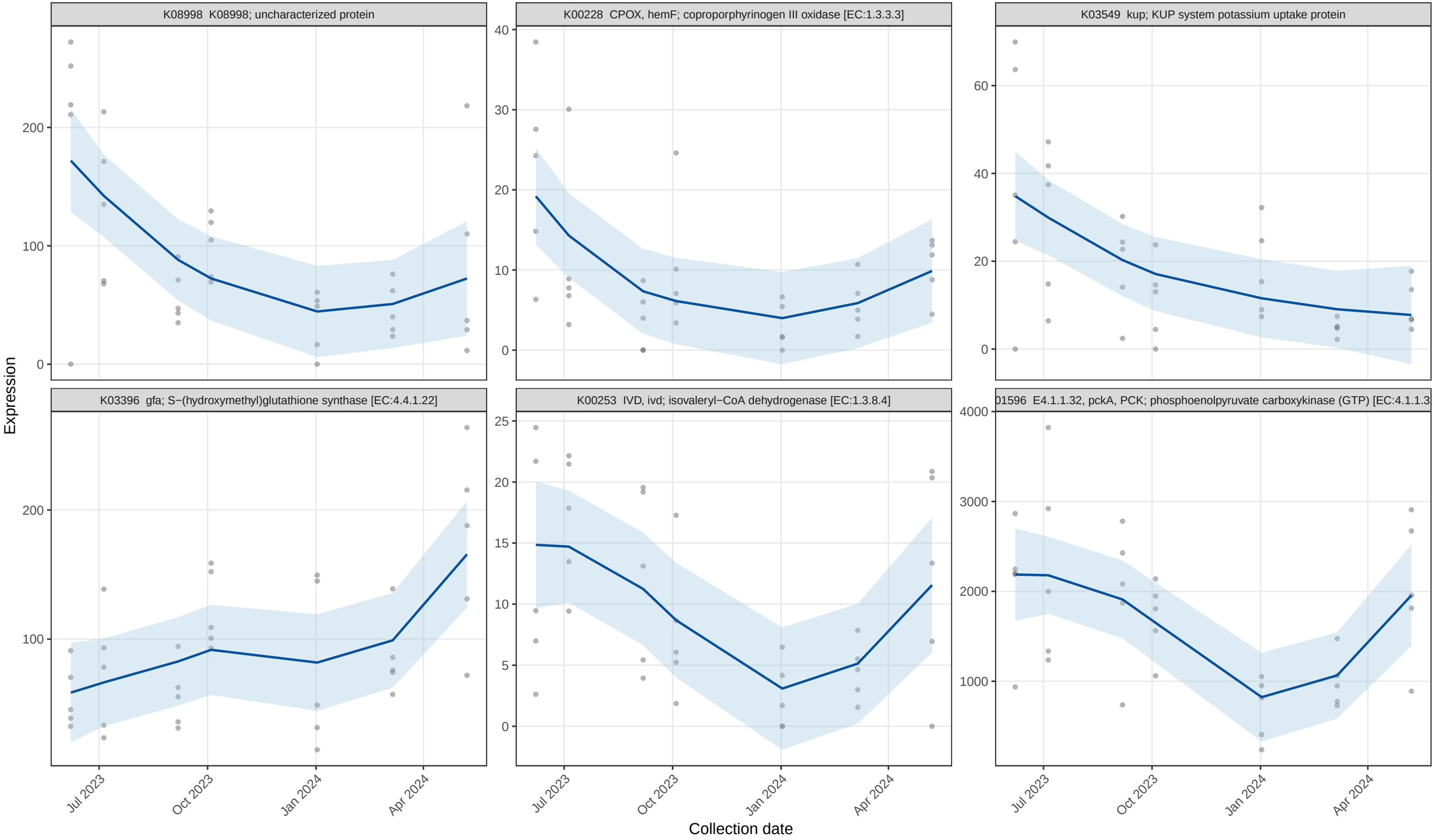

K19686 vapC28S; ribonuclease VapC28/30/34/36/42 [EC:3.1.−.−]
K13482 xdhB; xanthine dehydrogenase large subunit [EC:1.17.1.4]
00162 PDHB, pdhB; pyruvate dehydrogenase E1 component subunit beta [EC:1.2.4.
K01738 cysK; cysteine synthase [EC:2.5.1.47]
K01902 sucD; succinyl−CoA synthetase alpha subunit [EC:6.2.1.5]
K02046 cysU; sulfate/thiosulfate transport system permease protein
Expression
Collection date
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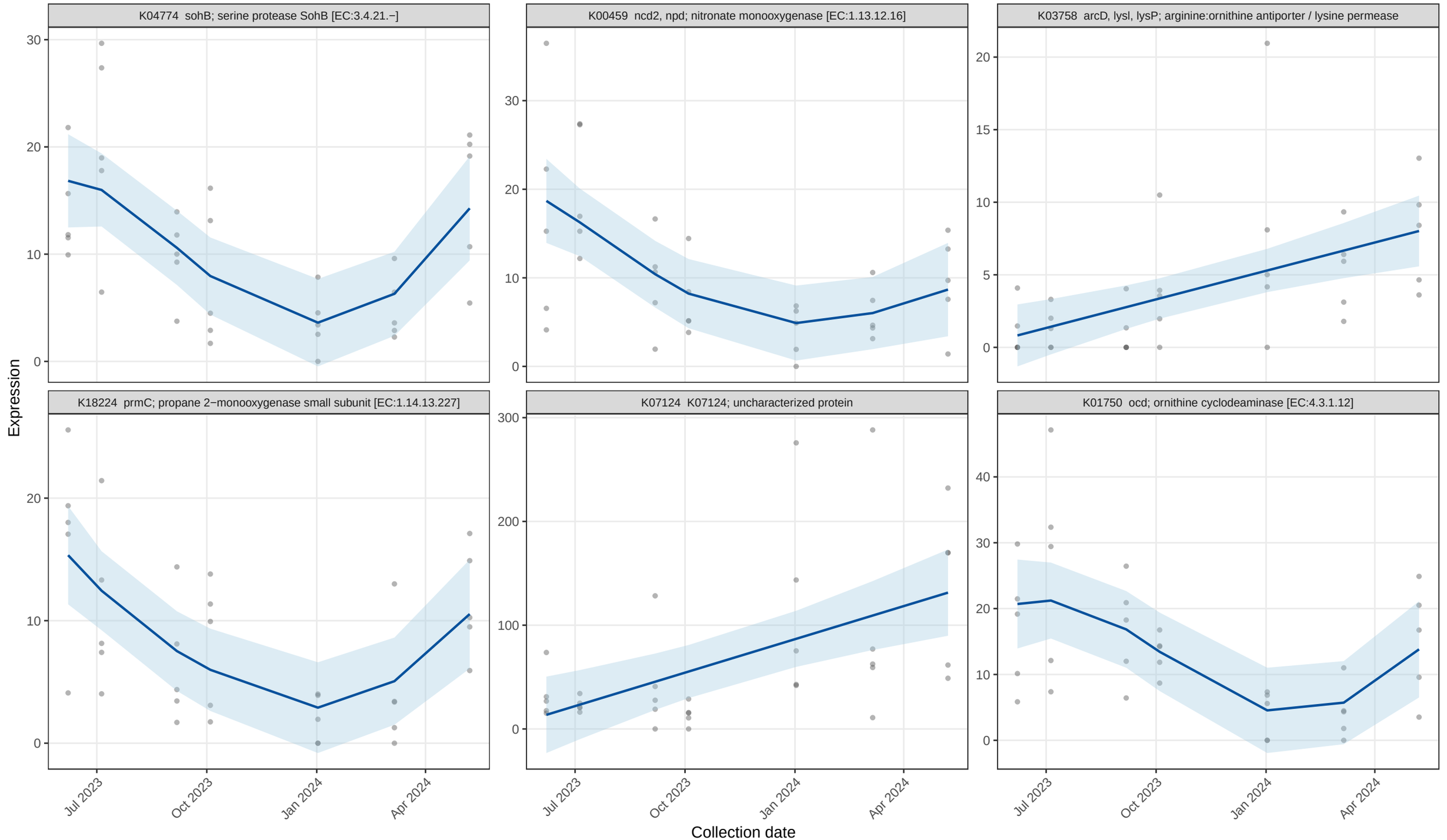

K00108 betA, CHDH; choline dehydrogenase [EC:1.1.99.1]
K03431 glmM; phosphoglucosamine mutase [EC:5.4.2.10]
K19804 lapB; lipopolysaccharide assembly protein B
K09883 cobT; cobaltochelatase CobT [EC:6.6.1.2]
K00297 metF, MTHFR; methylenetetrahydrofolate reductase (NADH) [EC:1.5.1.54]
09458 fabF, OXSM, CEM1; 3−oxoacyl−[acyl−carrier−protein] synthase II [EC:2.3.1.17
Expression
Collection date
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Apr 2024

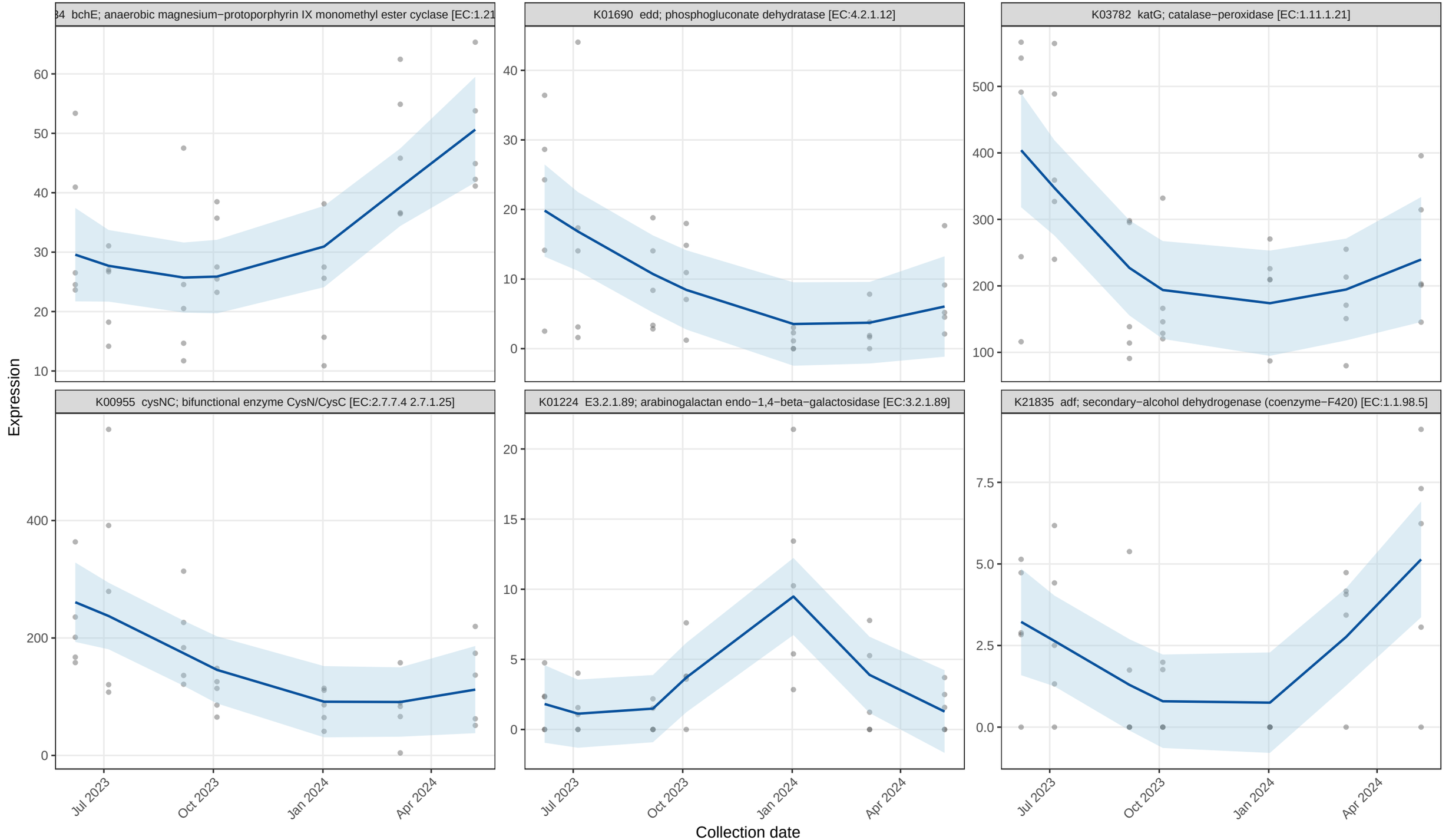

K16792 aksD; methanogen homoaconitase large subunit [EC:4.2.1.1.114]
ptsI; phosphoenolpyruvate−protein phosphotransferase (PTS system enzyme I) [EC:
K01932 capB, pgsB; gamma−polyglutamate synthase [EC:6.3.2.−]
K01577 oxc; oxalyl−CoA decarboxylase [EC:4.1.1.8]
K06942 ychF; ribosome−binding ATPase
K17225 soxC; sulfane dehydrogenase subunit SoxC
Expression
Collection date
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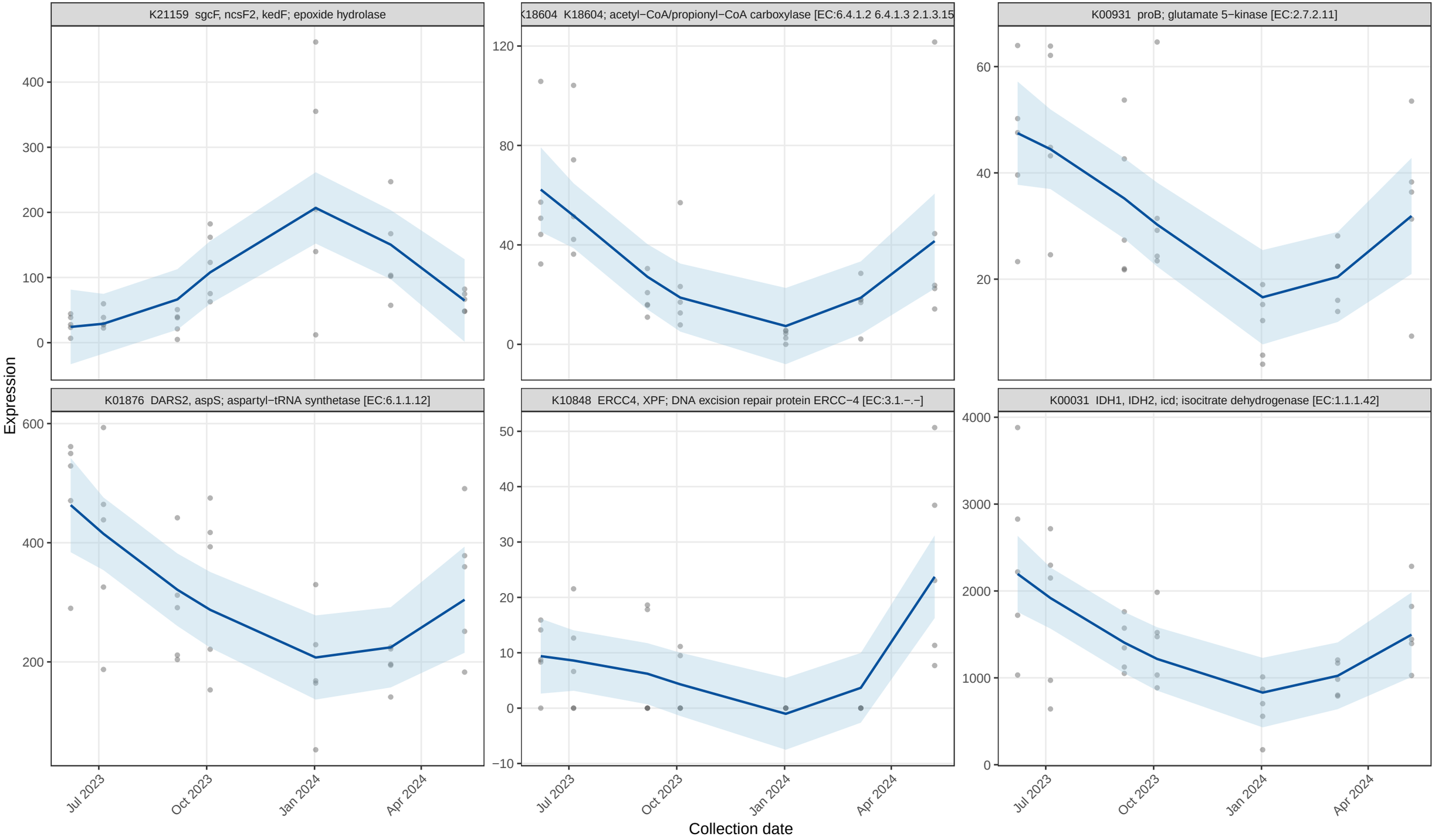

K22169 fpoN; F420H2 dehydrogenase subunit N [EC:1.5.98.3]
635 apgM; 2,3−bisphosphoglycerate−independent phosphoglycerate mutase [EC:5.4.2
K16514 galD; 4−oxalomesaconate tautomerase [EC:5.3.2.8]
K03727 helY; ATP−dependent RNA helicase HelY [EC:5.6.2.6]
K00332 nuoC; NADH−quinone oxidoreductase subunit C [EC:7.1.1.2]
K09820 ABC.MN.A; manganese/iron transport system ATP−binding protein
Expression
Collection date
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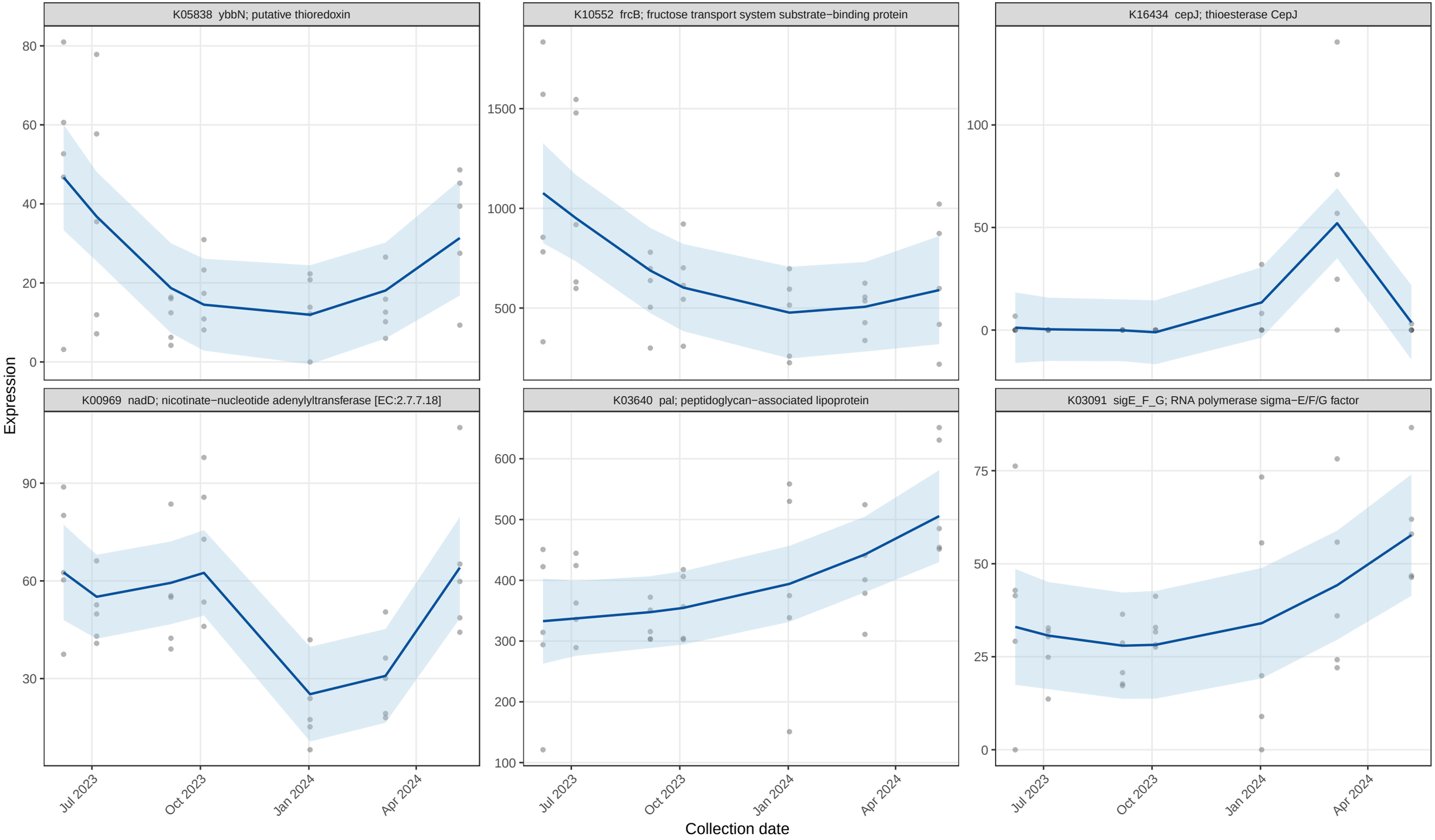

K07082 mltG; peptidoglycan lytic transglycosylase G [EC:4.2.2.29]
K19055 prdX, proX; Ala−tRNA(Pro) deacylase [EC:3.1.1.−]
K01908 ACSS3, prpE; propionyl−CoA synthetase [EC:6.2.1.17]
K07040 yceD, ylbN; DUF177 domain−containing protein
K00285 dadA; D−amino−acid dehydrogenase [EC:1.4.5.1]
3390 hdrC2; heterodisulfide reductase subunit C2 [EC:1.8.7.3 1.8.98.4 1.8.98.5 1.8.98
Expression
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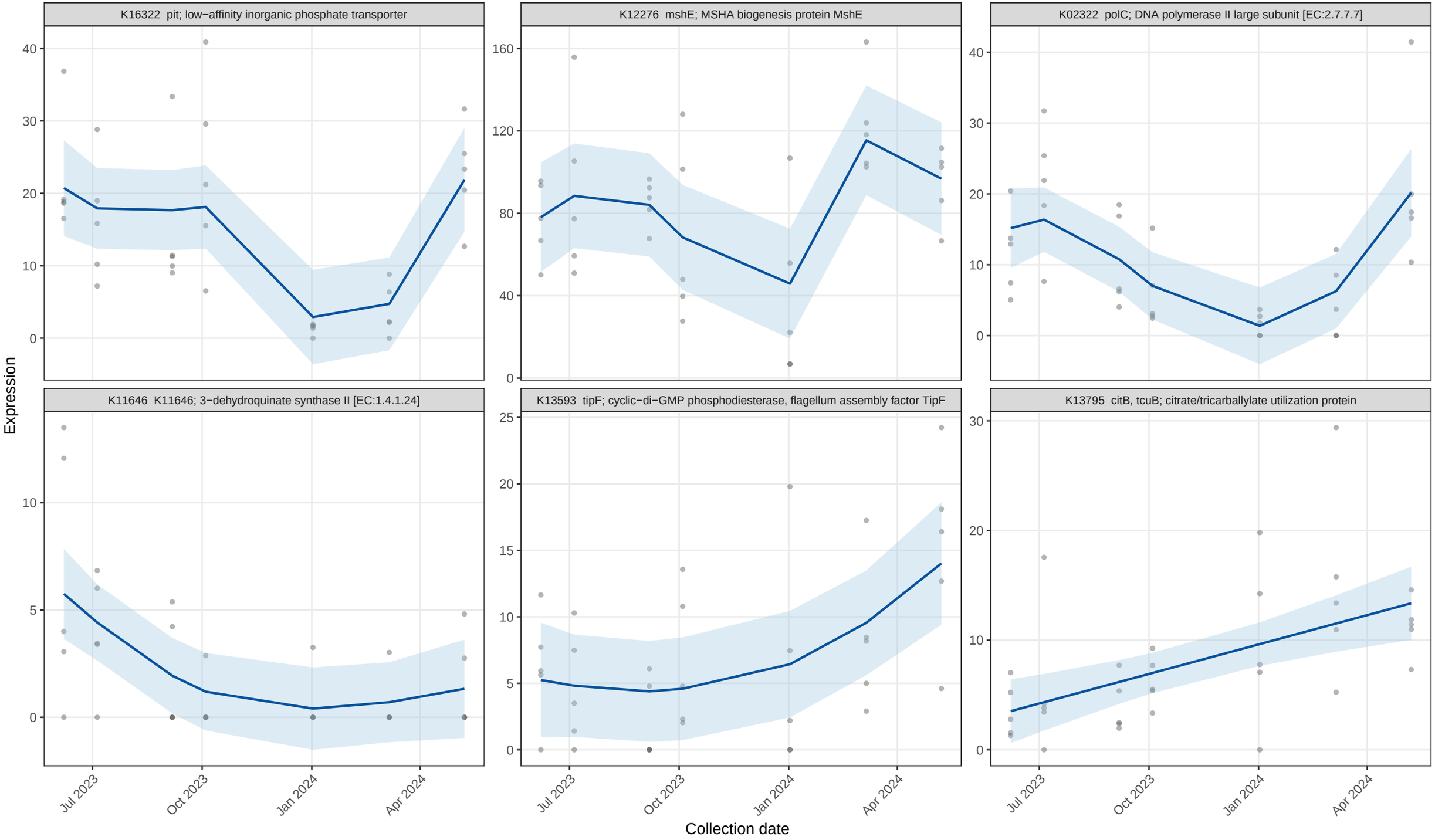

K08998 K08998; uncharacterized protein
K00228 CPOX, hemF; coproporphyrinogen III oxidase [EC:1.3.3.3]
K03549 kup; KUP system potassium uptake protein
K03396 gfa; S−(hydroxymethyl)glutathione synthase [EC:4.4.1.22]
K00253 IVD, ivd; isovaleryl−CoA dehydrogenase [EC:1.3.8.4]
01596 E4.1.1.32, pckA, PCK; phosphoenolpyruvate carboxykinase (GTP) [EC:4.1.1.3
Expression
Collection date
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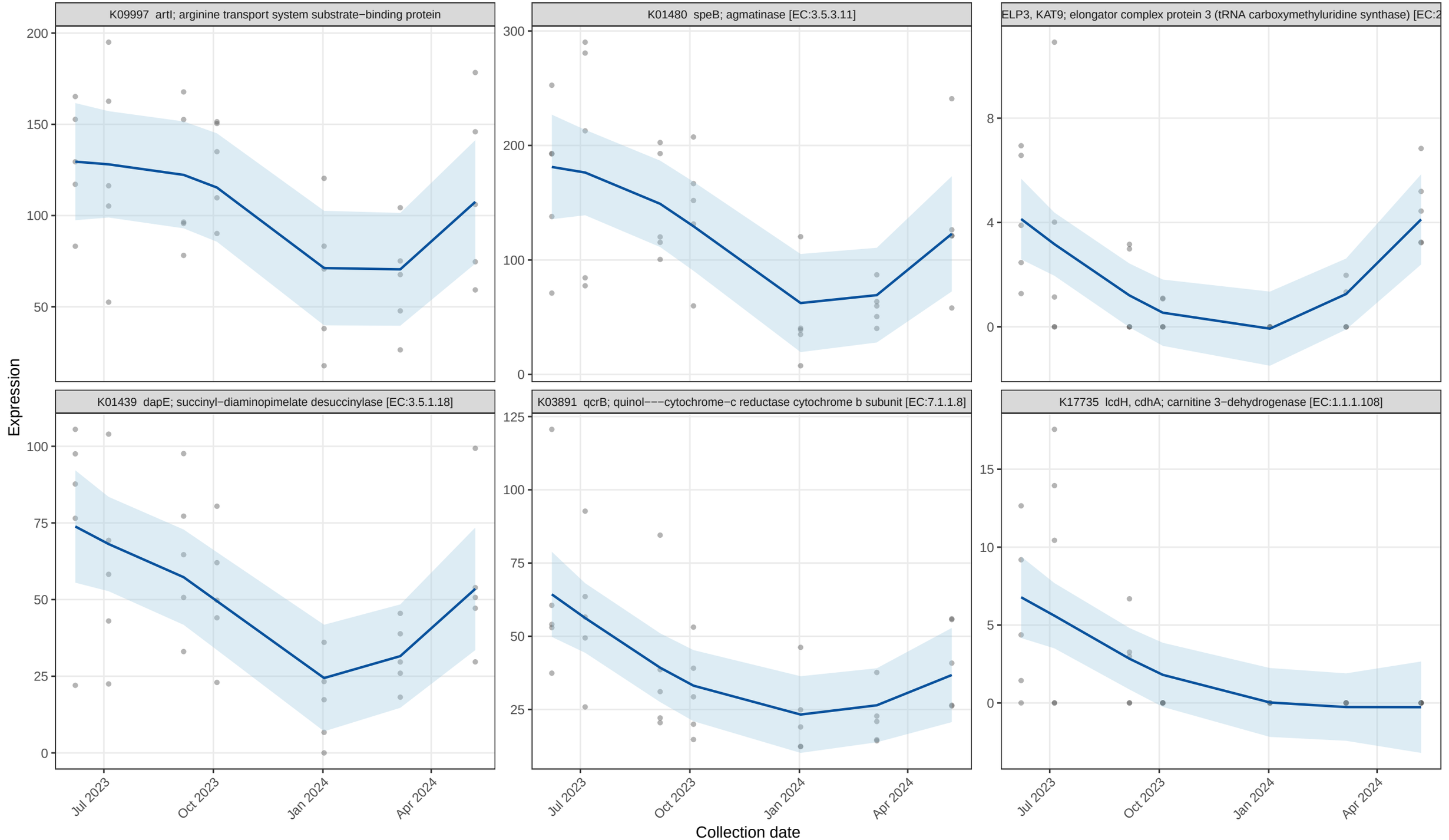

K04774 sohB; serine protease SohB [EC:3.4.21.−]
K00459 ncd2, npd; nitronate monooxygenase [EC:1.13.12.16]
K03758 arcD, lysI, lysP; arginine:ornithine antiporter / lysine permease
K18224 prmC; propane 2−monooxygenase small subunit [EC:1.14.13.227]
K07124 K07124; uncharacterized protein
K01750 ocd; ornithine cyclodeaminase [EC:4.3.1.12]
Expression
Collection date
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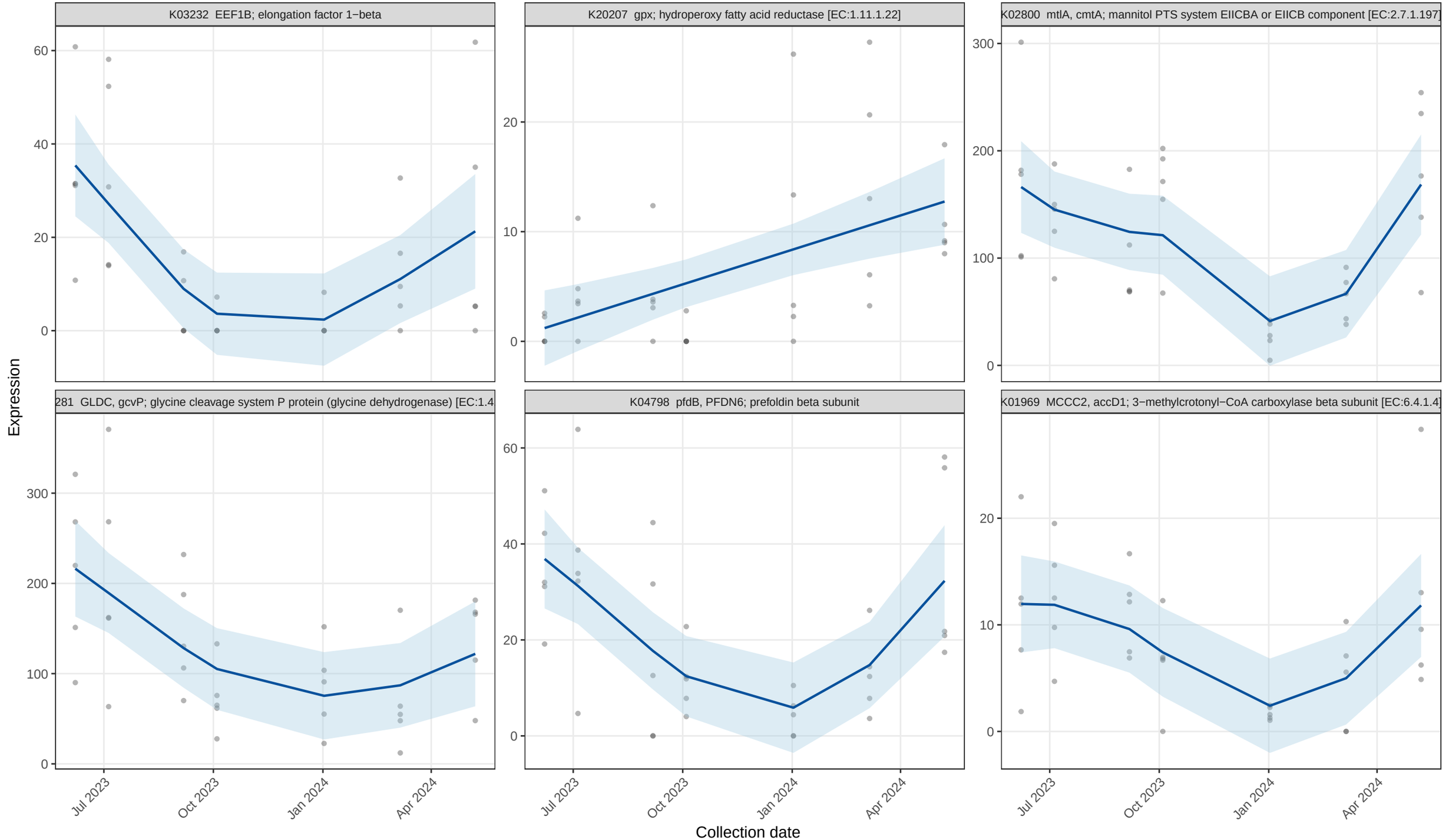

34 bchE; anaerobic magnesium−protoporphyrin IX monomethyl ester cyclase [EC:1.21
K01690 edd; phosphogluconate dehydratase [EC:4.2.1.12]
K03782 katG; catalase−peroxidase [EC:1.11.1.21]
K00955 cysNC; bifunctional enzyme CysN/CysC [EC:2.7.7.4 2.7.1.25]
K01224 E3.2.1.89; arabinogalactan endo−1,4−beta−galactosidase [EC:3.2.1.89]
K21835 adf; secondary−alcohol dehydrogenase (coenzyme−F420) [EC:1.1.98.5]
Expression
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Collection date

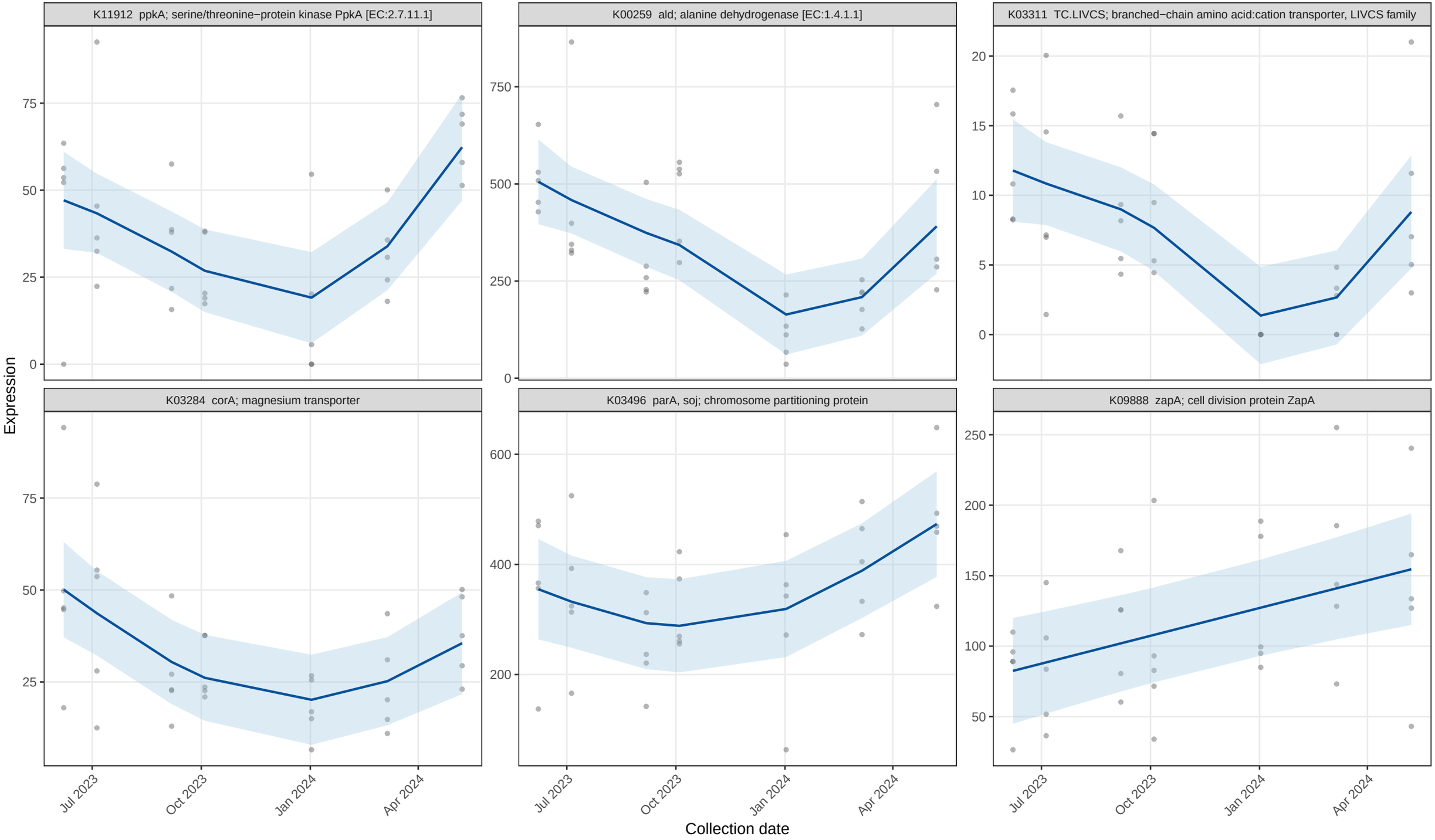

K21159 sgcF, ncsF2, kedF; epoxide hydrolase
K18604 K18604; acetyl−CoA/propionyl−CoA carboxylase [EC:6.4.1.2 6.4.1.3 2.1.3.15]
K00931 proB; glutamate 5−kinase [EC:2.7.2.11]
K01876 DARS2, aspS; aspartyl−tRNA synthetase [EC:6.1.1.12]
K10848 ERCC4, XPF; DNA excision repair protein ERCC−4 [EC:3.1.−.−]
K00031 IDH1, IDH2, icd; isocitrate dehydrogenase [EC:1.1.1.42]
Expression
Collection date
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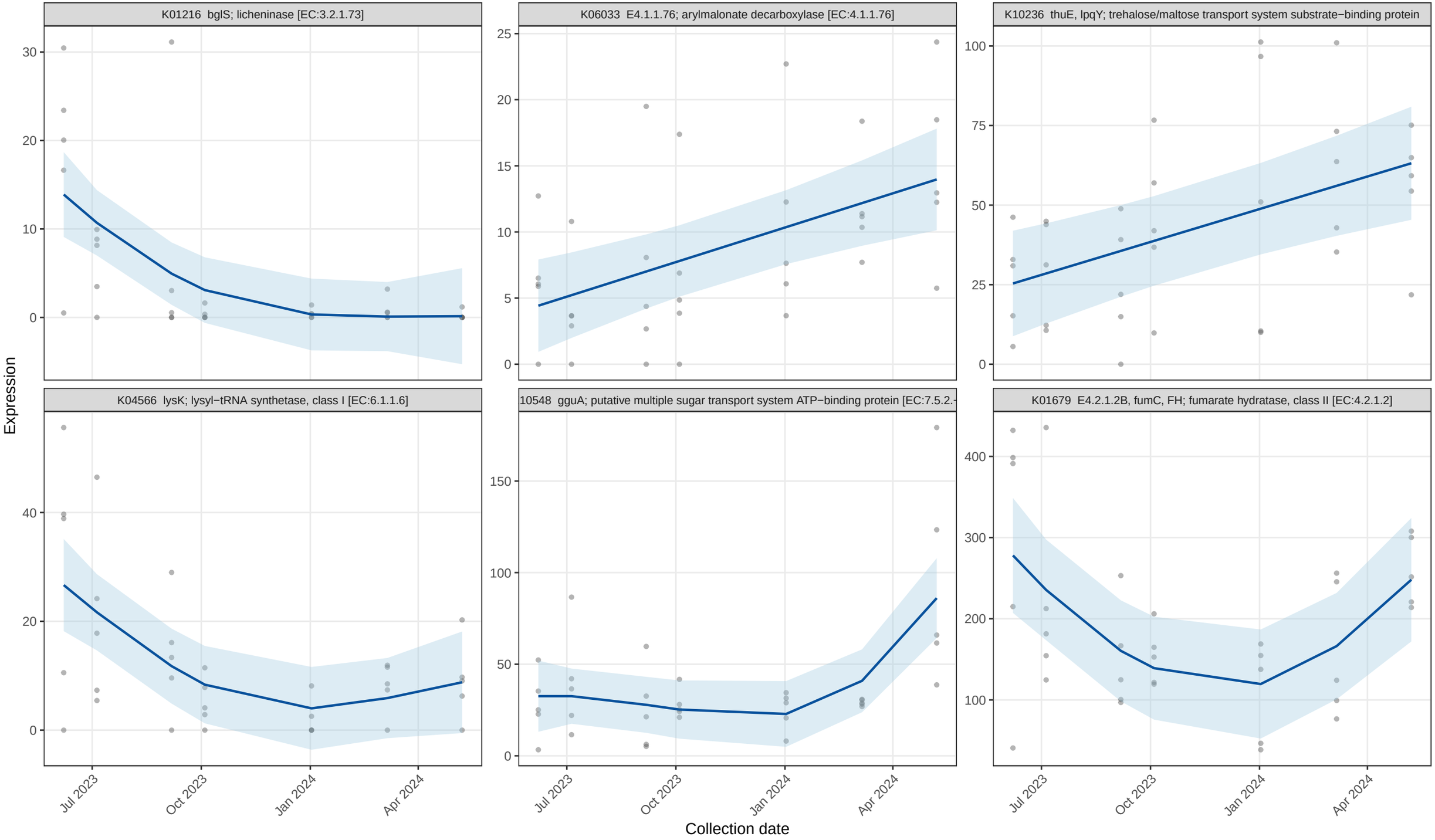

K05838 ybbN; putative thioredoxin
K10552 frcB; fructose transport system substrate−binding protein
K16434 cepJ; thioesterase CepJ
K00969 nadD; nicotinate−nucleotide adenylyltransferase [EC:2.7.7.18]
K03640 pal; peptidoglycan−associated lipoprotein
K03091 sigE_F_G; RNA polymerase sigma−E/F/G factor
Expression
Collection date
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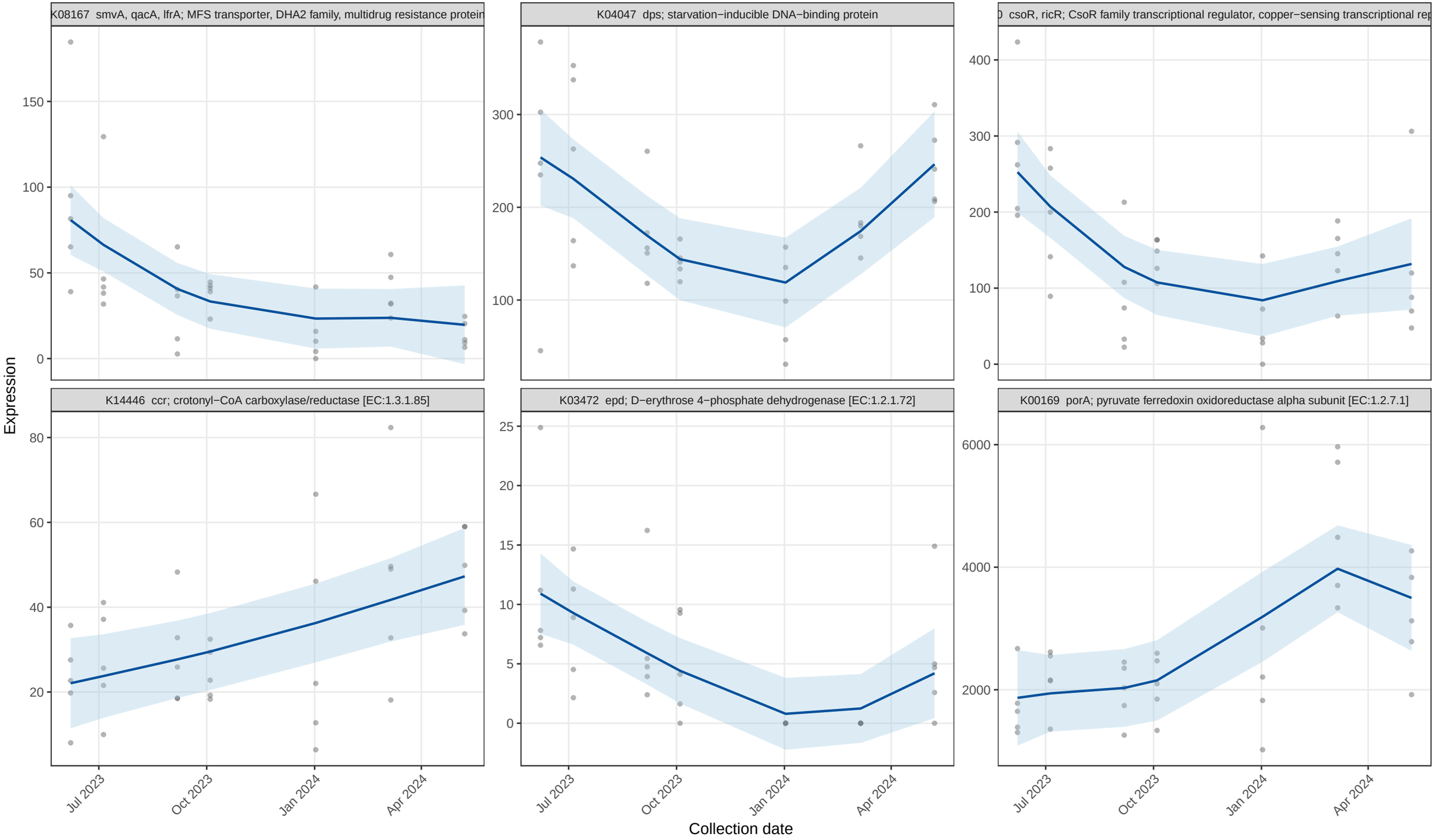

K16322 pit; low−affinity inorganic phosphate transporter
K12276 mshE; MSHA biogenesis protein MshE
K02322 polC; DNA polymerase II large subunit [EC:2.7.7.7]
K11646 K11646; 3−dehydroquinate synthase II [EC:1.4.1.24]
K13593 tipF; cyclic−di−GMP phosphodiesterase, flagellum assembly factor TipF
K13795 citB, tcuB; citrate/tricarballylate utilization protein
Expression
Collection date
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Apr 2024

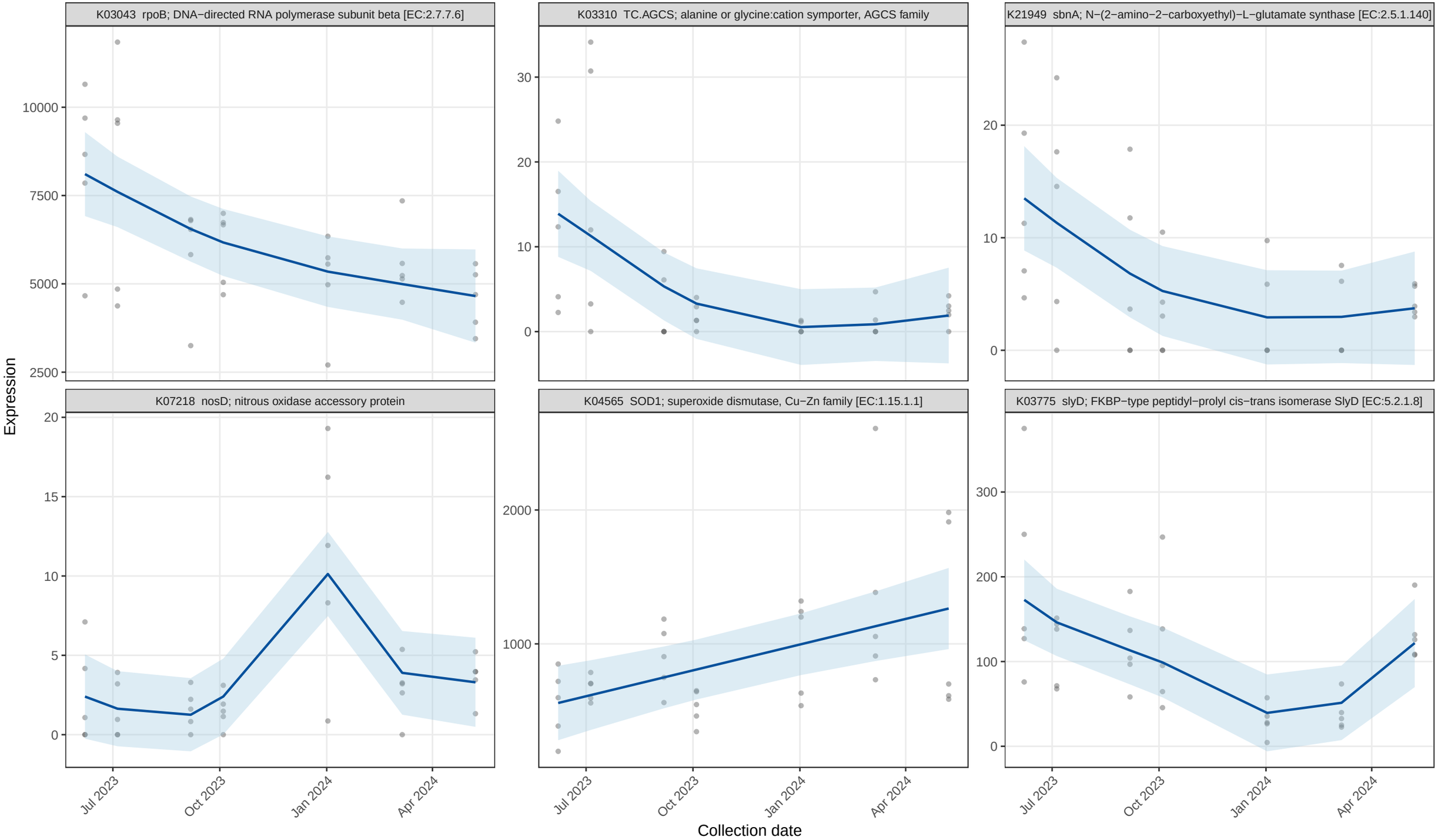

K09997 artI; arginine transport system substrate−binding protein
K01480 speB; agmatinase [EC:3.5.3.11]
ELP3, KAT9; elongator complex protein 3 (tRNA carboxymethyluridine synthase) [EC:2
K01439 dapE; succinyl−diaminopimelate desuccinylase [EC:3.5.1.18]
K03891 qcrB; quinol−−−cytochrome−c reductase cytochrome b subunit [EC:7.1.1.8]
K17735 lcdH, cdhA; carnitine 3−dehydrogenase [EC:1.1.1.108]
Expression
Collection date
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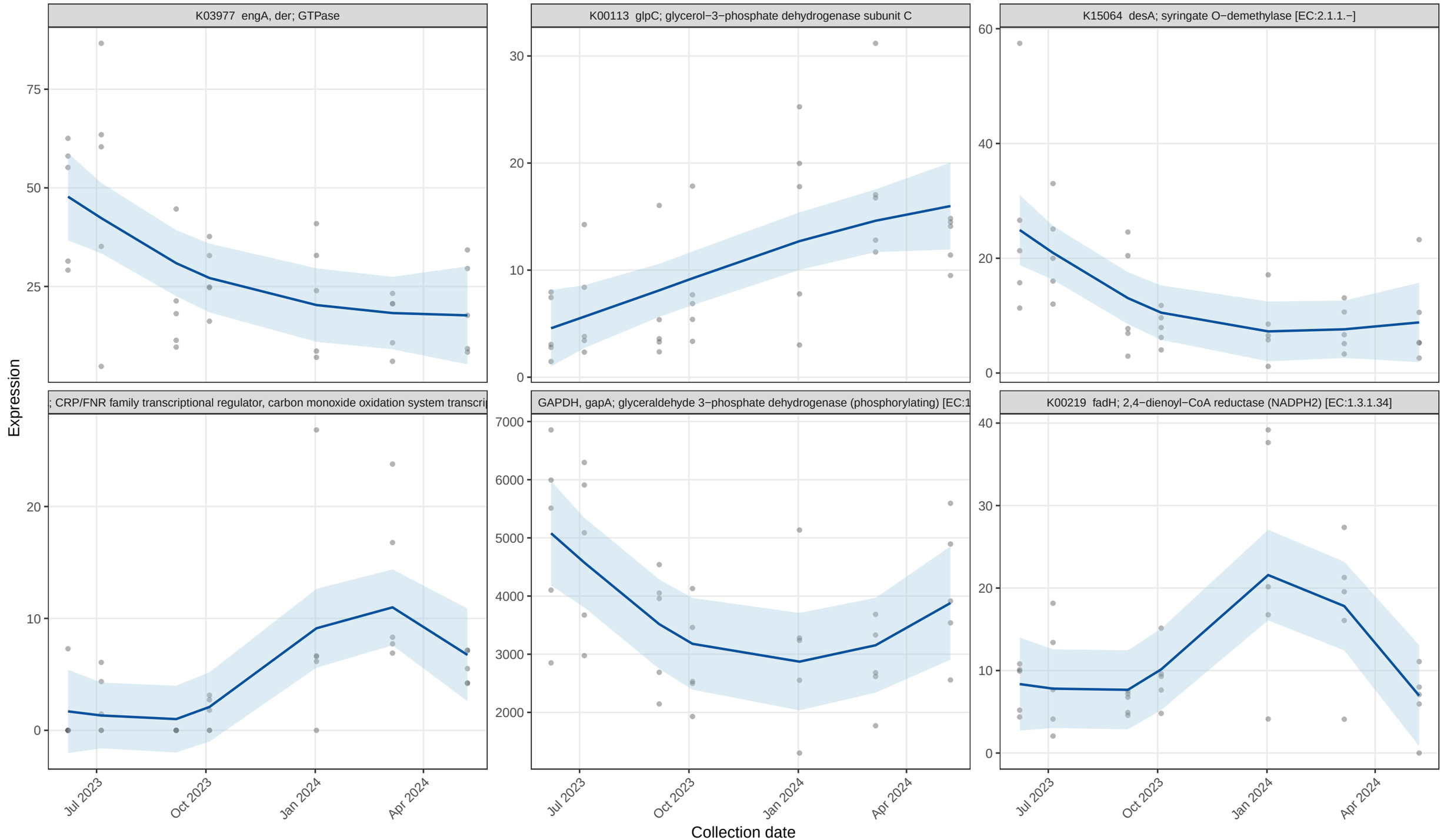

K03232 EEF1B; elongation factor 1−beta
K20207 gpx; hydroperoxy fatty acid reductase [EC:1.11.1.22]
K02800 mtlA, cmtA; mannitol PTS system EIICBA or EIICB component [EC:2.7.1.197]
281 GLDC, gcvP; glycine cleavage system P protein (glycine dehydrogenase) [EC:1.4
K04798 pfdB, PFDN6; prefoldin beta subunit
K01969 MCCC2, accD1; 3−methylcrotonyl−CoA carboxylase beta subunit [EC:6.4.1.4]
Expression
Collection date
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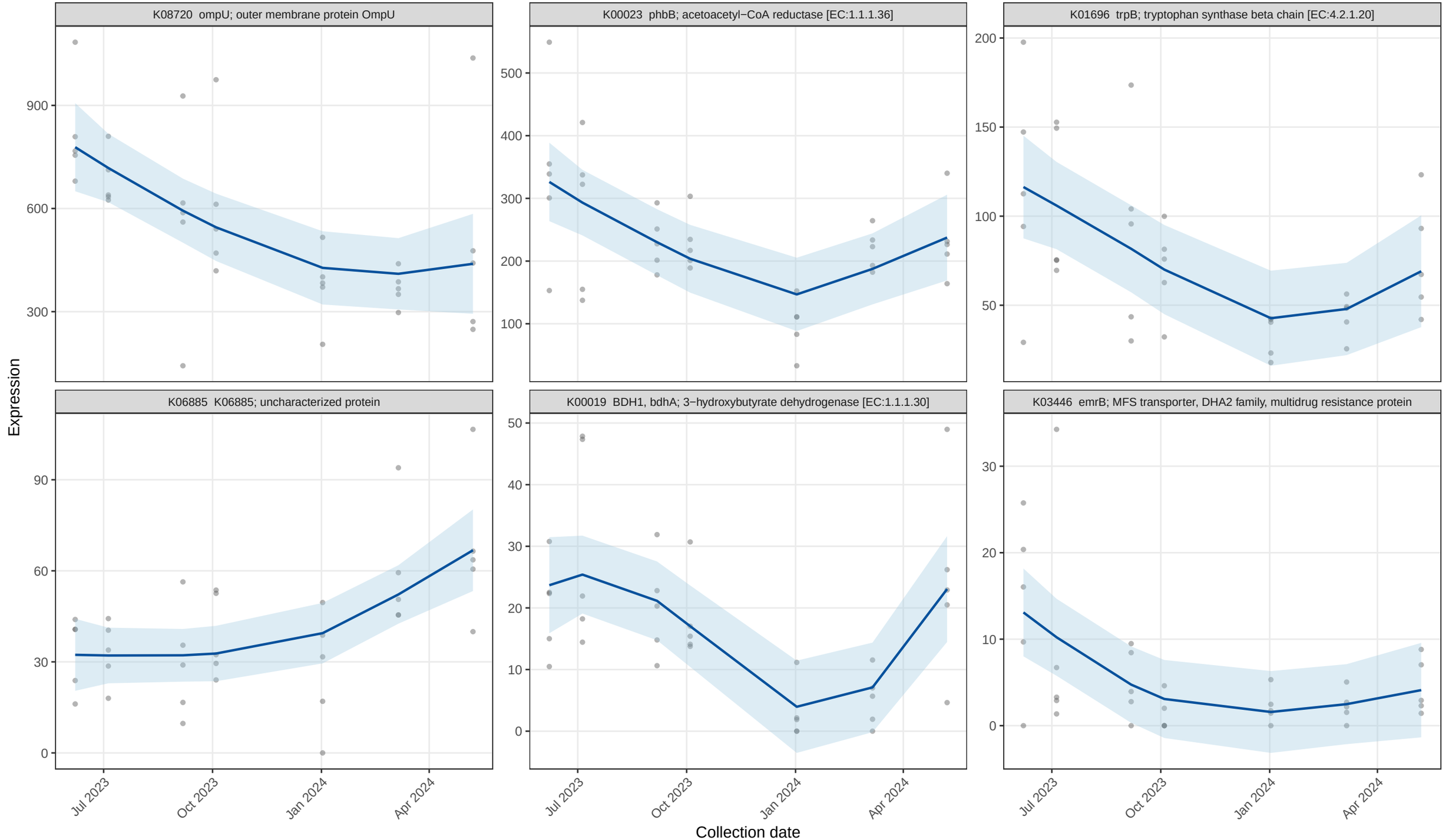

K11912 ppkA; serine/threonine−protein kinase PpkA [EC:2.7.11.1]
K00259 ald; alanine dehydrogenase [EC:1.4.1.1]
K03311 TC.LIVCS; branched−chain amino acid:cation transporter, LIVCS family
K03284 corA; magnesium transporter
K03496 parA, soj; chromosome partitioning protein
K09888 zapA; cell division protein ZapA
Expression
Collection date
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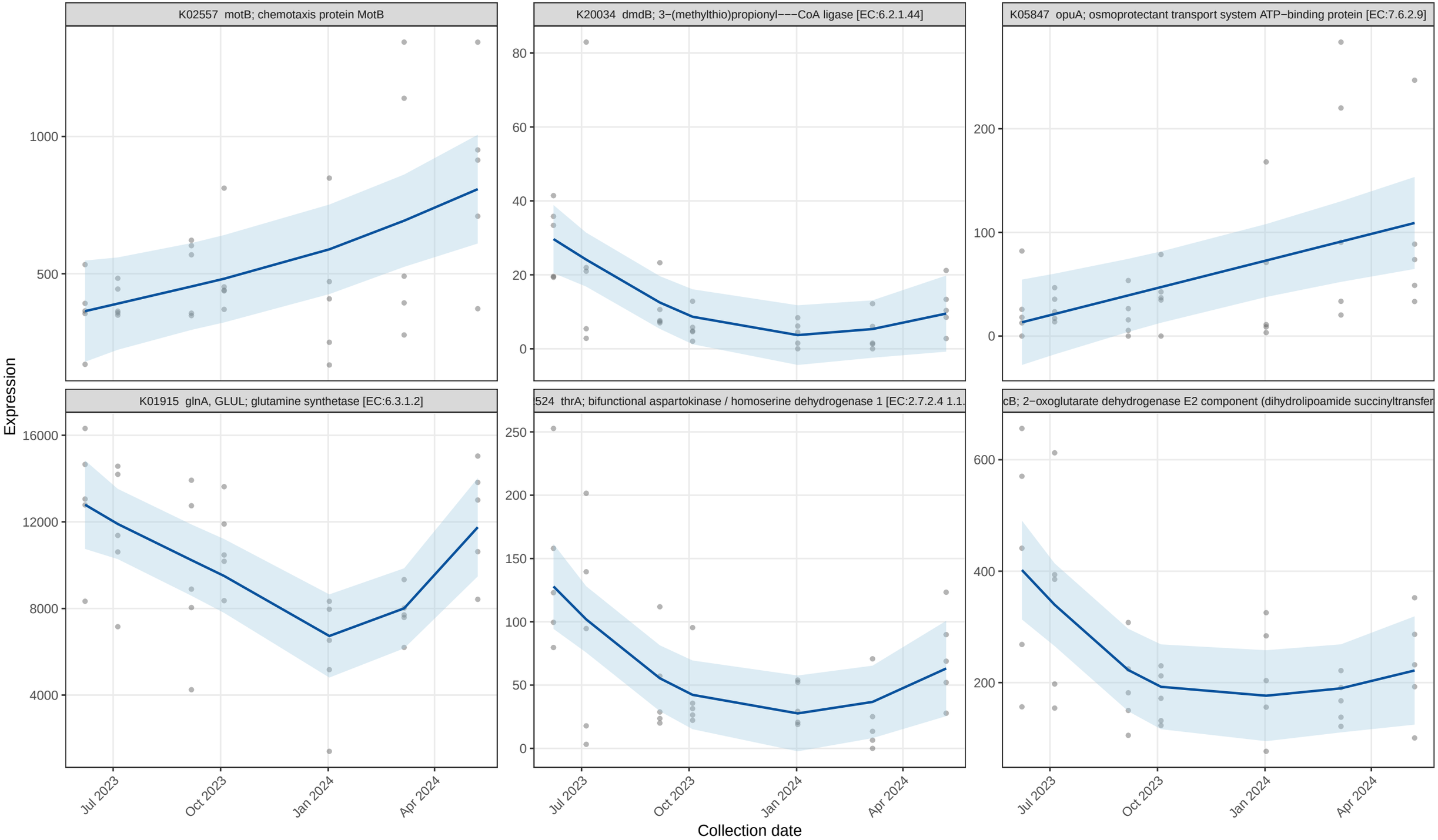

K01216 bglS; licheninase [EC:3.2.1.73]
K06033 E4.1.1.76; arylmalonate decarboxylase [EC:4.1.1.76]
K10236 thuE, lpqY; trehalose/maltose transport system substrate−binding protein
K04566 lysK; lysyl−tRNA synthetase, class I [EC:6.1.1.6]
10548 gguA; putative multiple sugar transport system ATP−binding protein [EC:7.5.2.-
K01679 E4.2.1.2B, fumC, FH; fumarate hydratase, class II [EC:4.2.1.2]
Expression
Collection date
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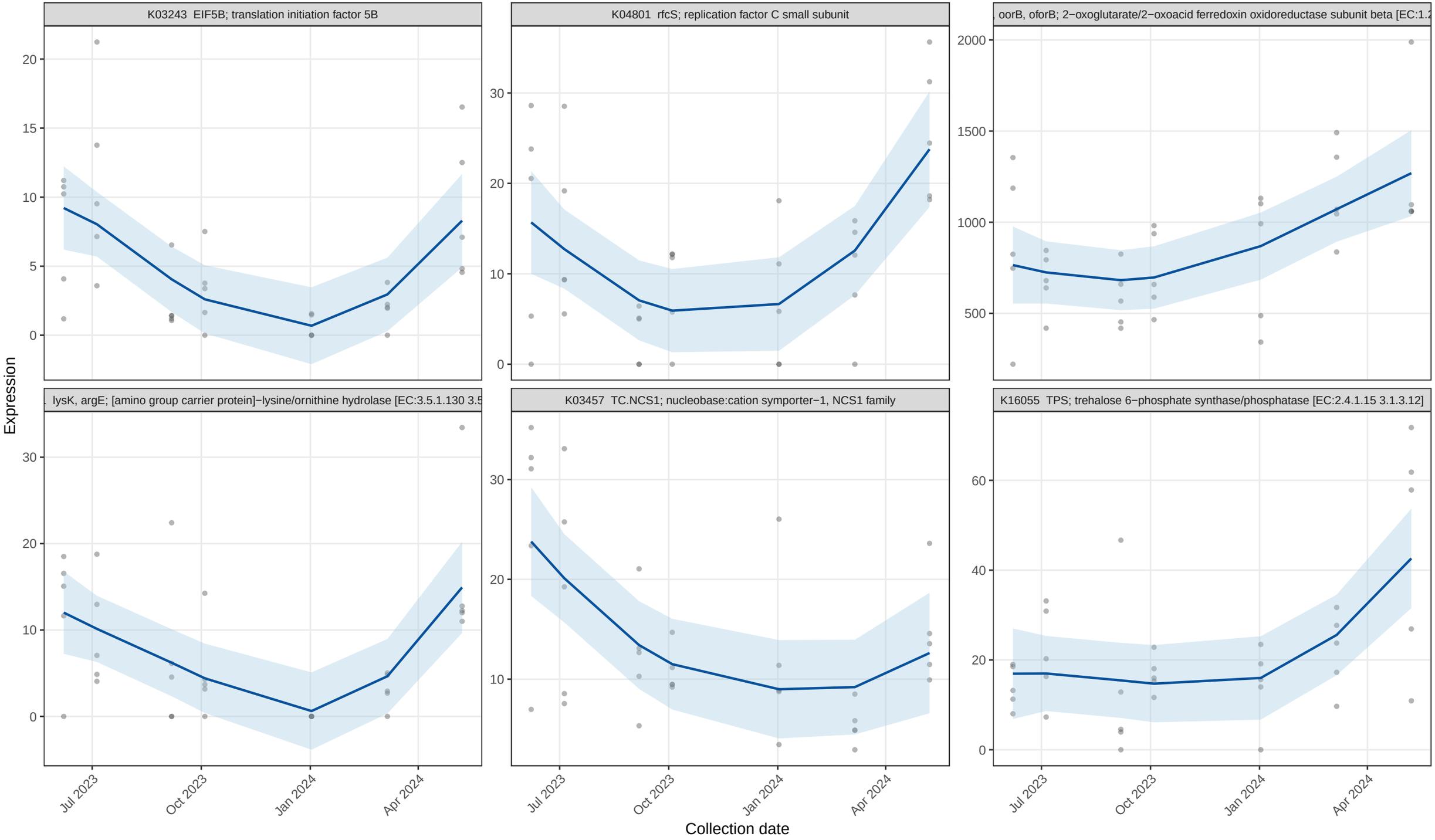

K08167 smvA, qacA, lfrA; MFS transporter, DHA2 family, multidrug resistance protein
K04047 dps; starvation−inducible DNA−binding protein
csoR, ricR; CsoR family transcriptional regulator, copper−sensing transcriptional rep
K14446 ccr; crotonyl−CoA carboxylase/reductase [EC:1.3.1.85]
K03472 epd; D−erythrose 4−phosphate dehydrogenase [EC:1.2.1.72]
K00169 porA; pyruvate ferredoxin oxidoreductase alpha subunit [EC:1.2.7.1]
Expression
Collection date
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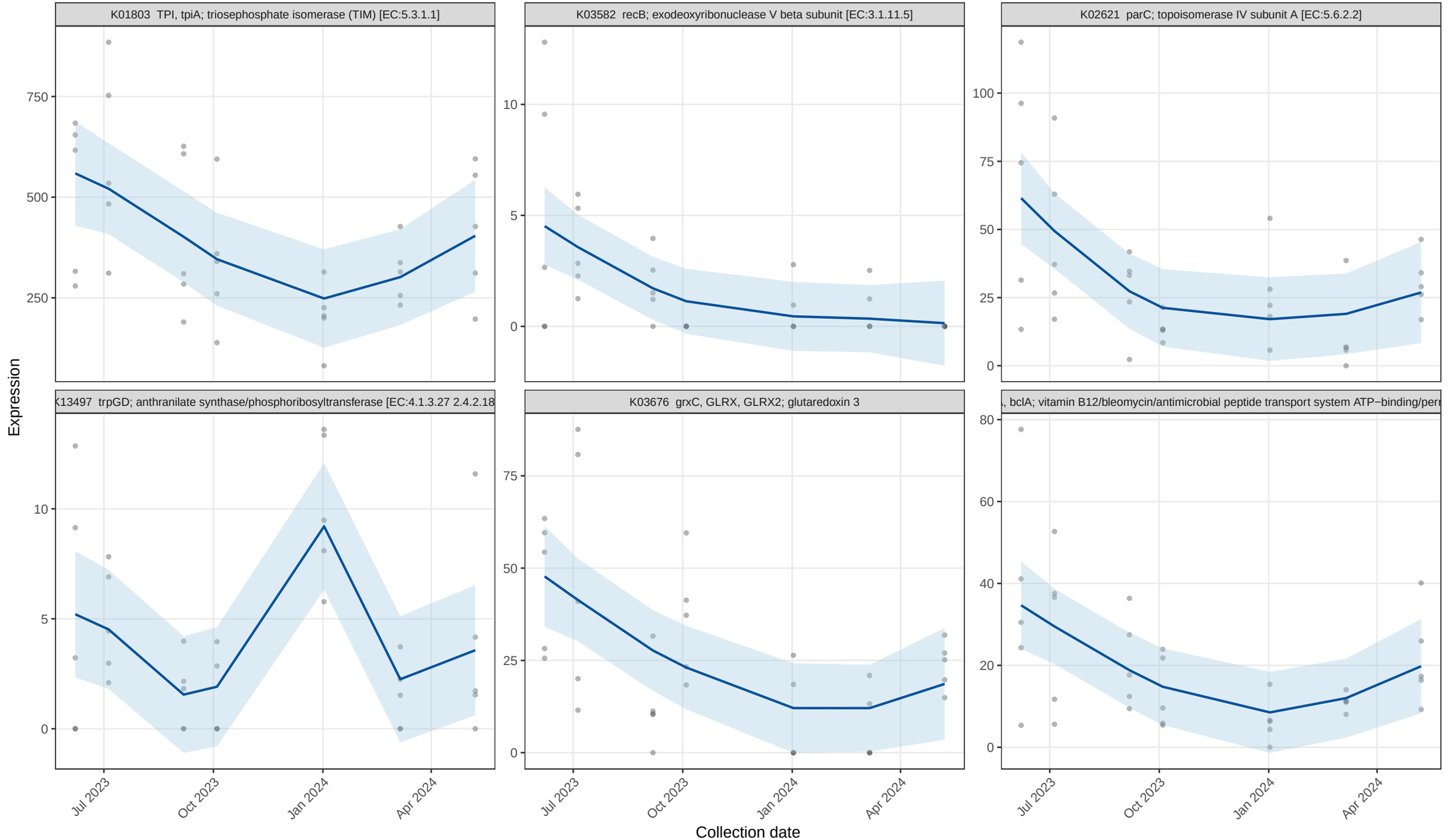

K03043 rpoB; DNA−directed RNA polymerase subunit beta [EC:2.7.7.6]
K03310 TC.AGCS; alanine or glycine:cation symporter, AGCS family
K21949 sbnA; N−(2−amino−2−carboxyethyl)−L−glutamate synthase [EC:2.5.1.140]
K07218 nosD; nitrous oxidase accessory protein
K04565 SOD1; superoxide dismutase, Cu−Zn family [EC:1.15.1.1]
K03775 slyD; FKBP−type peptidyl−prolyl cis−trans isomerase SlyD [EC:5.2.1.8]
Expression
Collection date
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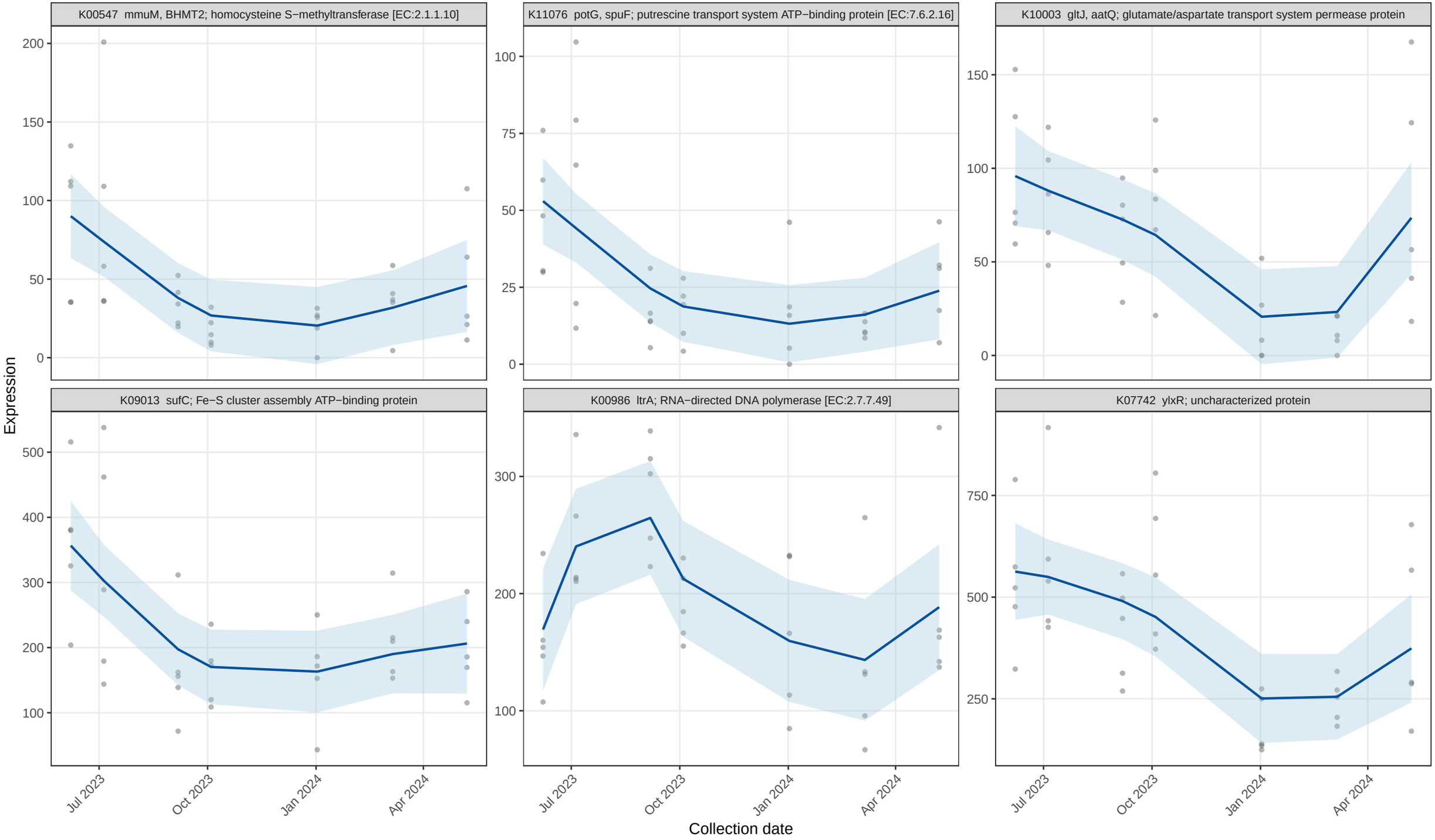

K03977 engA, der; GTPase
K00113 glpC; glycerol−3−phosphate dehydrogenase subunit C
K15064 desA; syringate O−demethylase [EC:2.1.1.−]
; CRP/FNR family transcriptional regulator, carbon monoxide oxidation system transcri
GAPDH, gapA; glyceraldehyde 3−phosphate dehydrogenase (phosphorylating) [EC:1
K00219 fadH; 2,4−dienoyl−CoA reductase (NADPH2) [EC:1.3.1.34]
Expression
Collection date
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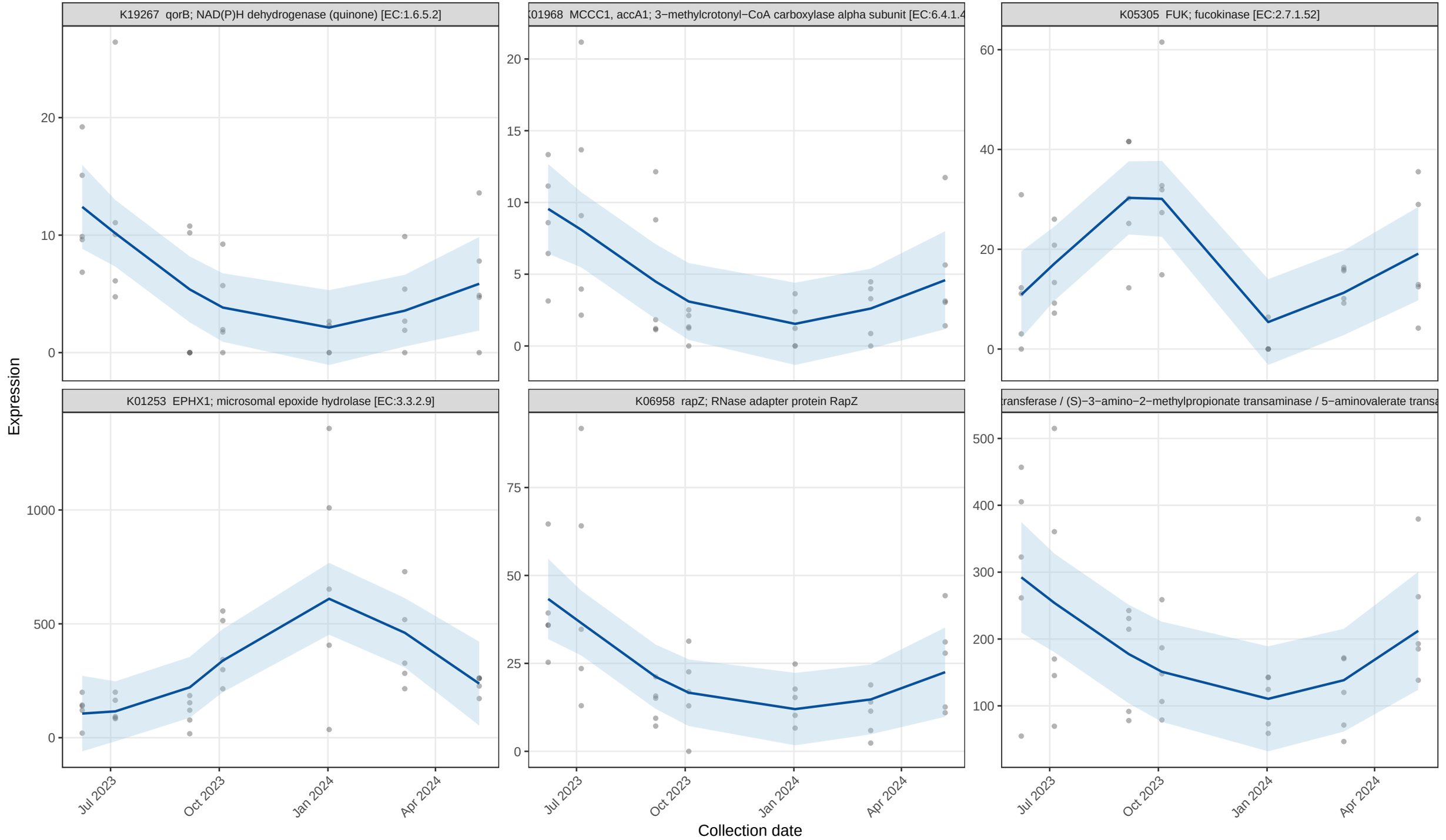

K08720 ompU; outer membrane protein OmpU
K00023 phbB; acetoacetyl−CoA reductase [EC:1.1.1.36]
K01696 trpB; tryptophan synthase beta chain [EC:4.2.1.20]
K06885 K06885; uncharacterized protein
K00019 BDH1, bdhA; 3−hydroxybutyrate dehydrogenase [EC:1.1.1.30]
K03446 emrB; MFS transporter, DHA2 family, multidrug resistance protein
Expression
Collection date
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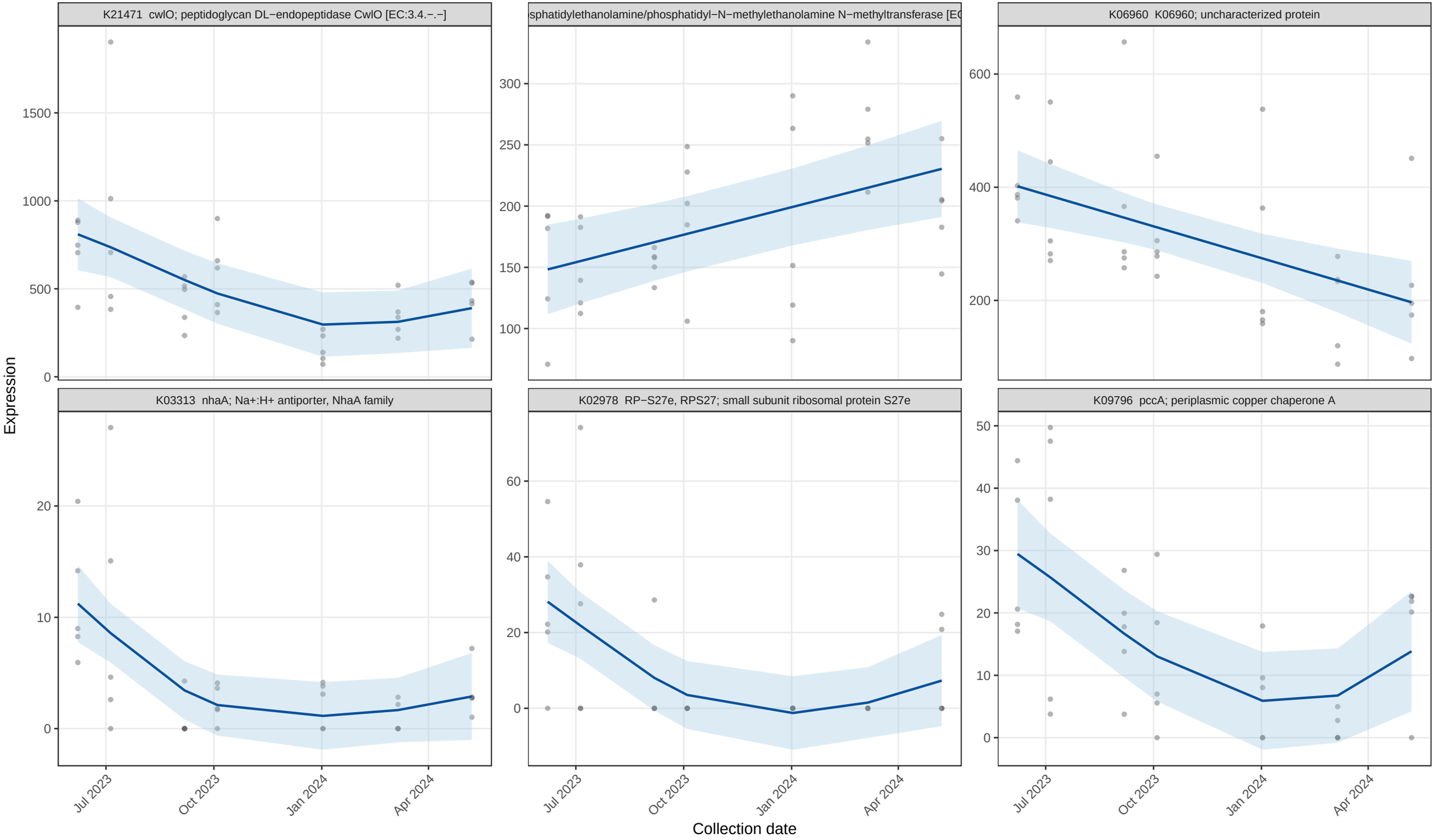

K02557 motB; chemotaxis protein MotB
K20034 dmdB; 3−(methylthio)propionyl−−−CoA ligase [EC:6.2.1.44]
K05847 opuA; osmoprotectant transport system ATP−binding protein [EC:7.6.2.9]
K01915 glnA, GLUL; glutamine synthetase [EC:6.3.1.2]
524 thrA; bifunctional aspartokinase / homoserine dehydrogenase 1 [EC:2.7.2.4 1.1.
cB; 2−oxoglutarate dehydrogenase E2 component (dihydrolipoamide succinyltransfer
Expression
Collection date
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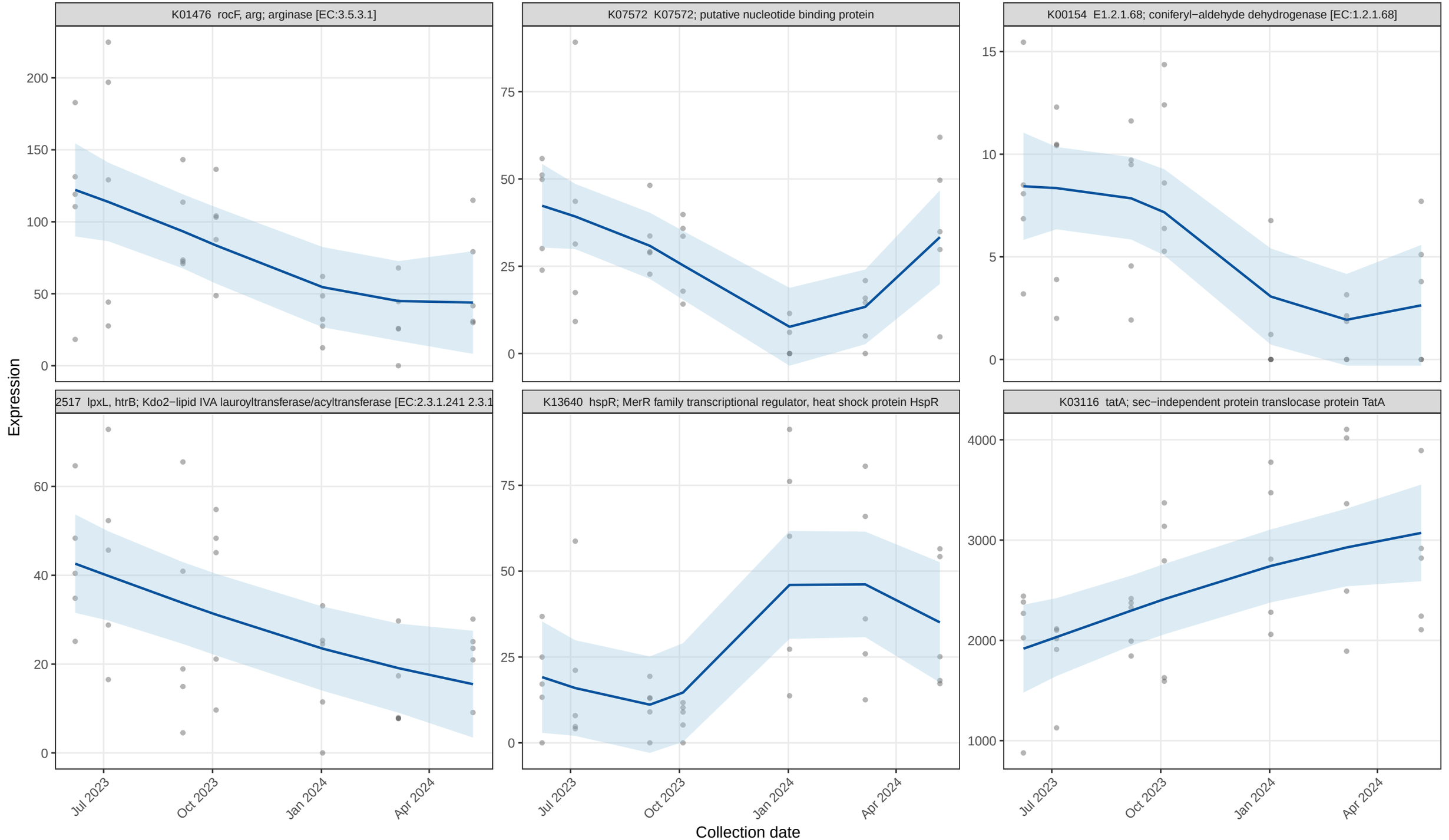

K03243 EIF5B; translation initiation factor 5B
K04801 rfcS; replication factor C small subunit
oorB, oforB; 2−oxoglutarate/2−oxoacid ferredoxin oxidoreductase subunit beta [EC:1.2
lysK, argE; [amino group carrier protein]−lysine/ornithine hydrolase [EC:3.5.1.130 3.5
K03457 TC.NCS1; nucleobase:cation symporter−1, NCS1 family
K16055 TPS; trehalose 6−phosphate synthase/phosphatase [EC:2.4.1.15 3.1.3.12]
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

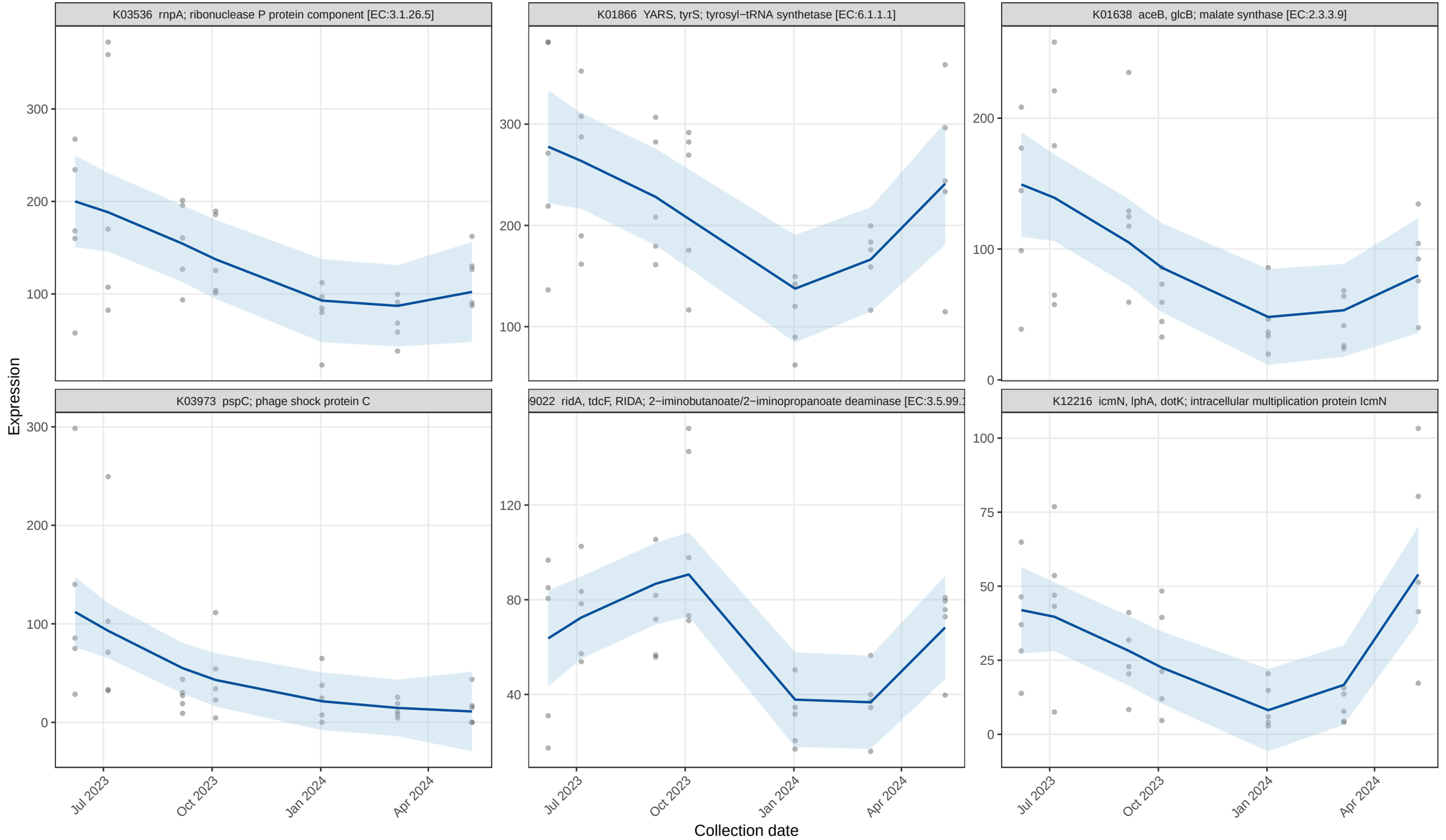

K01803 TPI, tpiA; triosephosphate isomerase (TIM) [EC:5.3.1.1]
K03582 recB; exodeoxyribonuclease V beta subunit [EC:3.1.11.5]
K02621 parC; topoisomerase IV subunit A [EC:5.6.2.2]
K13497 trpGD; anthranilate synthase/phosphoribosyltransferase [EC:4.1.3.27 2.4.2.18]
K03676 grxC, GLRX, GLRX2; glutaredoxin 3
, bclA; vitamin B12/bleomycin/antimicrobial peptide transport system ATP−binding/perm
Expression
Collection date
Jul 2023
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Jan 2024
Apr 2024

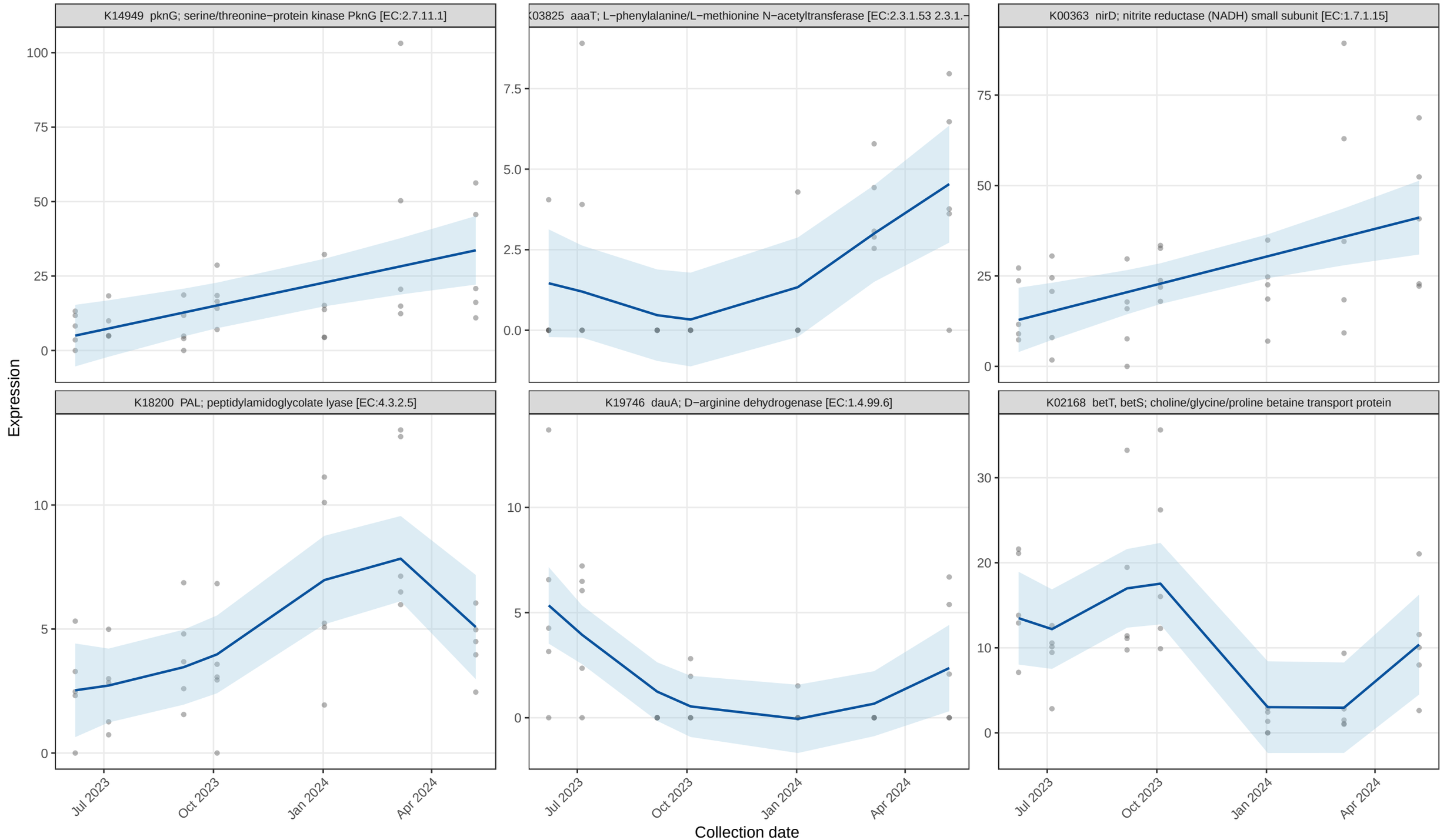

K00547 mmuM, BHMT2; homocysteine S−methyltransferase [EC:2.1.1.10]
K11076 potG, spuF; putrescine transport system ATP−binding protein [EC:7.6.2.16]
K10003 gltJ, aatQ; glutamate/aspartate transport system permease protein
K09013 sufC; Fe−S cluster assembly ATP−binding protein
K00986 ltrA; RNA−directed DNA polymerase [EC:2.7.7.49]
K07742 ylxR; uncharacterized protein
Expression
Collection date
Jul 2023
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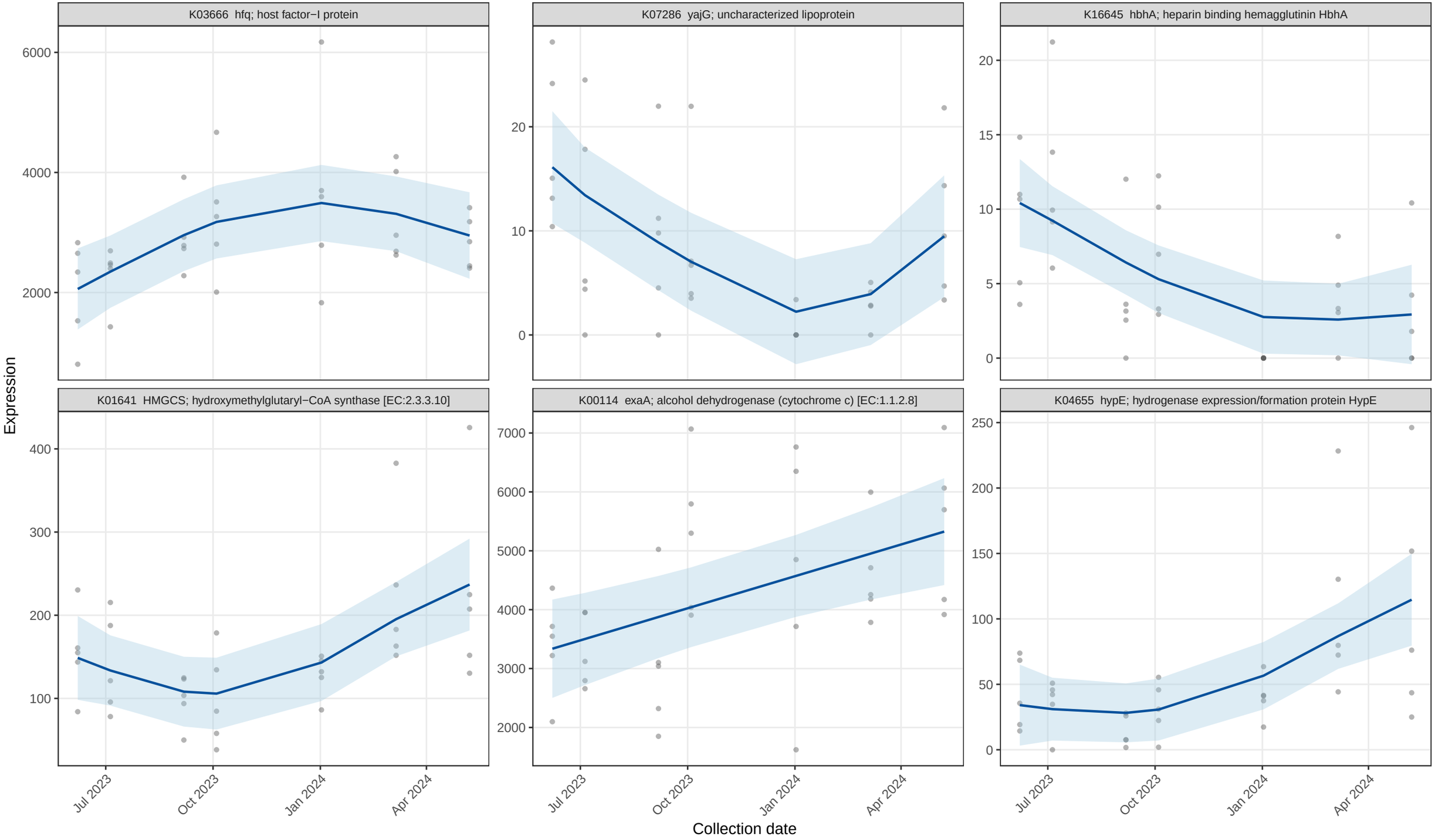

K19267 qorB; NAD(P)H dehydrogenase (quinone) [EC:1.6.5.2]
01968 MCCC1, accA1; 3−methylcrotonyl−CoA carboxylase alpha subunit [EC:6.4.1.4
K05305 FUK; fucokinase [EC:2.7.1.52]
K01253 EPHX1; microsomal epoxide hydrolase [EC:3.3.2.9]
K06958 rapZ; RNase adapter protein RapZ
ransferase / (S)−3−amino−2−methylpropionate transaminase / 5−aminovalerate transa
Expression
Collection date
Jul 2023
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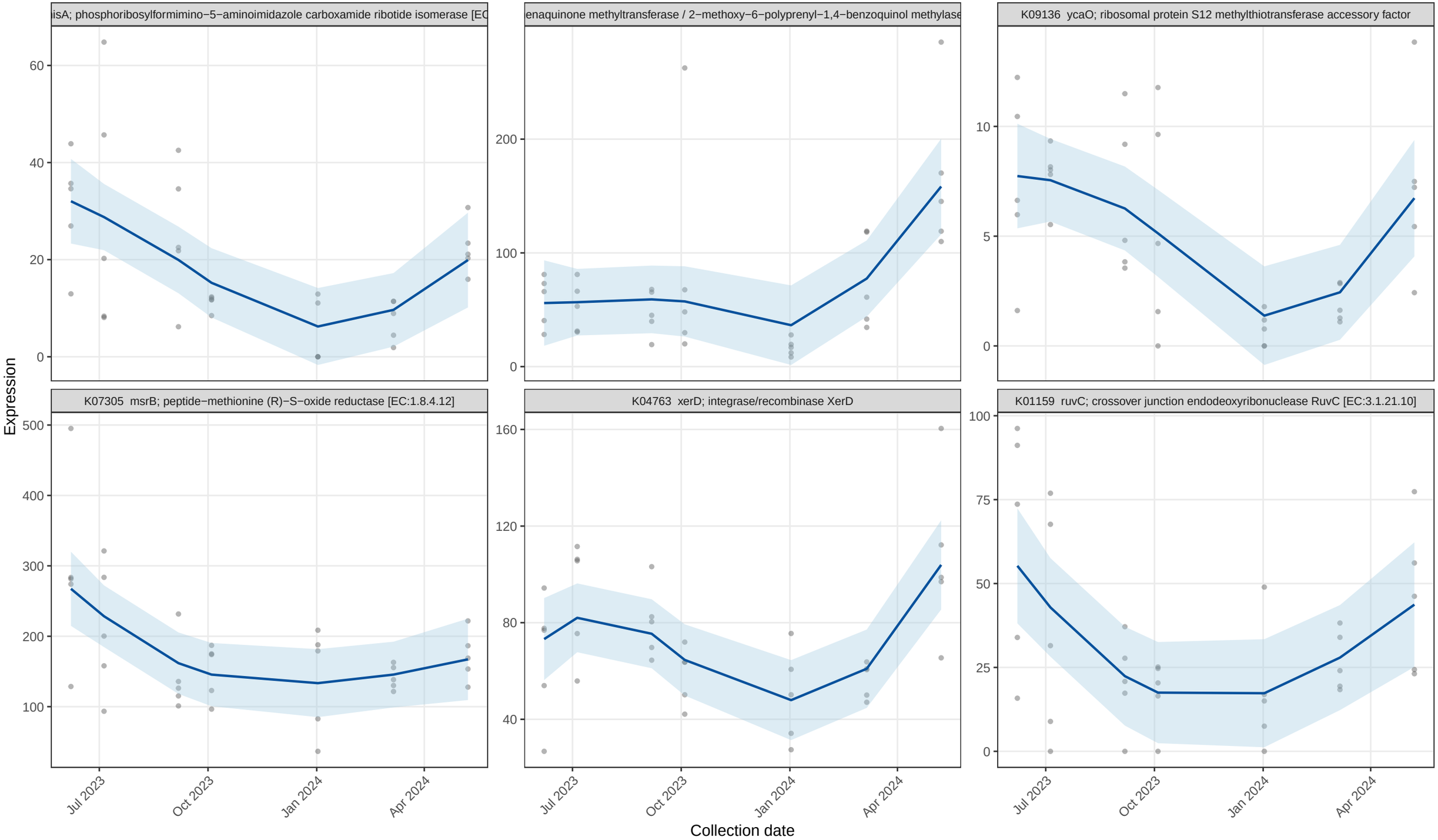

K21471 cwlO; peptidoglycan DL−endopeptidase CwlO [EC:3.4.−.−]
sphatidylethanolamine/phosphatidyl−N−methylethanolamine N−methyltransferase [E
K06960 K06960; uncharacterized protein
K03313 nhaA; Na+:H+ antiporter, NhaA family
K02978 RP−S27e, RPS27; small subunit ribosomal protein S27e
K09796 pccA; periplasmic copper chaperone A
Expression
Collection date
Jul 2023
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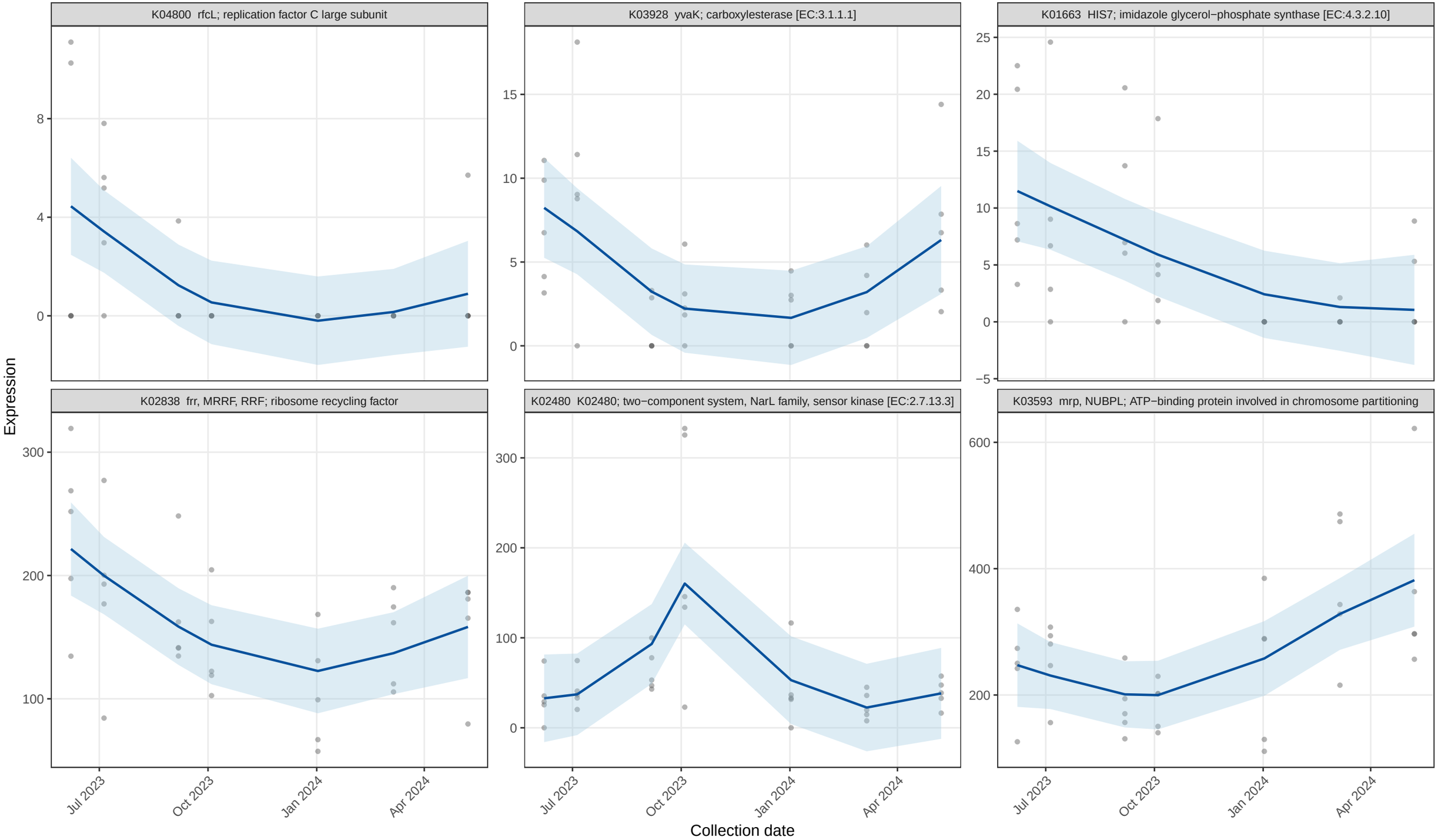

K01476 rocF, arg; arginase [EC:3.5.3.1]
K07572 K07572; putative nucleotide binding protein
K00154 E1.2.1.68; coniferyl−aldehyde dehydrogenase [EC:1.2.1.68]
2517 lpxL, htrB; Kdo2−lipid IVA lauroyltransferase/acyltransferase [EC:2.3.1.241 2.3.1
K13640 hspR; MerR family transcriptional regulator, heat shock protein HspR
K03116 tatA; sec−independent protein translocase protein TatA
Expression
Collection date
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Jan 2024
Apr 2024

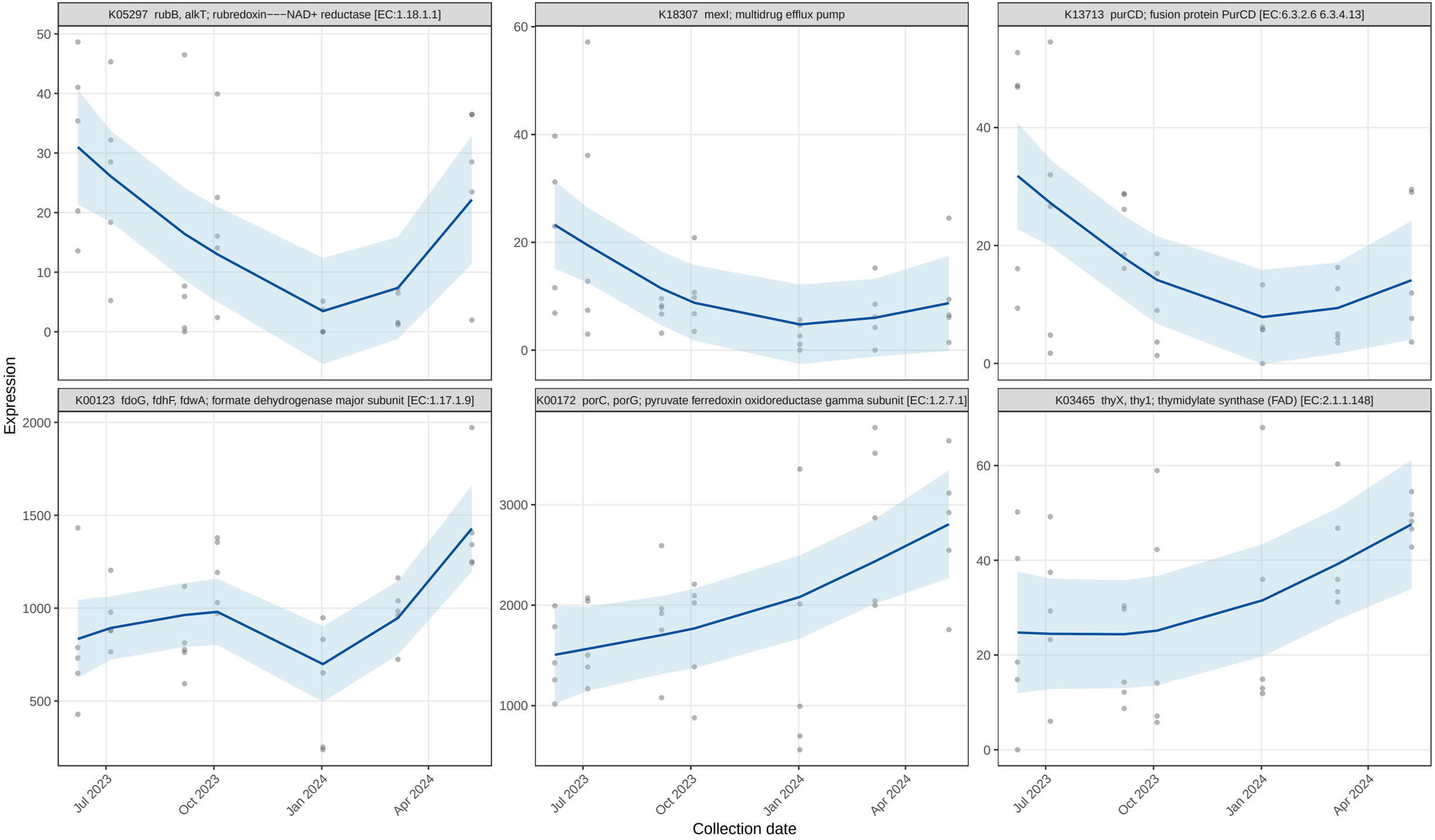

K03536 rnpA; ribonuclease P protein component [EC:3.1.26.5]
K01866 YARS, tyrS; tyrosyl−tRNA synthetase [EC:6.1.1.1]
K01638 aceB, glcB; malate synthase [EC:2.3.3.9]
K03973 pspC; phage shock protein C
9022 ridA, tdcF, RIDA; 2−iminobutanoate/2−iminopropanoate deaminase [EC:3.5.99.1
K12216 icmN, lphA, dotK; intracellular multiplication protein IcmN
Expression
Collection date
Jul 2023
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Apr 2024

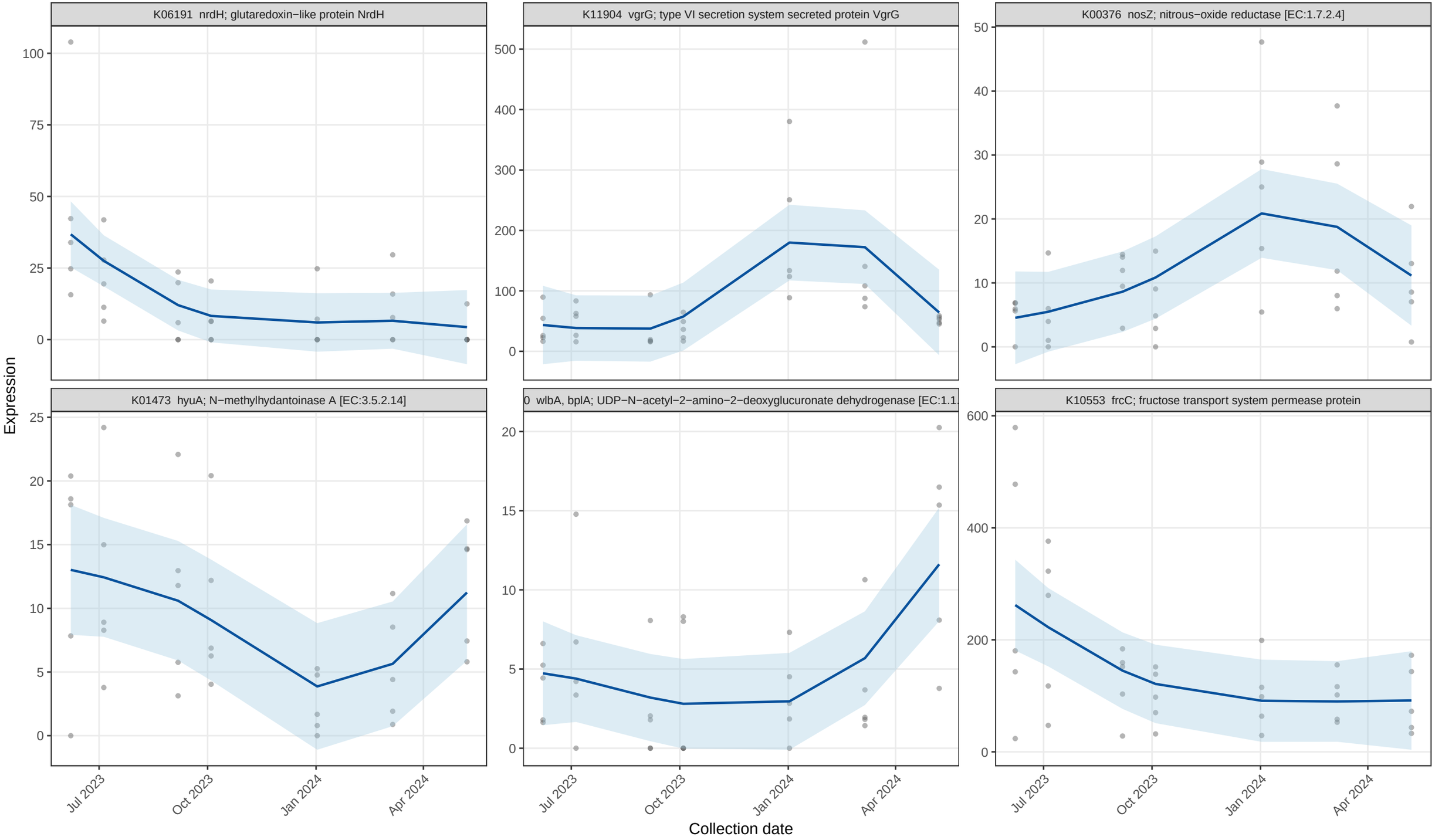

K14949 pknG; serine/threonine−protein kinase PknG [EC:2.7.11.1]
K03825 aaaT; L−phenylalanine/L−methionine N−acetyltransferase [EC:2.3.1.53 2.3.1.−
K00363 nirD; nitrite reductase (NADH) small subunit [EC:1.7.1.15]
K18200 PAL; peptidylamidoglycolate lyase [EC:4.3.2.5]
K19746 dauA; D−arginine dehydrogenase [EC:1.4.99.6]
K02168 betT, betS; choline/glycine/proline betaine transport protein
Expression
Collection date
Jul 2023
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Apr 2024

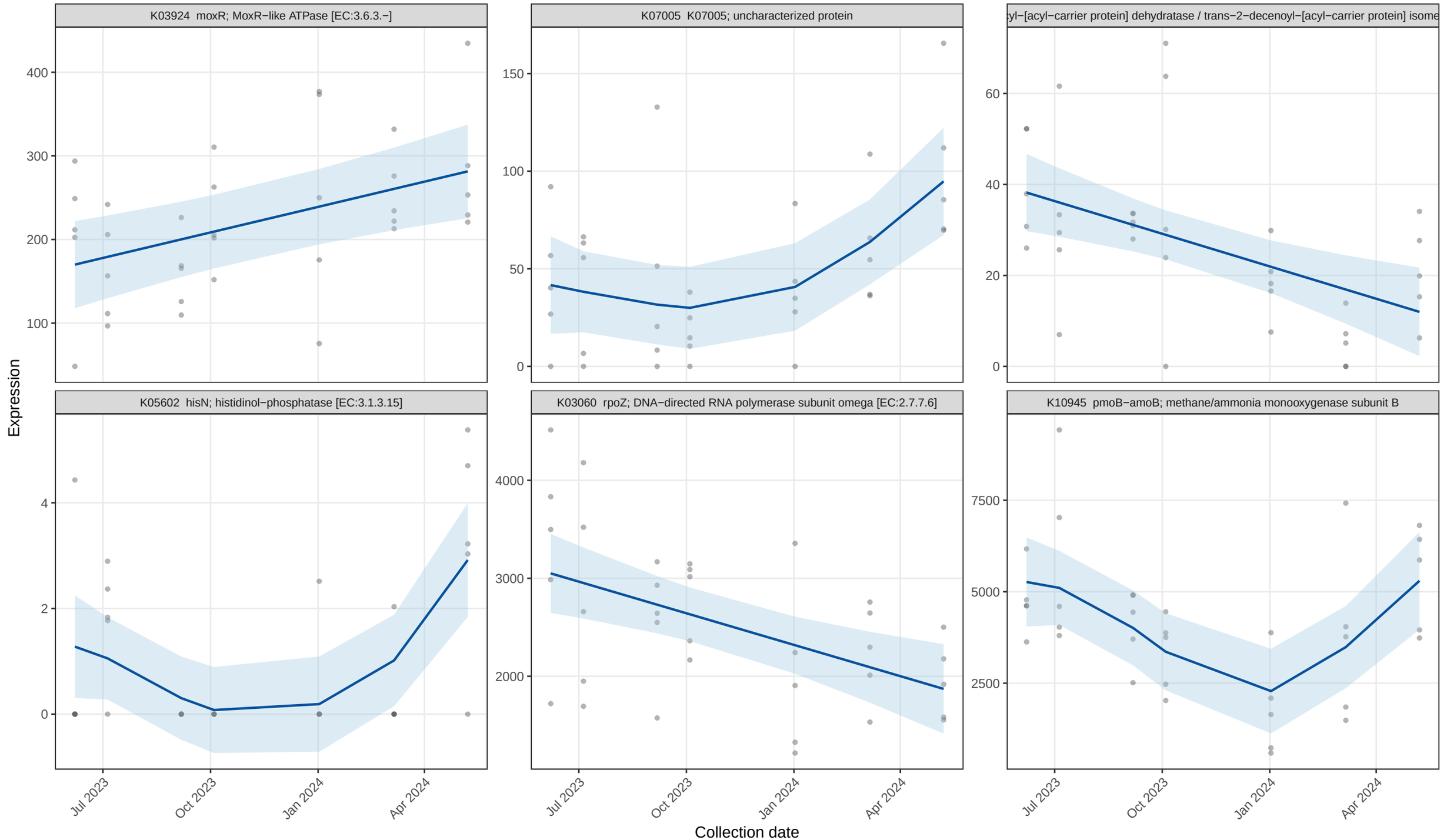

K03666 hfq; host factor−I protein
K07286 yajG; uncharacterized lipoprotein
K16645 hbhA; heparin binding hemagglutinin HbhA
K01641 HMGCS; hydroxymethylglutaryl−CoA synthase [EC:2.3.3.10]
K00114 exaA; alcohol dehydrogenase (cytochrome c) [EC:1.1.2.8]
K04655 hypE; hydrogenase expression/formation protein HypE
Expression
Collection date
Jul 2023
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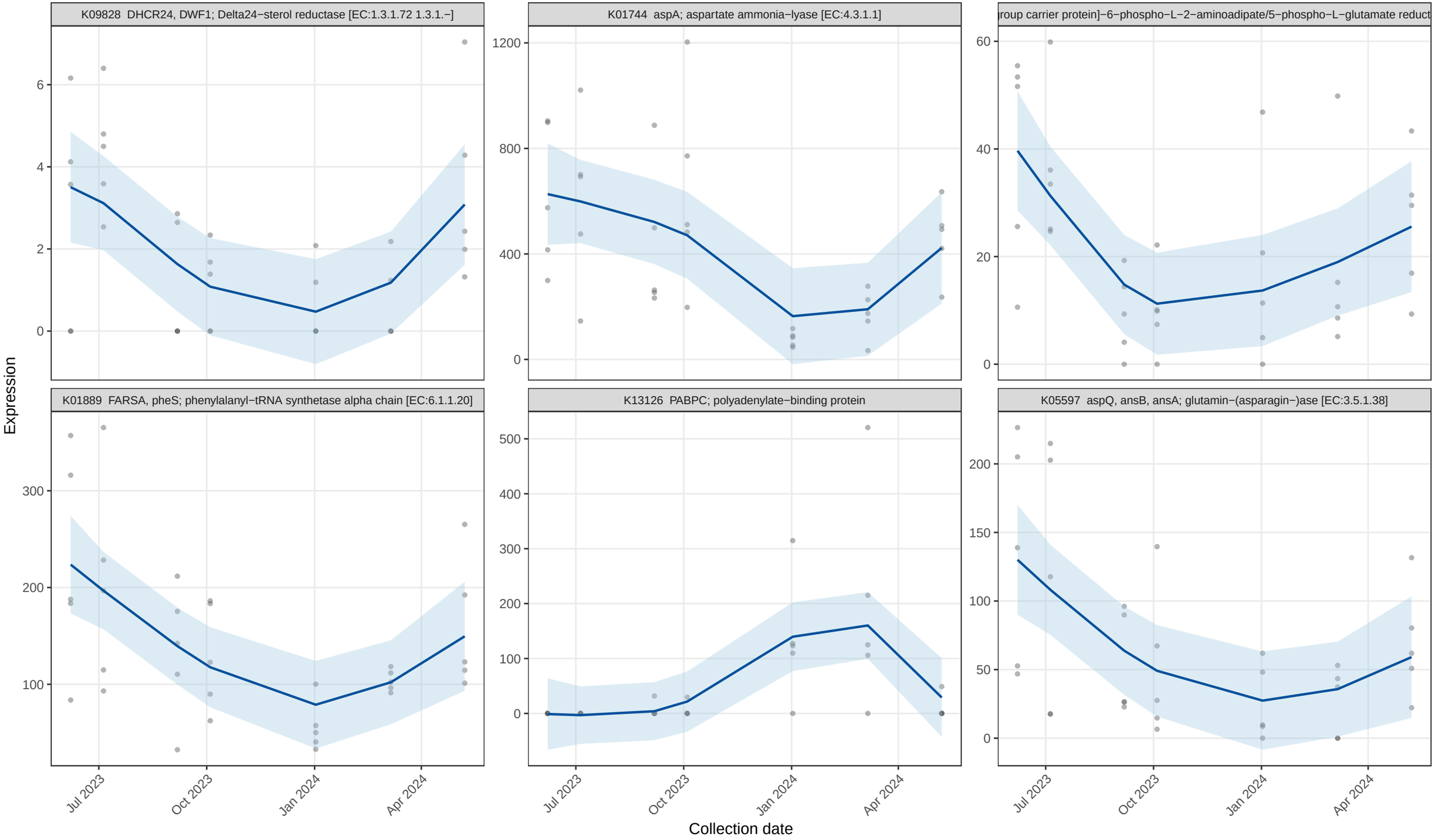

isA; phosphoribosylformimino−5−aminoimidazole carboxamide ribotide isomerase [EC
enaquinone methyltransferase / 2−methoxy−6−polyprenyl−1,4−benzoquinol methylase
K09136 ycaO; ribosomal protein S12 methylthiotransferase accessory factor
K07305 msrB; peptide−methionine (R)−S−oxide reductase [EC:1.8.4.12]
K04763 xerD; integrase/recombinase XerD
K01159 ruvC; crossover junction endodeoxyribonuclease RuvC [EC:3.1.21.10]
Expression
Collection date
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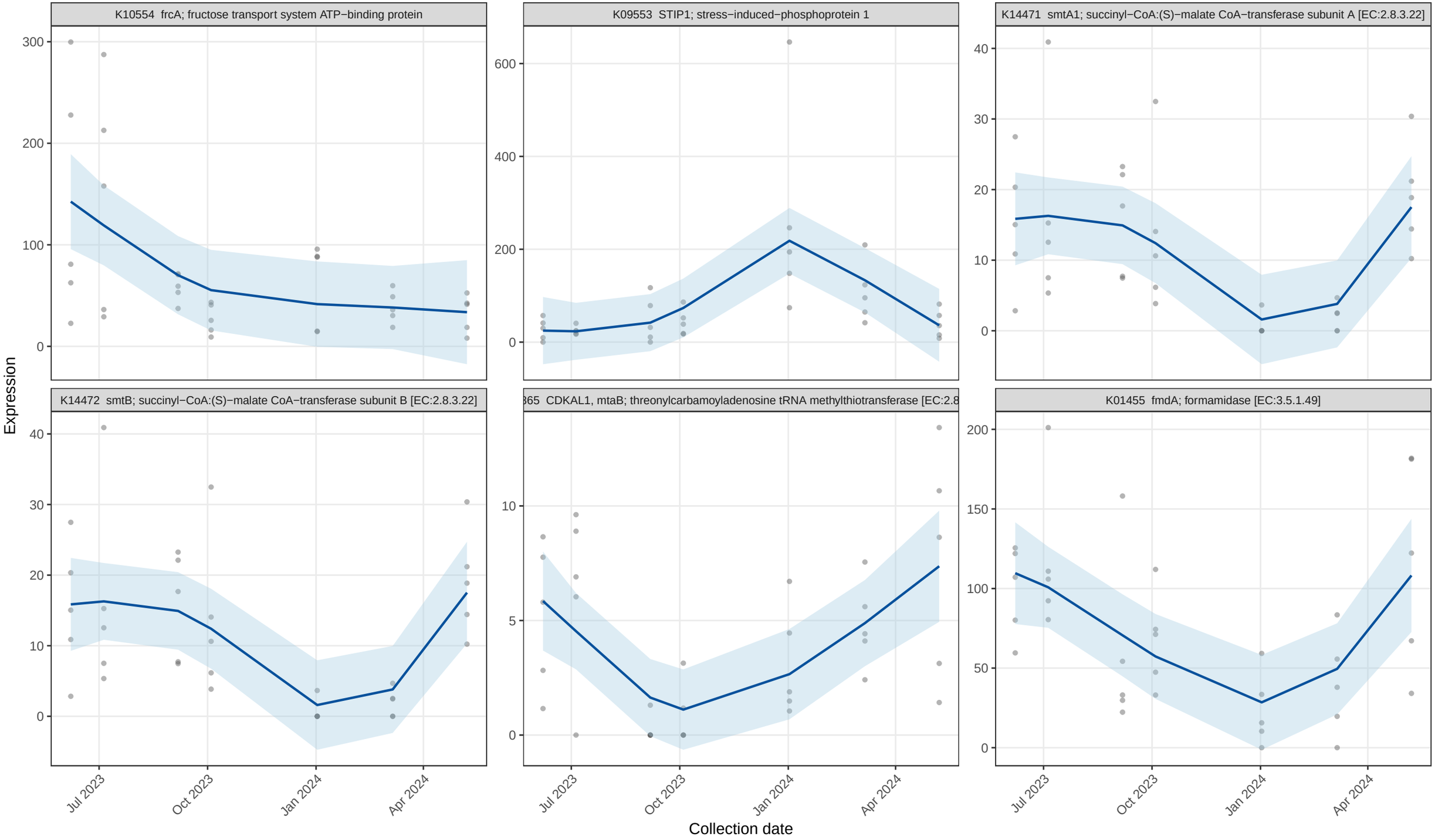

K04800 rfcL; replication factor C large subunit
K03928 yvaK; carboxylesterase [EC:3.1.1.1]
K01663 HIS7; imidazole glycerol−phosphate synthase [EC:4.3.2.10]
K02838 frr, MRRF, RRF; ribosome recycling factor
K02480 K02480; two−component system, NarL family, sensor kinase [EC:2.7.13.3]
K03593 mrp, NUBPL; ATP−binding protein involved in chromosome partitioning
Expression
Collection date
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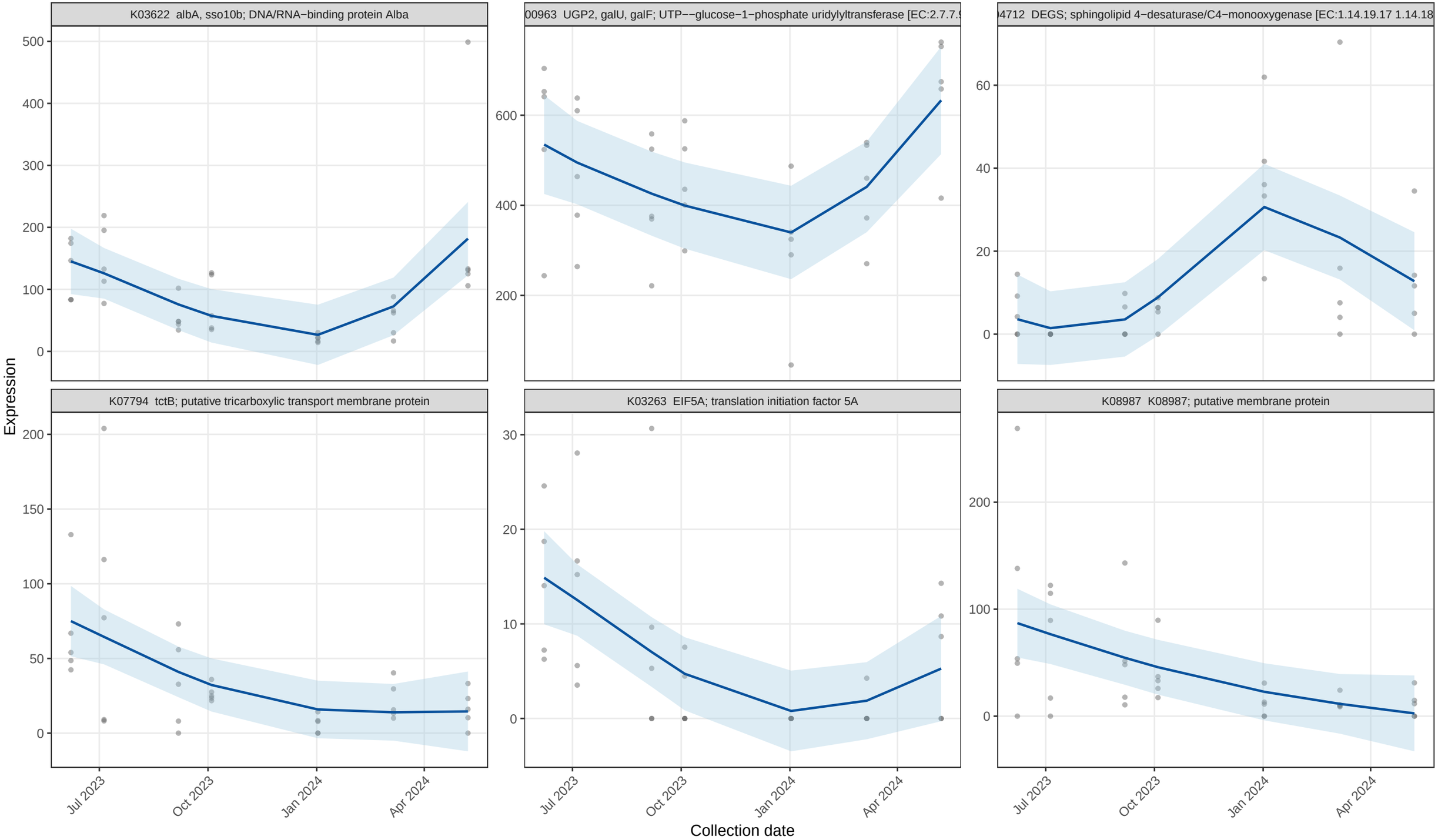

K05297 rubB, alkT; rubredoxin−−−NAD+ reductase [EC:1.18.1.1]
K18307 mexI; multidrug efflux pump
K13713 purCD; fusion protein PurCD [EC:6.3.2.6 6.3.4.13]
K00123 fdoG, fdhF, fdwA; formate dehydrogenase major subunit [EC:1.17.1.9]
K00172 porC, porG; pyruvate ferredoxin oxidoreductase gamma subunit [EC:1.2.7.1]
K03465 thyX, thy1; thymidylate synthase (FAD) [EC:2.1.1.148]
Expression
Collection date
Jul 2023
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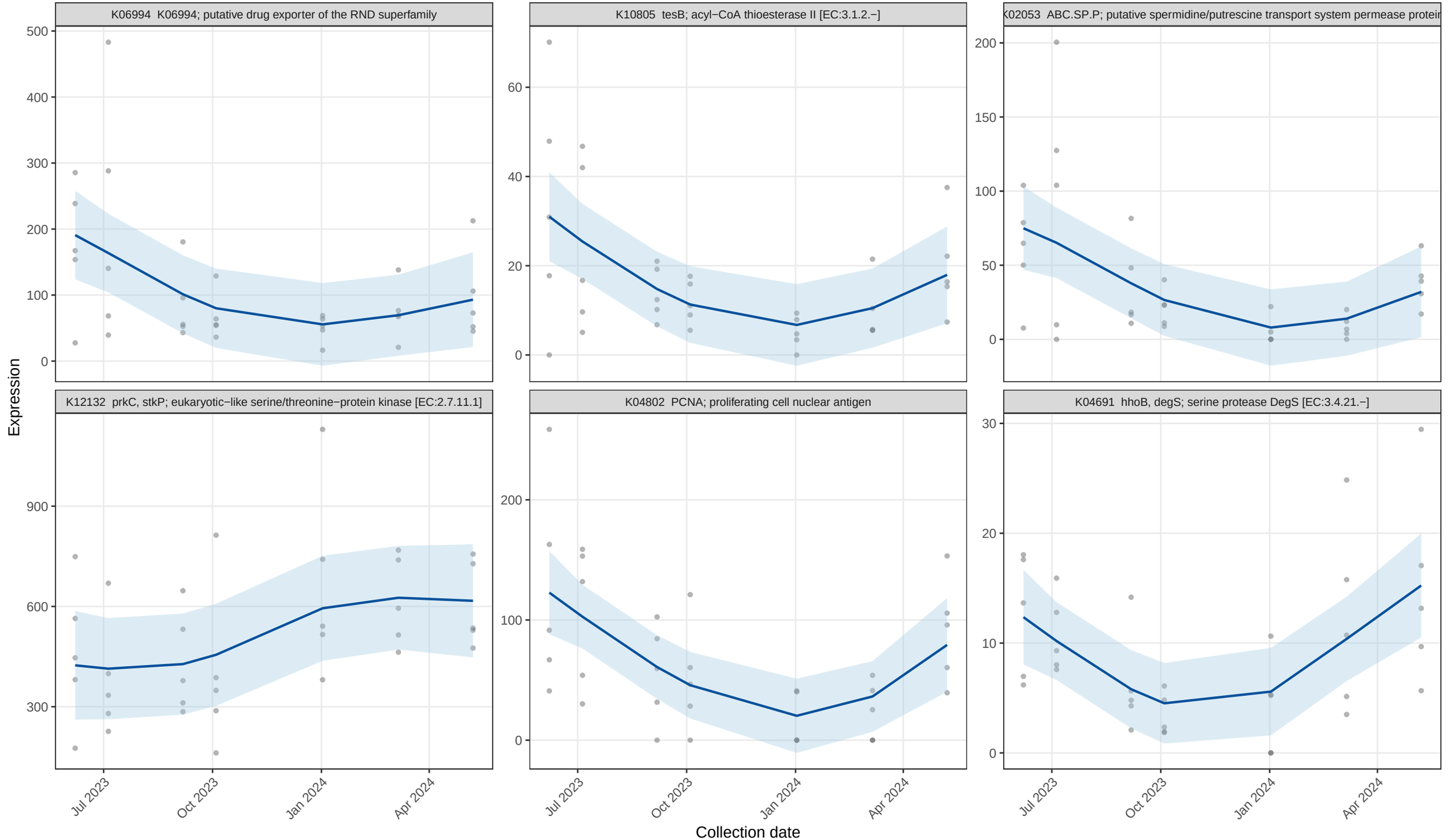

K06191 nrdH; glutaredoxin−like protein NrdH
K11904 vgrG; type VI secretion system secreted protein VgrG
K00376 nosZ; nitrous−oxide reductase [EC:1.7.2.4]
K01473 hyuA; N−methylhydantoinase A [EC:3.5.2.14]
0 wlbA, bplA; UDP−N−acetyl−2−amino−2−deoxyglucuronate dehydrogenase [EC:1.1.
K10553 frcC; fructose transport system permease protein
Expression
Collection date
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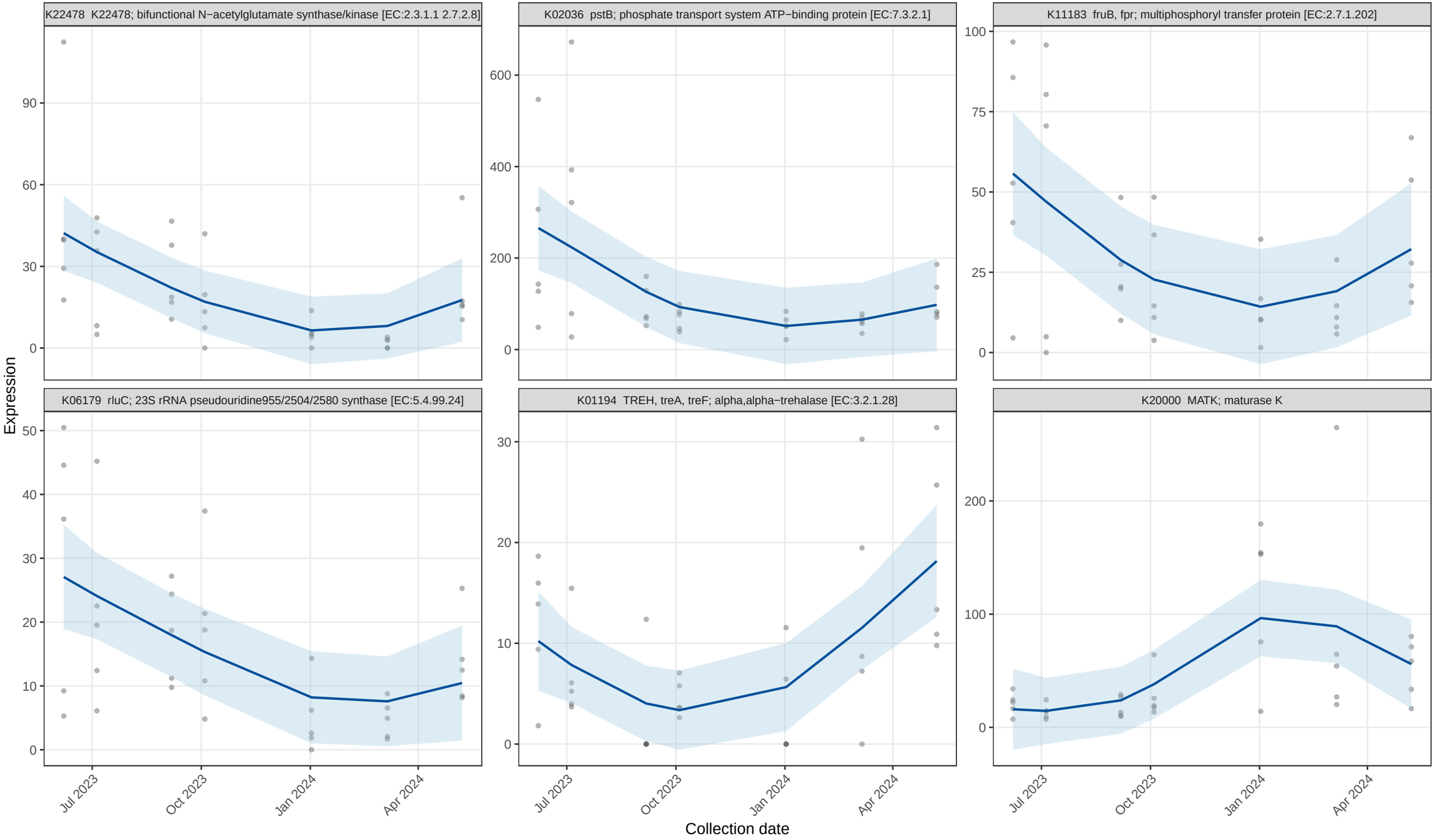

K03924 moxR; MoxR−like ATPase [EC:3.6.3.−]
K07005 K07005; uncharacterized protein
yl−[acyl−carrier protein] dehydratase / trans−2−decenoyl−[acyl−carrier protein] isome
K05602 hisN; histidinol−phosphatase [EC:3.1.3.15]
K03060 rpoZ; DNA−directed RNA polymerase subunit omega [EC:2.7.7.6]
K10945 pmoB−amoB; methane/ammonia monooxygenase subunit B
Expression
Collection date
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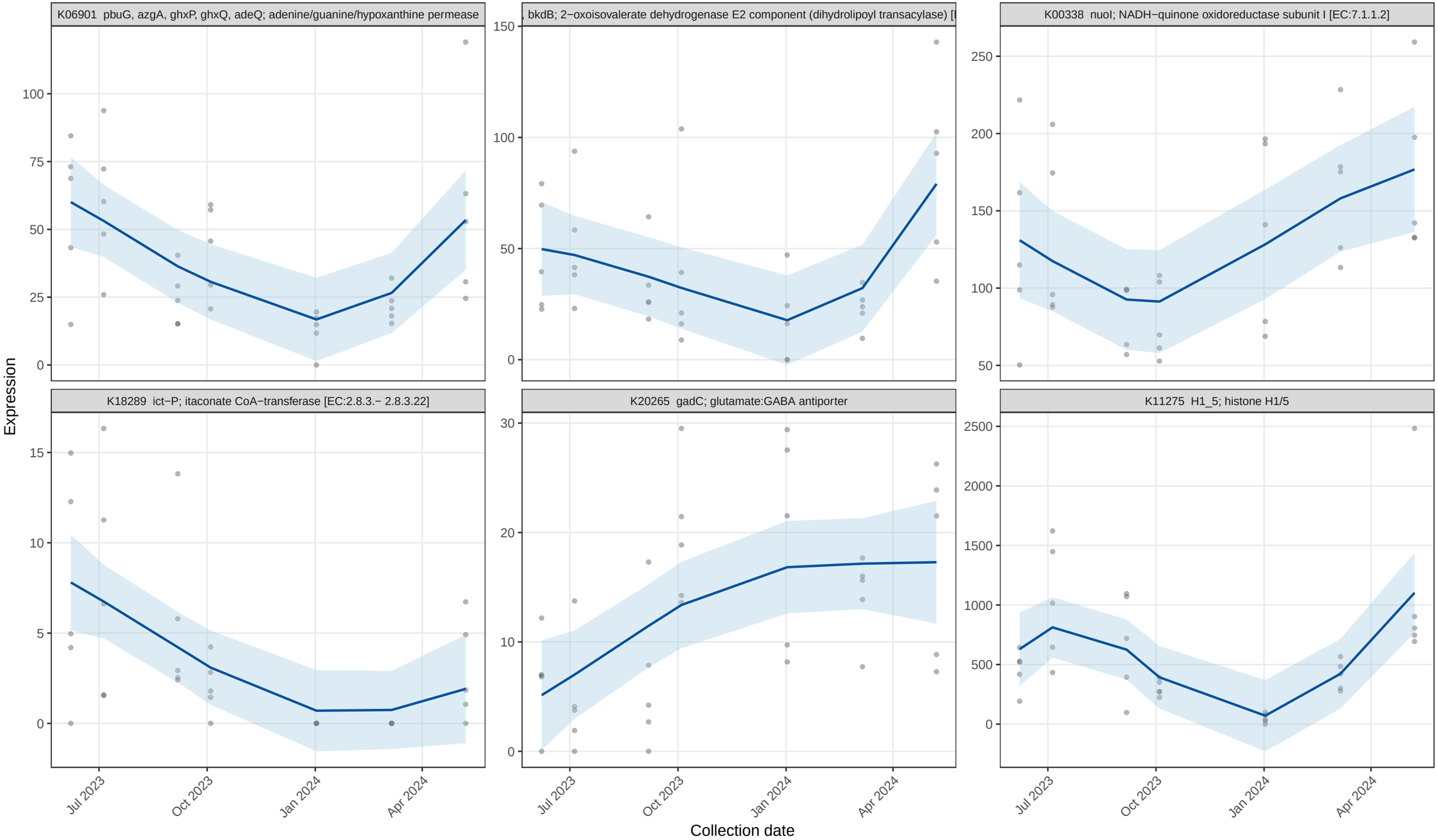

K09828 DHCR24, DWF1; Delta24−sterol reductase [EC:1.3.1.72 1.3.1.−]
K01744 aspA; aspartate ammonia−lyase [EC:4.3.1.1]
roup carrier protein]−6−phospho−L−2−aminoadipate/5−phospho−L−glutamate reduct
K01889 FARSA, pheS; phenylalanyl−tRNA synthetase alpha chain [EC:6.1.1.20]
K13126 PABPC; polyadenylate−binding protein
K05597 aspQ, ansB, ansA; glutamin−(asparagin−)ase [EC:3.5.1.38]
Expression
Collection date
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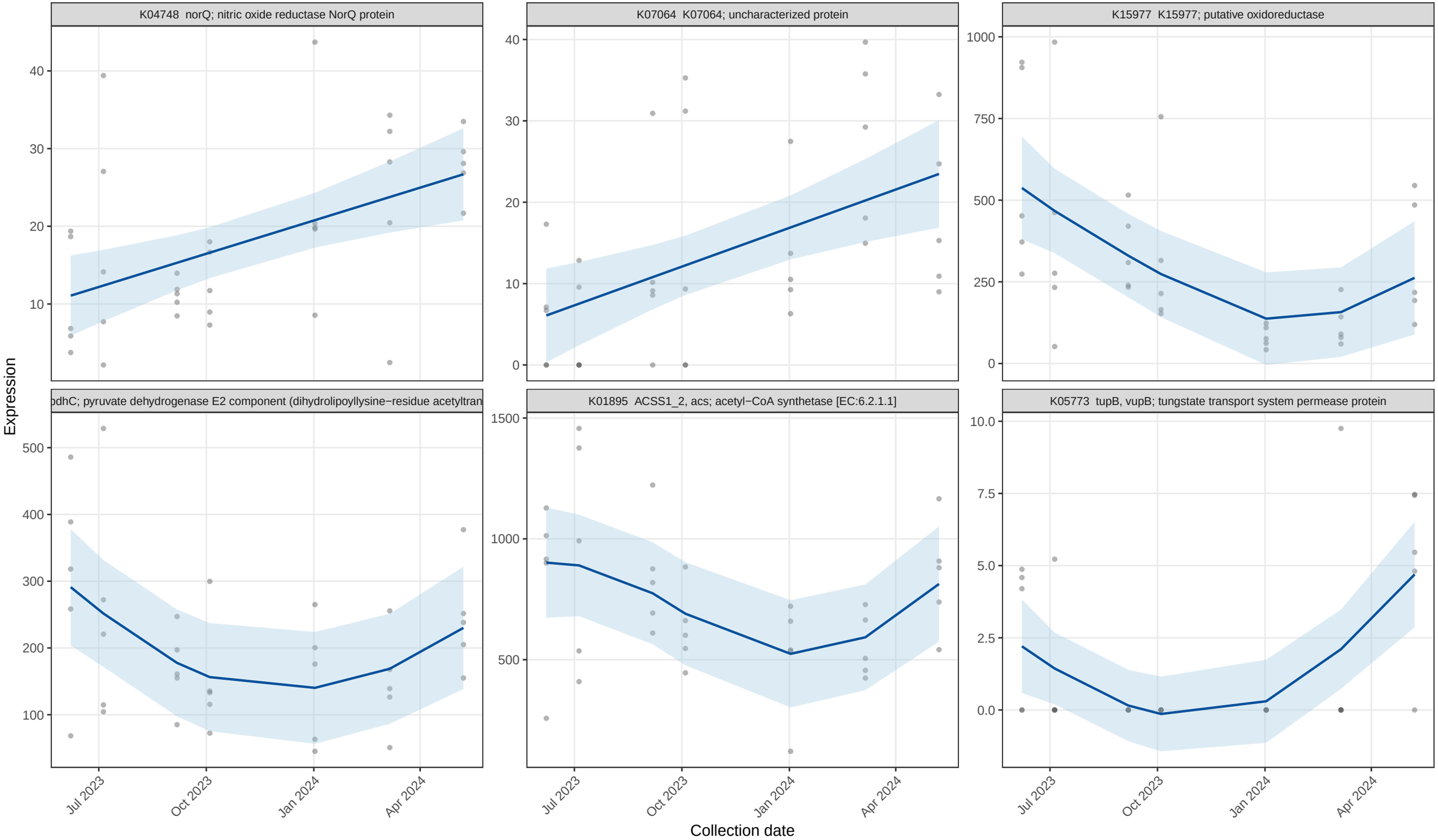

K10554 frcA; fructose transport system ATP−binding protein
K09553 STIP1; stress−induced−phosphoprotein 1
K14471 smtA1; succinyl−CoA:(S)−malate CoA−transferase subunit A [EC:2.8.3.22]
K14472 smtB; succinyl−CoA:(S)−malate CoA−transferase subunit B [EC:2.8.3.22]
365 CDKAL1, mtaB; threonylcarbamoyladenosine tRNA methylthiotransferase [EC:2.8
K01455 fmdA; formamidase [EC:3.5.1.49]
Expression
Collection date
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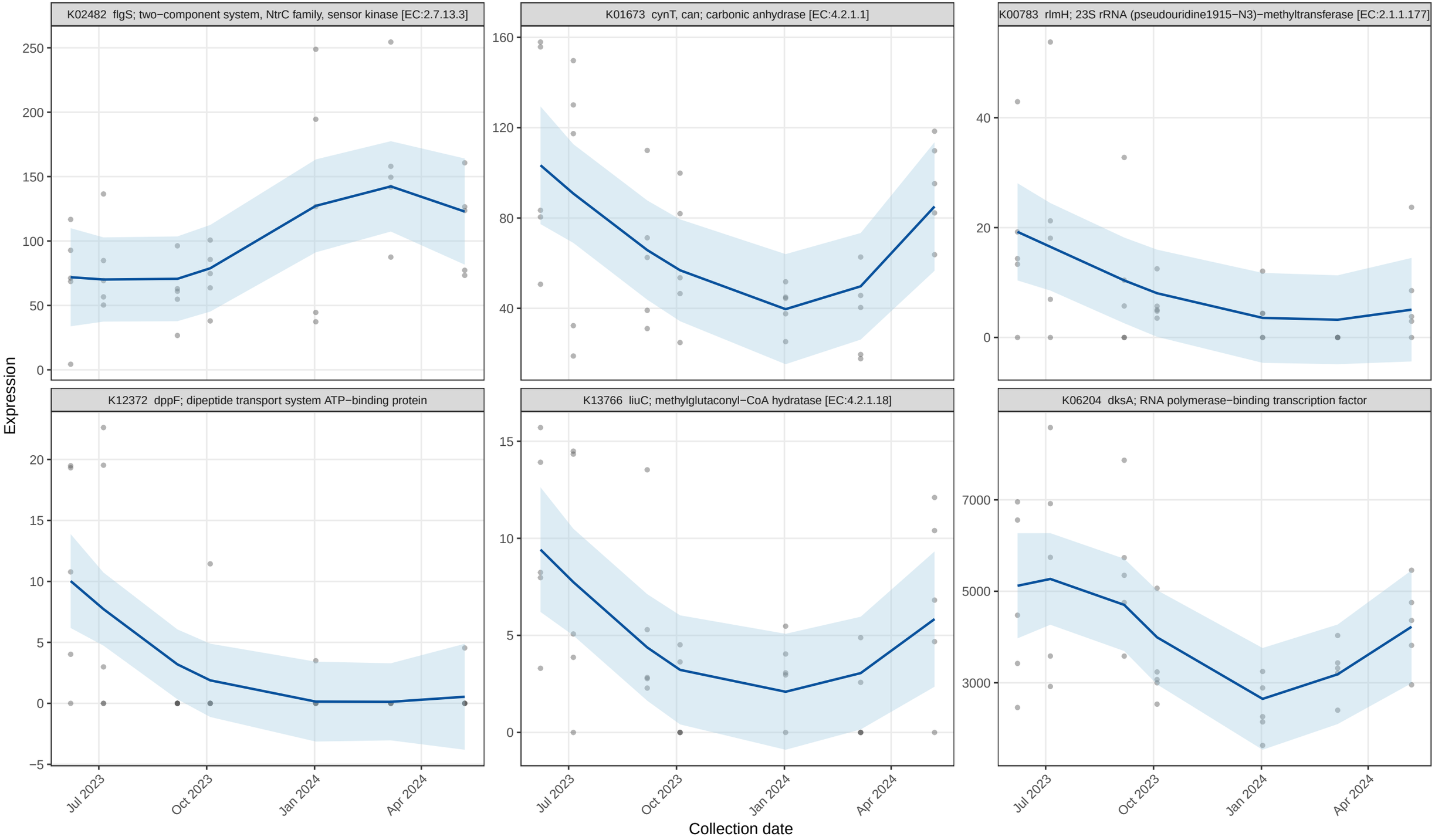

K03622 albA, sso10b; DNA/RNA−binding protein Alba
00963 UGP2, galU, galF; UTP−−glucose−1−phosphate uridylyltransferase [EC:2.7.7.9
4712 DEGS; sphingolipid 4−desaturase/C4−monooxygenase [EC:1.14.19.17 1.14.18
K07794 tctB; putative tricarboxylic transport membrane protein
K03263 EIF5A; translation initiation factor 5A
K08987 K08987; putative membrane protein
Expression
Collection date
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Apr 2024

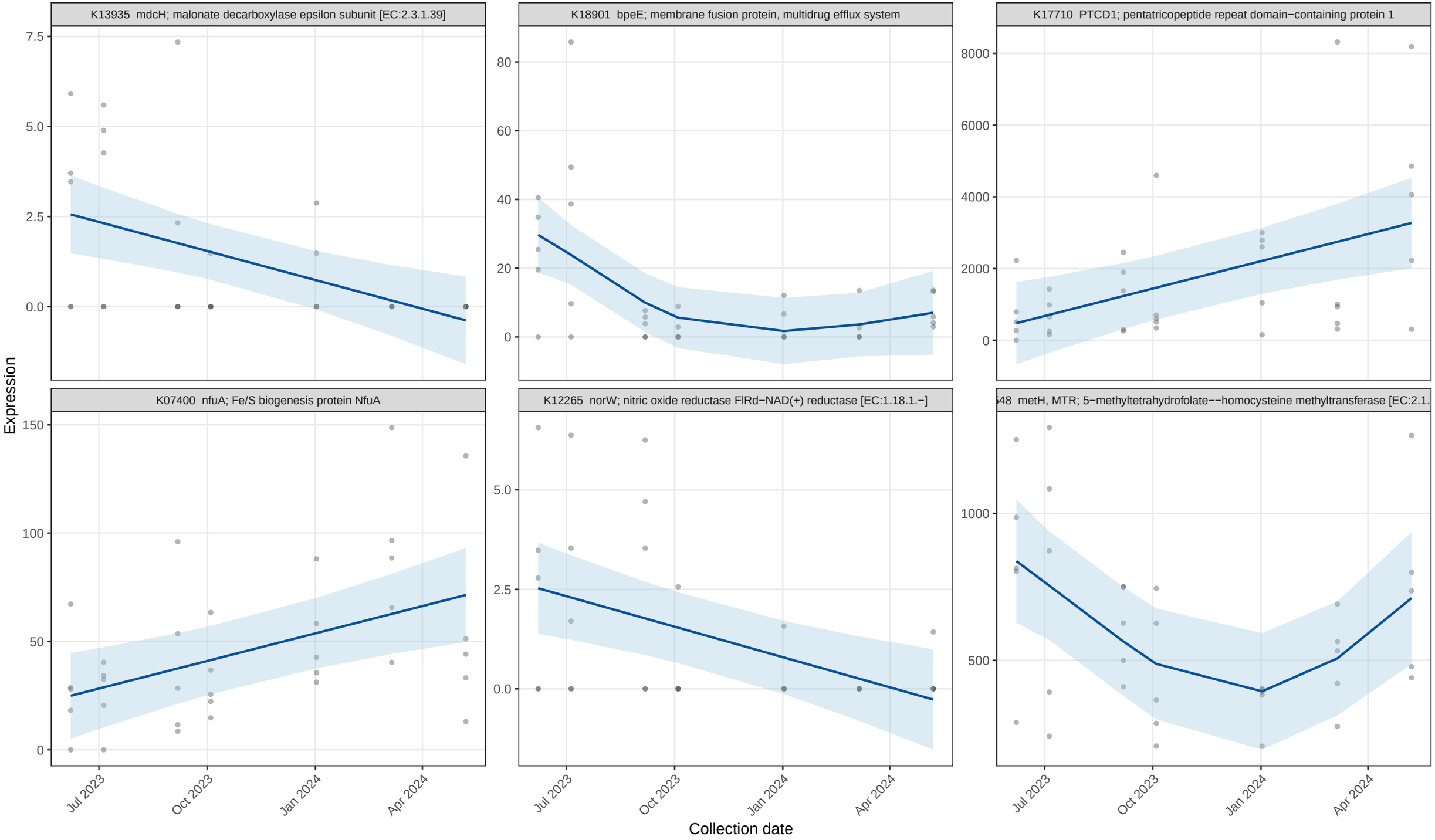

K06994 K06994; putative drug exporter of the RND superfamily
K10805 tesB; acyl−CoA thioesterase II [EC:3.1.2.−]
K02053 ABC.SP.P; putative spermidine/putrescine transport system permease protein
K12132 prkC, stkP; eukaryotic−like serine/threonine−protein kinase [EC:2.7.11.1]
K04802 PCNA; proliferating cell nuclear antigen
K04691 hhoB, degS; serine protease DegS [EC:3.4.21.−]
Expression
Collection date
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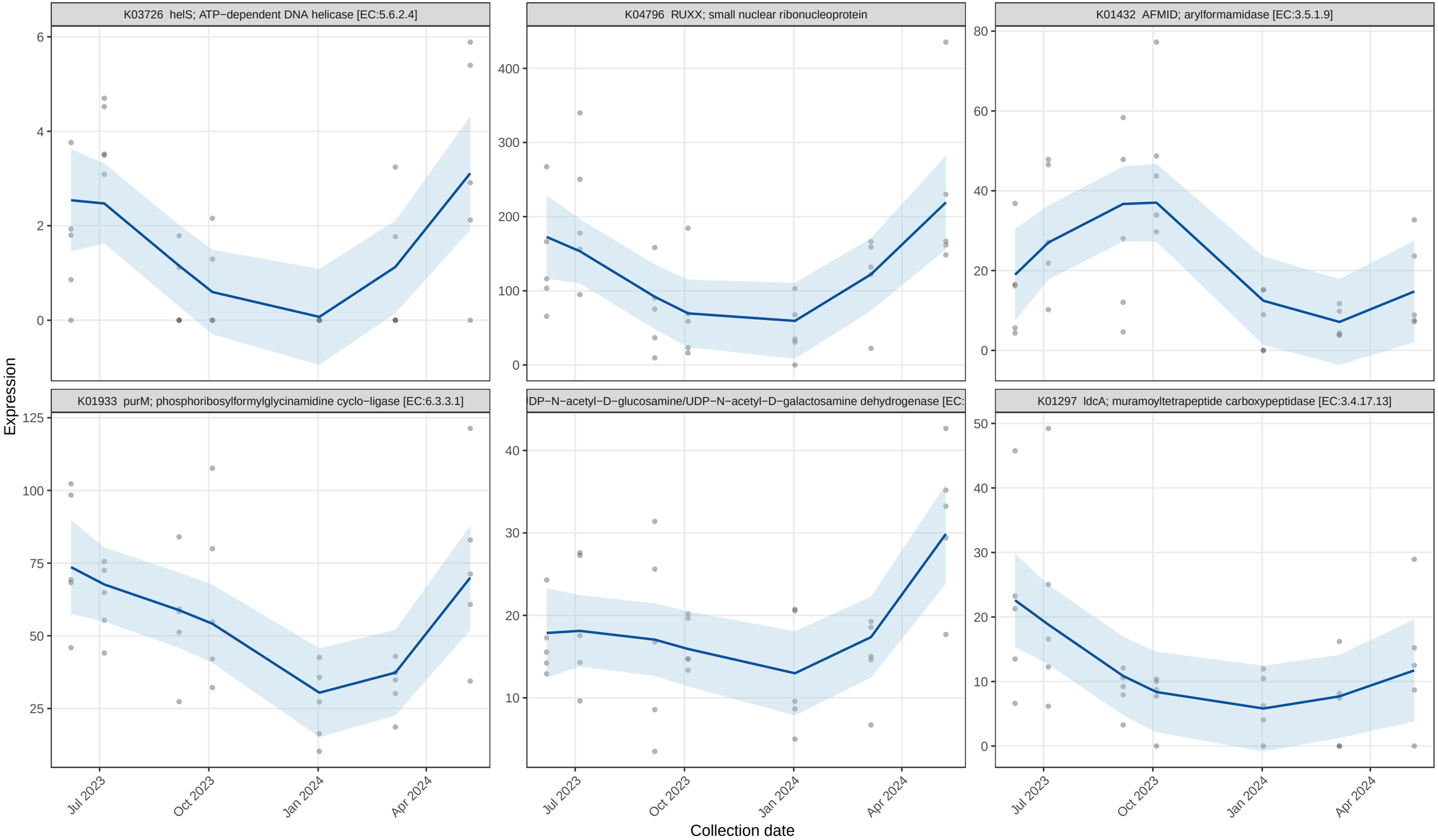

K22478 K22478; bifunctional N−acetylglutamate synthase/kinase [EC:2.3.1.1 2.7.2.8]
K02036 pstB; phosphate transport system ATP−binding protein [EC:7.3.2.1]
K11183 fruB, fpr; multiphosphoryl transfer protein [EC:2.7.1.202]
K06179 rluC; 23S rRNA pseudouridine955/2504/2580 synthase [EC:5.4.99.24]
K01194 TREH, treA, treF; alpha,alpha−trehalase [EC:3.2.1.28]
K20000 MATK; maturase K
Expression
Collection date
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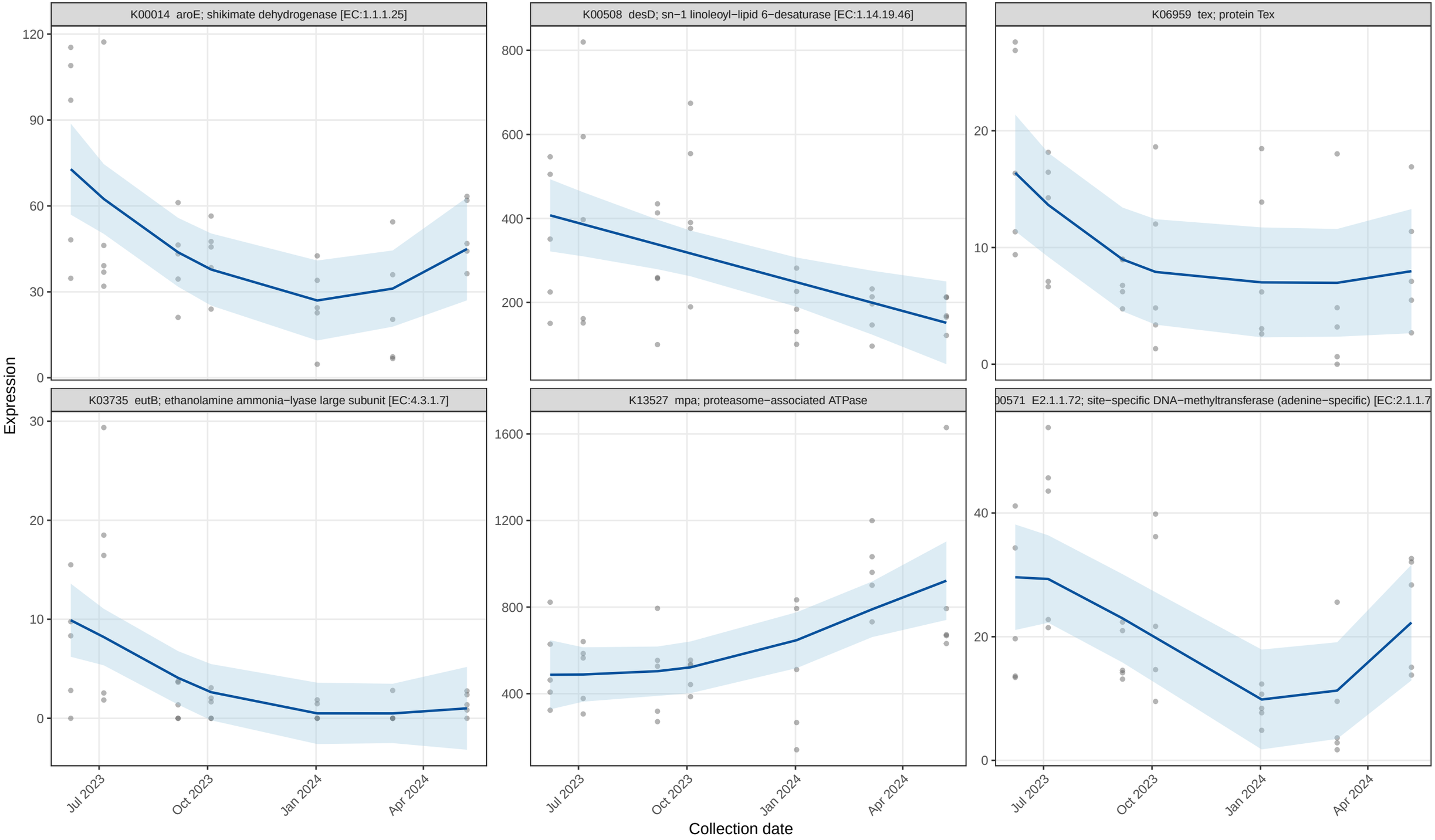

K06901 pbuG, azgA, ghxP, ghxQ, adeQ; adenine/guanine/hypoxanthine permease
, bkdB; 2−oxoisovalerate dehydrogenase E2 component (dihydrolipoyl transacylase) [
K00338 nuoI; NADH−quinone oxidoreductase subunit I [EC:7.1.1.2]
K18289 ict−P; itaconate CoA−transferase [EC:2.8.3.− 2.8.3.22]
K20265 gadC; glutamate:GABA antiporter
K11275 H1_5; histone H1/5
Expression
Collection date
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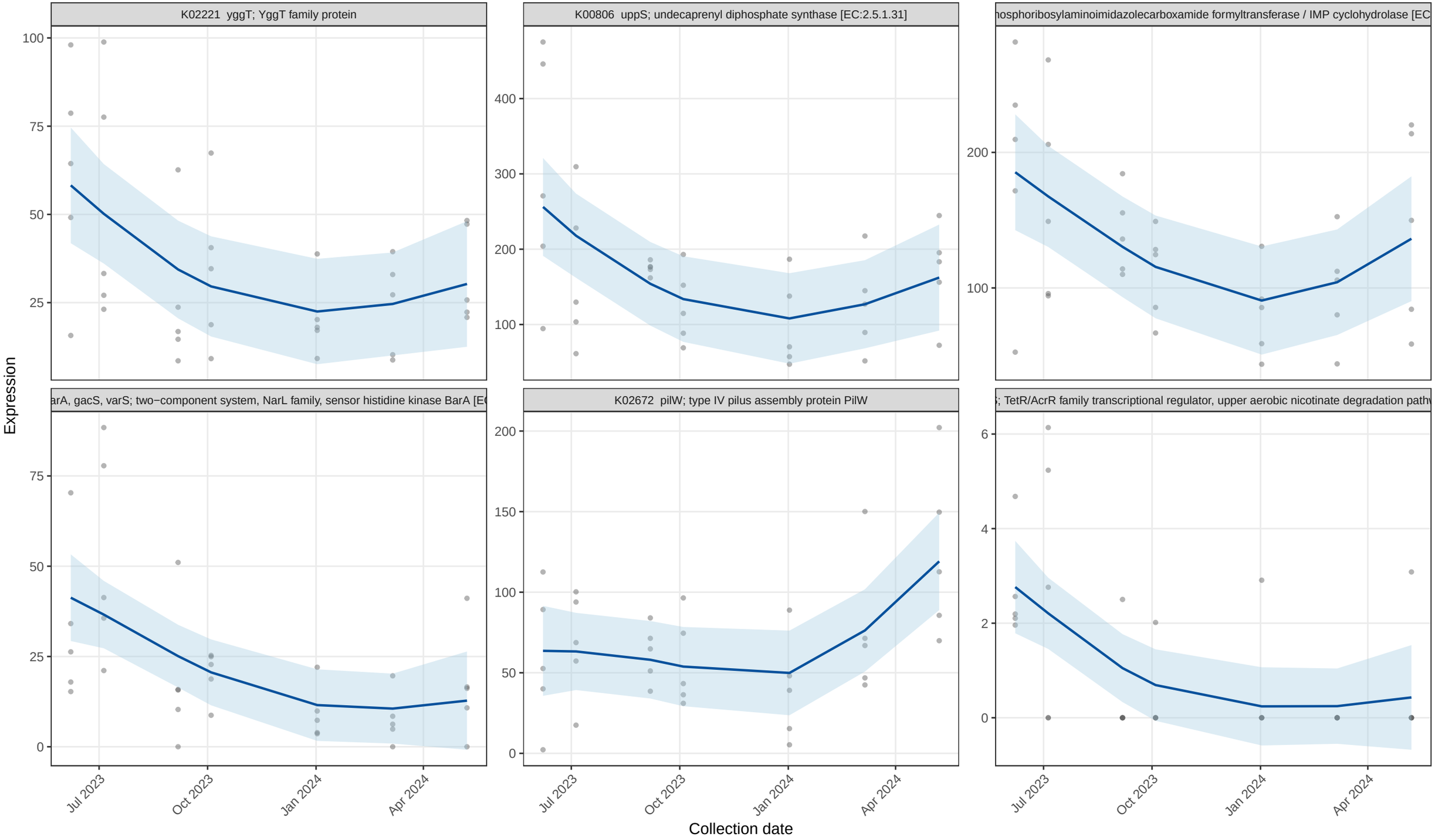

K04748 norQ; nitric oxide reductase NorQ protein
K07064 K07064; uncharacterized protein
K15977 K15977; putative oxidoreductase
odhC; pyruvate dehydrogenase E2 component (dihydrolipoyllysine−residue acetyltran
K01895 ACSS1_2, acs; acetyl−CoA synthetase [EC:6.2.1.1]
K05773 tupB, vupB; tungstate transport system permease protein
Expression
Collection date
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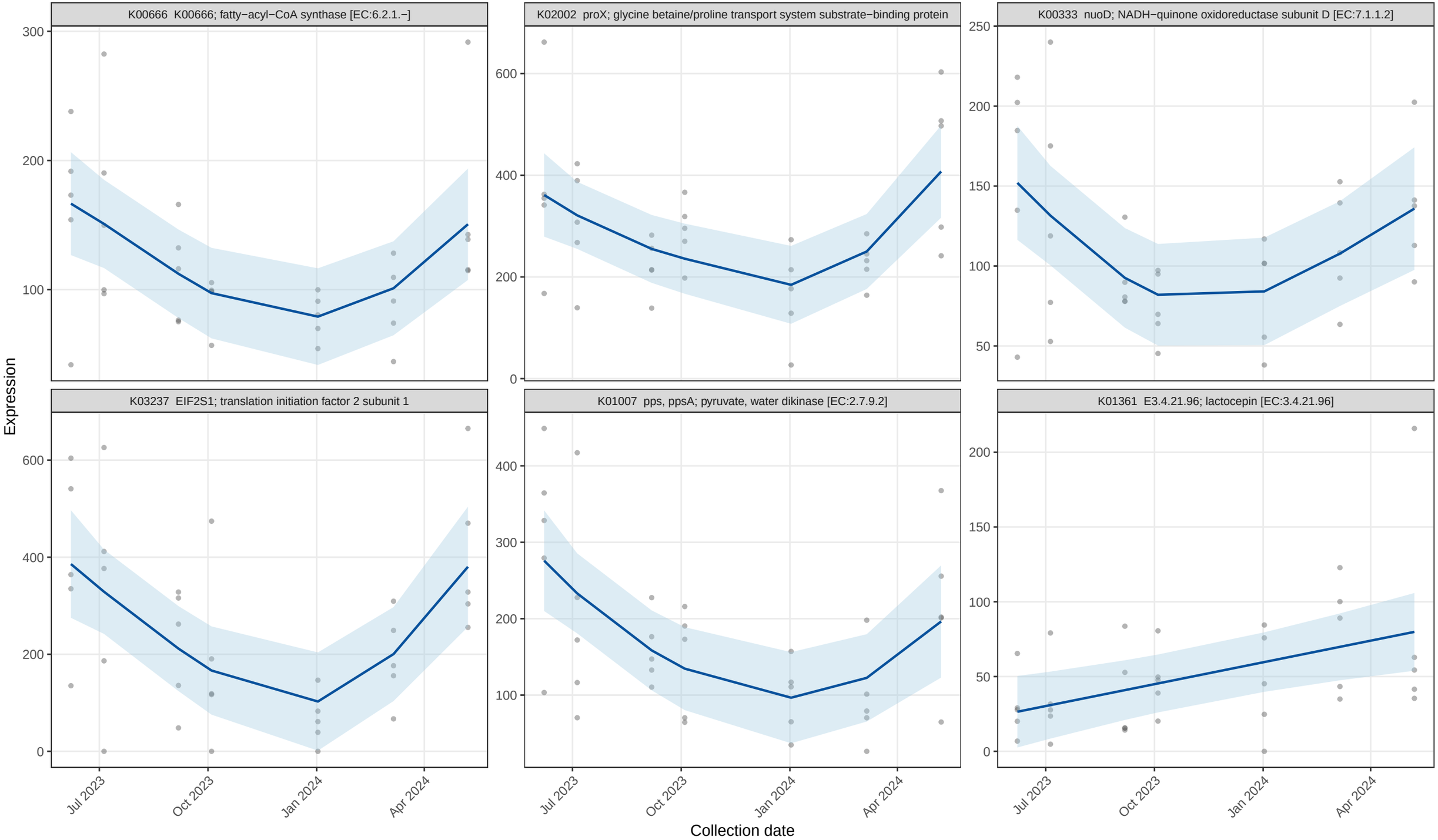

K02482 flgS; two−component system, NtrC family, sensor kinase [EC:2.7.13.3]
K01673 cynT, can; carbonic anhydrase [EC:4.2.1.1]
K00783 rlmH; 23S rRNA (pseudouridine1915−N3)−methyltransferase [EC:2.1.1.177]
K12372 dppF; dipeptide transport system ATP−binding protein
K13766 liuC; methylglutaconyl−CoA hydratase [EC:4.2.1.18]
K06204 dksA; RNA polymerase−binding transcription factor
Expression
Collection date
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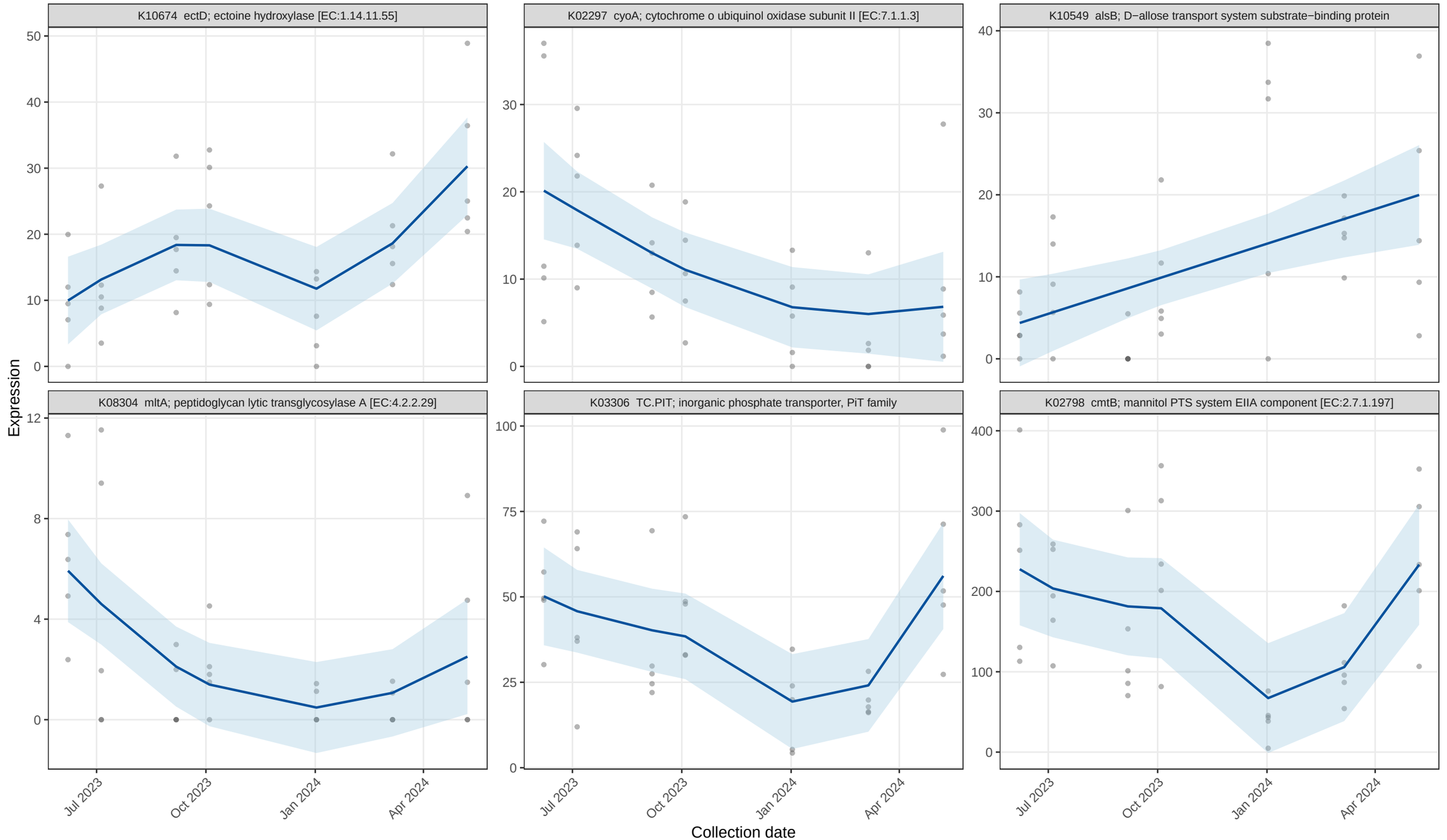

K13935 mdcH; malonate decarboxylase epsilon subunit [EC:2.3.1.39]
K18901 bpeE; membrane fusion protein, multidrug efflux system
K17710 PTCD1; pentatricopeptide repeat domain−containing protein 1
K07400 nfuA; Fe/S biogenesis protein NfuA
K12265 norW; nitric oxide reductase FlRd−NAD(+) reductase [EC:1.18.1.−]
48 metH, MTR; 5−methyltetrahydrofolate−−homocysteine methyltransferase [EC:2.1.
Expression
Collection date
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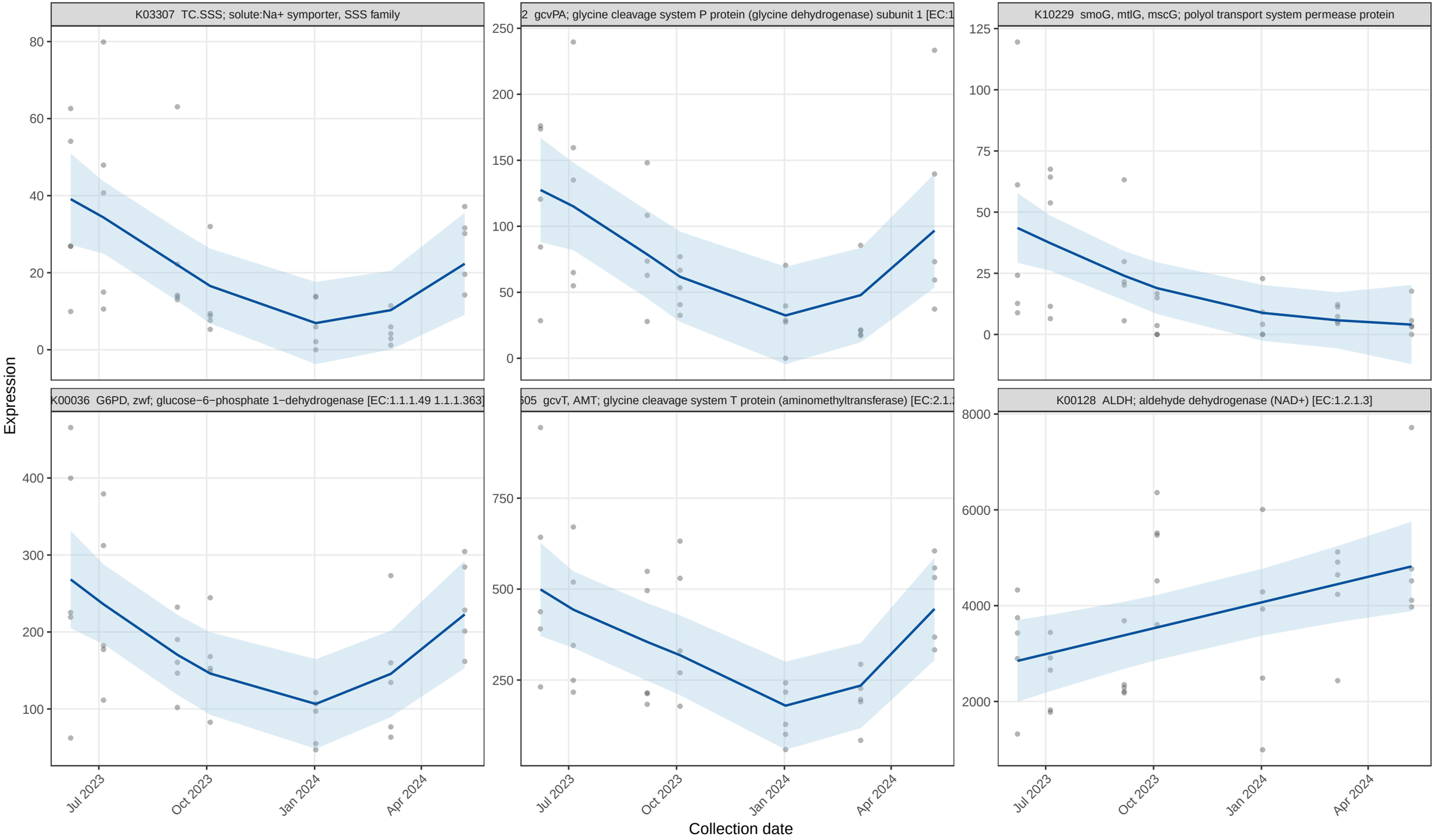

K03726 helS; ATP−dependent DNA helicase [EC:5.6.2.4]
K04796 RUXX; small nuclear ribonucleoprotein
K01432 AFMID; arylformamidase [EC:3.5.1.9]
K01933 purM; phosphoribosylformylglycinamidine cyclo−ligase [EC:6.3.3.1]
DP−N−acetyl−D−glucosamine/UDP−N−acetyl−D−galactosamine dehydrogenase [EC:
K01297 ldcA; muramoyltetrapeptide carboxypeptidase [EC:3.4.17.13]
Expression
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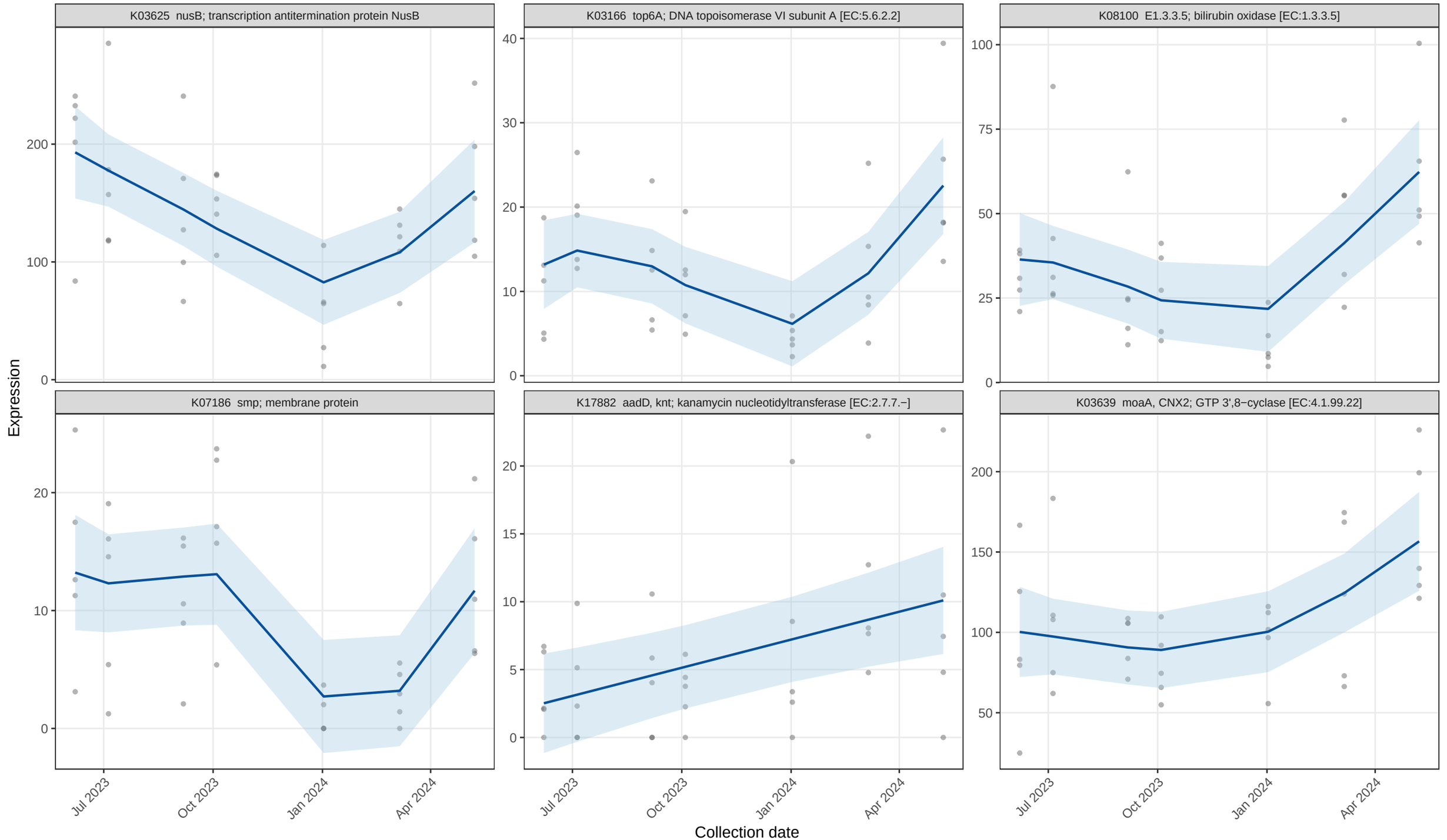

K00014 aroE; shikimate dehydrogenase [EC:1.1.1.25]
K00508 desD; sn−1 linoleoyl−lipid 6−desaturase [EC:1.14.19.46]
K06959 tex; protein Tex
K03735 eutB; ethanolamine ammonia−lyase large subunit [EC:4.3.1.7]
K13527 mpa; proteasome−associated ATPase
00571 E2.1.1.72; site−specific DNA−methyltransferase (adenine−specific) [EC:2.1.1.7
Expression
Collection date
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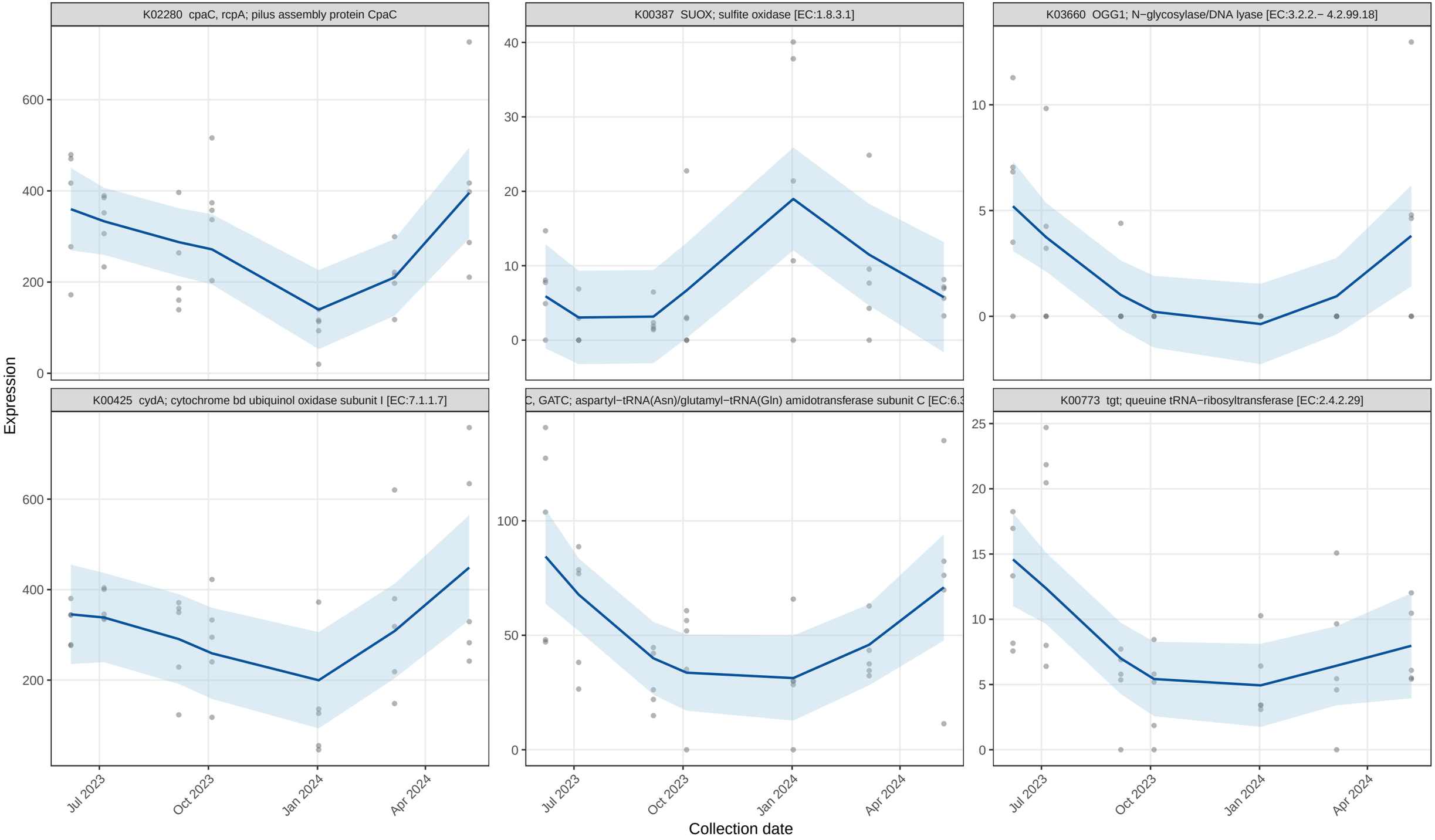

K02221 yggT; YggT family protein
K00806 uppS; undecaprenyl diphosphate synthase [EC:2.5.1.31]
hosphoribosylaminoimidazolecarboxamide formyltransferase / IMP cyclohydrolase [EC
arA, gacS, varS; two−component system, NarL family, sensor histidine kinase BarA [E
K02672 pilW; type IV pilus assembly protein PilW
; TetR/AcrR family transcriptional regulator, upper aerobic nicotinate degradation pathv
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

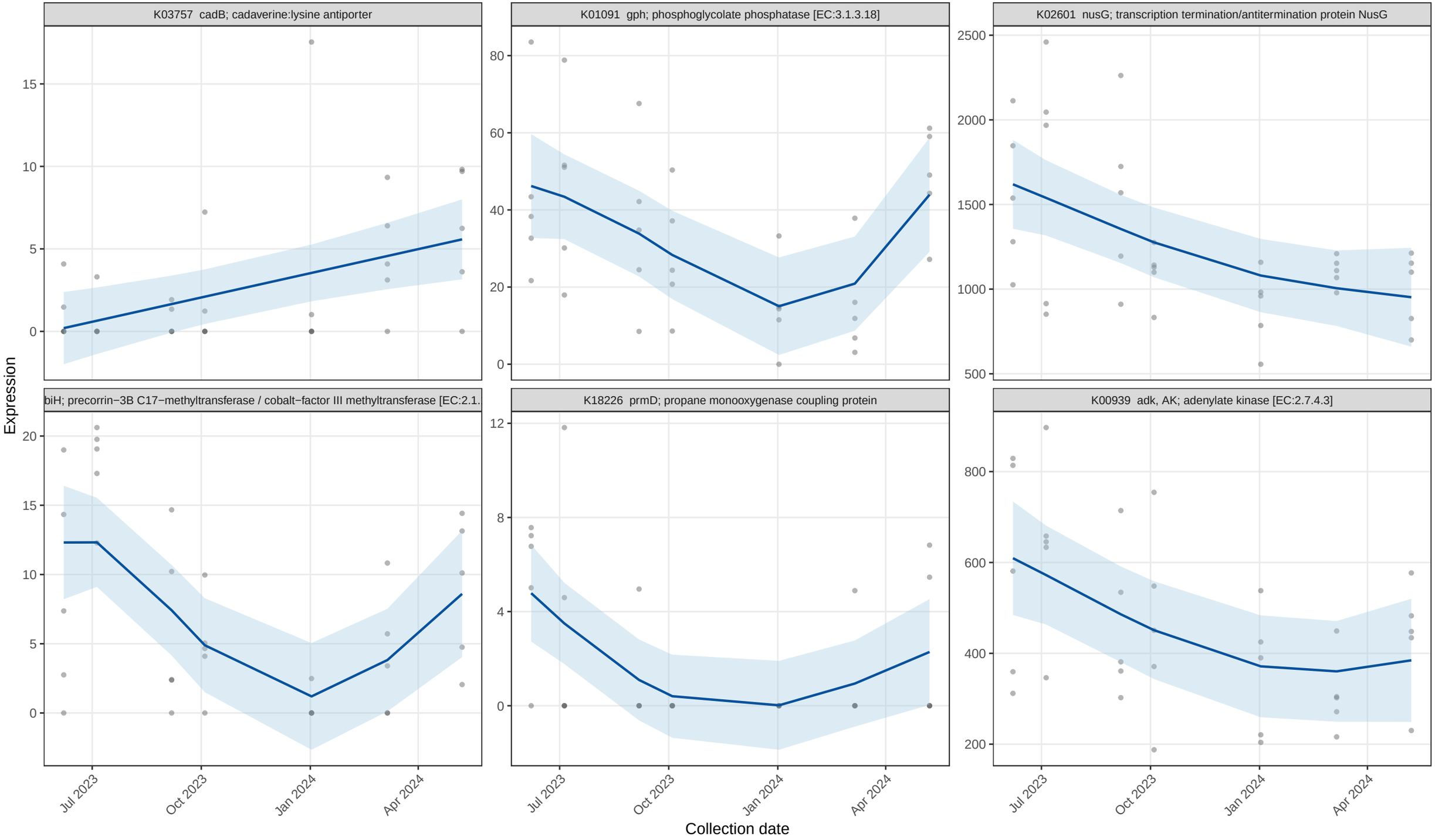

K00666 K00666; fatty−acyl−CoA synthase [EC:6.2.1.−]
K02002 proX; glycine betaine/proline transport system substrate−binding protein
K00333 nuoD; NADH−quinone oxidoreductase subunit D [EC:7.1.1.2]
K03237 EIF2S1; translation initiation factor 2 subunit 1
K01007 pps, ppsA; pyruvate, water dikinase [EC:2.7.9.2]
K01361 E3.4.21.96; lactocepin [EC:3.4.21.96]
Expression
Collection date
Jul 2023
Oct 2023
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Apr 2024

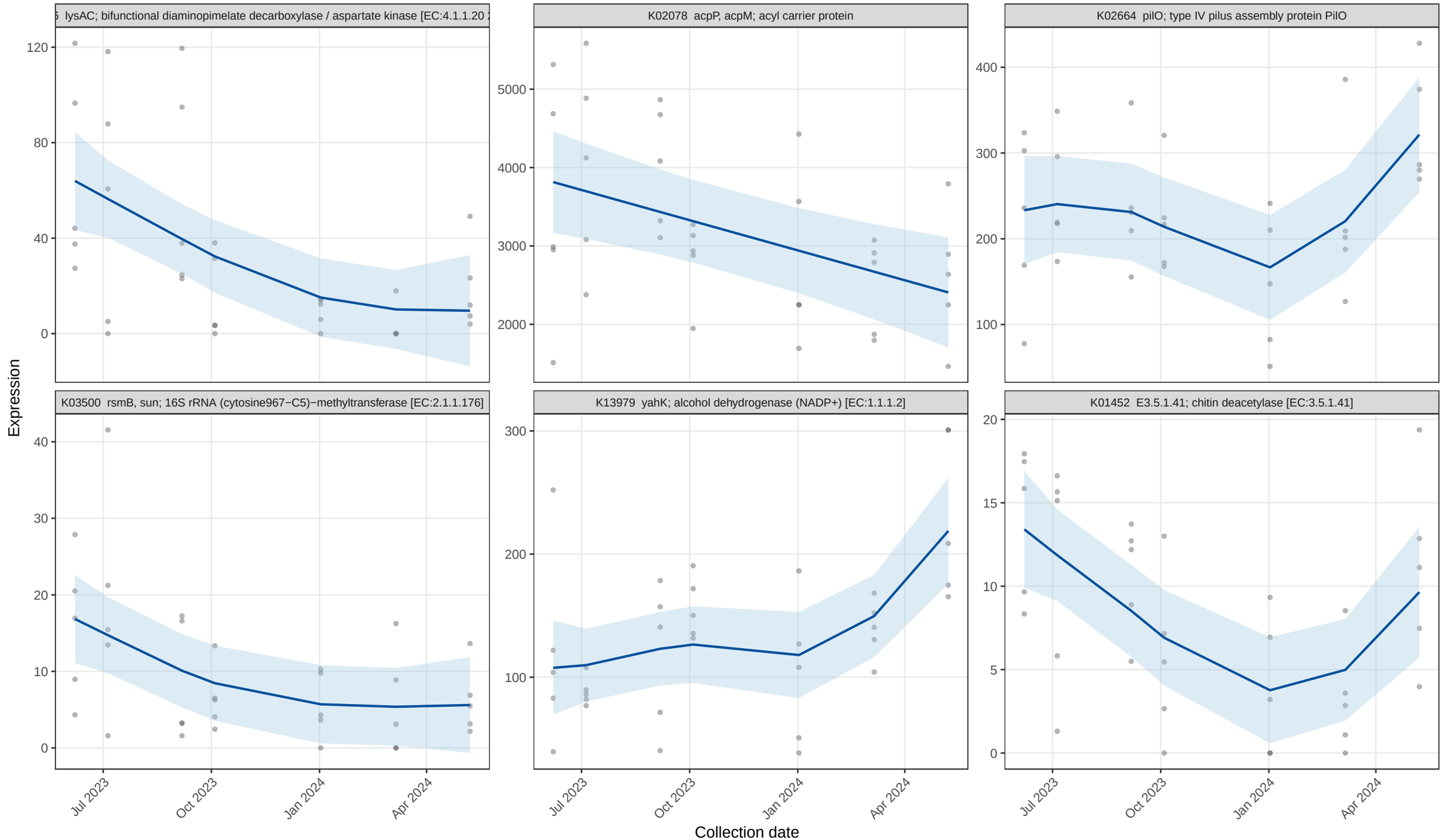

K10674 ectD; ectoine hydroxylase [EC:1.14.11.55]
K02297 cyoA; cytochrome o ubiquinol oxidase subunit II [EC:7.1.1.3]
K10549 alsB; D−allose transport system substrate−binding protein
K08304 mltA; peptidoglycan lytic transglycosylase A [EC:4.2.2.29]
K03306 TC.PIT; inorganic phosphate transporter, PiT family
K02798 cmtB; mannitol PTS system EIIA component [EC:2.7.1.197]
Expression
Collection date
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Apr 2024

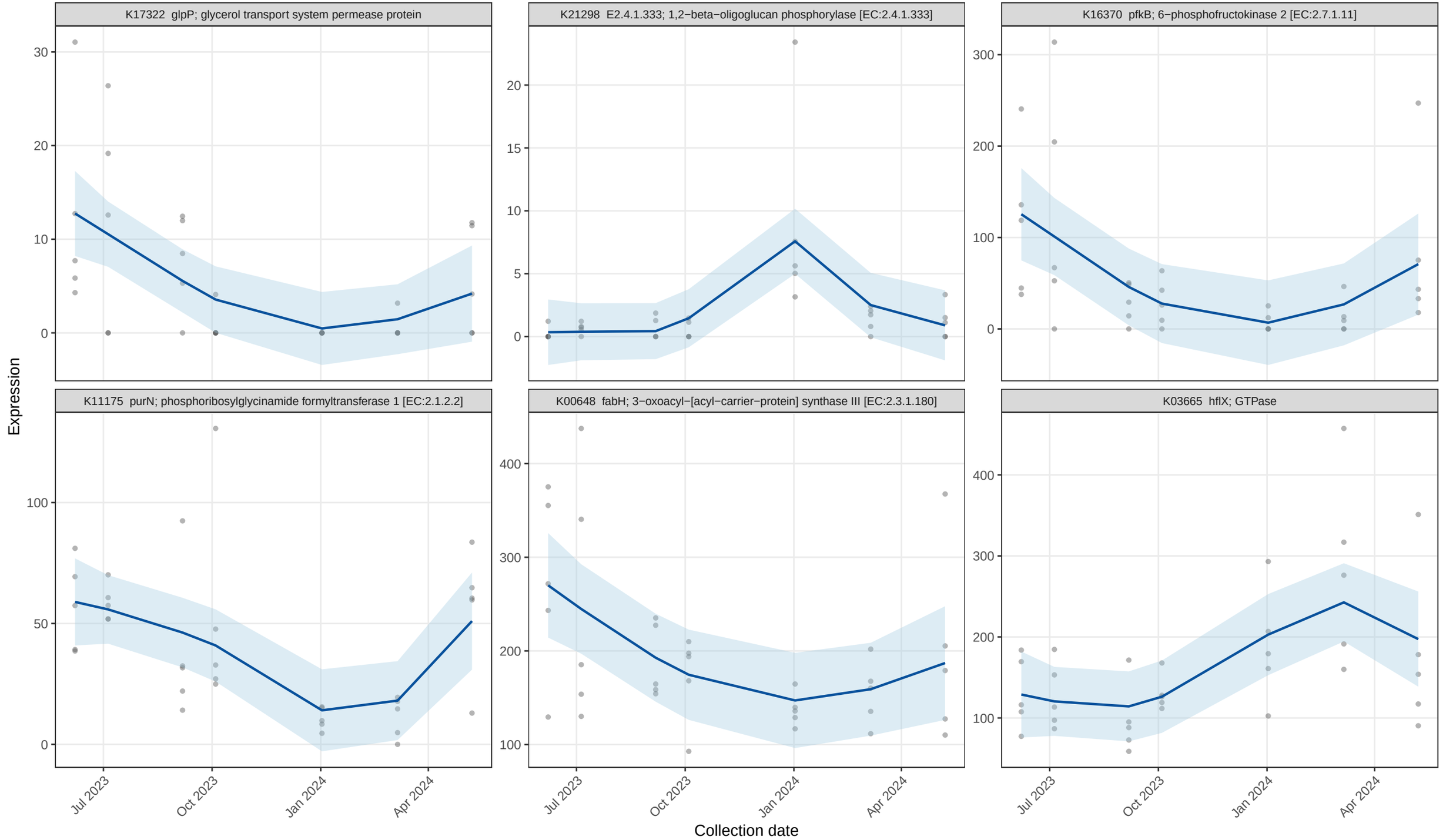

K03307 TC.SSS; solute:Na+ symporter, SSS family
2 gcvPA; glycine cleavage system P protein (glycine dehydrogenase) subunit 1 [EC:1
K10229 smoG, mtlG, mscG; polyol transport system permease protein
K00036 G6PD, zwf; glucose−6−phosphate 1−dehydrogenase [EC:1.1.1.49 1.1.1.363]
505 gcvT, AMT; glycine cleavage system T protein (aminomethyltransferase) [EC:2.1.
K00128 ALDH; aldehyde dehydrogenase (NAD+) [EC:1.2.1.3]
Expression
Collection date
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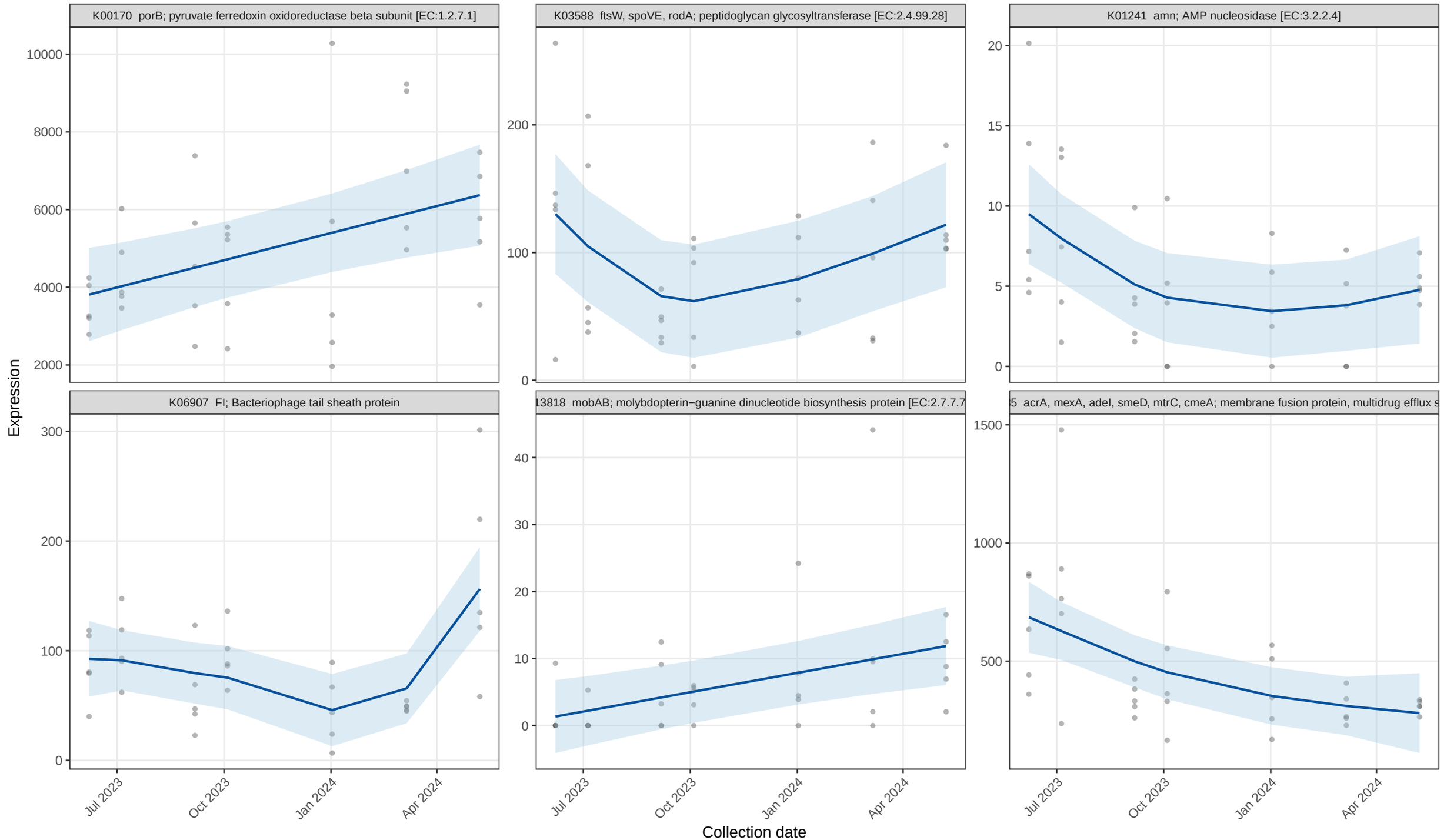

K03625 nusB; transcription antitermination protein NusB
K03166 top6A; DNA topoisomerase VI subunit A [EC:5.6.2.2]
K08100 E1.3.3.5; bilirubin oxidase [EC:1.3.3.5]
K07186 smp; membrane protein
K17882 aadD, knt; kanamycin nucleotidyltransferase [EC:2.7.7.−]
K03639 moaA, CNX2; GTP 3',8−cyclase [EC:4.1.99.22]
Expression
Collection date
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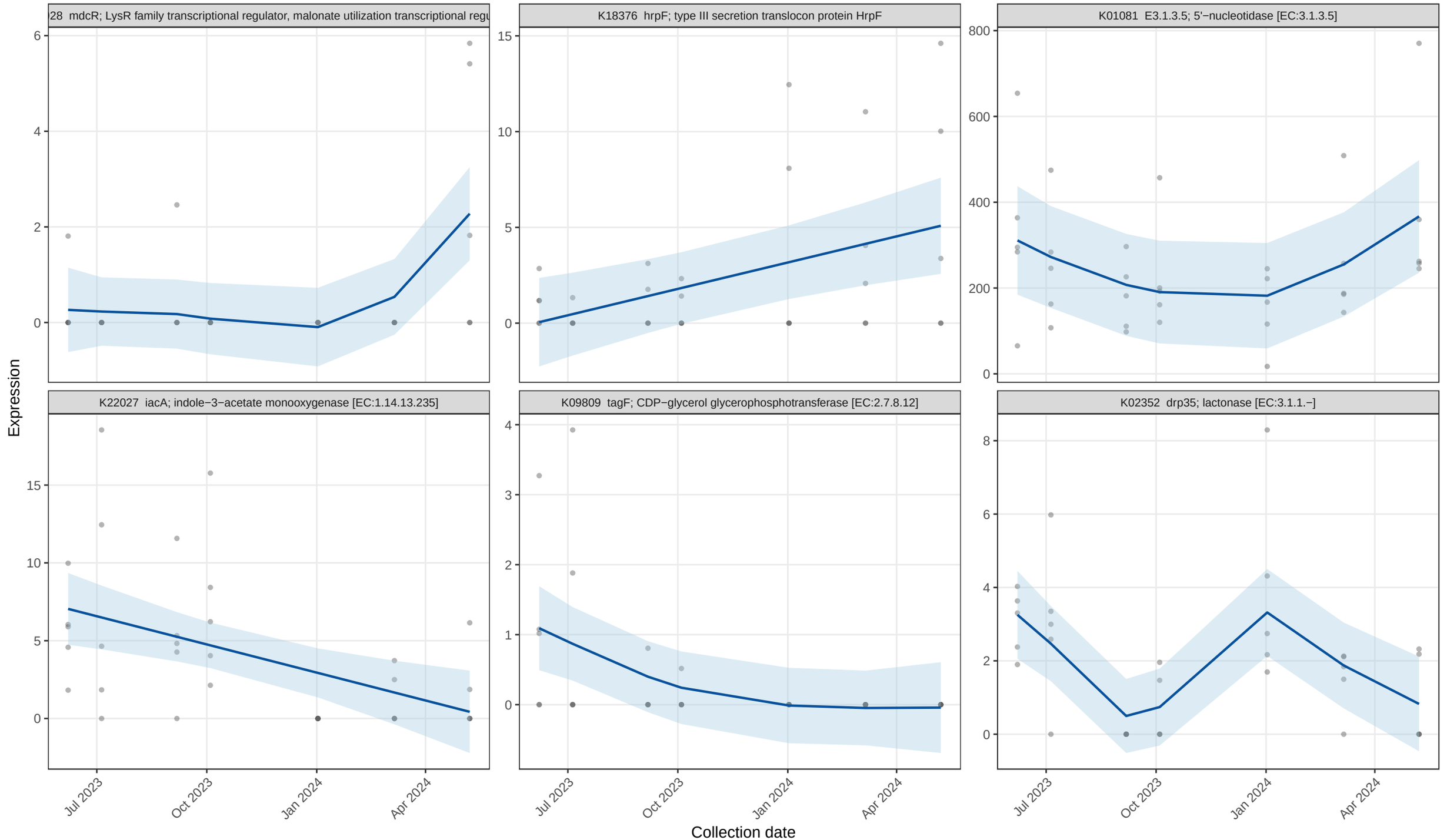

K02280 cpaC, rcpA; pilus assembly protein CpaC
K00387 SUOX; sulfite oxidase [EC:1.8.3.1]
K03660 OGG1; N−glycosylase/DNA lyase [EC:3.2.2.− 4.2.99.18]
K00425 cydA; cytochrome bd ubiquinol oxidase subunit I [EC:7.1.1.7]
C, GATC; aspartyl−tRNA(Asn)/glutamyl−tRNA(Gln) amidotransferase subunit C [EC:6.3
K00773 tgt; queuine tRNA−ribosyltransferase [EC:2.4.2.29]
Expression
Collection date
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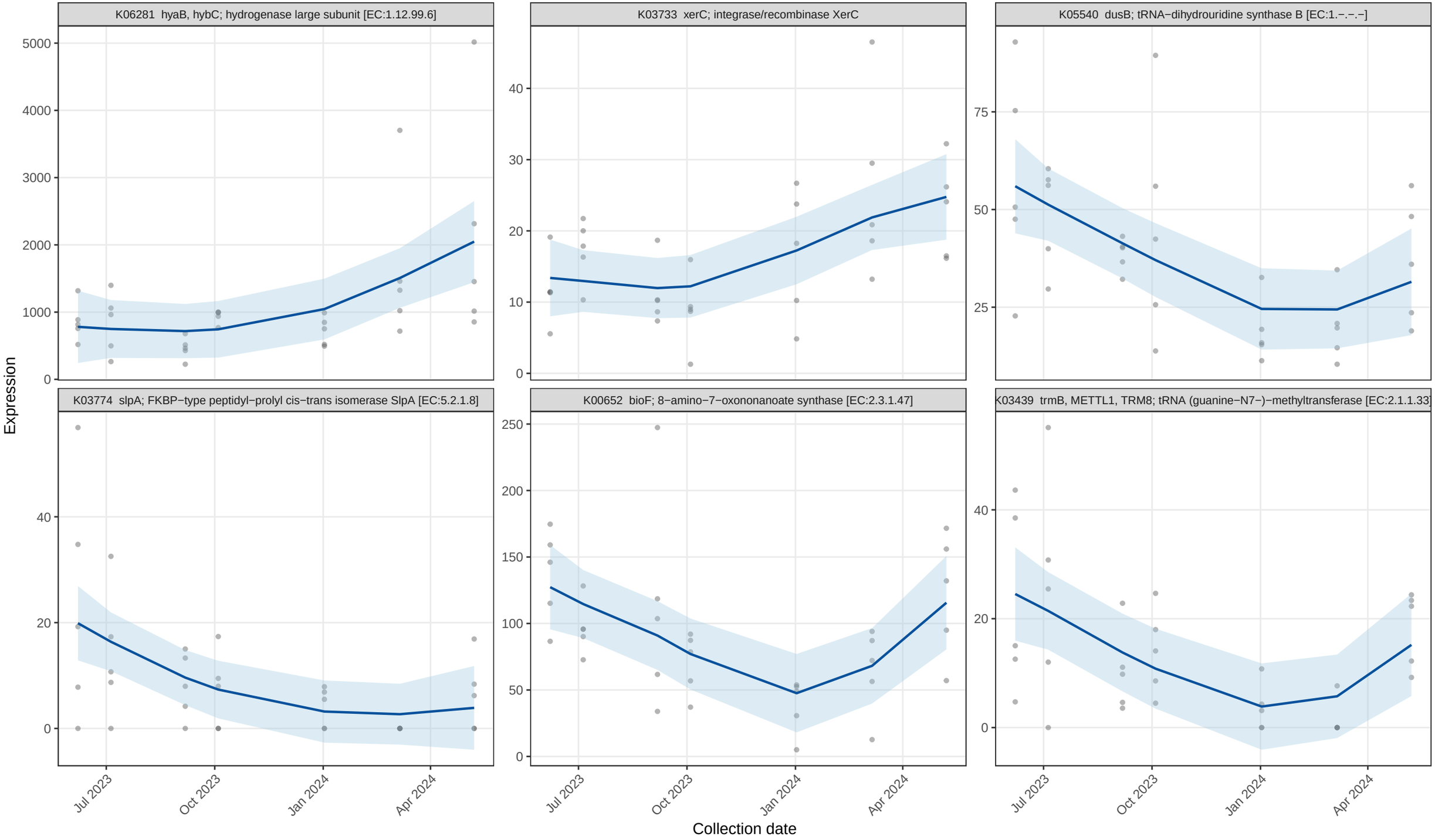

K03757 cadB; cadaverine:lysine antiporter
K01091 gph; phosphoglycolate phosphatase [EC:3.1.3.18]
K02601 nusG; transcription termination/antitermination protein NusG
biH; precorrin−3B C17−methyltransferase / cobalt−factor III methyltransferase [EC:2.1.
K18226 prmD; propane monooxygenase coupling protein
K00939 adk, AK; adenylate kinase [EC:2.7.4.3]
Expression
Collection date
Jul 2023
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Jan 2024
Apr 2024

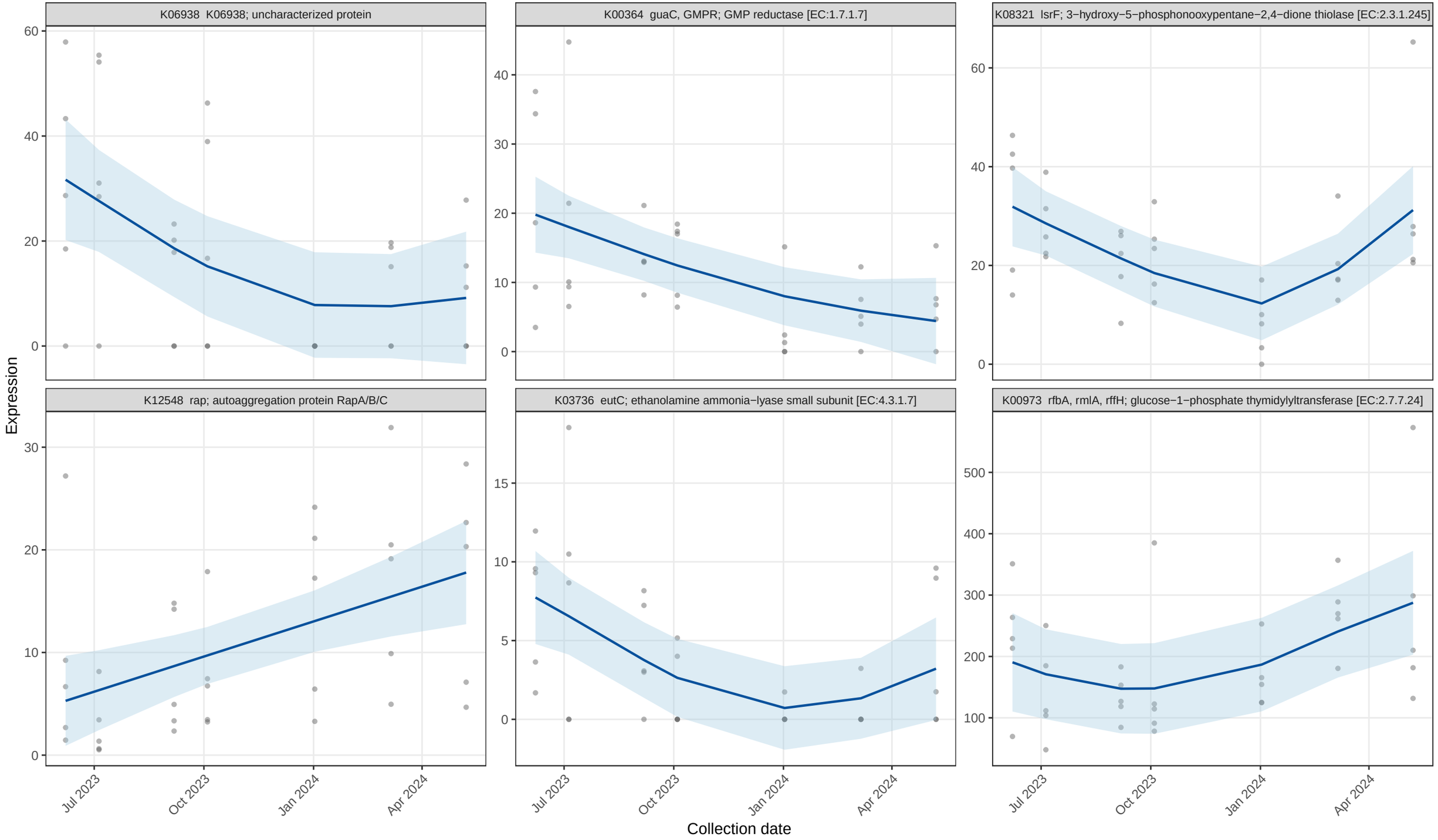

lysAC; bifunctional diaminopimelate decarboxylase / aspartate kinase [EC:4.1.1.20
K02078 acpP, acpM; acyl carrier protein
K02664 pilO; type IV pilus assembly protein PilO
K03500 rsmB, sun; 16S rRNA (cytosine967−C5)−methyltransferase [EC:2.1.1.176]
K13979 yahK; alcohol dehydrogenase (NADP+) [EC:1.1.1.2]
K01452 E3.5.1.41; chitin deacetylase [EC:3.5.1.41]
Expression
Collection date
Jul 2023
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Jan 2024
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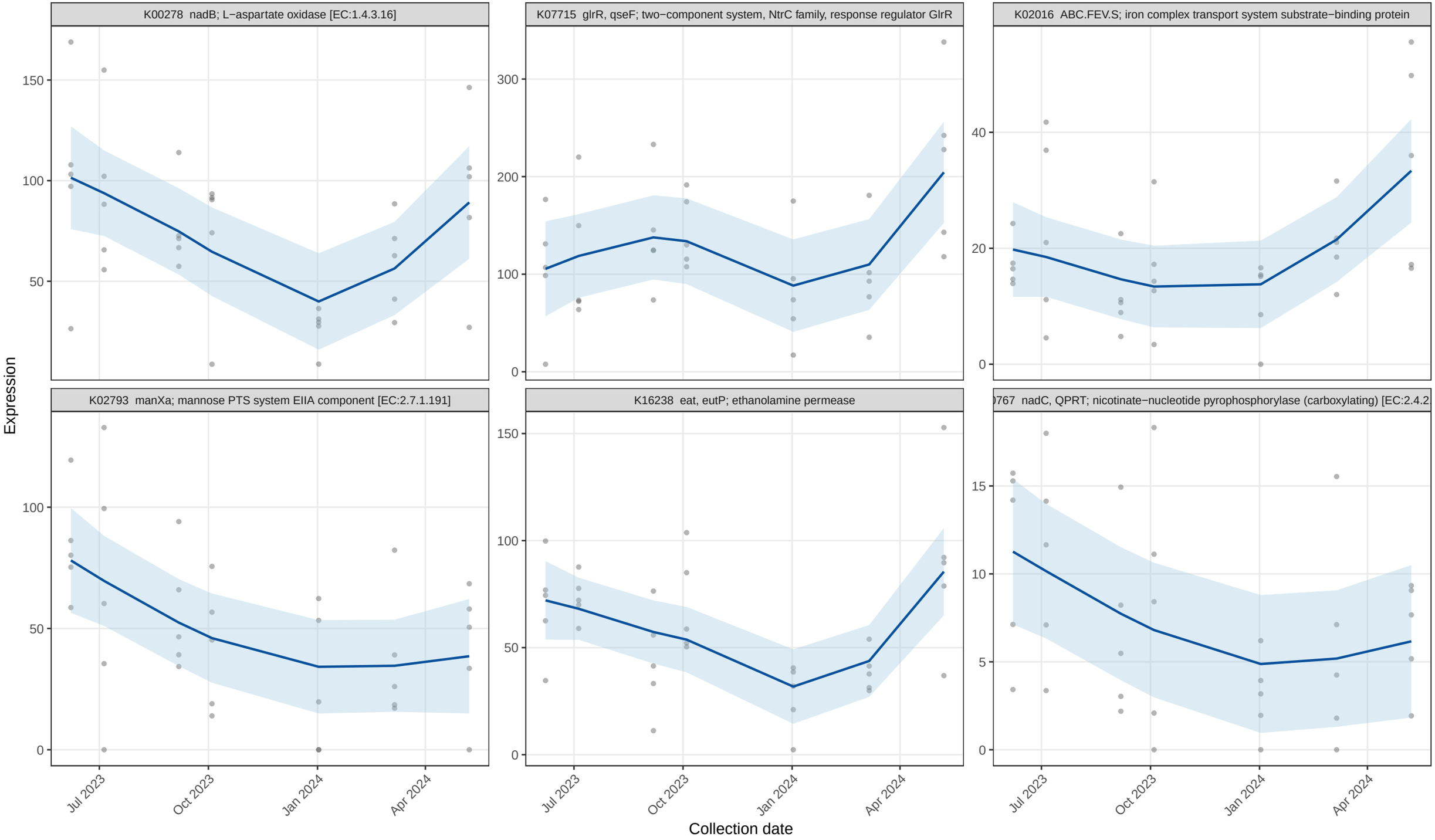

K17322 glpP; glycerol transport system permease protein
K21298 E2.4.1.333; 1,2−beta−oligoglucan phosphorylase [EC:2.4.1.333]
K16370 pfkB; 6−phosphofructokinase 2 [EC:2.7.1.11]
K11175 purN; phosphoribosylglycinamide formyltransferase 1 [EC:2.1.2.2]
K00648 fabH; 3−oxoacyl−[acyl−carrier−protein] synthase III [EC:2.3.1.180]
K03665 hflX; GTPase
Expression
Collection date
Jul 2023
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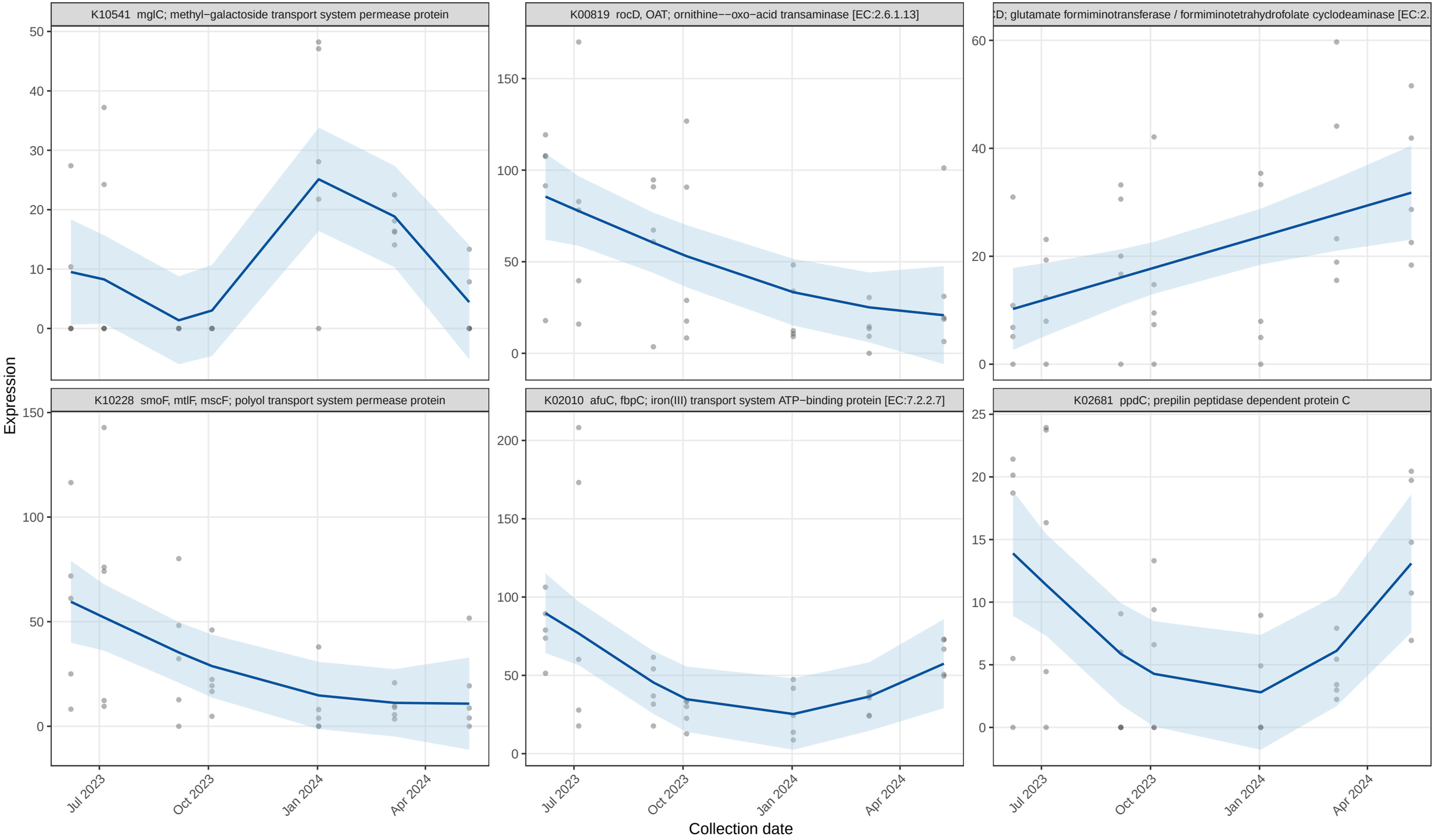

K00170 porB; pyruvate ferredoxin oxidoreductase beta subunit [EC:1.2.7.1]
K03588 ftsW, spoVE, rodA; peptidoglycan glycosyltransferase [EC:2.4.99.28]
K01241 amn; AMP nucleosidase [EC:3.2.2.4]
K06907 FI; Bacteriophage tail sheath protein
13818 mobAB; molybdopterin−guanine dinucleotide biosynthesis protein [EC:2.7.7.7
5 acrA, mexA, adeI, smeD, mtrC, cmeA; membrane fusion protein, multidrug efflux s
Expression
Collection date
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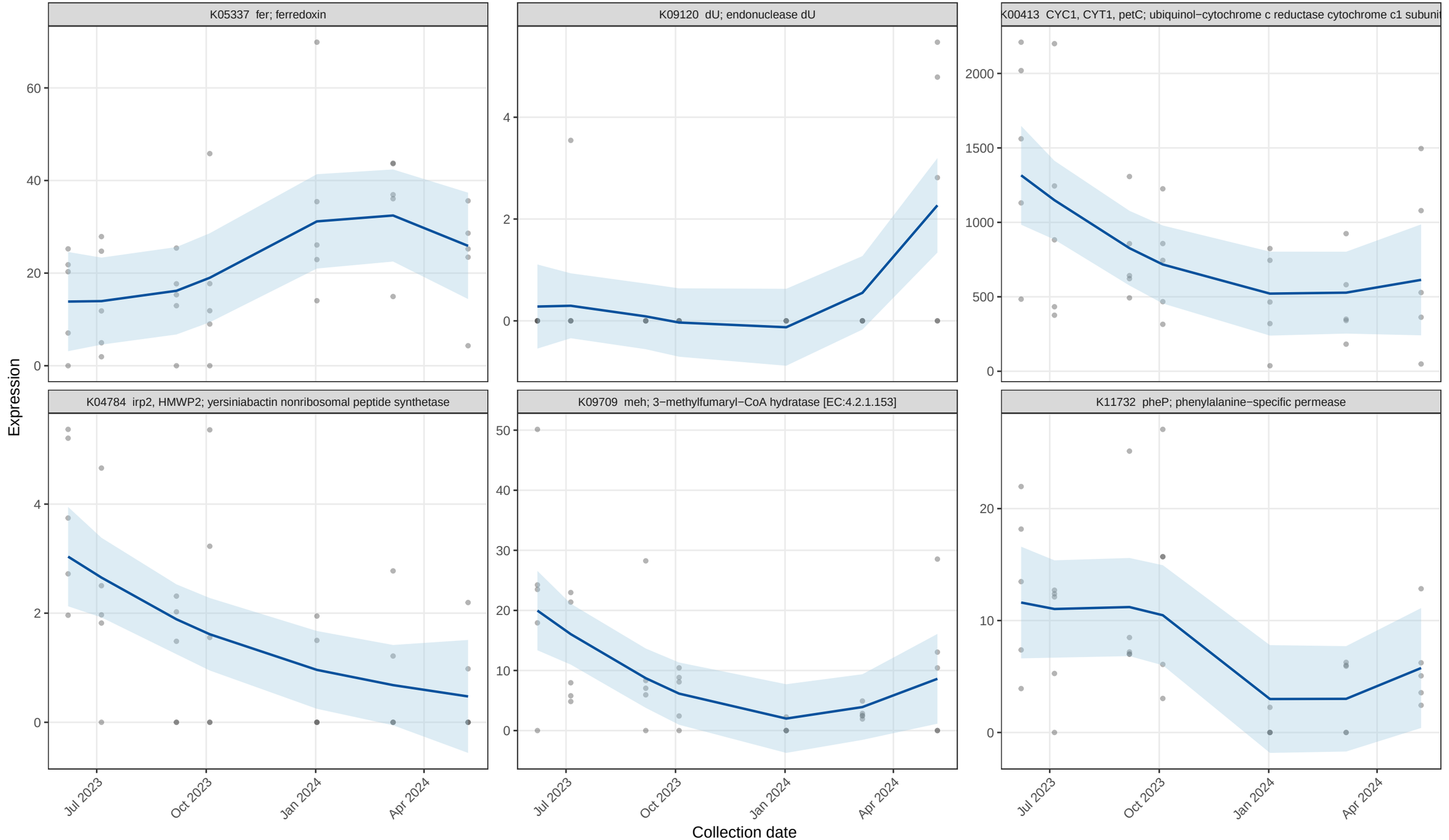

28 mdcR; LysR family transcriptional regulator, malonate utilization transcriptional regu
K18376 hrpF; type III secretion translocon protein HrpF
K01081 E3.1.3.5; 5'−nucleotidase [EC:3.1.3.5]
K22027 iacA; indole−3−acetate monooxygenase [EC:1.14.13.235]
K09809 tagF; CDP−glycerol glycerophosphotransferase [EC:2.7.8.12]
K02352 drp35; lactonase [EC:3.1.1.−]
Expression
Collection date
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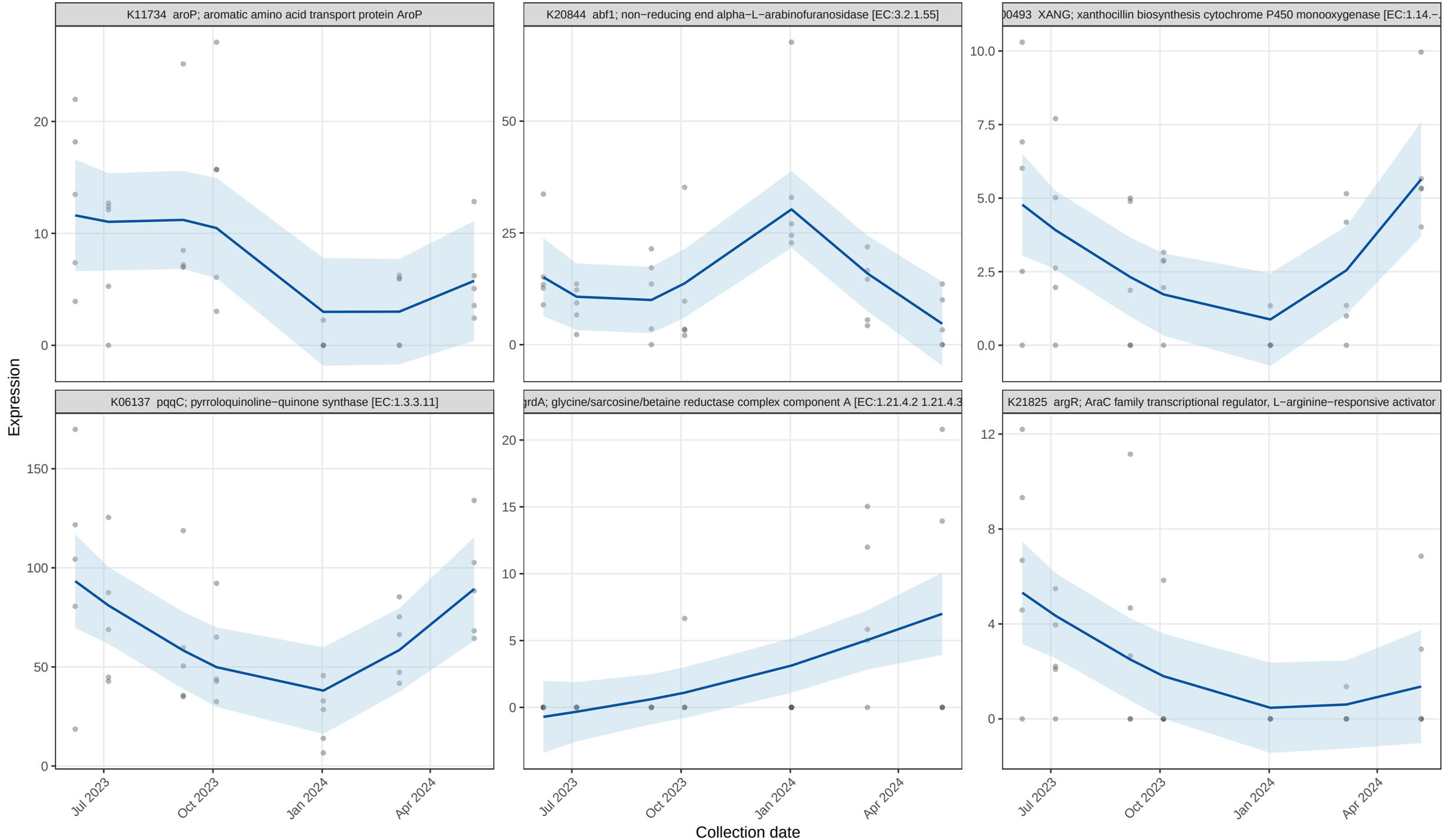

K06281 hyaB, hybC; hydrogenase large subunit [EC:1.12.99.6]
K03733 xerC; integrase/recombinase XerC
K05540 dusB; tRNA−dihydrouridine synthase B [EC:1.−.−.−]
K03774 slpA; FKBP−type peptidyl−prolyl cis−trans isomerase SlpA [EC:5.2.1.8]
K00652 bioF; 8−amino−7−oxononanoate synthase [EC:2.3.1.47]
K03439 trmB, METTL1, TRM8; tRNA (guanine−N7−)−methyltransferase [EC:2.1.1.33]
Expression
Collection date
Jul 2023
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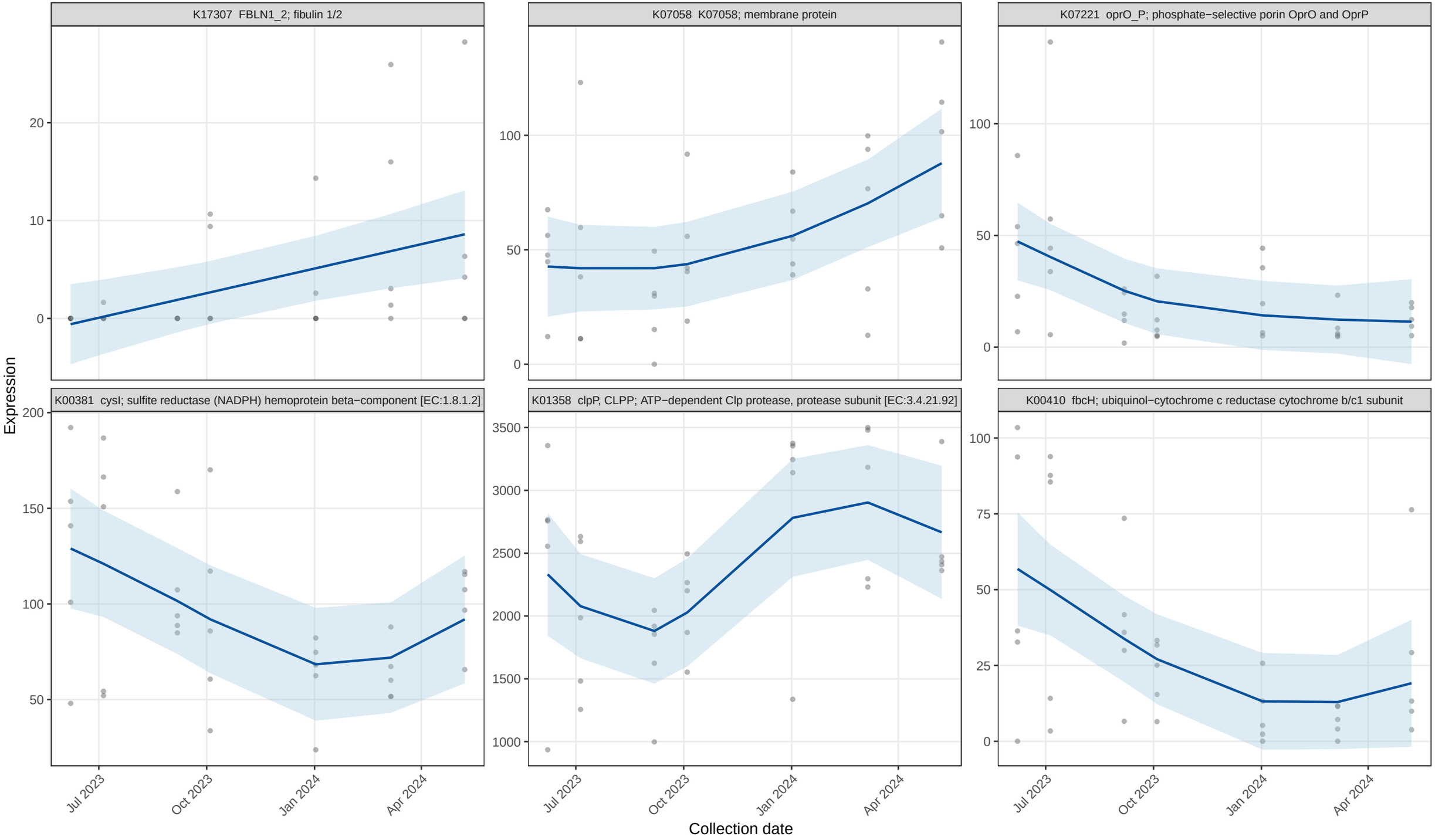

K06938 K06938; uncharacterized protein
K00364 guaC, GMPR; GMP reductase [EC:1.7.1.7]
K08321 lsrF; 3−hydroxy−5−phosphonooxypentane−2,4−dione thiolase [EC:2.3.1.245]
K12548 rap; autoaggregation protein RapA/B/C
K03736 eutC; ethanolamine ammonia−lyase small subunit [EC:4.3.1.7]
K00973 rfbA, rmlA, rffH; glucose−1−phosphate thymidylyltransferase [EC:2.7.7.24]
Expression
Collection date
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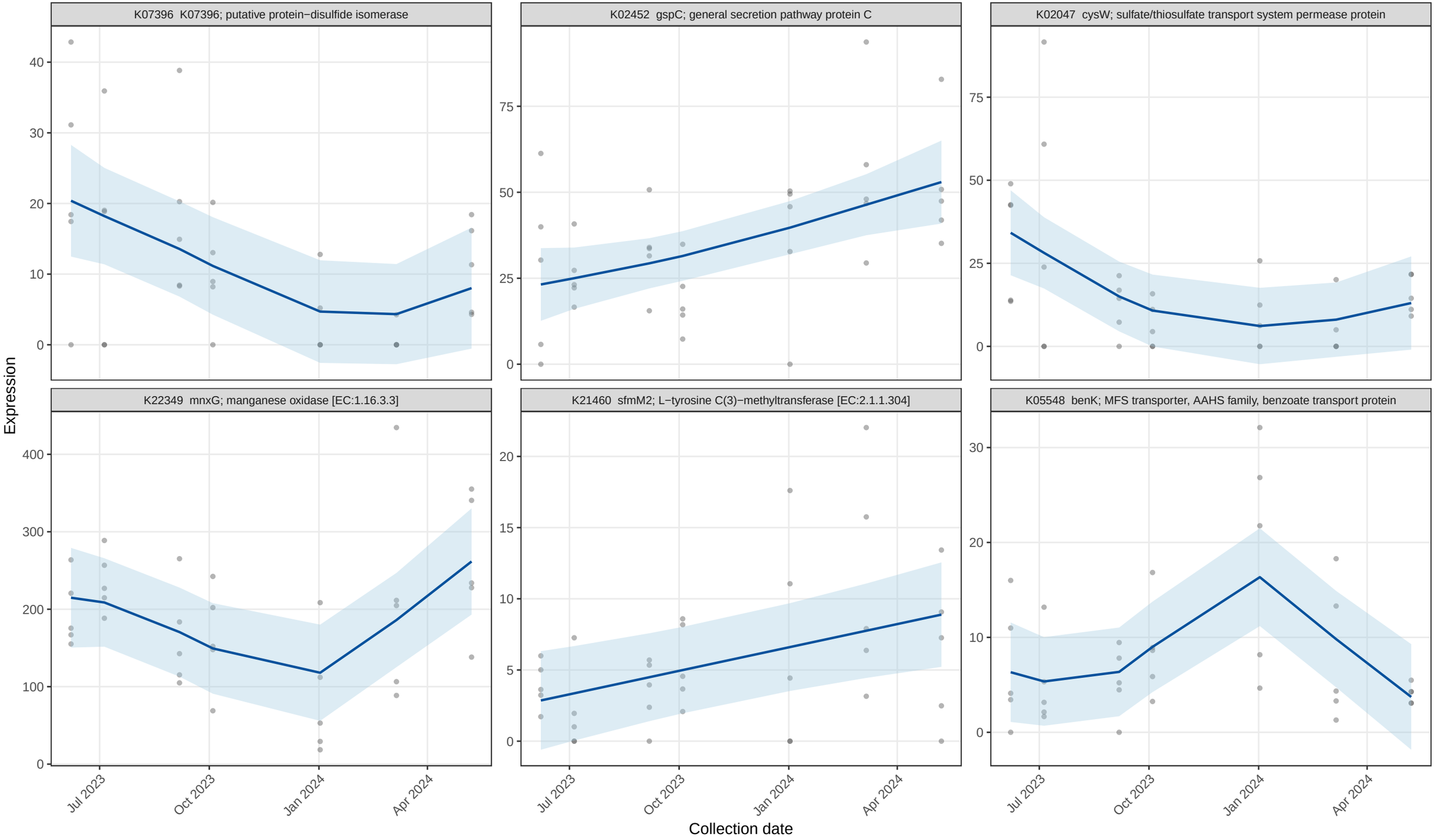

K00278 nadB; L−aspartate oxidase [EC:1.4.3.16]
K07715 glrR, qseF; two−component system, NtrC family, response regulator GlrR
K02016 ABC.FEV.S; iron complex transport system substrate−binding protein
K02793 manXa; mannose PTS system EIIA component [EC:2.7.1.191]
K16238 eat, eutP; ethanolamine permease
0767 nadC, QPRT; nicotinate−nucleotide pyrophosphorylase (carboxylating) [EC:2.4.2
Expression
Collection date
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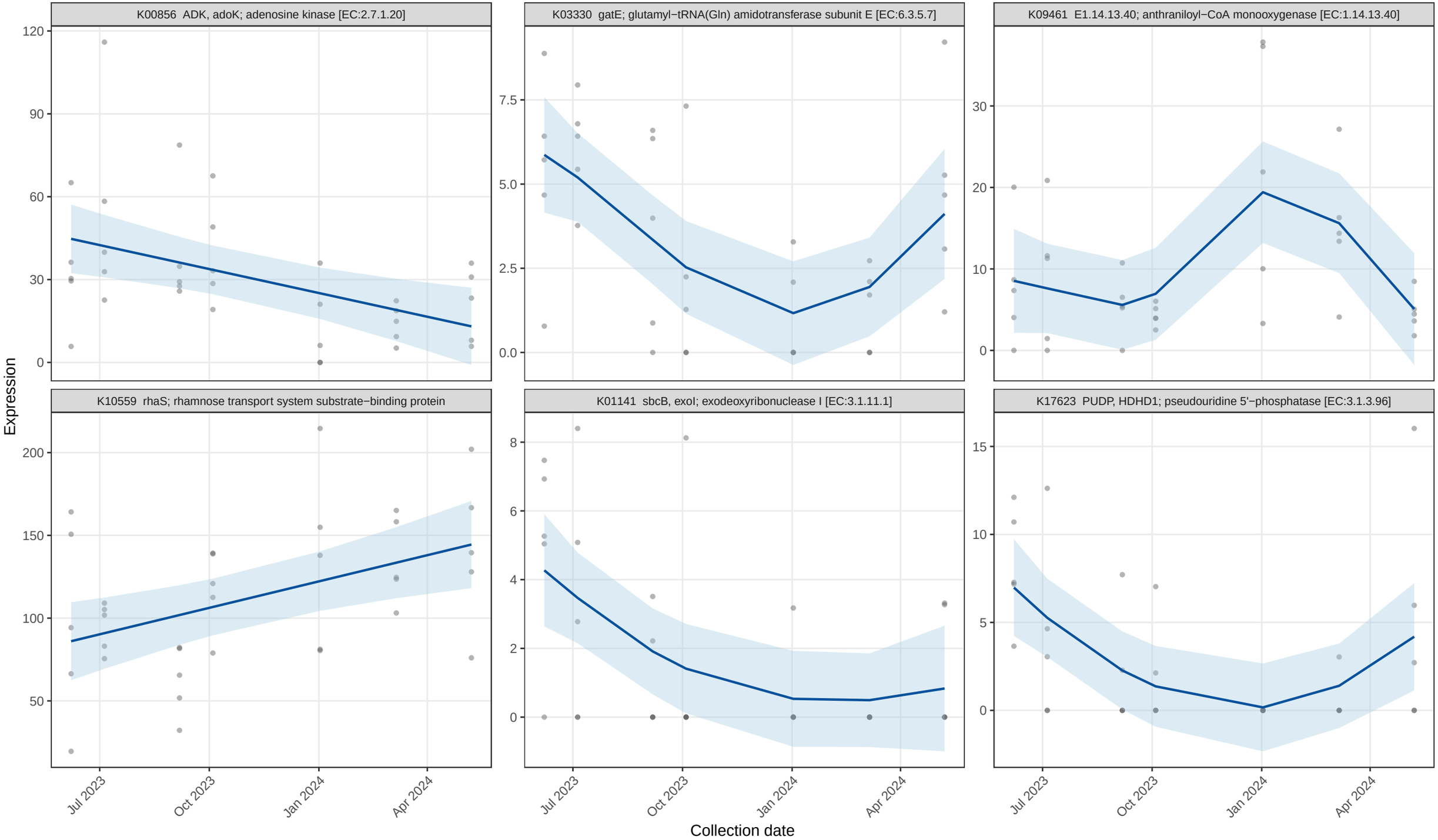

K10541 mglC; methyl−galactoside transport system permease protein
K00819 rocD, OAT; ornithine−−oxo−acid transaminase [EC:2.6.1.13]
D; glutamate formiminotransferase / formiminotetrahydrofolate cyclodeaminase [EC:2.
K10228 smoF, mtlF, mscF; polyol transport system permease protein
K02010 afuC, fbpC; iron(III) transport system ATP−binding protein [EC:7.2.2.7]
K02681 ppdC; prepilin peptidase dependent protein C
Expression
Collection date
Jul 2023
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Apr 2024

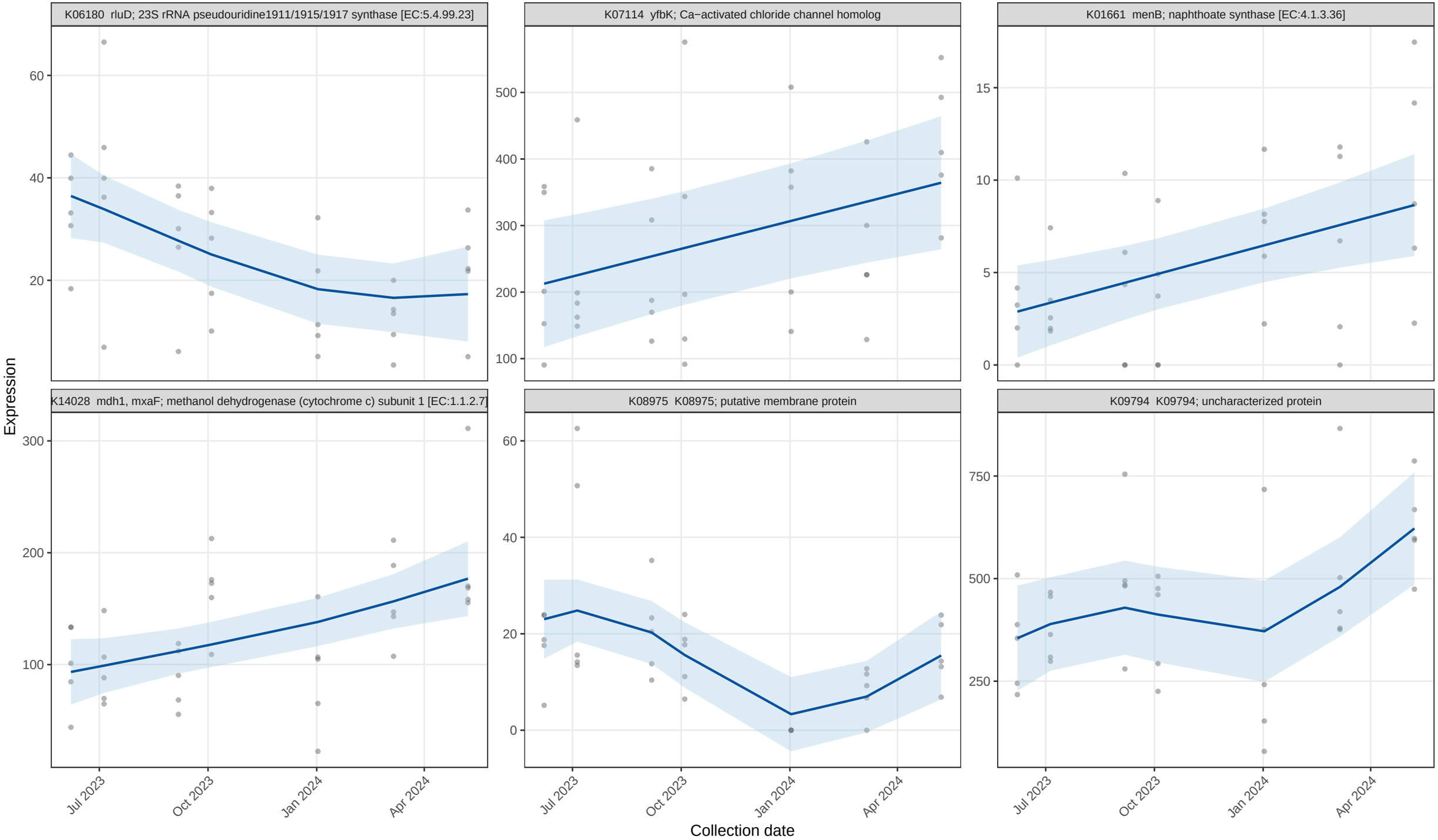

K05337 fer; ferredoxin
K09120 dU; endonuclease dU
K00413 CYC1, CYT1, petC; ubiquinol−cytochrome c reductase cytochrome c1 subunit
K04784 irp2, HMWP2; yersiniabactin nonribosomal peptide synthetase
K09709 meh; 3−methylfumaryl−CoA hydratase [EC:4.2.1.153]
K11732 pheP; phenylalanine−specific permease
Expression
Collection date
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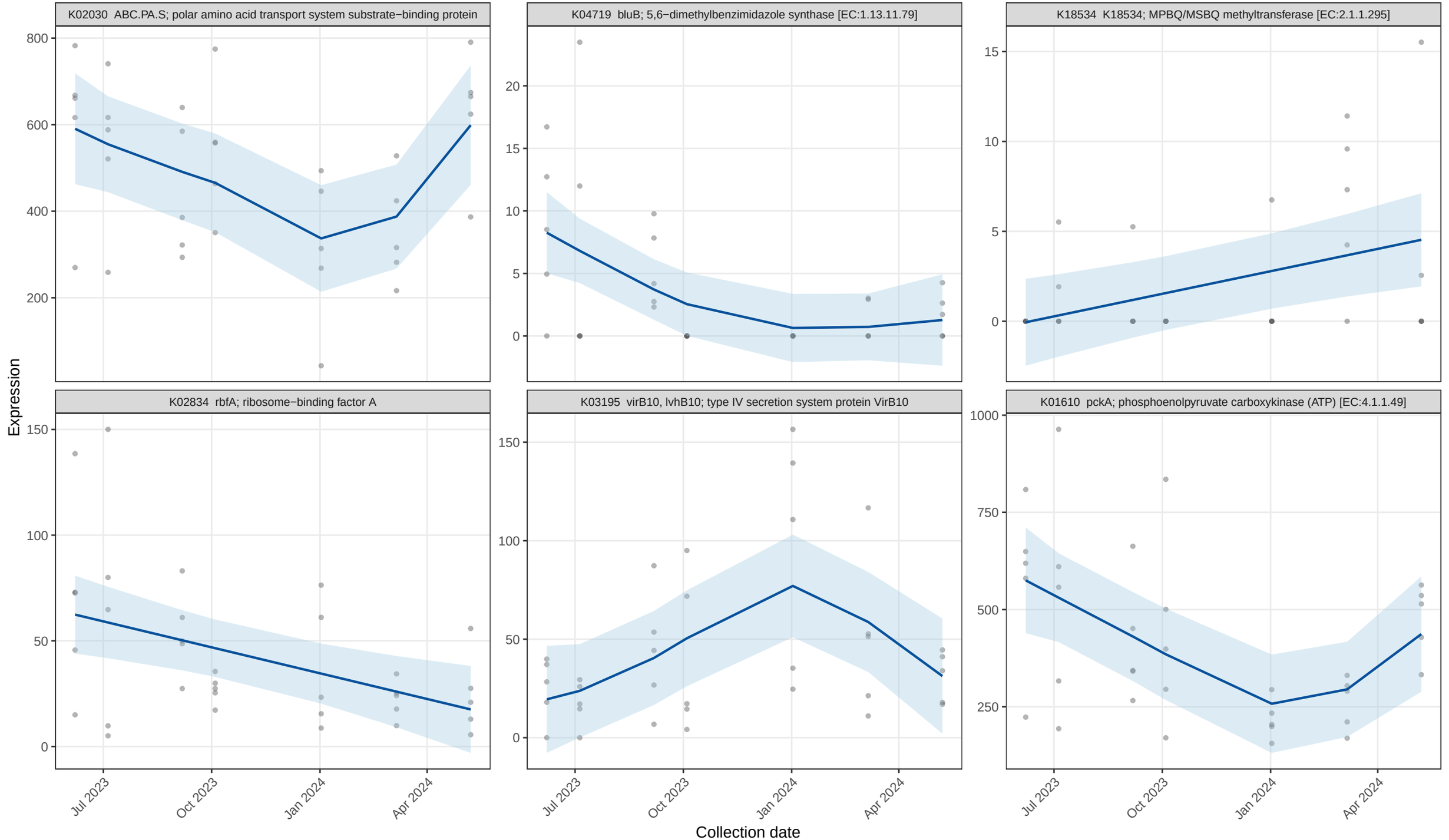

K11734 aroP; aromatic amino acid transport protein AroP
K20844 abf1; non−reducing end alpha−L−arabinofuranosidase [EC:3.2.1.55]
00493 XANG; xanthocillin biosynthesis cytochrome P450 monooxygenase [EC:1.14.−
K06137 pqqC; pyrroloquinoline−quinone synthase [EC:1.3.3.11]
grdA; glycine/sarcosine/betaine reductase complex component A [EC:1.21.4.2 1.21.4.3
K21825 argR; AraC family transcriptional regulator, L−arginine−responsive activator
Expression
Collection date
Jul 2023
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Apr 2024

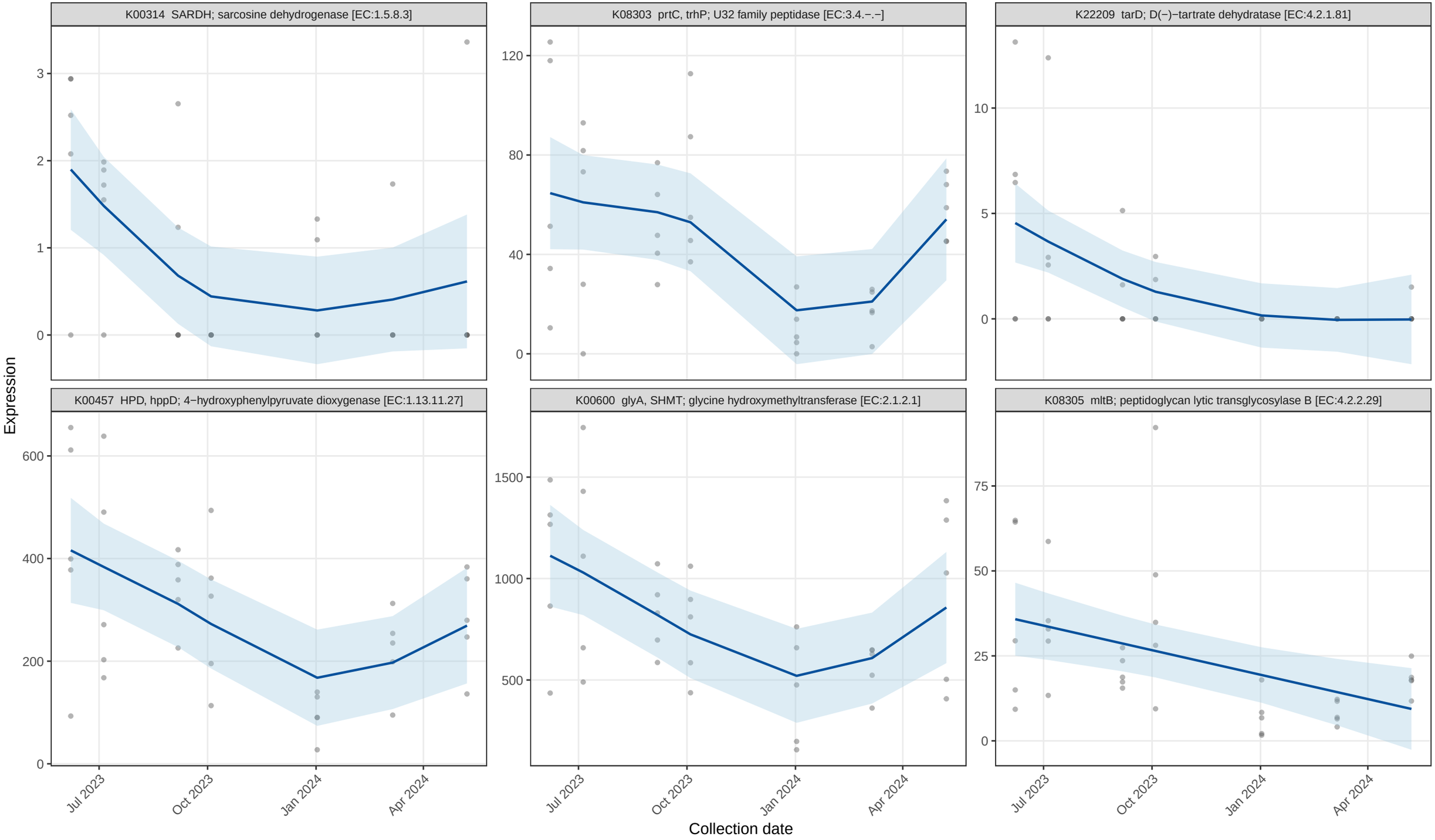

K17307 FBLN1_2; fibulin 1/2
K07058 K07058; membrane protein
K07221 oprO_P; phosphate−selective porin OprO and OprP
K00381 cysI; sulfite reductase (NADPH) hemoprotein beta−component [EC:1.8.1.2]
K01358 clpP, CLPP; ATP−dependent Clp protease, protease subunit [EC:3.4.21.92]
K00410 fbcH; ubiquinol−cytochrome c reductase cytochrome b/c1 subunit
Expression
Collection date
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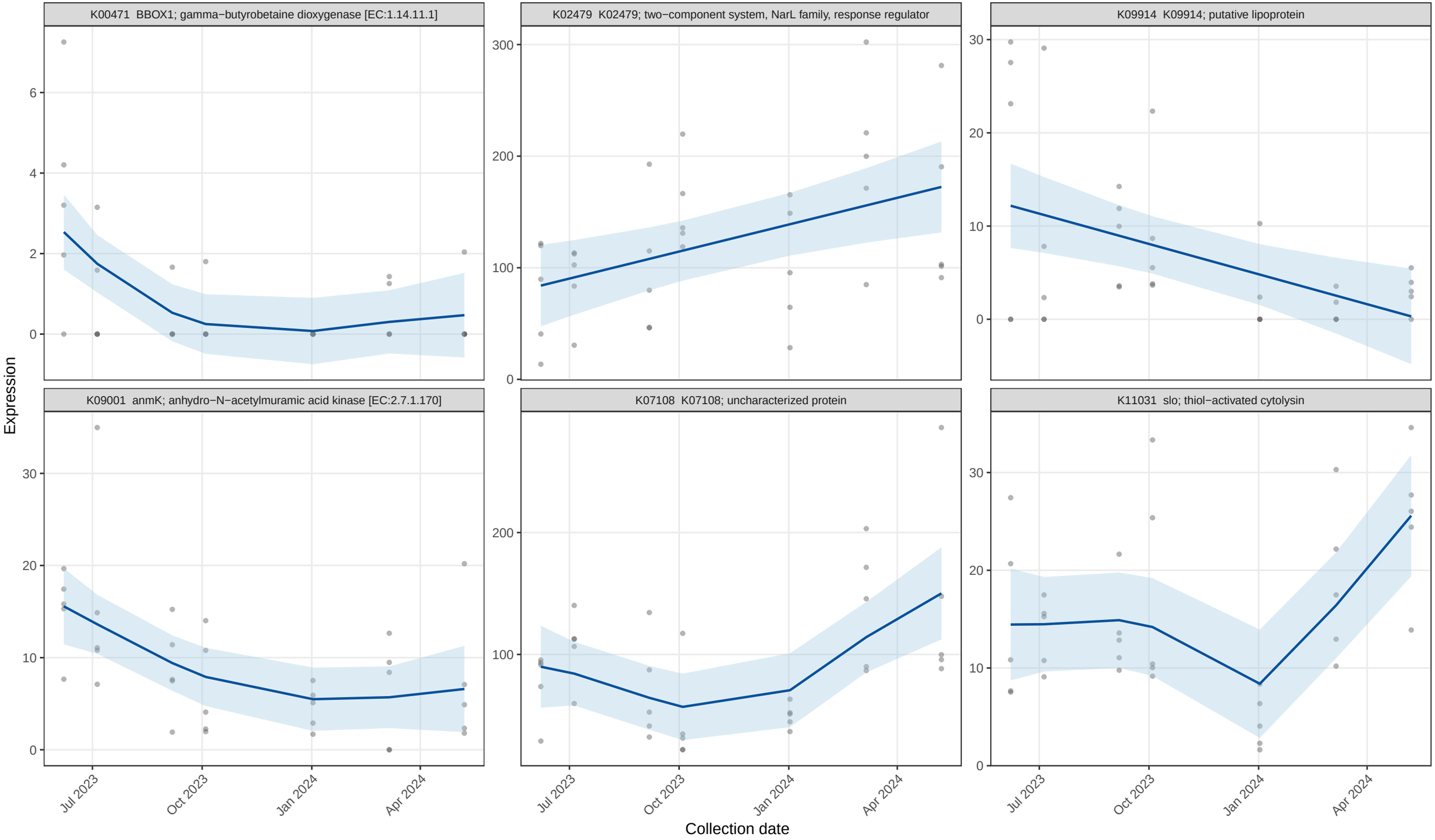

K07396 K07396; putative protein−disulfide isomerase
K02452 gspC; general secretion pathway protein C
K02047 cysW; sulfate/thiosulfate transport system permease protein
K22349 mnxG; manganese oxidase [EC:1.16.3.3]
K21460 sfmM2; L−tyrosine C(3)−methyltransferase [EC:2.1.1.304]
K05548 benK; MFS transporter, AAHS family, benzoate transport protein
Expression
Collection date
Jul 2023
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Apr 2024

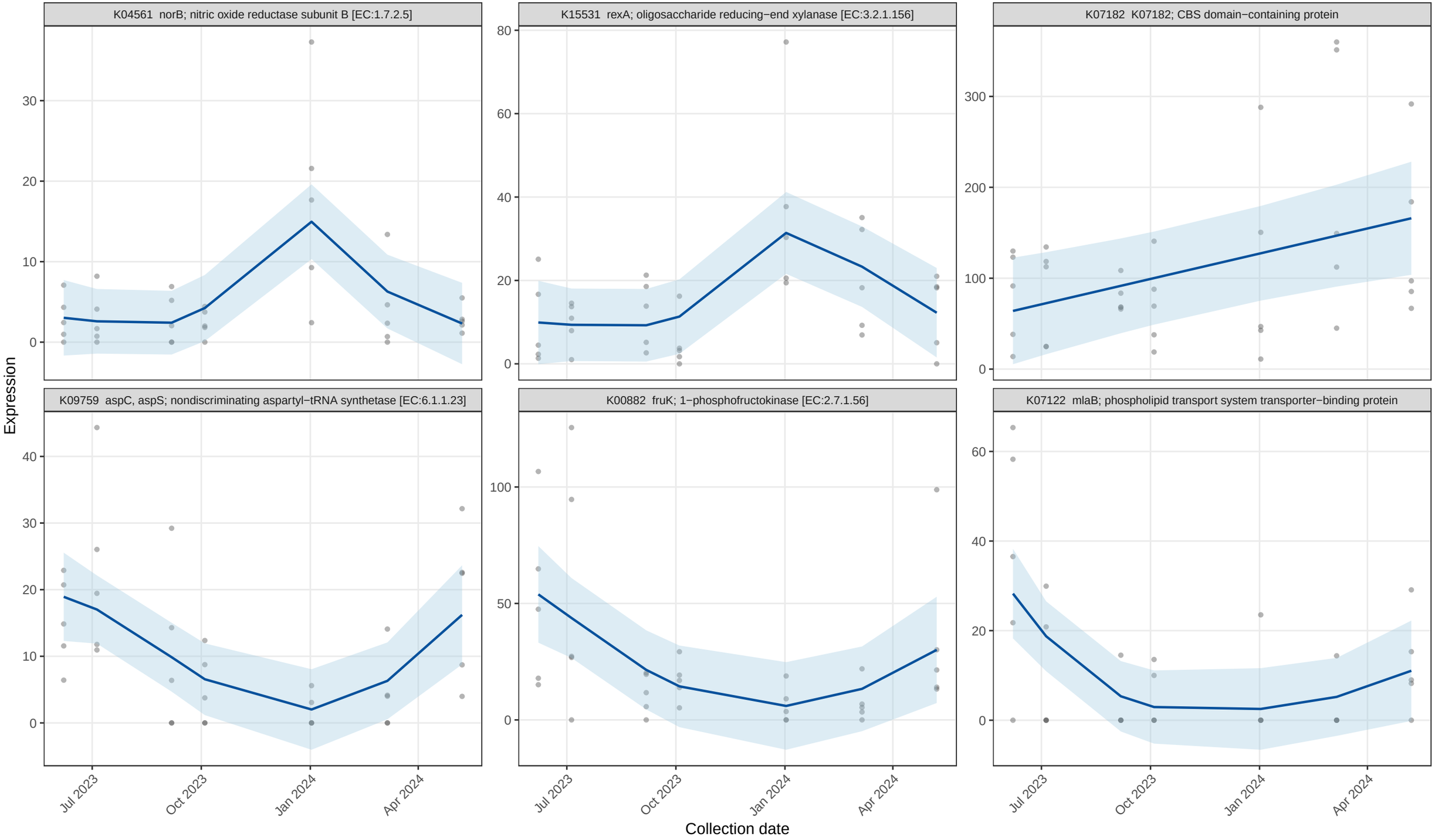

K00856 ADK, adoK; adenosine kinase [EC:2.7.1.20]
K03330 gatE; glutamyl−tRNA(Gln) amidotransferase subunit E [EC:6.3.5.7]
K09461 E1.14.13.40; anthraniloyl−CoA monooxygenase [EC:1.14.13.40]
K10559 rhaS; rhamnose transport system substrate−binding protein
K01141 sbcB, exoI; exodeoxyribonuclease I [EC:3.1.11.1]
K17623 PUDP, HDHD1; pseudouridine 5'−phosphatase [EC:3.1.3.96]
Expression
Collection date
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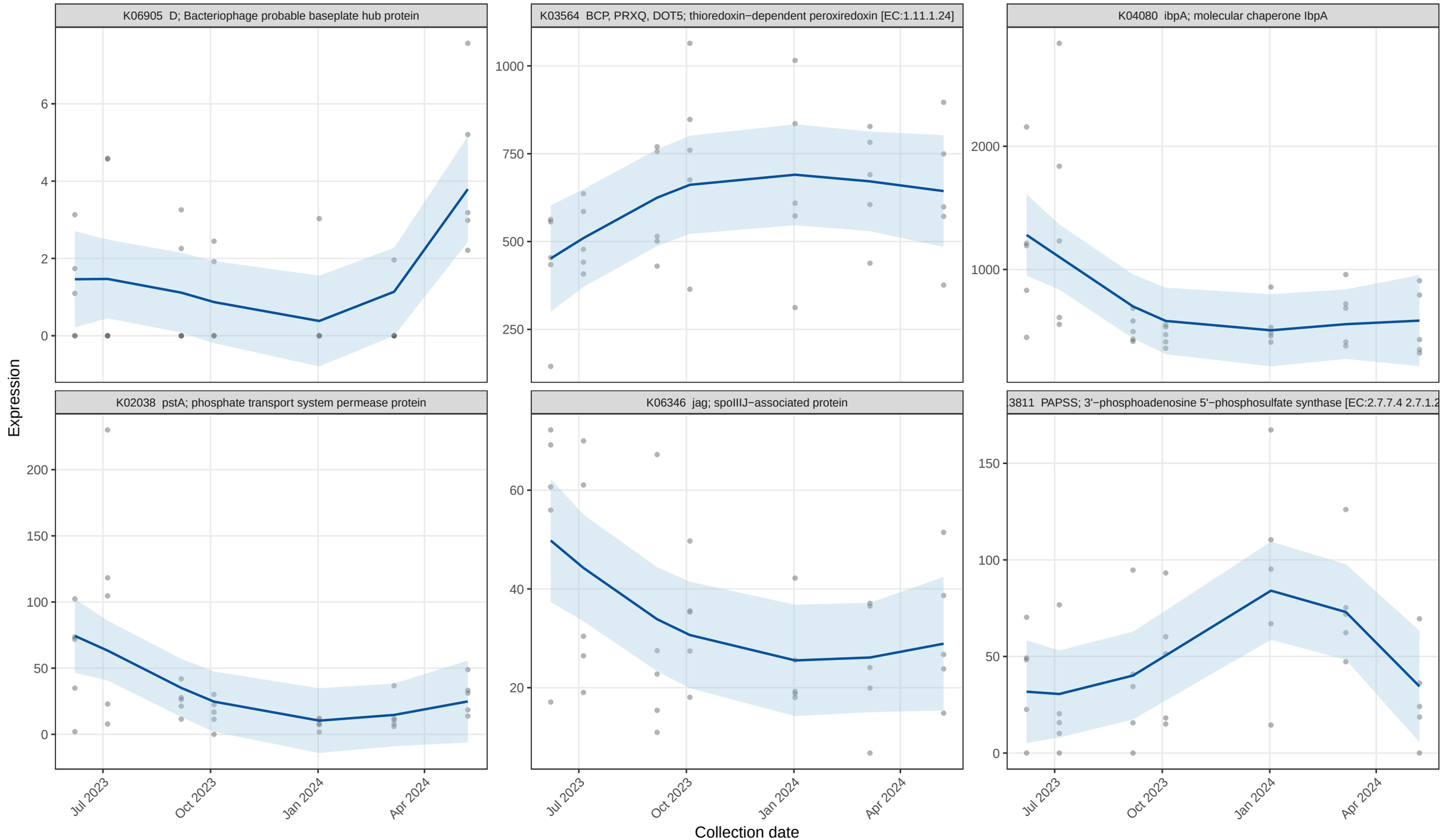

K06180 rluD; 23S rRNA pseudouridine1911/1915/1917 synthase [EC:5.4.99.23]
K07114 yfbK; Ca−activated chloride channel homolog
K01661 menB; naphthoate synthase [EC:4.1.3.36]
K14028 mdh1, mxaF; methanol dehydrogenase (cytochrome c) subunit 1 [EC:1.1.2.7]
K08975 K08975; putative membrane protein
K09794 K09794; uncharacterized protein
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024


K02030 ABC.PA.S; polar amino acid transport system substrate−binding protein
K04719 bluB; 5,6−dimethylbenzimidazole synthase [EC:1.13.11.79]
K18534 K18534; MPBQ/MSBQ methyltransferase [EC:2.1.1.295]
K02834 rbfA; ribosome−binding factor A
K03195 virB10, lvhB10; type IV secretion system protein VirB10
K01610 pckA; phosphoenolpyruvate carboxykinase (ATP) [EC:4.1.1.49]
Expression
Collection date
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Apr 2024

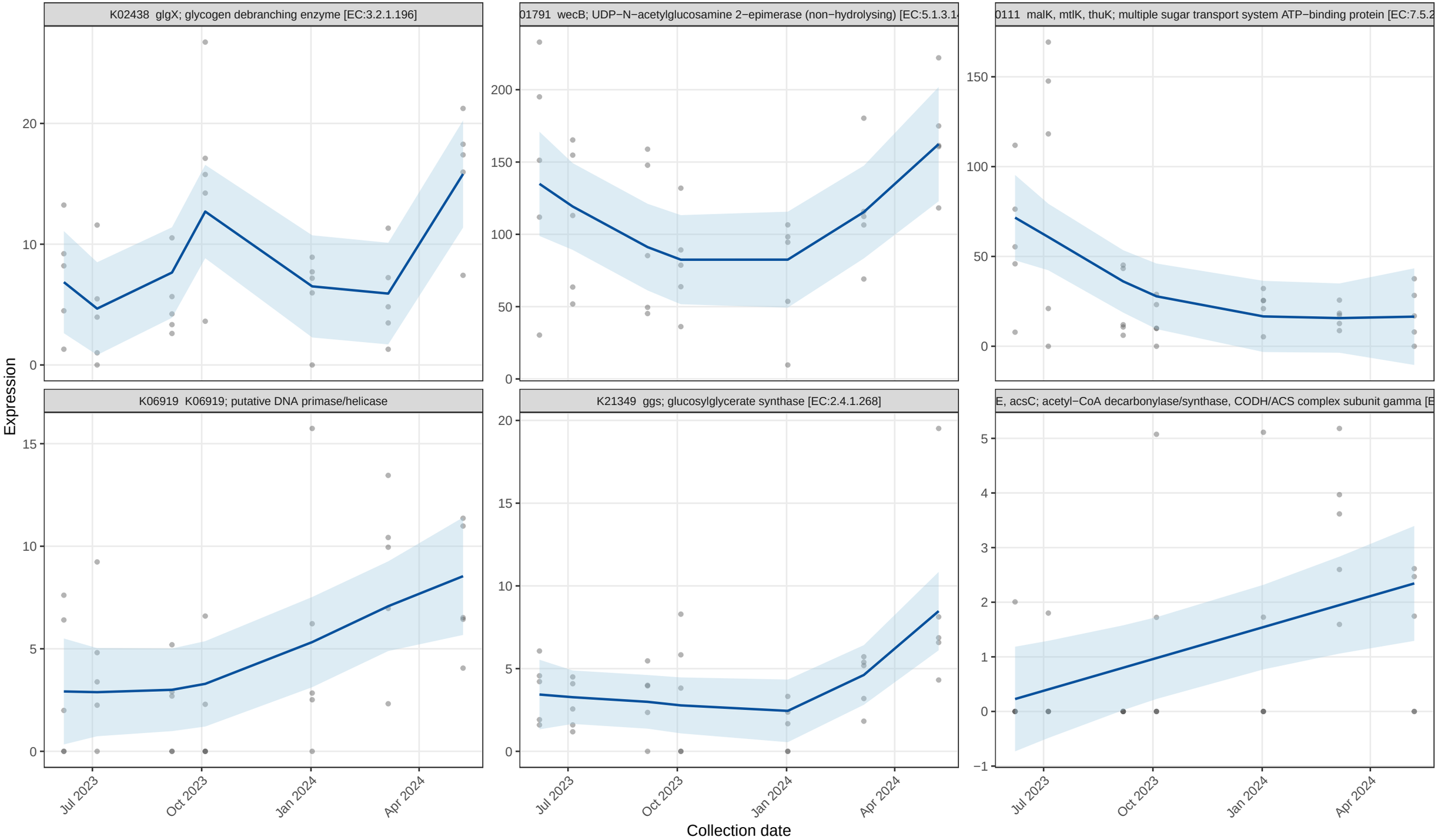

K00314 SARDH; sarcosine dehydrogenase [EC:1.5.8.3]
K08303 prtC, trhP; U32 family peptidase [EC:3.4.−.−]
K22209 tarD; D(−)−tartrate dehydratase [EC:4.2.1.81]
K00457 HPD, hppD; 4−hydroxyphenylpyruvate dioxygenase [EC:1.13.11.27]
K00600 glyA, SHMT; glycine hydroxymethyltransferase [EC:2.1.2.1]
K08305 mltB; peptidoglycan lytic transglycosylase B [EC:4.2.2.29]
Expression
Collection date
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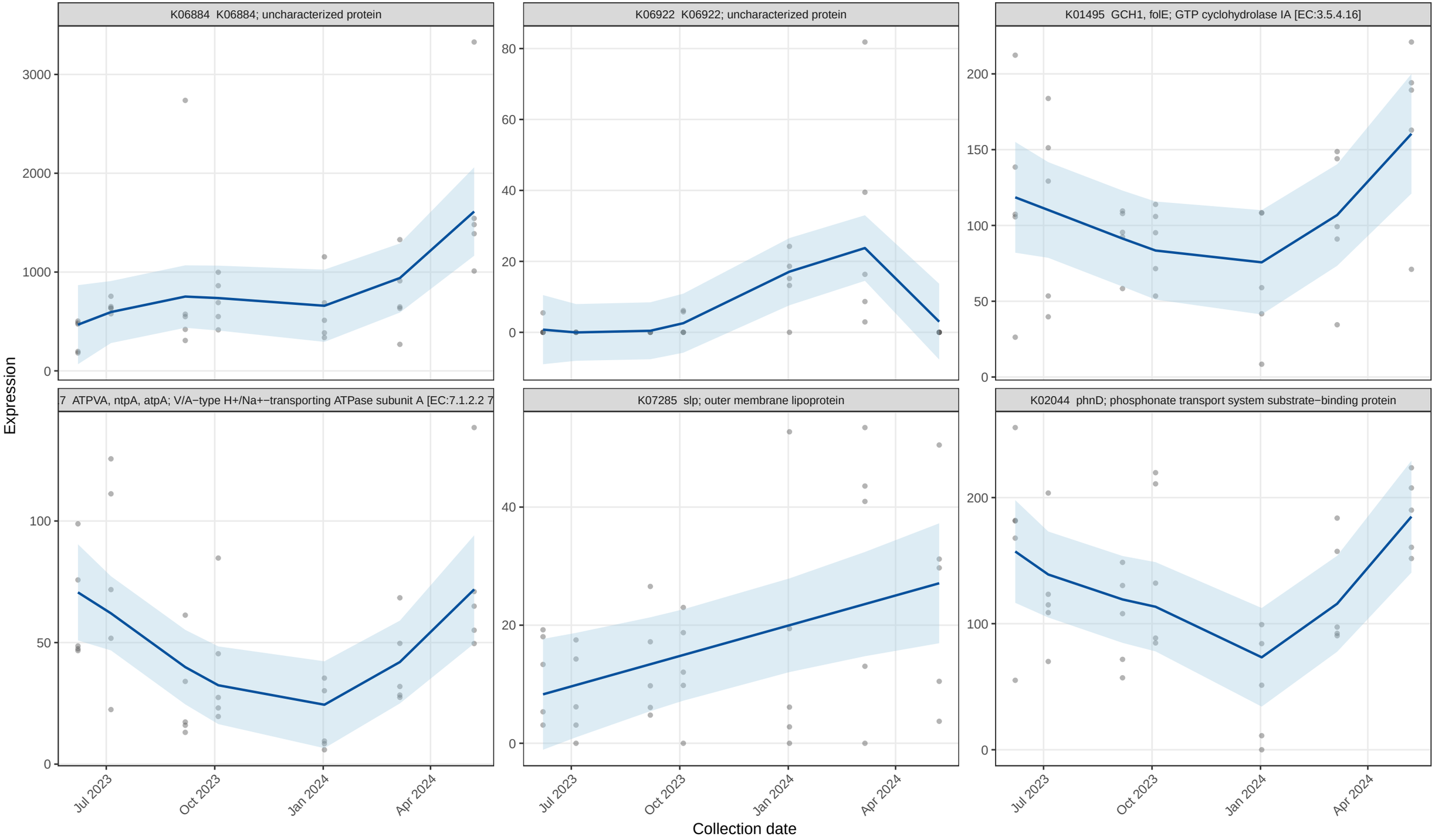

K00471 BBOX1; gamma−butyrobetaine dioxygenase [EC:1.14.11.1]
K02479 K02479; two−component system, NarL family, response regulator
K09914 K09914; putative lipoprotein
K09001 anmK; anhydro−N−acetylmuramic acid kinase [EC:2.7.1.170]
K07108 K07108; uncharacterized protein
K11031 slo; thiol−activated cytolysin
Expression
Collection date
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Apr 2024

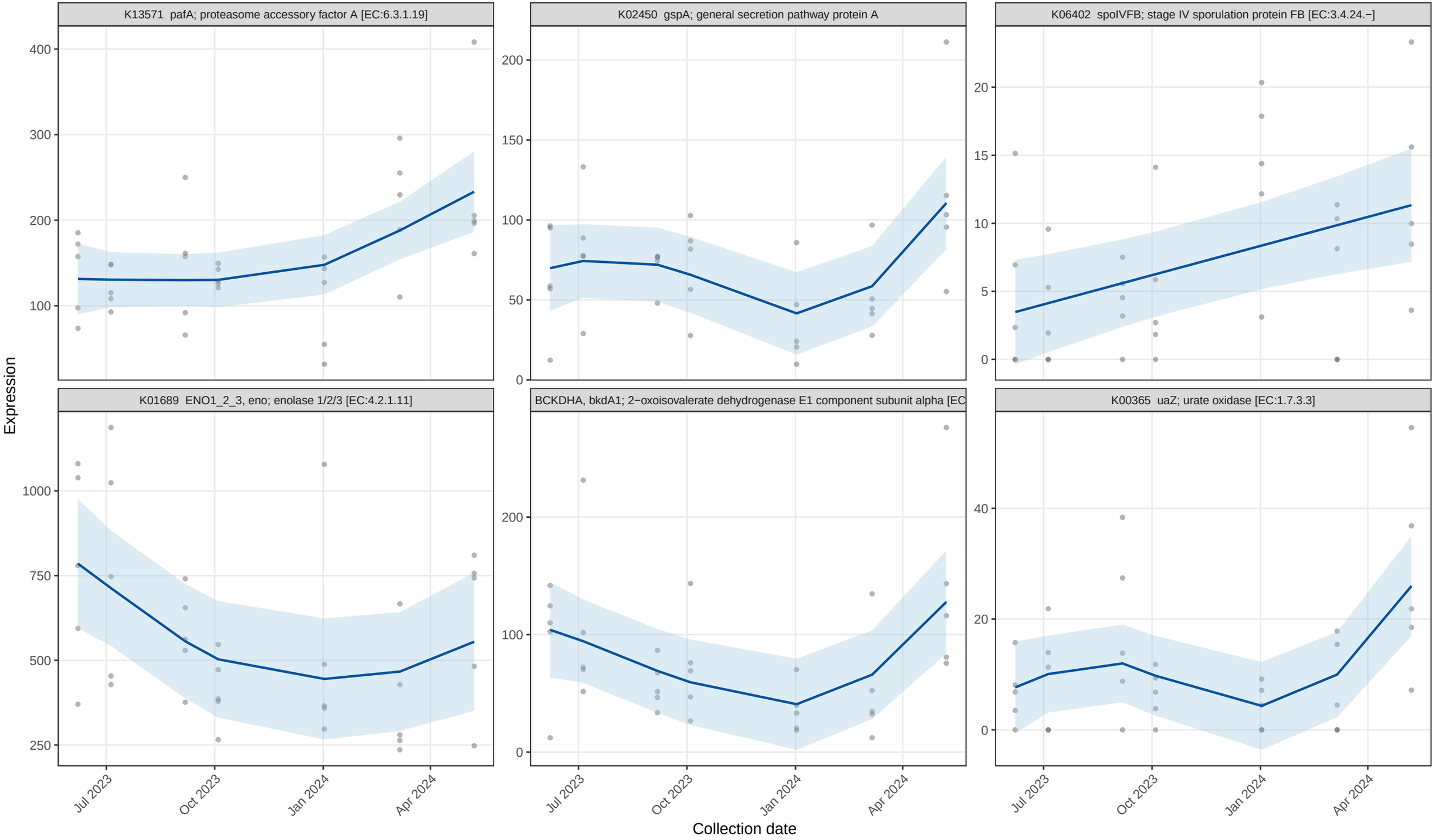

K04561 norB; nitric oxide reductase subunit B [EC:1.7.2.5]
K15531 rexA; oligosaccharide reducing−end xylanase [EC:3.2.1.156]
K07182 K07182; CBS domain−containing protein
K09759 aspC, aspS; nondiscriminating aspartyl−tRNA synthetase [EC:6.1.1.23]
K00882 fruK; 1−phosphofructokinase [EC:2.7.1.56]
K07122 mlaB; phospholipid transport system transporter−binding protein
Expression
Collection date
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Apr 2024

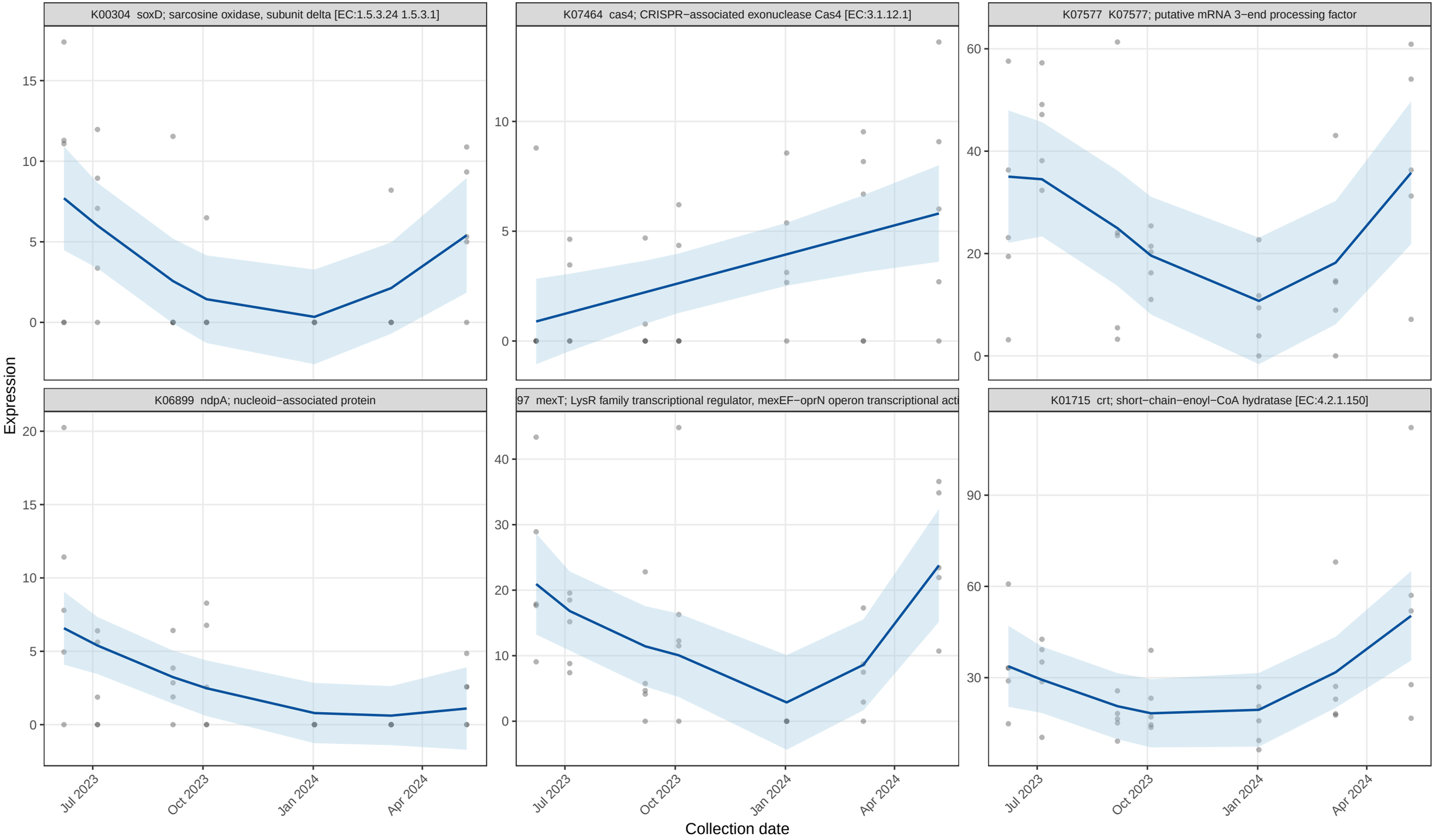

K06905 D; Bacteriophage probable baseplate hub protein
K03564 BCP, PRXQ, DOT5; thioredoxin−dependent peroxiredoxin [EC:1.11.1.24]
K04080 ibpA; molecular chaperone IbpA
K02038 pstA; phosphate transport system permease protein
K06346 jag; spoIIIJ−associated protein
3811 PAPSS; 3'−phosphoadenosine 5'−phosphosulfate synthase [EC:2.7.7.4 2.7.1.2
Expression
Collection date
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Apr 2024

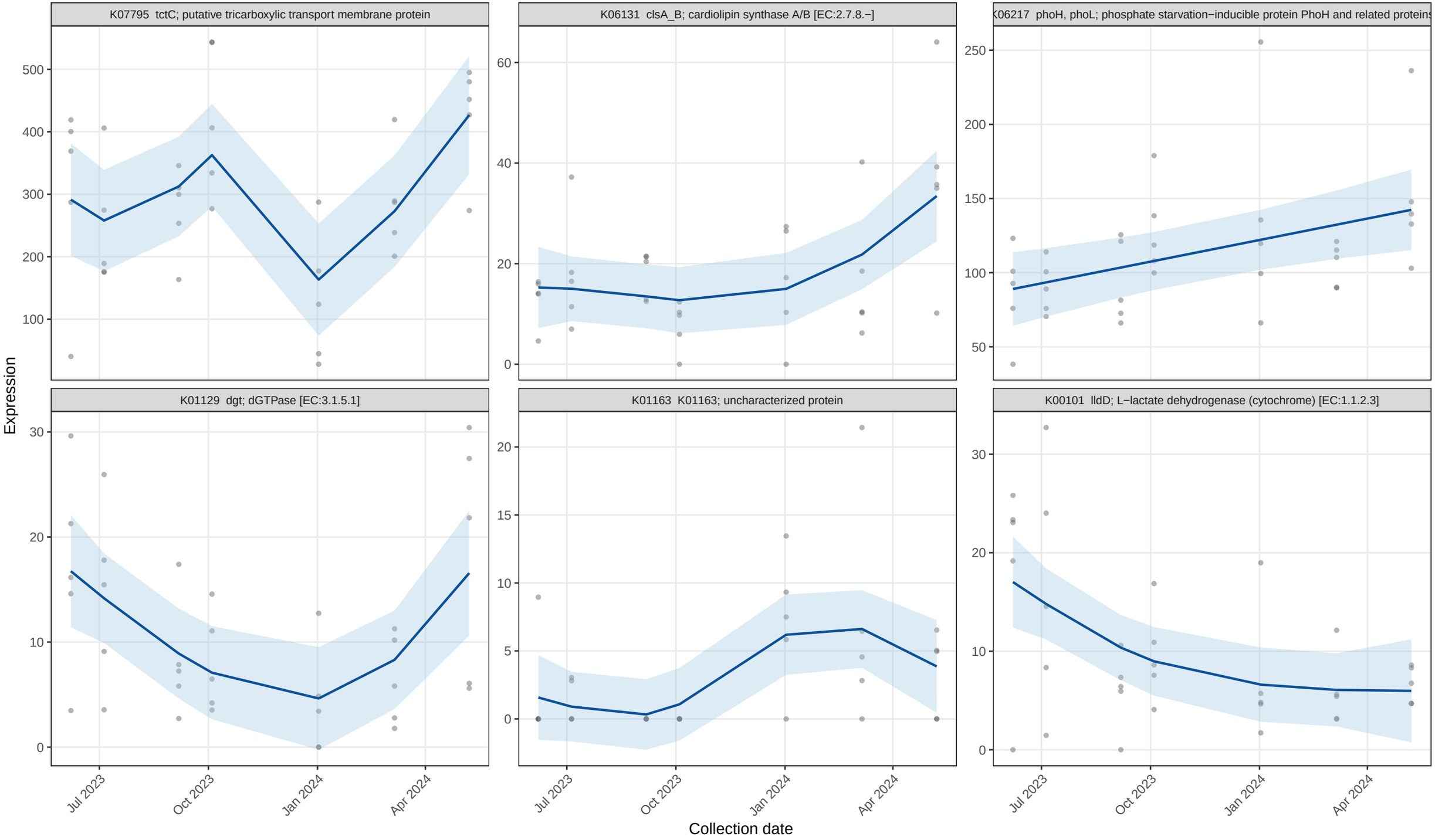
brite_D: BRITE classes with significant temporal GAM smooths, page 154 of 181

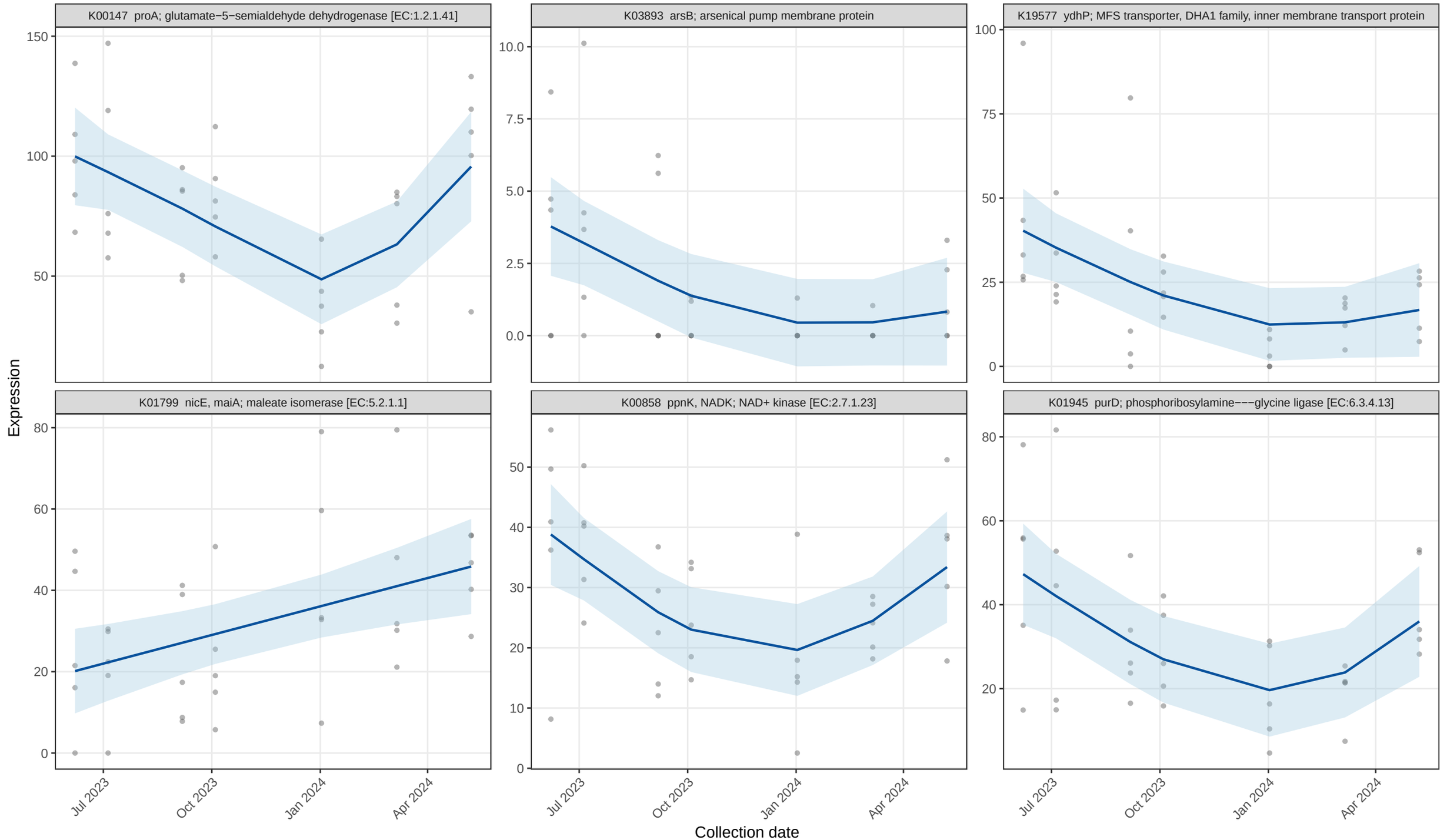


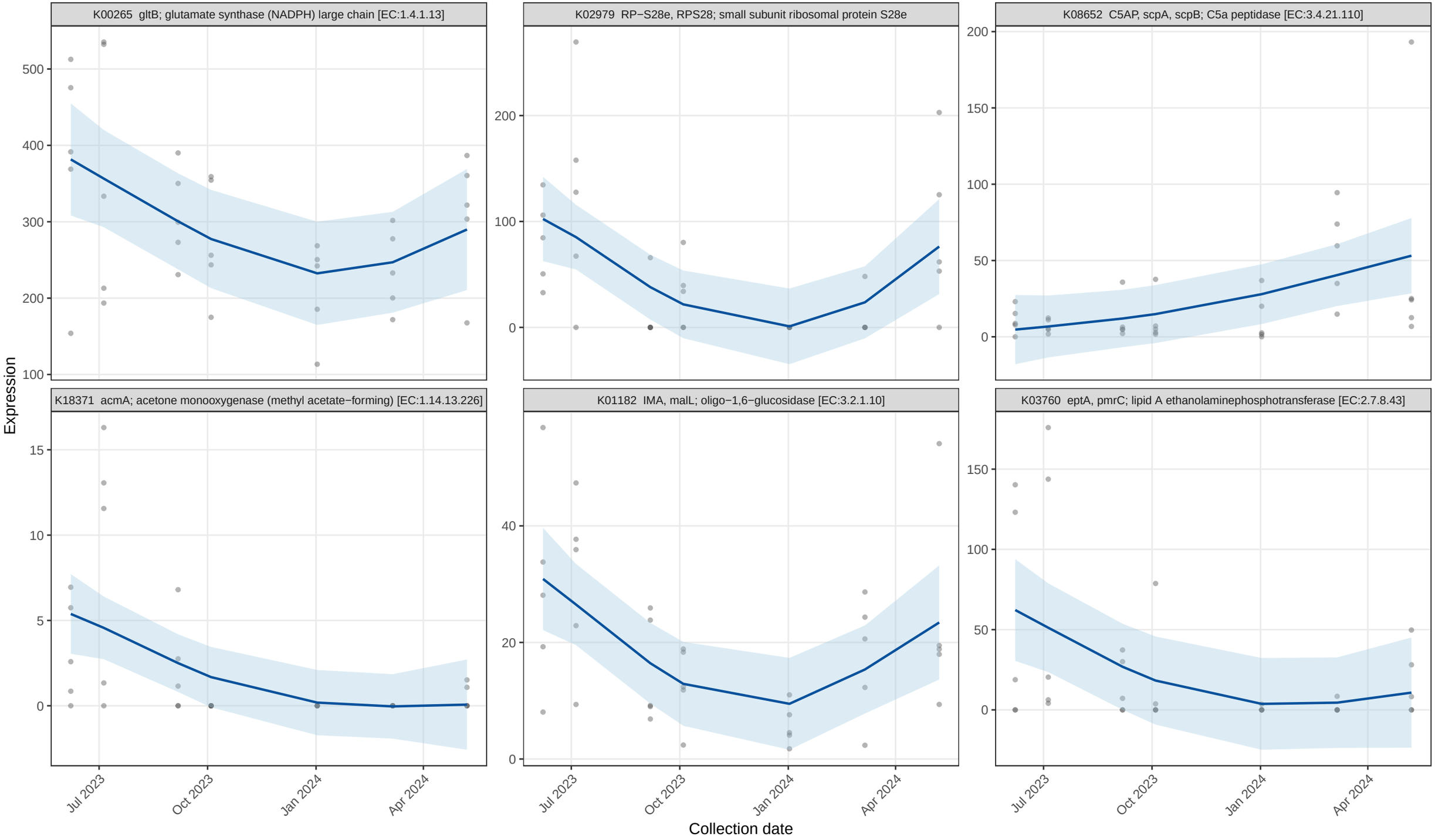

K06884 K06884; uncharacterized protein
K06922 K06922; uncharacterized protein
K01495 GCH1, folE; GTP cyclohydrolase IA [EC:3.5.4.16]
7 ATPVA, ntpA, atpA; V/A−type H+/Na+−transporting ATPase subunit A [EC:7.1.2.2 7
K07285 slp; outer membrane lipoprotein
K02044 phnD; phosphonate transport system substrate−binding protein
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

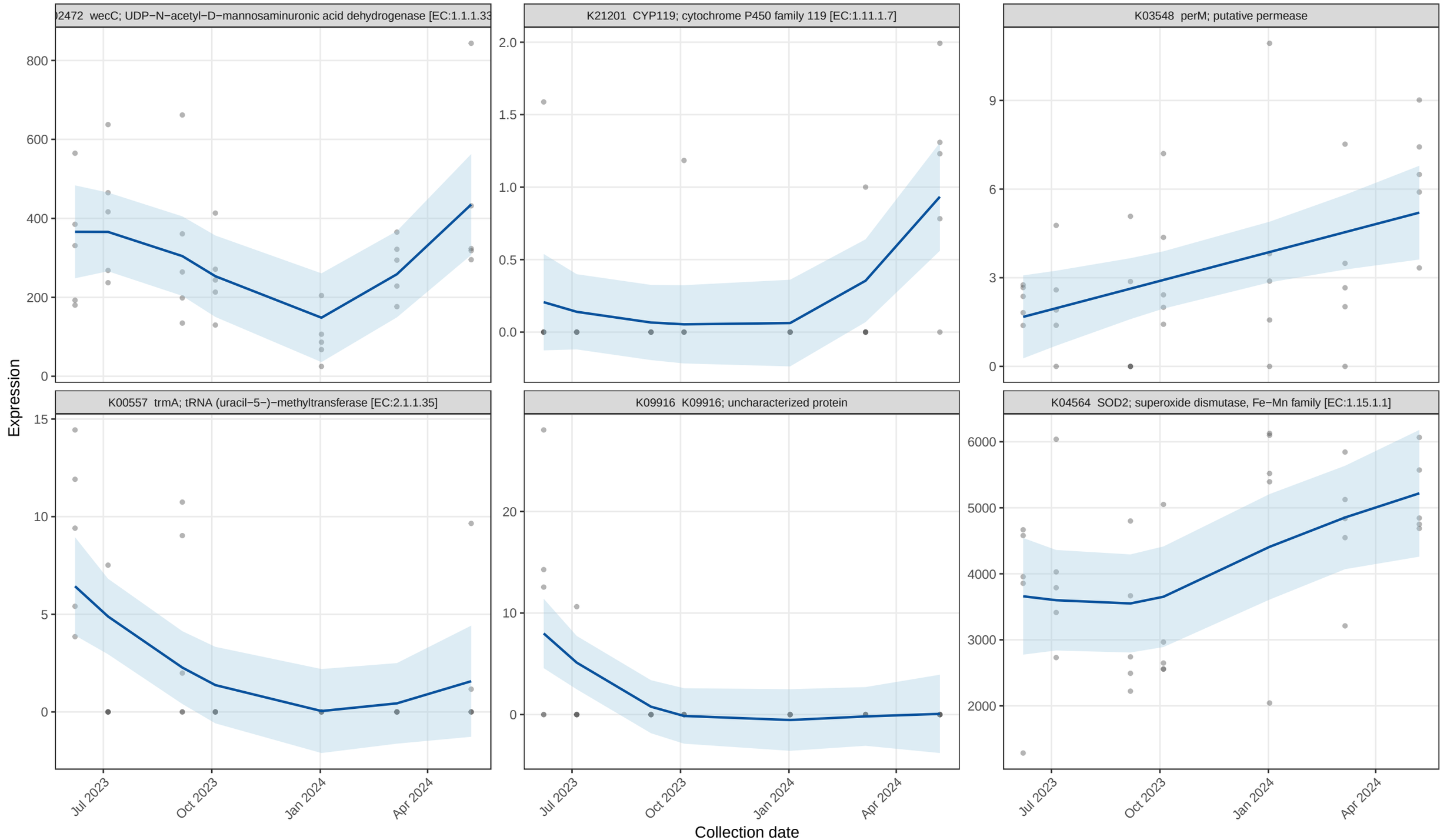

K13571 pafA; proteasome accessory factor A [EC:6.3.1.19]
K02450 gspA; general secretion pathway protein A
K06402 spoIVFB; stage IV sporulation protein FB [EC:3.4.24.−]
K01689 ENO1_2_3, eno; enolase 1/2/3 [EC:4.2.1.11]
BCKDHA, bkdA1; 2−oxoisovalerate dehydrogenase E1 component subunit alpha [EC
K00365 uaZ; urate oxidase [EC:1.7.3.3]
Expression
Collection date
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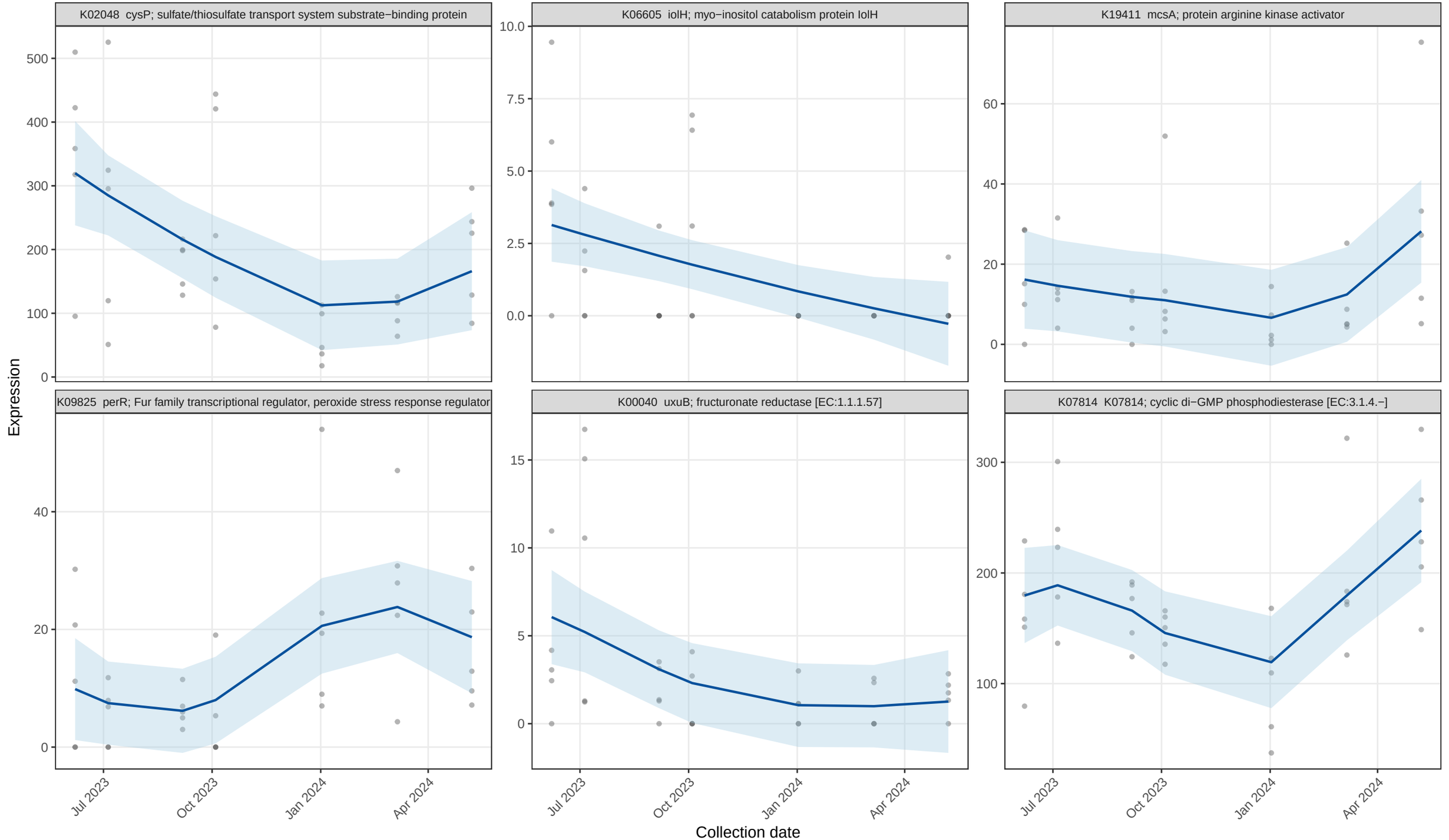

K00304 soxD; sarcosine oxidase, subunit delta [EC:1.5.3.24 1.5.3.1]
K07464 cas4; CRISPR−associated exonuclease Cas4 [EC:3.1.12.1]
K07577 K07577; putative mRNA 3−end processing factor
K06899 ndpA; nucleoid−associated protein
97 mexT; LysR family transcriptional regulator, mexEF−oprN operon transcriptional acti
K01715 crt; short−chain−enoyl−CoA hydratase [EC:4.2.1.150]
Expression
Collection date
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Apr 2024

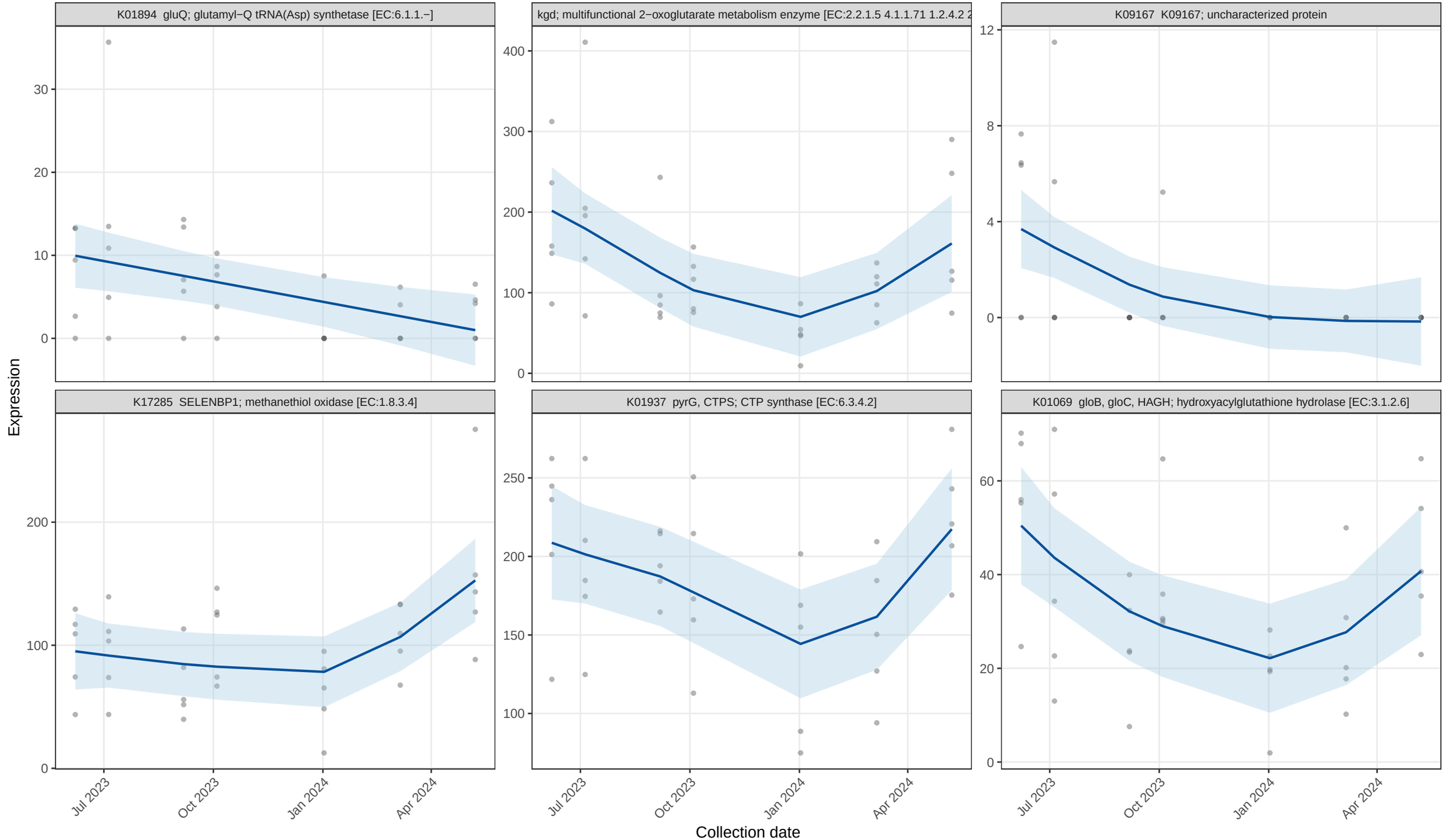

K07795 tctC; putative tricarboxylic transport membrane protein
K06131 clsA_B; cardiolipin synthase A/B [EC:2.7.8.−]
K06217 phoH, phoL; phosphate starvation−inducible protein PhoH and related proteins
K01129 dgt; dGTPase [EC:3.1.5.1]
K01163 K01163; uncharacterized protein
K00101 lldD; L−lactate dehydrogenase (cytochrome) [EC:1.1.2.3]
Expression
Collection date
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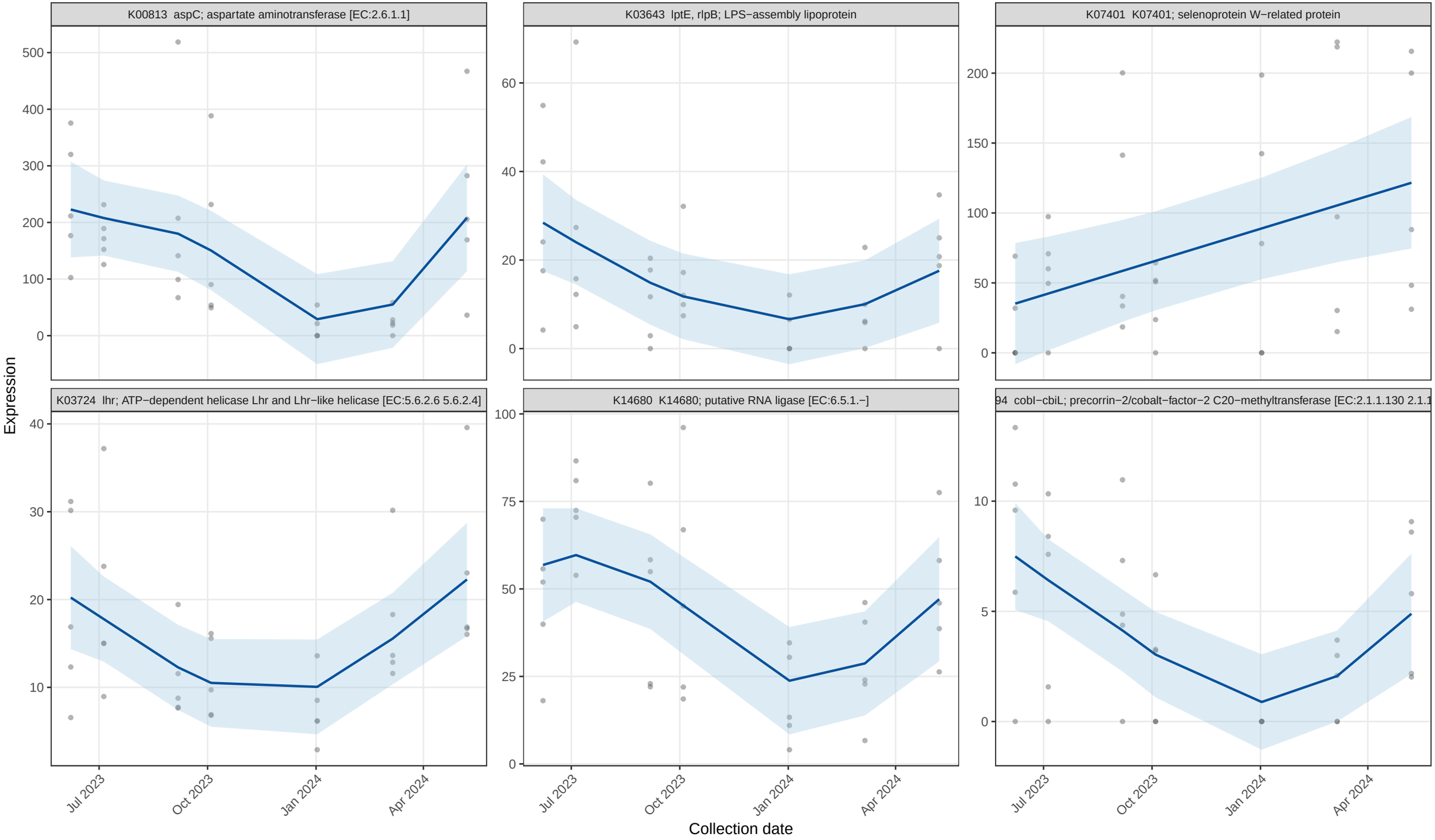

K00147 proA; glutamate−5−semialdehyde dehydrogenase [EC:1.2.1.41]
K03893 arsB; arsenical pump membrane protein
K19577 ydhP; MFS transporter, DHA1 family, inner membrane transport protein
K01799 nicE, maiA; maleate isomerase [EC:5.2.1.1]
K00858 ppnK, NADK; NAD+ kinase [EC:2.7.1.23]
K01945 purD; phosphoribosylamine−−−glycine ligase [EC:6.3.4.13]
Expression
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K00265 gltB; glutamate synthase (NADPH) large chain [EC:1.4.1.13]
K02979 RP−S28e, RPS28; small subunit ribosomal protein S28e
K08652 C5AP, scpA, scpB; C5a peptidase [EC:3.4.21.110]
K18371 acmA; acetone monooxygenase (methyl acetate−forming) [EC:1.14.13.226]
K01182 IMA, malL; oligo−1,6−glucosidase [EC:3.2.1.10]
K03760 eptA, pmrC; lipid A ethanolaminephosphotransferase [EC:2.7.8.43]
Expression
Collection date
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Apr 2024

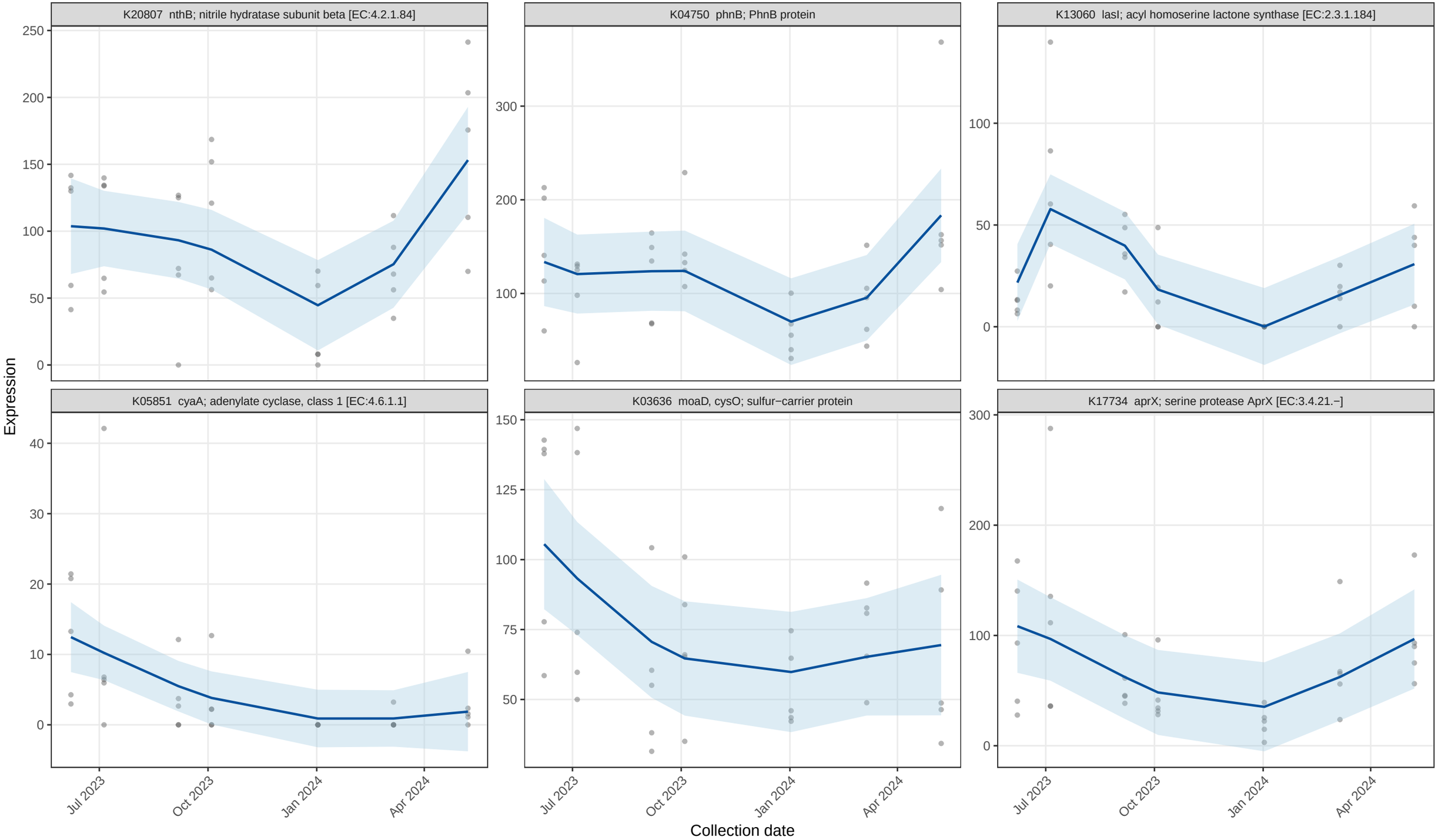

02472 wecC; UDP−N−acetyl−D−mannosaminuronic acid dehydrogenase [EC:1.1.1.33
K21201 CYP119; cytochrome P450 family 119 [EC:1.11.1.7]
K03548 perM; putative permease
K00557 trmA; tRNA (uracil−5−)−methyltransferase [EC:2.1.1.35]
K09916 K09916; uncharacterized protein
K04564 SOD2; superoxide dismutase, Fe−Mn family [EC:1.15.1.1]
Expression
Collection date
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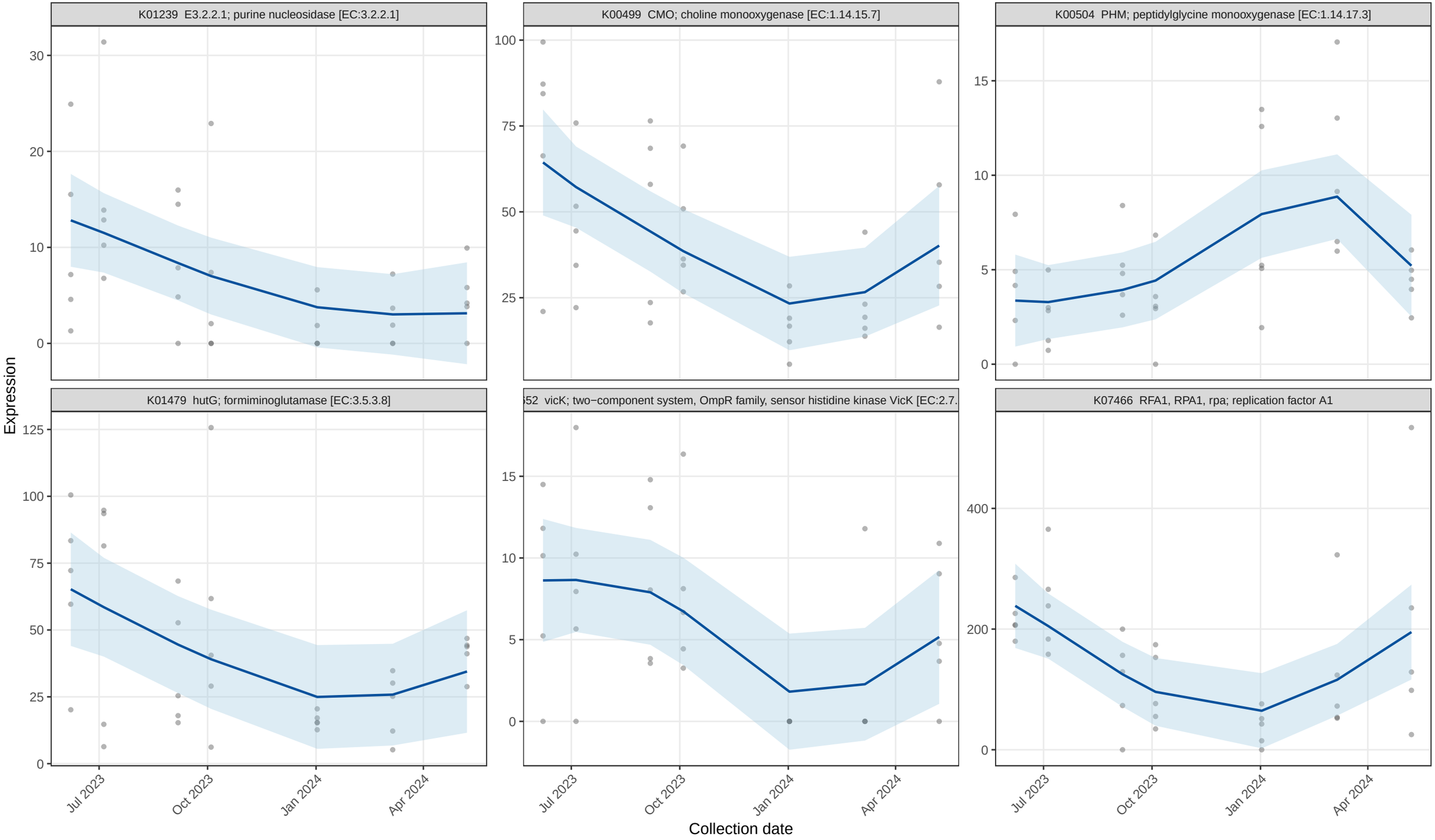

K02048 cysP; sulfate/thiosulfate transport system substrate−binding protein
K06605 iolH; myo−inositol catabolism protein IolH
K19411 mcsA; protein arginine kinase activator
K09825 perR; Fur family transcriptional regulator, peroxide stress response regulator
K00040 uxuB; fructuronate reductase [EC:1.1.1.57]
K07814 K07814; cyclic di−GMP phosphodiesterase [EC:3.1.4.−]
Expression
Collection date
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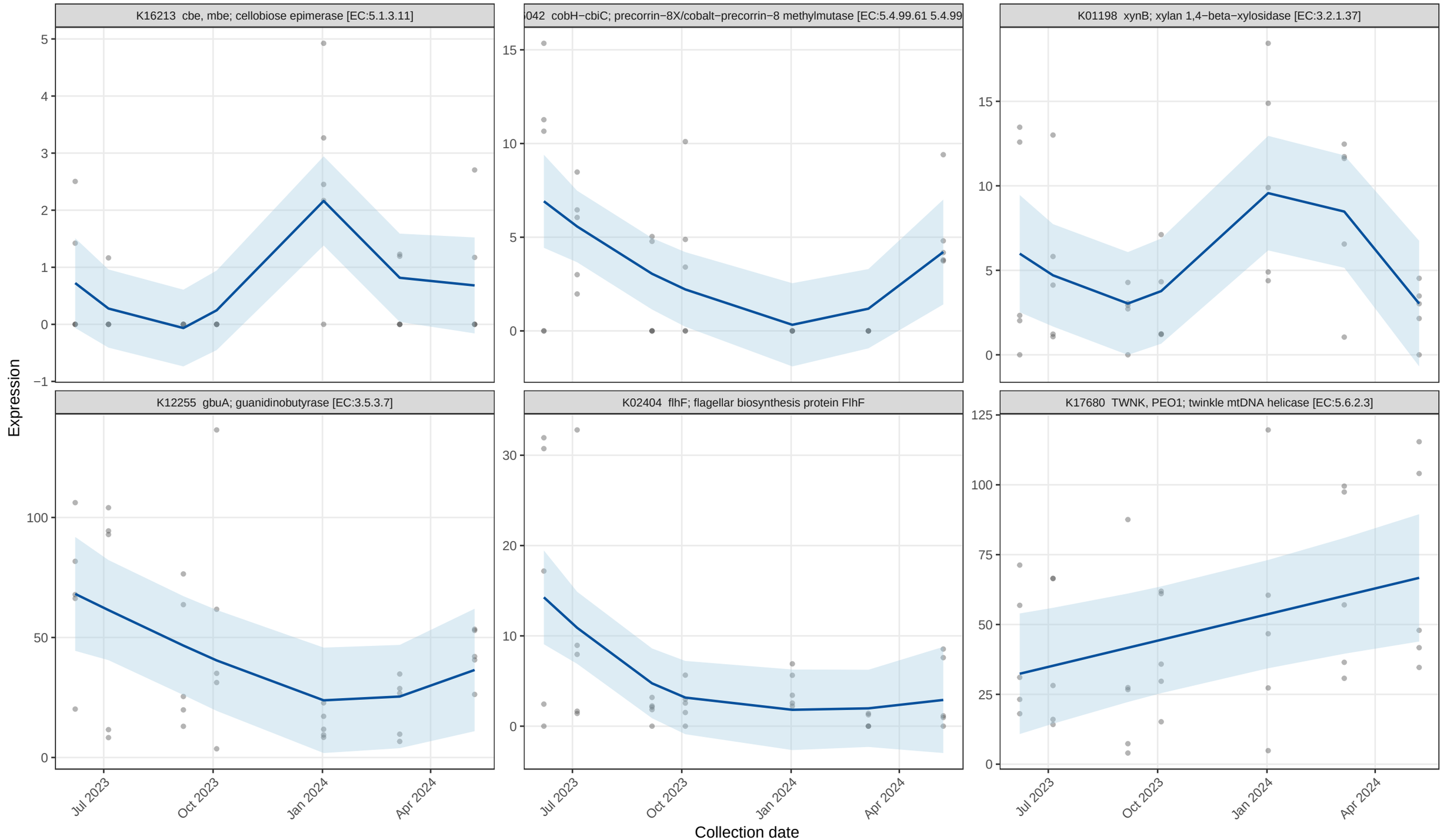

K01894 gluQ; glutamyl−Q tRNA(Asp) synthetase [EC:6.1.1.−]
kgd; multifunctional 2−oxoglutarate metabolism enzyme [EC:2.2.1.5 4.1.1.71 1.2.4.2 2
K09167 K09167; uncharacterized protein
K17285 SELENBP1; methanethiol oxidase [EC:1.8.3.4]
K01937 pyrG, CTPS; CTP synthase [EC:6.3.4.2]
K01069 gloB, gloC, HAGH; hydroxyacylglutathione hydrolase [EC:3.1.2.6]
Expression
Collection date
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Apr 2024

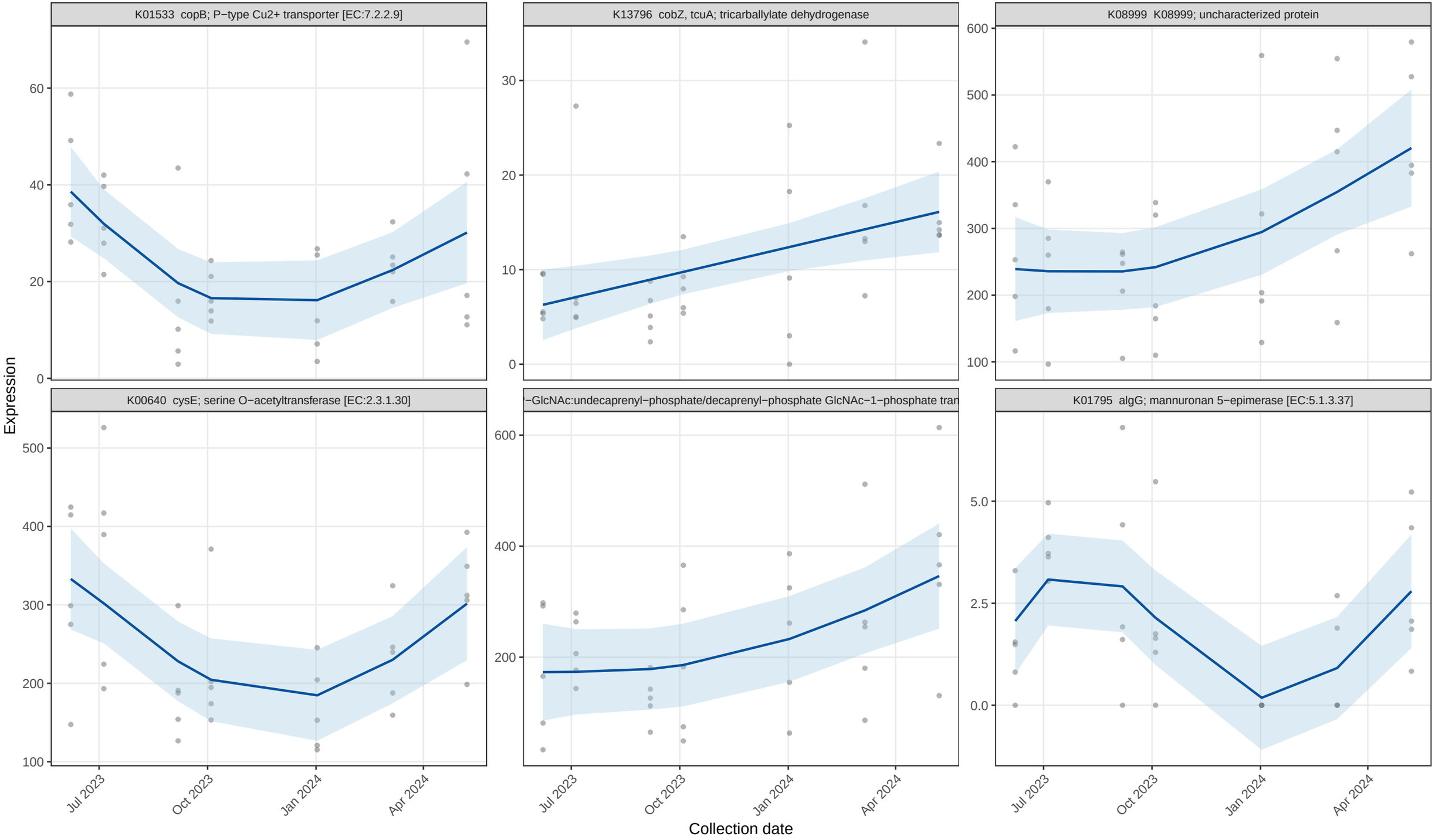

K00813 aspC; aspartate aminotransferase [EC:2.6.1.1]
K03643 lptE, rlpB; LPS−assembly lipoprotein
K07401 K07401; selenoprotein W−related protein
K03724 lhr; ATP−dependent helicase Lhr and Lhr−like helicase [EC:5.6.2.6 5.6.2.4]
K14680 K14680; putative RNA ligase [EC:6.5.1.−]
94 cobI−cbiL; precorrin−2/cobalt−factor−2 C20−methyltransferase [EC:2.1.1.130 2.1.1
Expression
Collection date
Jul 2023
Oct 2023
Jan 2024
Apr 2024

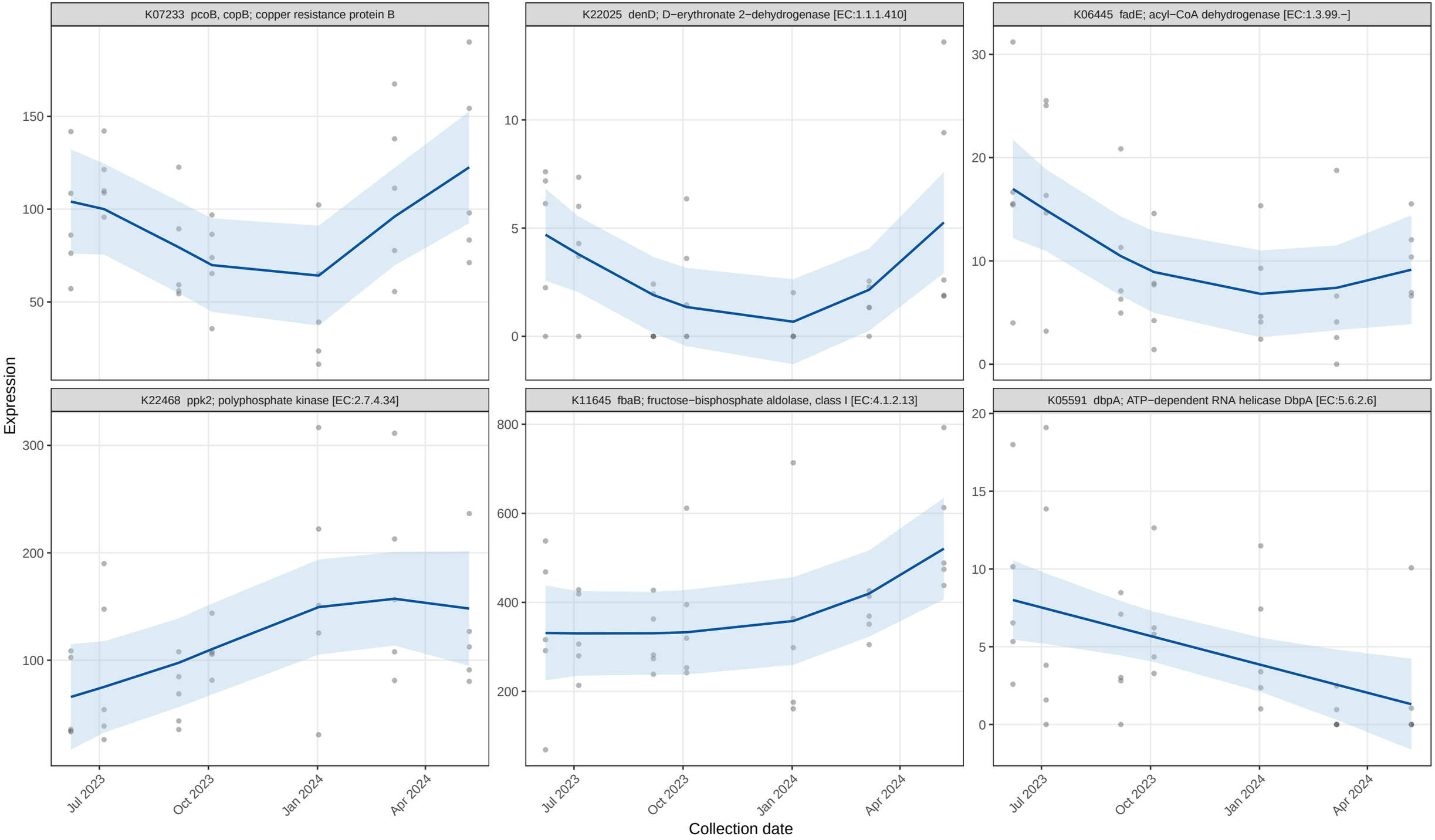
brite_D: BRITE classes with significant temporal GAM smooths, page 165 of 181

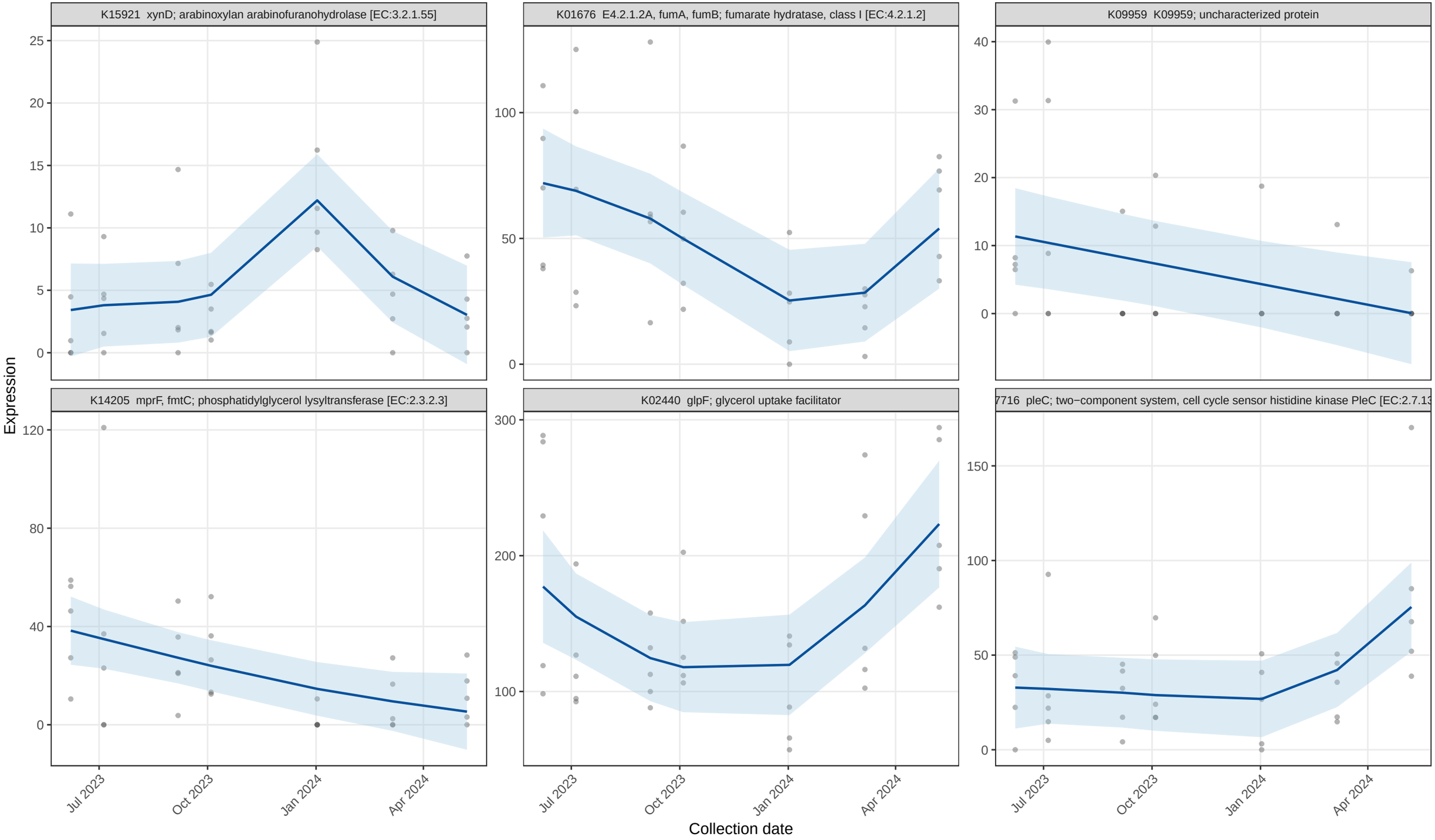


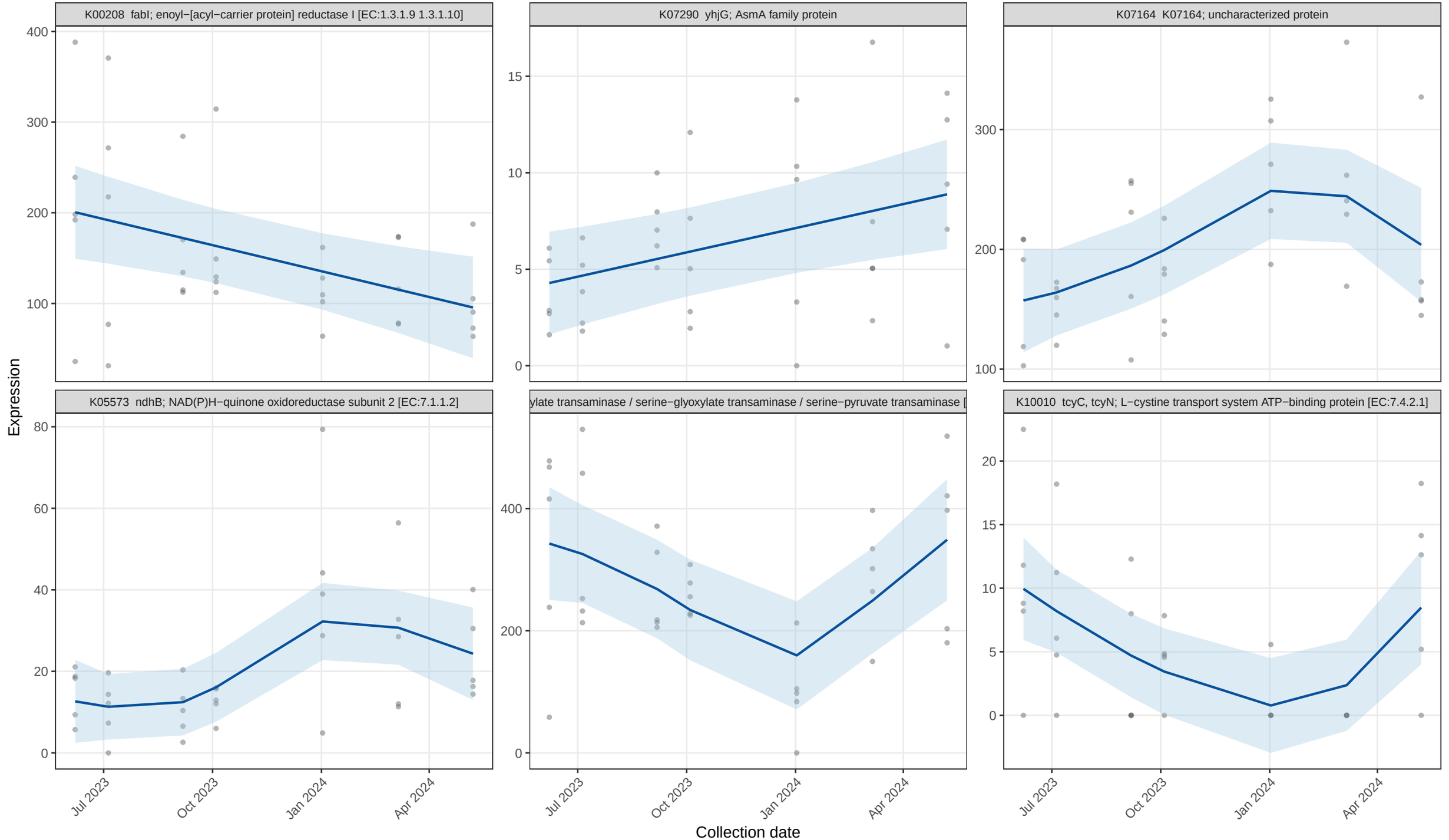

K01239 E3.2.2.1; purine nucleosidase [EC:3.2.2.1]
K00499 CMO; choline monooxygenase [EC:1.14.15.7]
K00504 PHM; peptidylglycine monooxygenase [EC:1.14.17.3]
K01479 hutG; formiminoglutamase [EC:3.5.3.8]
52 vicK; two−component system, OmpR family, sensor histidine kinase VicK [EC:2.7.
K07466 RFA1, RPA1, rpa; replication factor A1
Expression
Collection date
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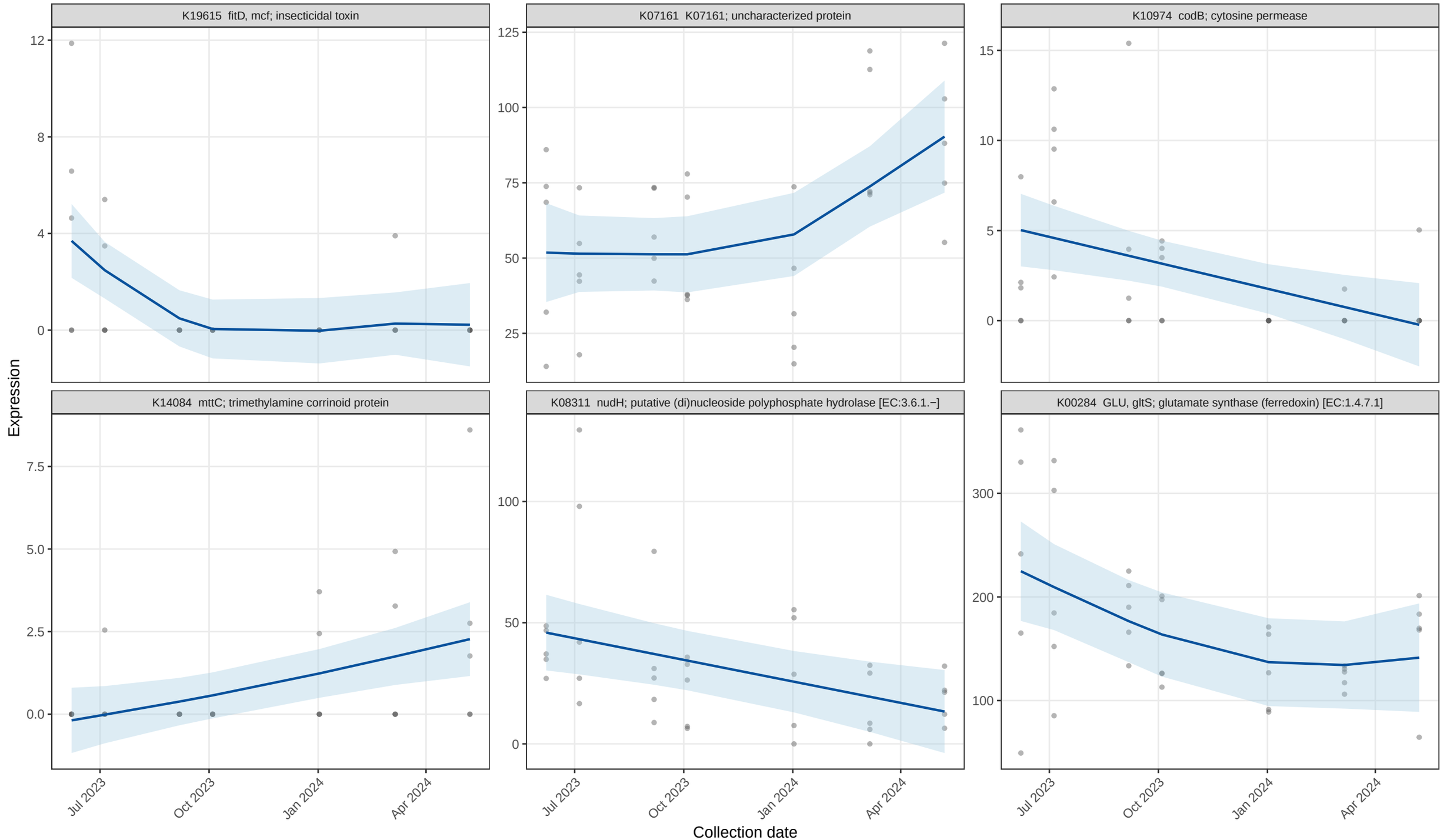

K16213 cbe, mbe; cellobiose epimerase [EC:5.1.3.11]
042 cobH−cbiC; precorrin−8X/cobalt−precorrin−8 methylmutase [EC:5.4.99.61 5.4.99
K01198 xynB; xylan 1,4−beta−xylosidase [EC:3.2.1.37]
K12255 gbuA; guanidinobutyrase [EC:3.5.3.7]
K02404 flhF; flagellar biosynthesis protein FlhF
K17680 TWNK, PEO1; twinkle mtDNA helicase [EC:5.6.2.3]
Expression
Collection date
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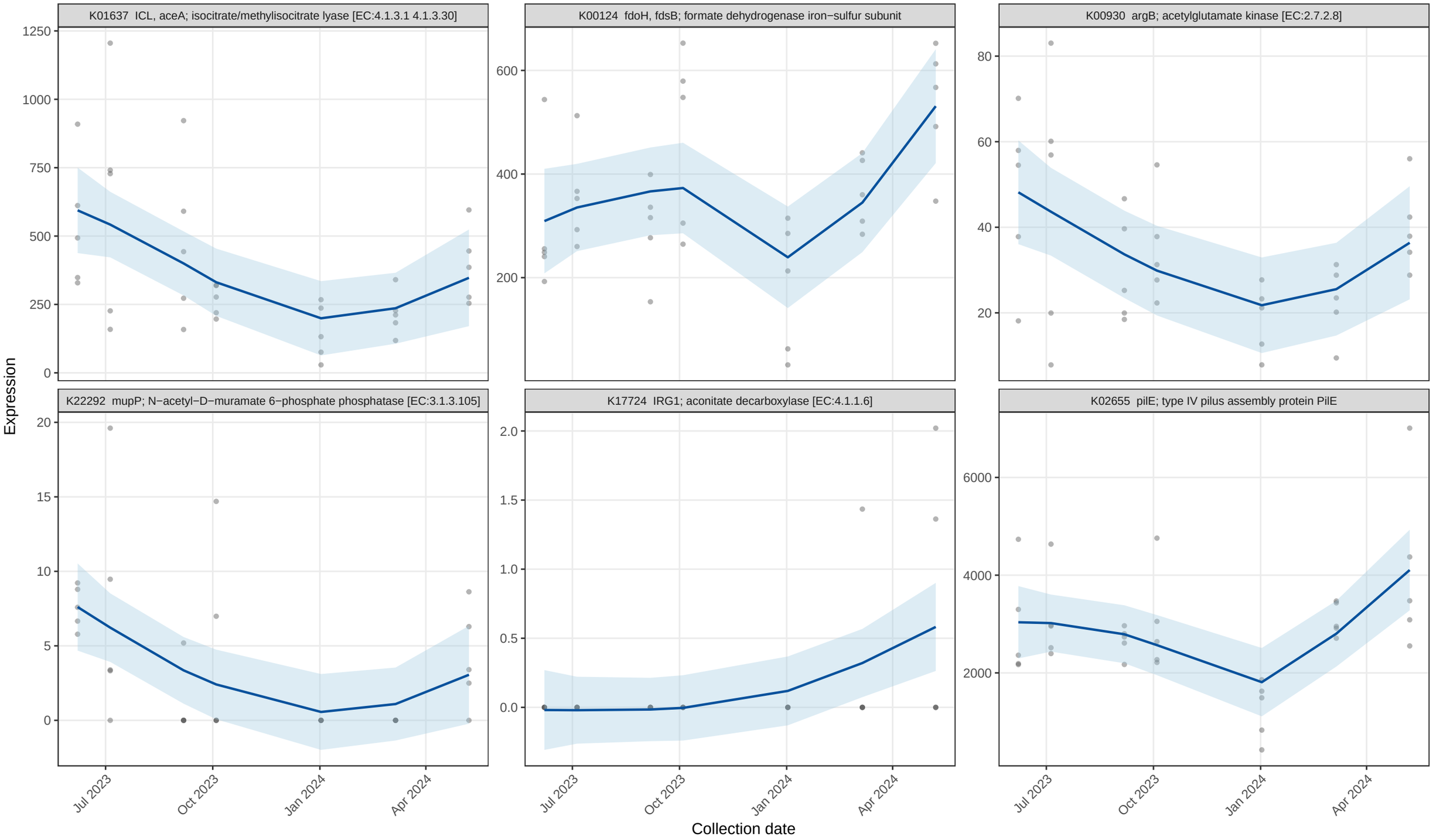

K01533 copB; P−type Cu2+ transporter [EC:7.2.2.9]
K13796 cobZ, tcuA; tricarballylate dehydrogenase
K08999 K08999; uncharacterized protein
K00640 cysE; serine O−acetyltransferase [EC:2.3.1.30]
−GlcNAc:undecaprenyl−phosphate/decaprenyl−phosphate GlcNAc−1−phosphate tran
K01795 algG; mannuronan 5−epimerase [EC:5.1.3.37]
Expression
Collection date
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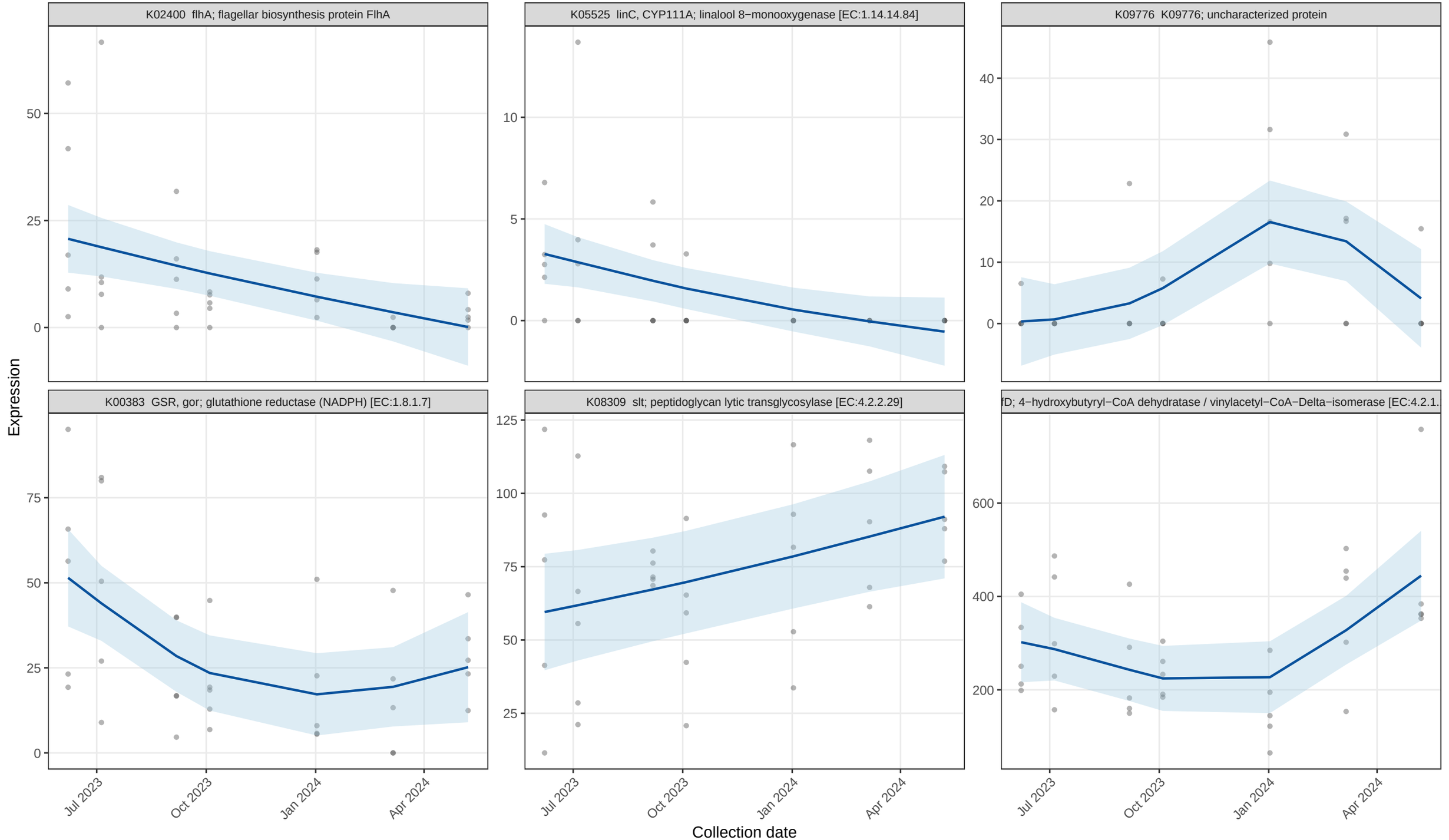

K07233 pcoB, copB; copper resistance protein B
K22025 denD; D−erythronate 2−dehydrogenase [EC:1.1.1.410]
K06445 fadE; acyl−CoA dehydrogenase [EC:1.3.99.−]
K22468 ppk2; polyphosphate kinase [EC:2.7.4.34]
K11645 fbaB; fructose−bisphosphate aldolase, class I [EC:4.1.2.13]
K05591 dbpA; ATP−dependent RNA helicase DbpA [EC:5.6.2.6]
Expression
Collection date
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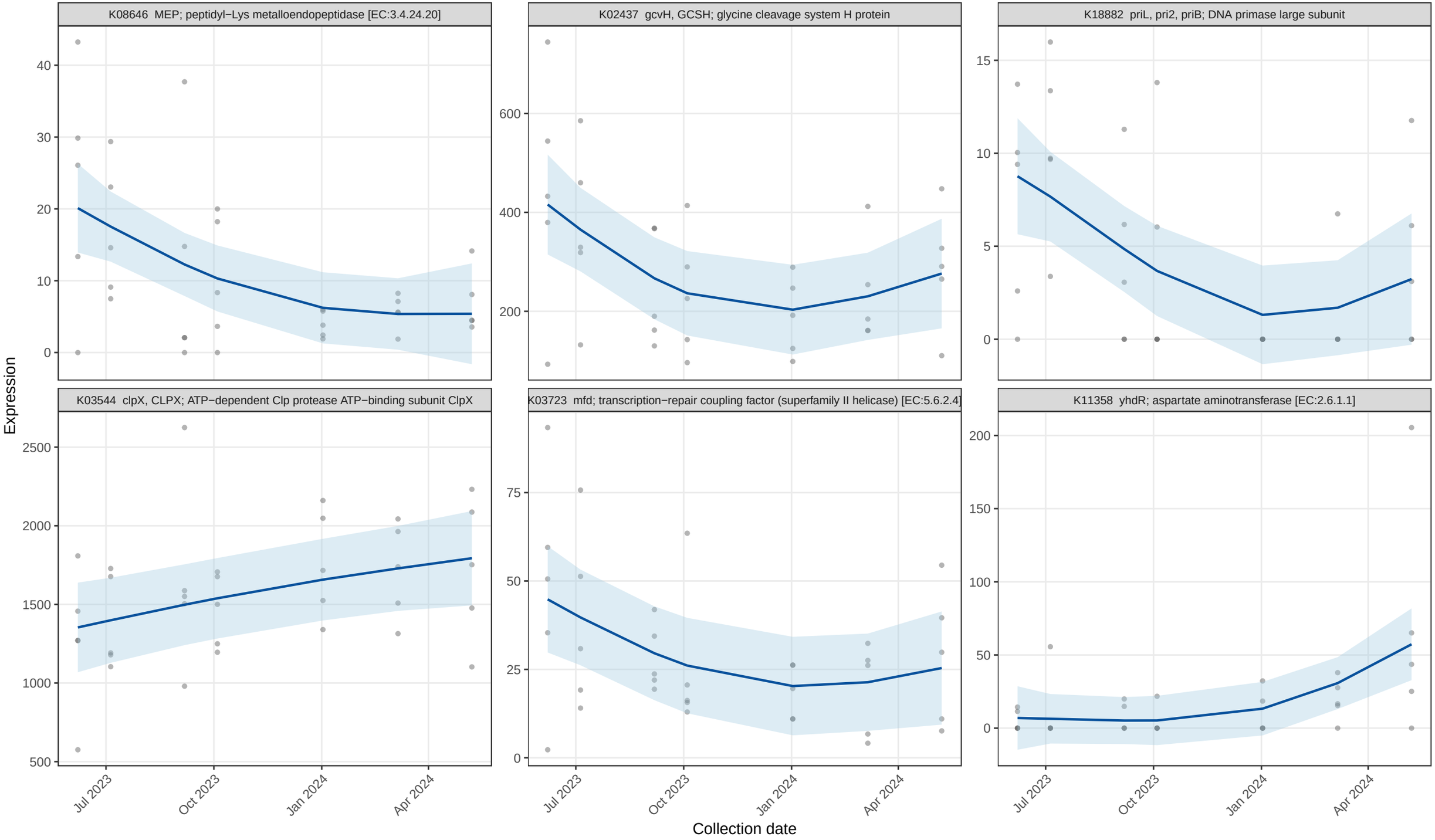

K15921 xynD; arabinoxylan arabinofuranohydrolase [EC:3.2.1.55]
K01676 E4.2.1.2A, fumA, fumB; fumarate hydratase, class I [EC:4.2.1.2]
K09959 K09959; uncharacterized protein
K14205 mprF, fmtC; phosphatidylglycerol lysyltransferase [EC:2.3.2.3]
K02440 glpF; glycerol uptake facilitator
7716 pleC; two−component system, cell cycle sensor histidine kinase PleC [EC:2.7.13
Expression
Collection date
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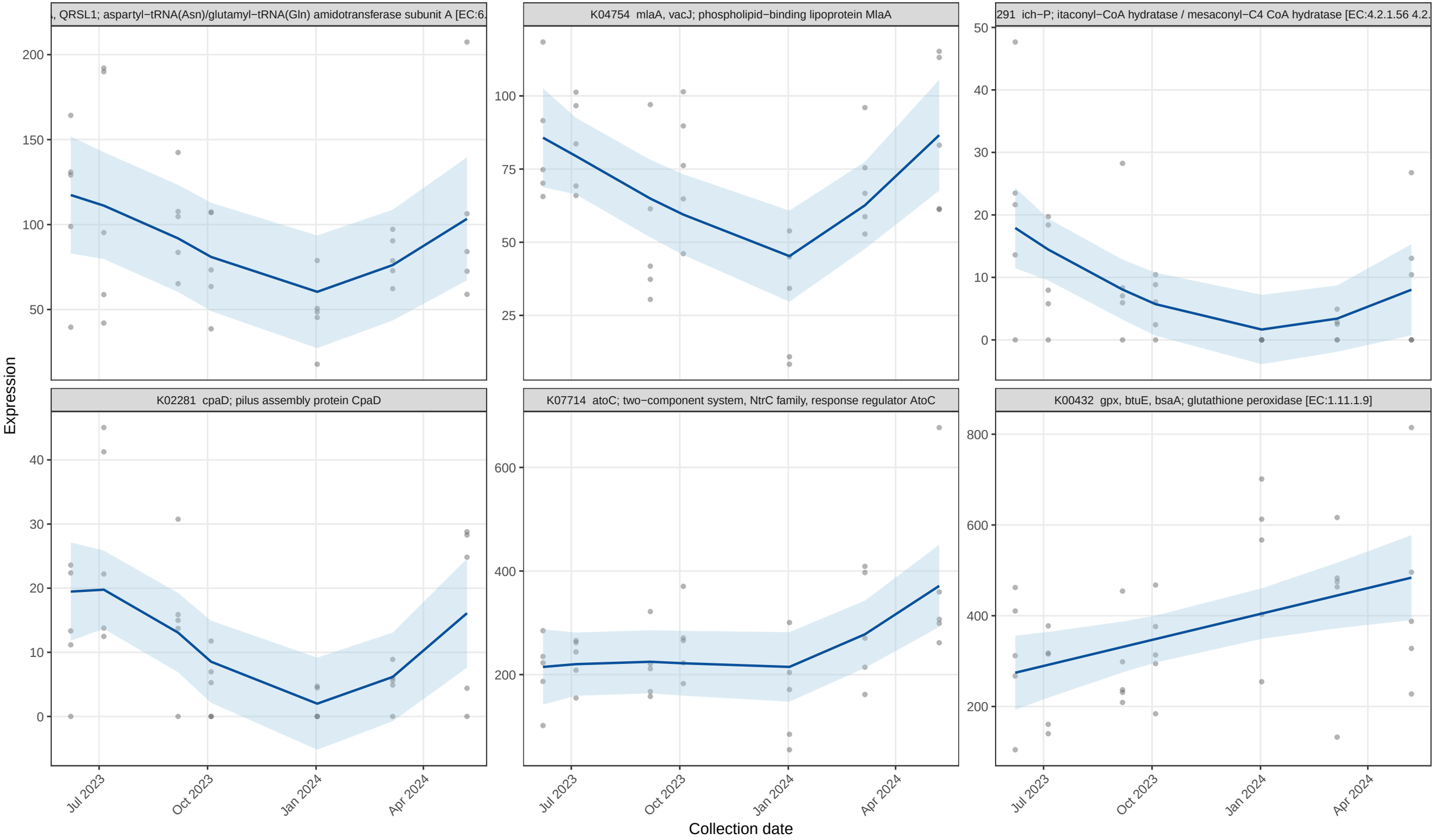

K00208 fabI; enoyl−[acyl−carrier protein] reductase I [EC:1.3.1.9 1.3.1.10]
K07290 yhjG; AsmA family protein
K07164 K07164; uncharacterized protein
K05573 ndhB; NAD(P)H−quinone oxidoreductase subunit 2 [EC:7.1.1.2]
ylate transaminase / serine−glyoxylate transaminase / serine−pyruvate transaminase [
K10010 tcyC, tcyN; L−cystine transport system ATP−binding protein [EC:7.4.2.1]
Expression
Collection date
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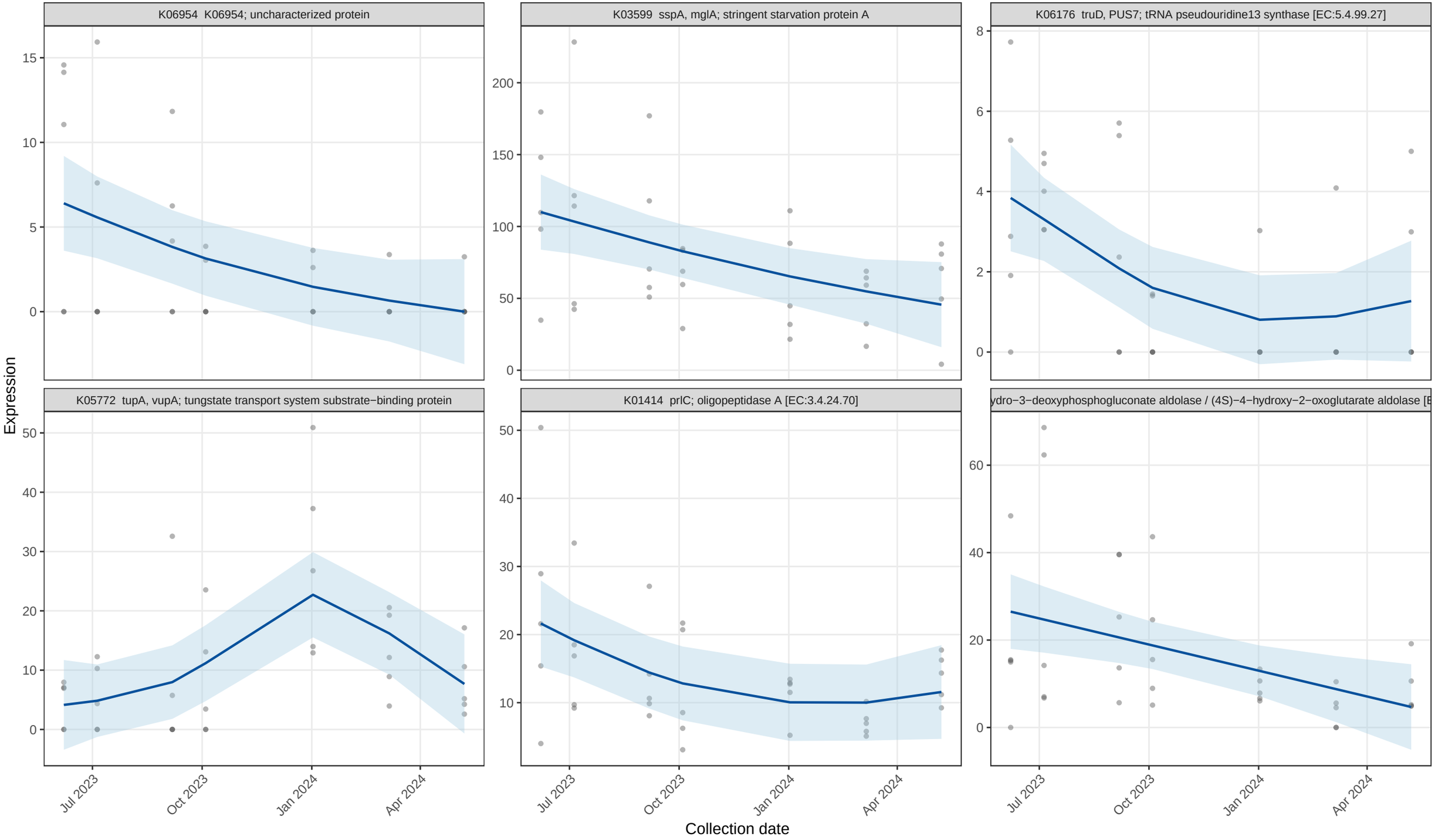

K19615 fitD, mcf; insecticidal toxin
K07161 K07161; uncharacterized protein
K10974 codB; cytosine permease
K14084 mttC; trimethylamine corrinoid protein
K08311 nudH; putative (di)nucleoside polyphosphate hydrolase [EC:3.6.1.−]
K00284 GLU, gltS; glutamate synthase (ferredoxin) [EC:1.4.7.1]
Expression
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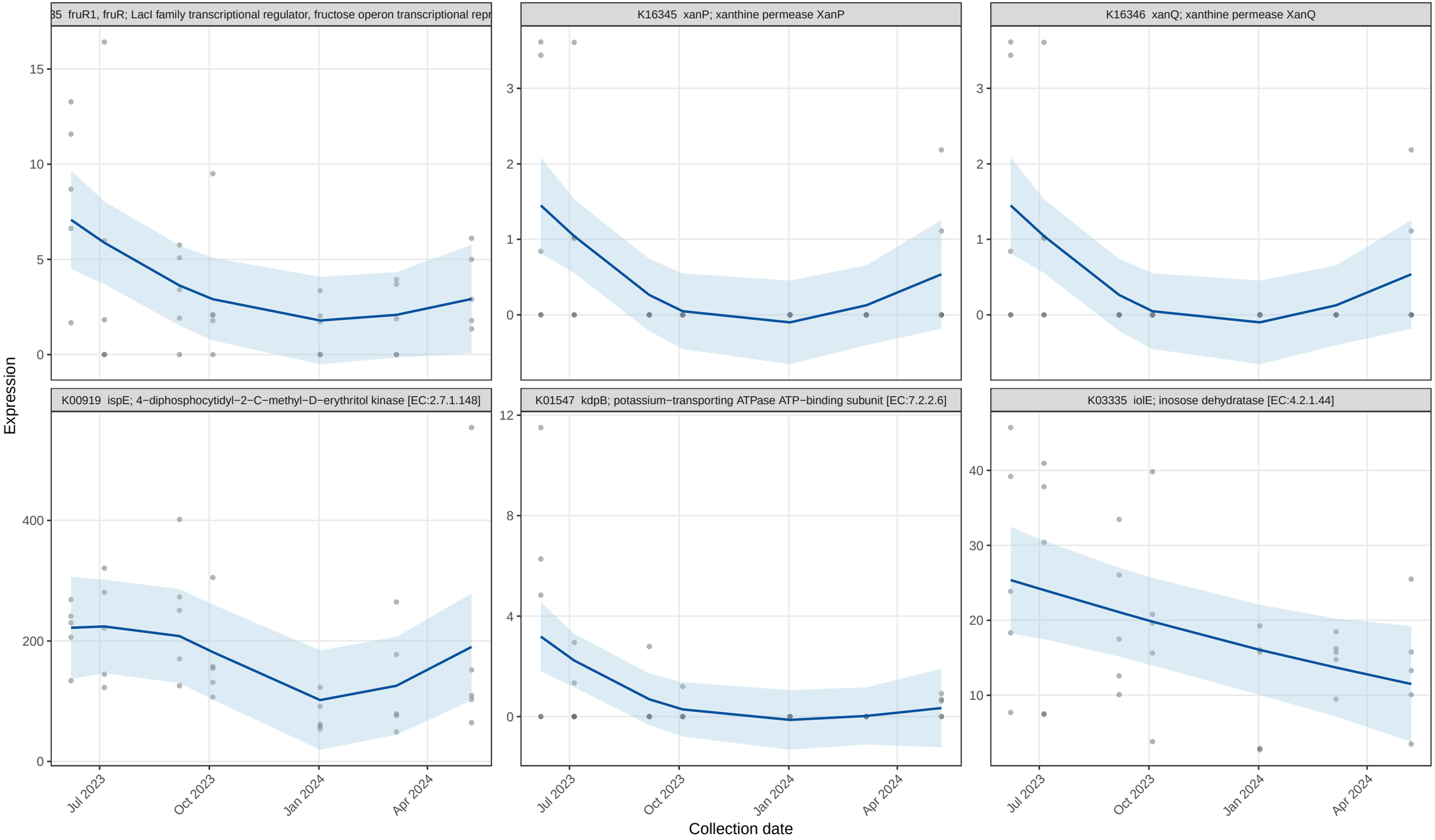

K01637 ICL, aceA; isocitrate/methylisocitrate lyase [EC:4.1.3.1 4.1.3.30]
K00124 fdoH, fdsB; formate dehydrogenase iron−sulfur subunit
K00930 argB; acetylglutamate kinase [EC:2.7.2.8]
K22292 mupP; N−acetyl−D−muramate 6−phosphate phosphatase [EC:3.1.3.105]
K17724 IRG1; aconitate decarboxylase [EC:4.1.1.6]
K02655 pilE; type IV pilus assembly protein PilE
Expression
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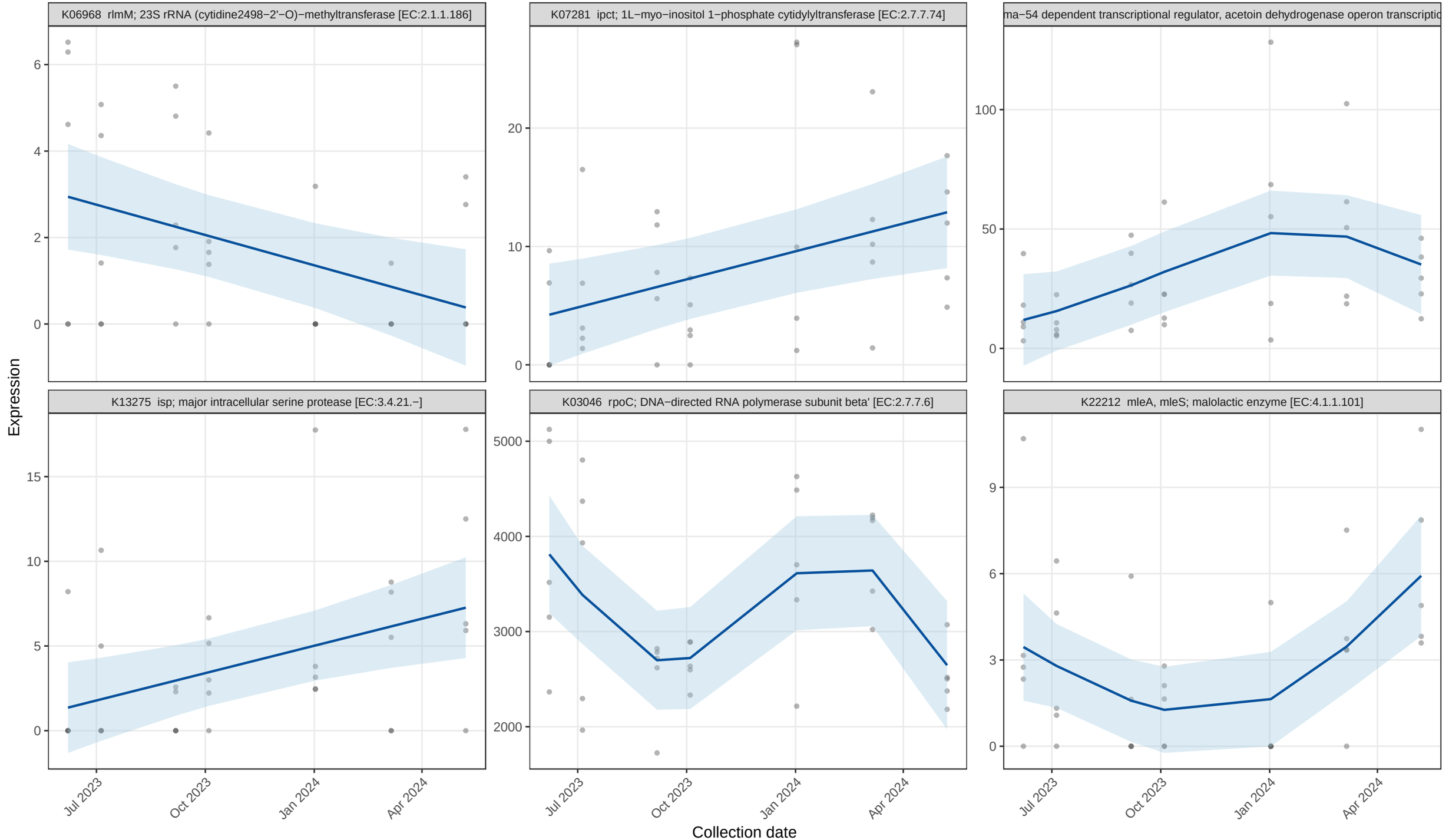

K02400 flhA; flagellar biosynthesis protein FlhA
K05525 linC, CYP111A; linalool 8−monooxygenase [EC:1.14.14.84]
K09776 K09776; uncharacterized protein
K00383 GSR, gor; glutathione reductase (NADPH) [EC:1.8.1.7]
K08309 slt; peptidoglycan lytic transglycosylase [EC:4.2.2.29]
fD; 4−hydroxybutyryl−CoA dehydratase / vinylacetyl−CoA−Delta−isomerase [EC:4.2.1.
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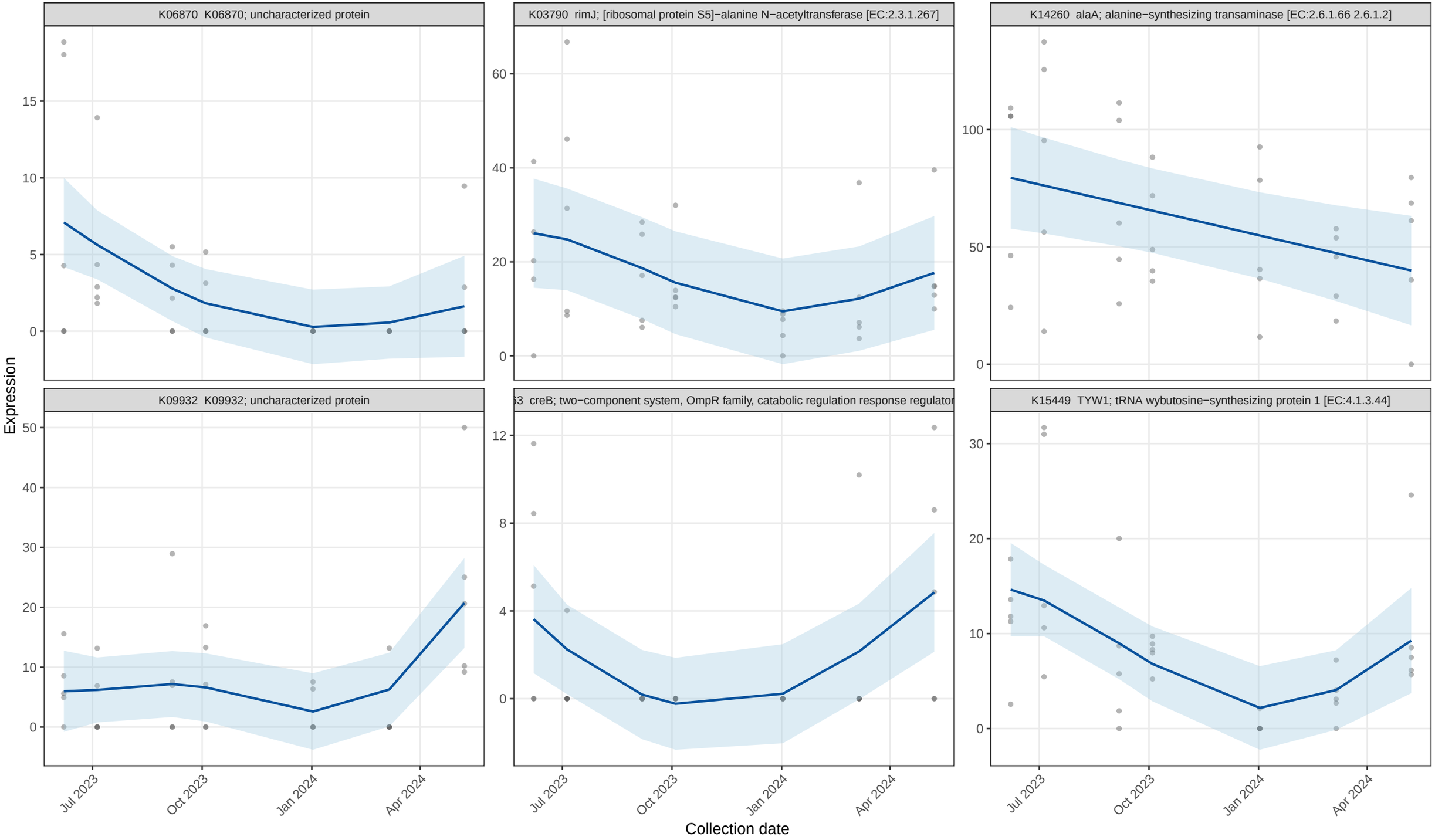

K08646 MEP; peptidyl−Lys metalloendopeptidase [EC:3.4.24.20]
K02437 gcvH, GCSH; glycine cleavage system H protein
K18882 priL, pri2, priB; DNA primase large subunit
K03544 clpX, CLPX; ATP−dependent Clp protease ATP−binding subunit ClpX
K03723 mfd; transcription−repair coupling factor (superfamily II helicase) [EC:5.6.2.4]
K11358 yhdR; aspartate aminotransferase [EC:2.6.1.1]
Expression
Collection date
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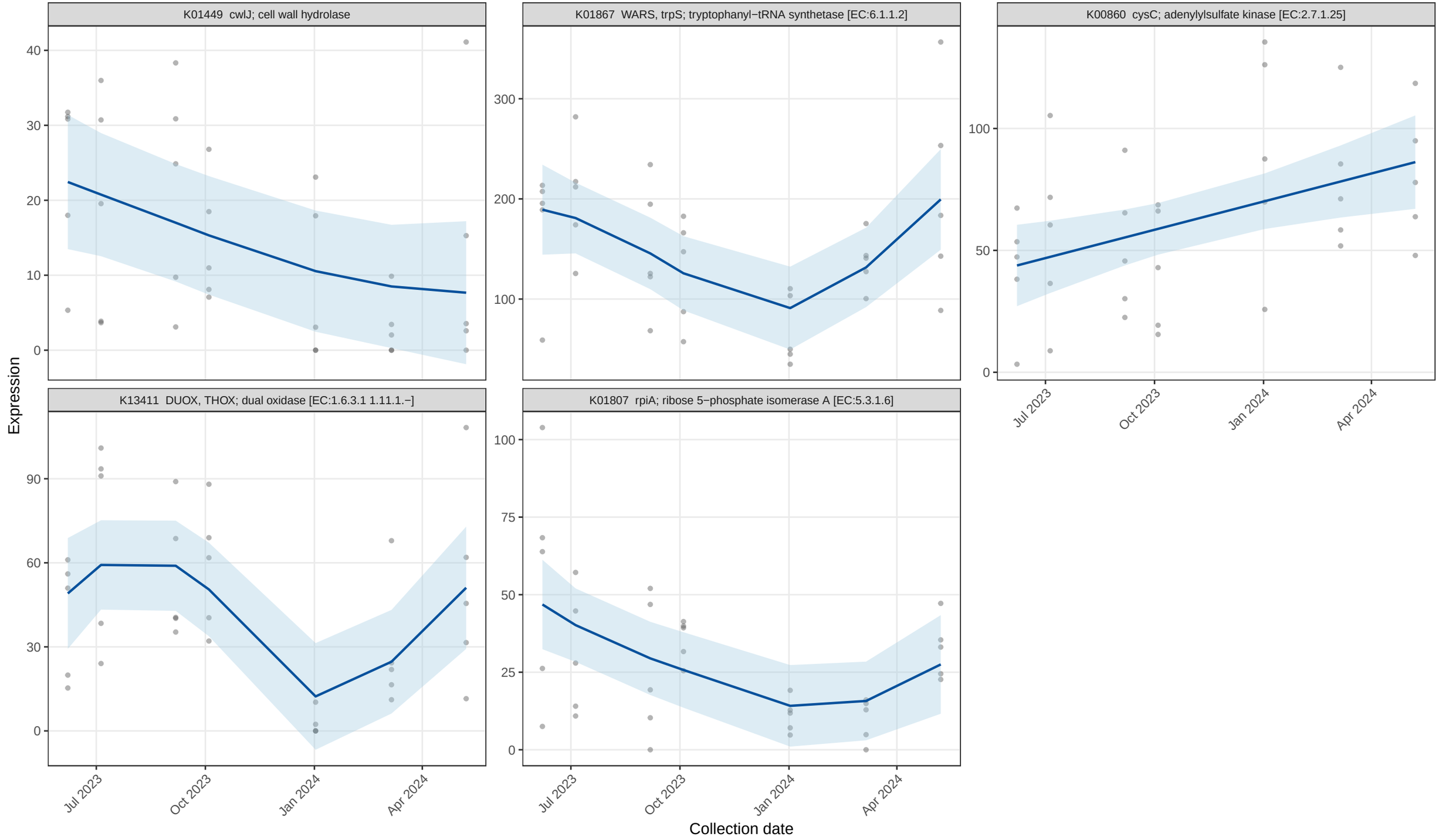

, QRSL1; aspartyl−tRNA(Asn)/glutamyl−tRNA(Gln) amidotransferase subunit A [EC:6.
K04754 mlaA, vacJ; phospholipid−binding lipoprotein MlaA
291 ich−P; itaconyl−CoA hydratase / mesaconyl−C4 CoA hydratase [EC:4.2.1.56 4.2.
K02281 cpaD; pilus assembly protein CpaD
K07714 atoC; two−component system, NtrC family, response regulator AtoC
K00432 gpx, btuE, bsaA; glutathione peroxidase [EC:1.11.1.9]
Expression
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Jan 2024
Apr 2024
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K06954 K06954; uncharacterized protein
K03599 sspA, mglA; stringent starvation protein A
K06176 truD, PUS7; tRNA pseudouridine13 synthase [EC:5.4.99.27]
K05772 tupA, vupA; tungstate transport system substrate−binding protein
K01414 prlC; oligopeptidase A [EC:3.4.24.70]
ydro−3−deoxyphosphogluconate aldolase / (4S)−4−hydroxy−2−oxoglutarate aldolase [E
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35 fruR1, fruR; LacI family transcriptional regulator, fructose operon transcriptional repr
K16345 xanP; xanthine permease XanP
K16346 xanQ; xanthine permease XanQ
K00919 ispE; 4−diphosphocytidyl−2−C−methyl−D−erythritol kinase [EC:2.7.1.148]
K01547 kdpB; potassium−transporting ATPase ATP−binding subunit [EC:7.2.2.6]
K03335 iolE; inosose dehydratase [EC:4.2.1.44]
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K06968 rlmM; 23S rRNA (cytidine2498−2'−O)−methyltransferase [EC:2.1.1.186]
K07281 ipct; 1L−myo−inositol 1−phosphate cytidylyltransferase [EC:2.7.7.74]
ma−54 dependent transcriptional regulator, acetoin dehydrogenase operon transcriptio
K13275 isp; major intracellular serine protease [EC:3.4.21.−]
K03046 rpoC; DNA−directed RNA polymerase subunit beta' [EC:2.7.7.6]
K22212 mleA, mleS; malolactic enzyme [EC:4.1.1.101]
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Jan 2024
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K06870 K06870; uncharacterized protein
K03790 rimJ; [ribosomal protein S5]−alanine N−acetyltransferase [EC:2.3.1.267]
K14260 alaA; alanine−synthesizing transaminase [EC:2.6.1.66 2.6.1.2]
K09932 K09932; uncharacterized protein
creB; two−component system, OmpR family, catabolic regulation response regulator
K15449 TYW1; tRNA wybutosine−synthesizing protein 1 [EC:4.1.3.44]
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K01449 cwlJ; cell wall hydrolase
K01867 WARS, trpS; tryptophanyl−tRNA synthetase [EC:6.1.1.2]
K00860 cysC; adenylylsulfate kinase [EC:2.7.1.25]
K13411 DUOX, THOX; dual oxidase [EC:1.6.3.1 1.11.1.−]
K01807 rpiA; ribose 5−phosphate isomerase A [EC:5.3.1.6]
Expression
Collection date
Jul 2023
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